Mayhem Firmware Manual

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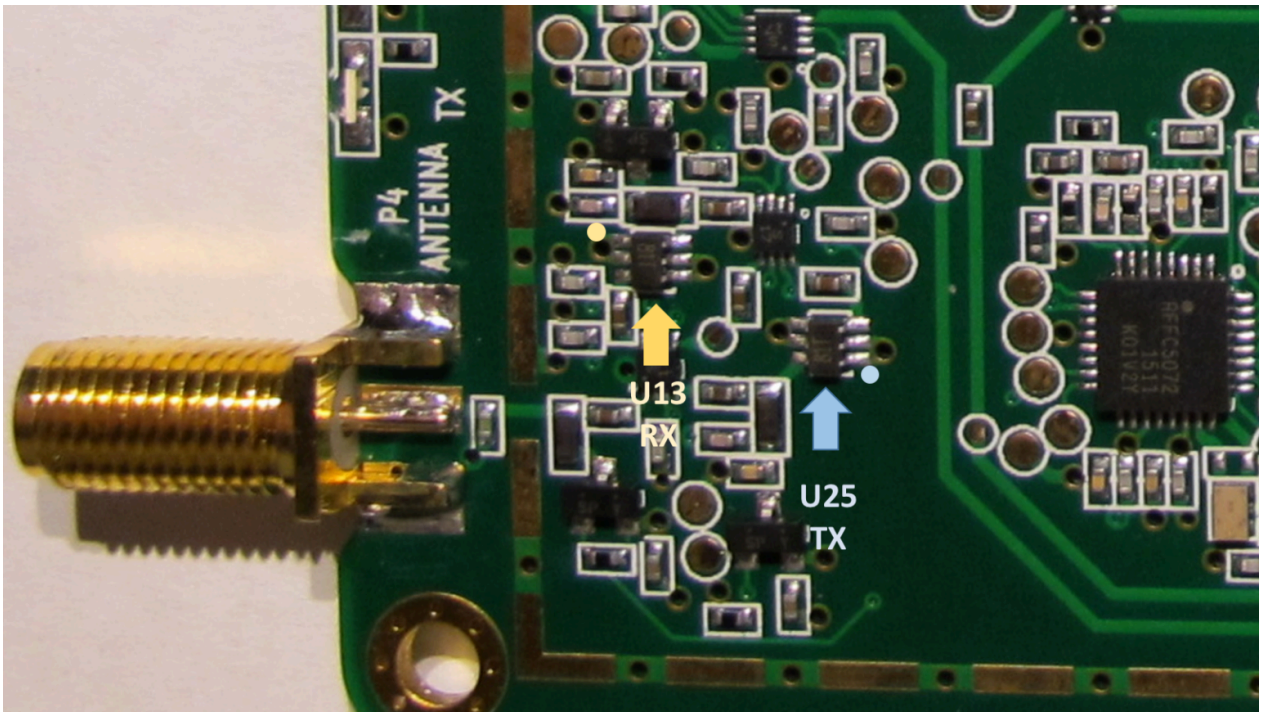
1 Repairs

1.1 Replace AMP

The MGA-81563 can be damaged easily, but you can buy (relatively cheap: https://s.click.aliexpress.com/e/_Dl4U7Yl) and replace them yourself. For more information about this procedure, please check https://www.onesdr.com/2020/04/09/how-to-fix-a-broken-hackrf/

The previous link has a very good reference picture:



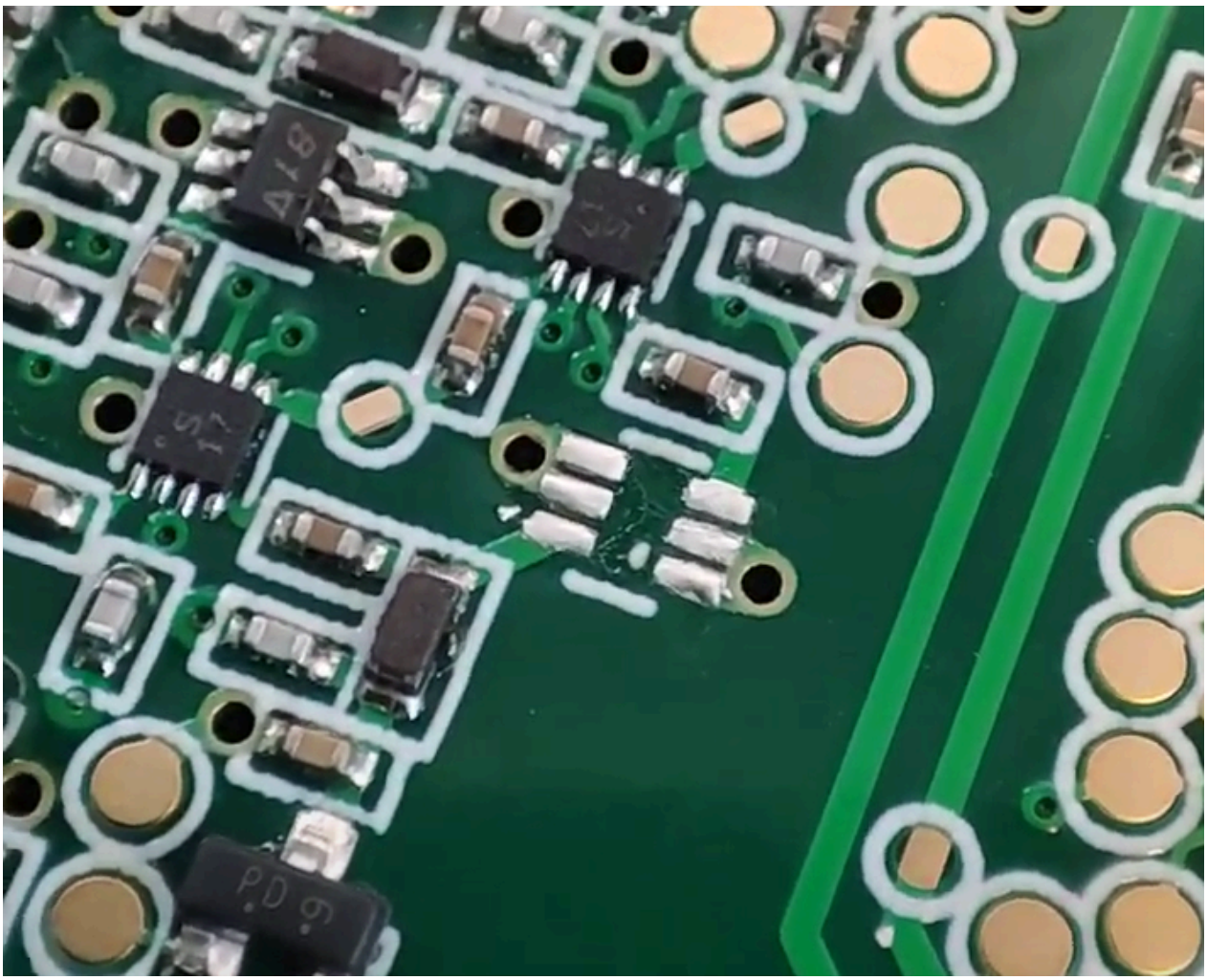
2 RX problems

If you find on a strong signal such as an broadcast FM station that you select Amp as “0” and you get a signal and then select Amp “1” and you do not then this is a good sign that the RF Preamplifier u13 (MGA-81563) is damaged. This can happen if you use a stronger transmitter near to your HackRF.

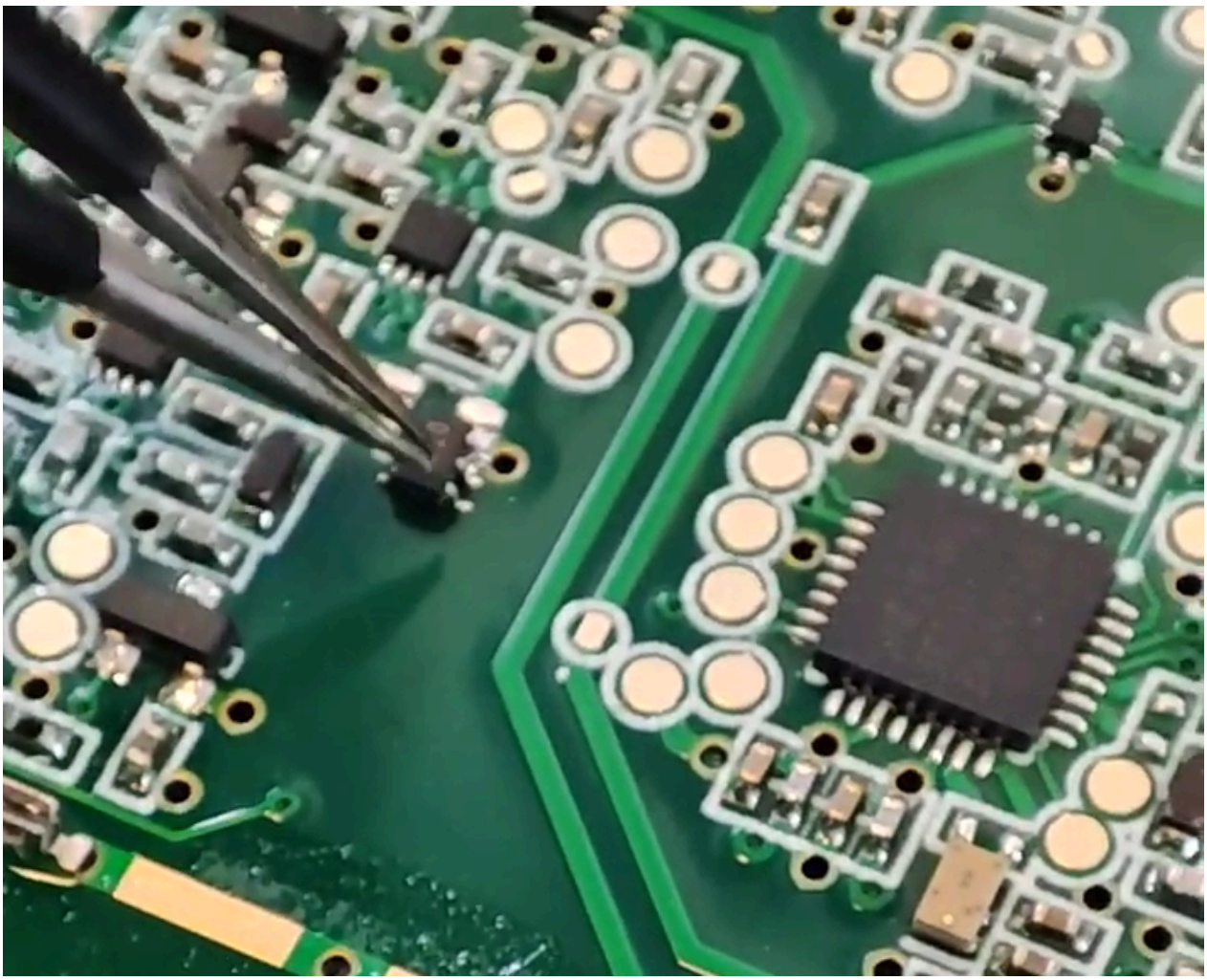
It is possible to replace the damaged IC with simple tools (soldering iron, solder wick). First solder flood the damaged component to remove it, preferably protecting the surrounding. Clean the pads and replace the component:

3 TX problems

The same component is used for TX, which also can be damaged. Below is an example of the replacement procedure. First remove and clean the pads. This can be done with hot air or normal soldering iron:



For the orientation, you could use the text of the chip itself. It should be “up”, pointing to the antenna jack of the HackRF:



3.1 SMD buttons and caps

4 Push-buttons

Current PortaPack models have five push buttons on the front (UP, DOWN, LEFT, RIGHT, and SELECT) and two on the top (DFU and RESET). In most cases, a long press of a directional button will repeat automatically after a short delay (firmware may define additional behavior for a long press of any button).

If the push buttons "stick", check to make sure that the buttons are centered in the holes in the case, and are not rubbing against the inside of the holes.

If the push-buttons are intermittently unresponsive, first make sure that the Portapack firmware is upgraded to version 1.7.1. or later, which improves handling of glitchy switch contacts. The type of push-buttons used on the Portapack is SMD tactile switches of 6 mm per side and 7 mm of height. You can buy replacements here: https://a.aliexpress.com/_dZqx4vv

The easiest way to remove them is to clip the legs off, clean the pads carefully with solder wick and solder the new buttons.

You can also find replacement caps for the tactile switches: https://www.aliexpress.com/item/1005001700437278.html

Alternative caps for the switches: https://www.ebay.com/itm/284831256446

4.1 Encoder knob

NOTE This section is mostly talking about H2

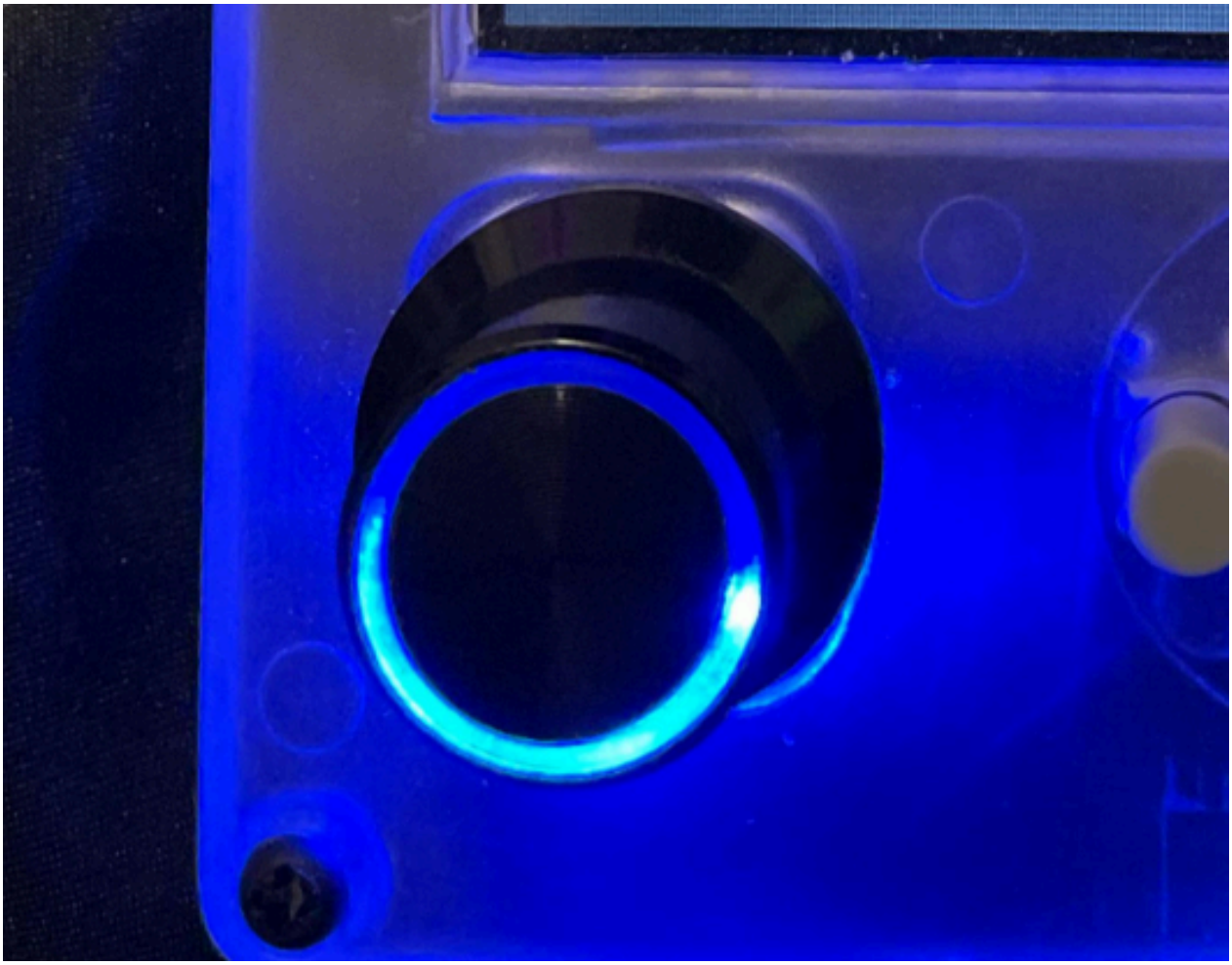
Encoder for H2/H2+ uses a 30-step EC11 encoder with push button and shaft length of 12.5mm.
(For H4M click wheel, try ANO Directional Navigation and Scroll Wheel Rotary Encoder)

A short press of the push button turns on the PortaPack; a long press or two short presses turns off the PortaPack. If no click is felt when the knob is pressed, there may be insufficient clearance between the knob and the case, in which case some padding may be added inside the knob shaft.

A malfunctioning rotary encoder may sometimes work better if the knob is pressed slightly to the left when rotating. Make sure that the Portapack is running firmware version 1.7.1. or later, which improves handling of glitchy switch contacts. Encoder dial sensitivity can also be modified in Settings.

4.2 Plastic knob

Depending on the shaft of your encoder, you can use diverse knobs:



4.3 Encoder Replacement on PortaPack H2

4.3.1 Disassemble the device

1. Unscrew the screws and SMA connector rings.
2. Pull apart the case and carefully separate the PortaPack from the HackRF by gently pulling upwards to disconnect the pins.
3. Disconnect the speaker and battery.

4.3.2 Removing the Encoder

The encoder is mounted in a way that makes it difficult to desolder due to pin placement. If you are having trouble with this, try the following technique:

1. Use a pair of tin snips to cut the encoder's leads and remove the encoder.
2. Apply flux to the solder pads.
3. Carefully heat each pad with a soldering iron, one by one, to remove the clipped leads. This approach reduces strain on the pads and lowers the risk of tearing up the traces.

WARNING The upper-most pin is close to the screen, take care not to scorch it with your soldering iron.

4.3.3 Installing the Replacement Encoder

The replacement rotary encoder will likely have long leads designed for through-hole mounting. However, the PortaPack uses surface mounting for this component, so the leads will need to be trimmed down. Here's how to do it:

NOTE Correct lead length is important to avoid clearance issues when reassembling the case.

1. **Trim the Leads:** Shorten the encoder leads so that they are approximately flush with the body of the encoder. This will ensure that the encoder sits stably and doesn't increase the height of the assembly.
2. After trimming, solder the encoder on the PCB, taking care to align it correctly before soldering.
3. After soldering each pin, do a visual inspection to ensure there is good electrical contact.

4.3.4 Reassembly

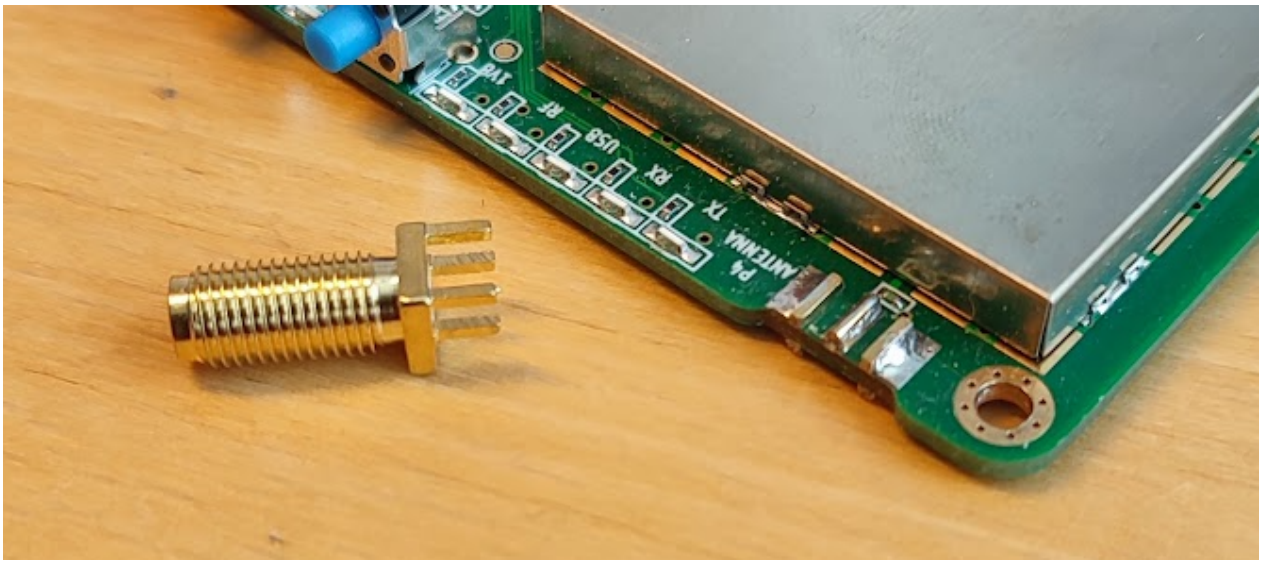
Once the new encoder is in place, follow the disassembly steps in reverse to reassemble your PortaPack.

4.4 SMA Connector replacement or removal

NOTE It is a good idea to protect the antenna connectors with a silicone cover

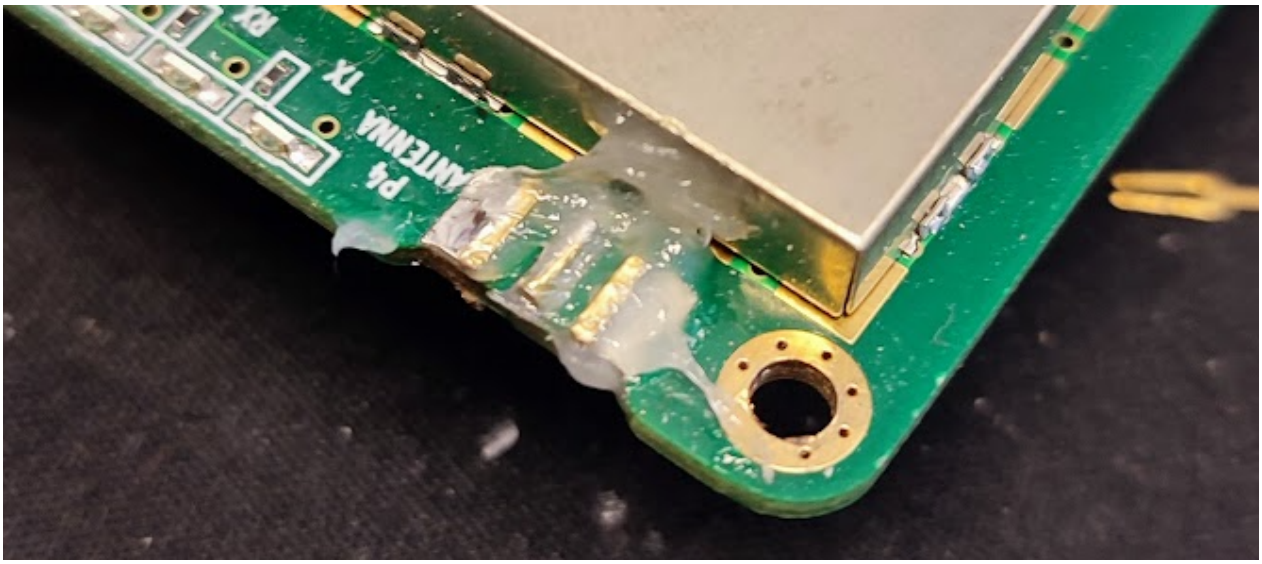
The antenna connectors on the HackRF board are the `18.5 mm` variant available here.

Notice that you *could* fit the `SMA-KE 18.5mm` variant, but it will be not the perfect fit for the type of pcb mount used in originally.

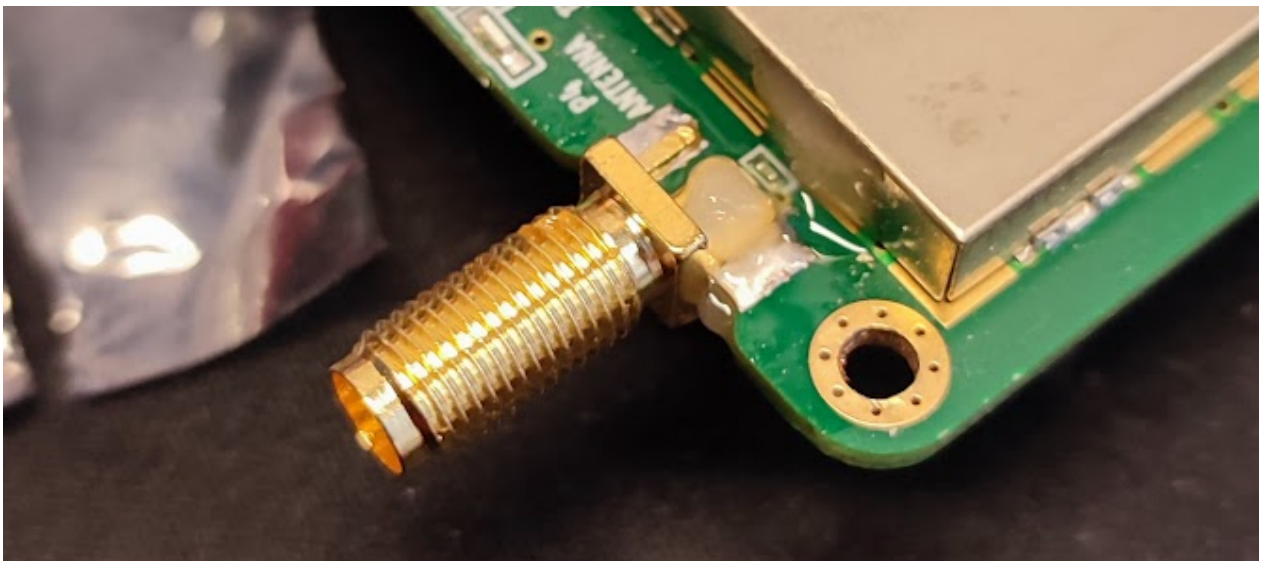


5 Replacing a broken connector

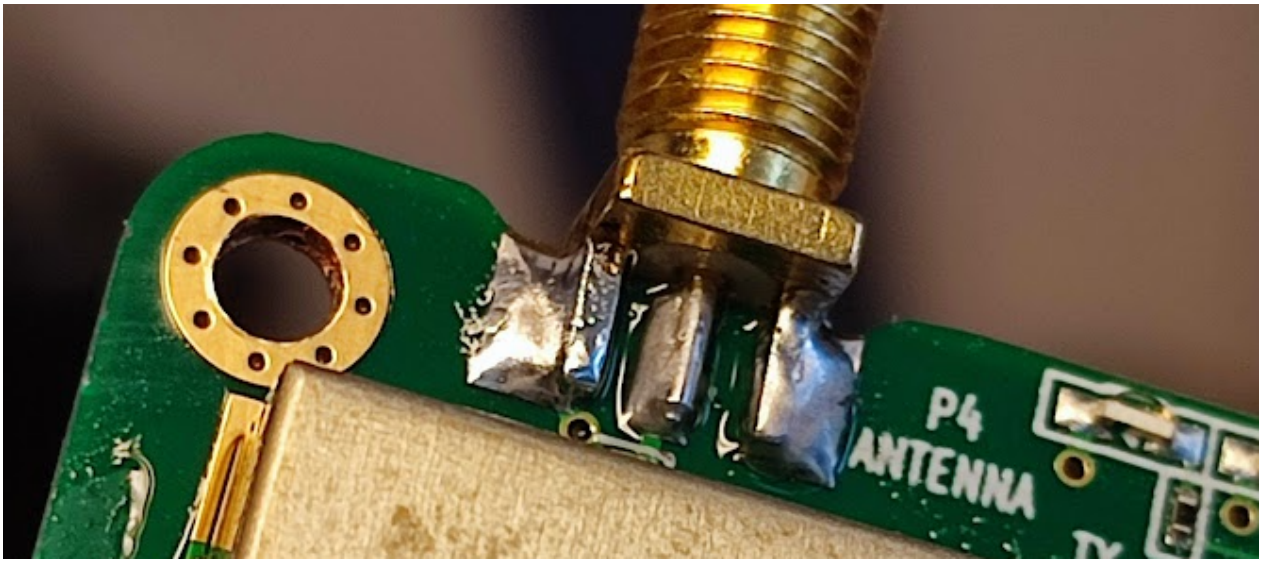
1. Desolder the original, use enough flux (i.e. this "self cleaning" variant) and fresh solder. If there is parts of the SMA connector still there, it might be more difficult to remove because the thermal mass.



2. Clean the pads
3. Place the new one. It will be very tight. Align it correctly



4. Solder and clean the residual flux



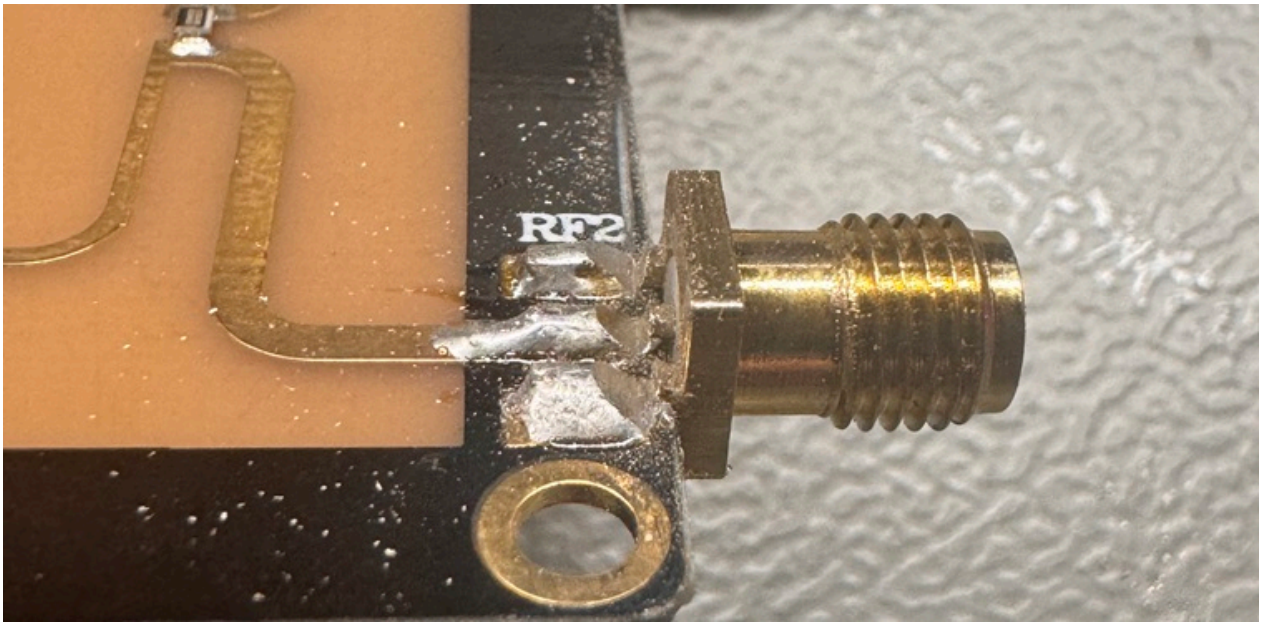
6 Removing a connector

You may want to remove the bottom sma connectors if you dont use them, or you may have a damaged connector that you need to remove before replacing it. This is a simple way to do that without risk of damaging your PCB.

If you only want to remove the connector, all you need is a small file, preferably one that has a knife-like edge, it will make it easier.

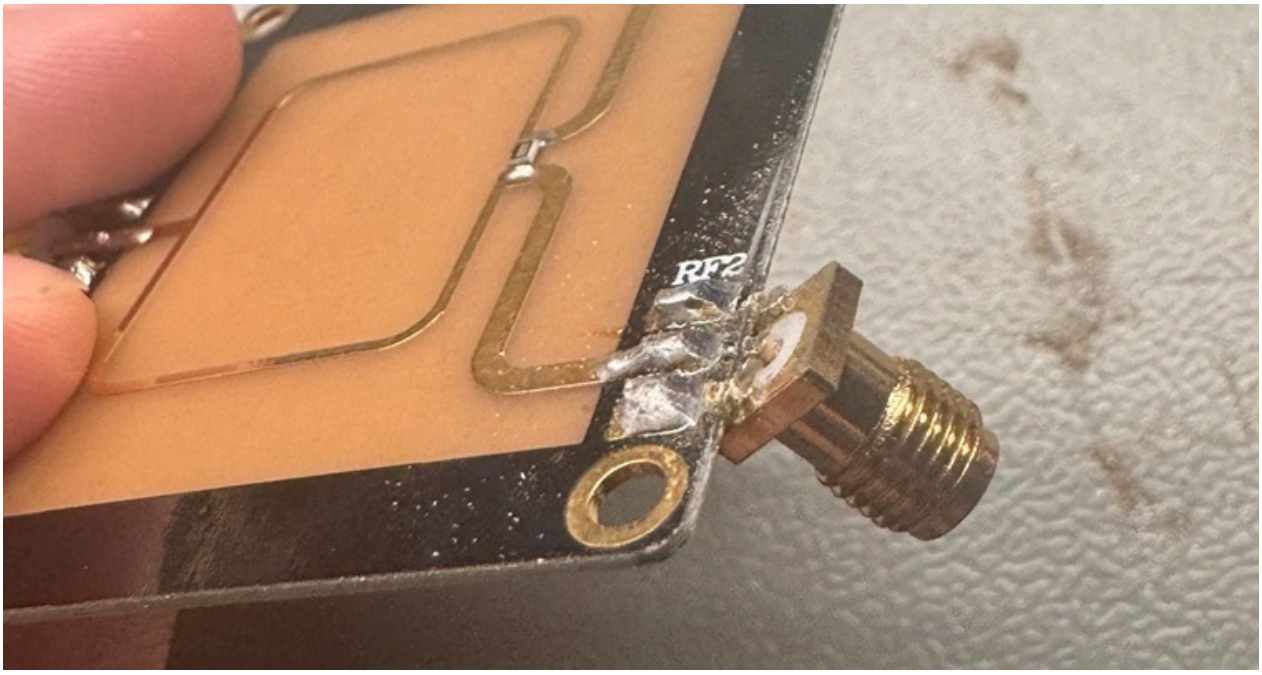
Note that im using a random pcb i had lying around for this demonstration.

1. Use your file to file down the pins holding the connector almost all the way down.

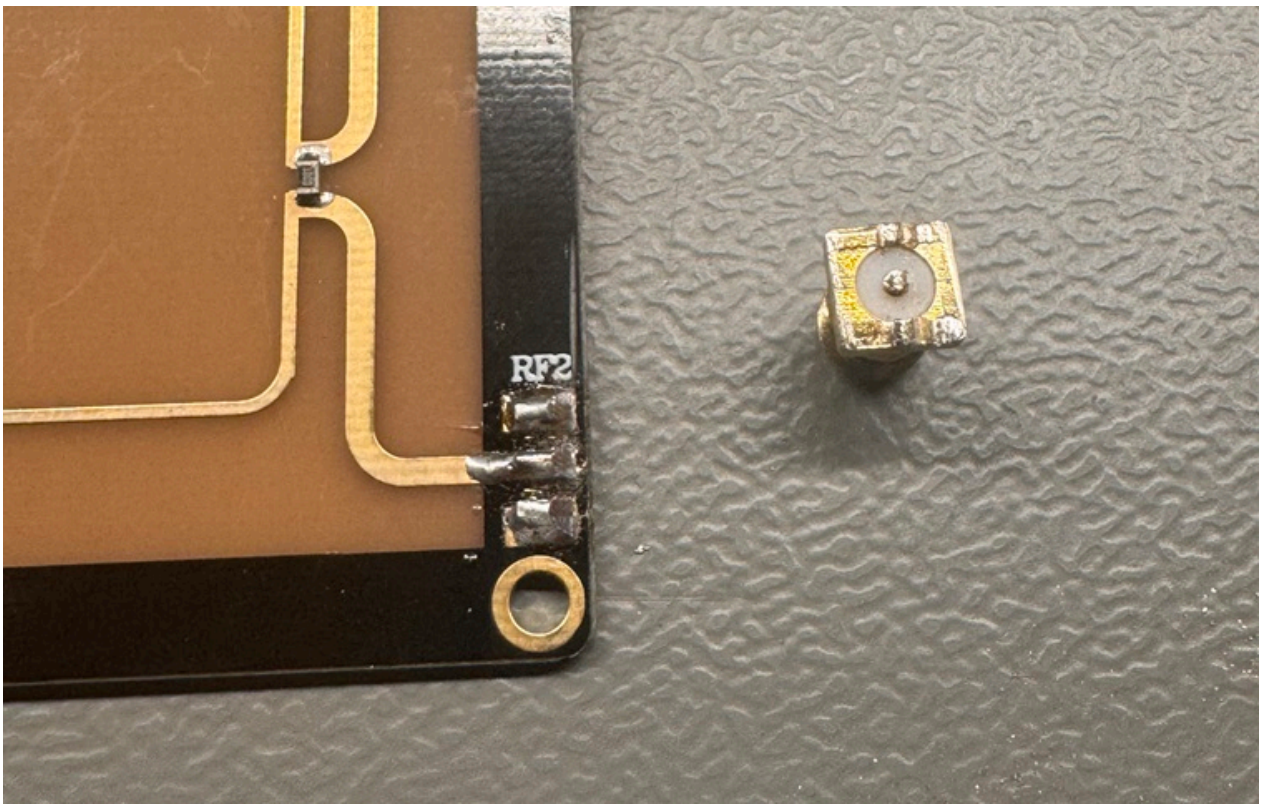


2. Now flip the board over and do the same on the other side.

3. At this point you want to rock the plug up and down lightly, it should break off. You may have to file the pins a little bit more if it doesnt break off with very light force.



4. Once the connector is removed, you can de-solder the remaining pins one by one, or just clean up the dust and leave it as it is, if you are just removing the CLCK IN/OUT connectors.



7 Manual

7.1 Intended use and Legality

WARNING This is not intended to be legal advice and should not be construed as authoritative, accurate or complete for your or any jurisdiction. You should seek the advice of a local legal professional to clarify the legality of your usage of HackRF One, Portapack and Mayhem for your jurisdiction.

8 Background

8.1 HackRF One

The HackRF One was originally conceived of and created by Security Researcher and Hardware Hacker Michael Ossmann. Previously Ossmann had created other devices primarily for security research in the area of RF. These devices include the YARD Stick One, a general purpose Software Designed Transceiver capable of transmitting and receiving signals below 1GHz, and the UberTooth One a device made specifically for Bluetooth Penetration Testing and Security Research.

The HackRF One can be seen as a logical progression from these devices, extending both functionality and frequency coverage.

8.2 Portapack

The Portapack was originally designed and created by Security Researcher and Hardware Hacker Jared Boone as a way to provide portable operating capabilities for the HackRF One.

9 Open Source Hardware and Software

All of the original source code, documentation and design files for HackRF One are copyright Michael Ossmann and licensed under the GNU GPL 2.0.

All of the original source code, documentation and design files for Portapack H1 are copyright Jared Boone and licensed under the GNU GPL 2.0.

The implications of the above are that anyone can take any of these source files and add, modify, extend and release their work, providing they make available on request, any additions, modifications and extensions they have made and license this under the GPL 2.0.

There is no restriction on using any of this for commercial gain. That being said, there is also no implication or warranty of fitness made by either the original authors or any subsequent contributors or vendors. Do your research and Caveat Emptor.

10 Intended Use

The HackRF One was originally designed as a tool for RF analysis and research, particularly in the domain of Information Security.

The Portapack was originally designed to facilitate the portable use of HackRF One for RF analysis and research, particularly in the domain of Information Security.

However, the HackRF One is fundamentally a general purpose Software Defined Radio. As such, it can be put to any use that a radio with similar characteristics can be. The Portapack is fundamentally a control surface for the HackRF One, and as such can be used for any general control of the SDR.

11 Unauthorised Access

Since much of the software and use-cases for the HackRF One and Portapack centres around Information Security Research, there is plenty of opportunity for getting into legal hot-water. The area of Security Research is as yet a legal grey-area, with many of the laws governing it being quite antiquated, and often not relevant to the technology of today. As such, it is advised that you only ever transmit where explicitly permitted, and when doing Security Research, that you only access or act upon devices you own or are explicitly permitted to access or act upon.

12 Transmitting

In general, it should be assumed that transmitting radio signals is not legal. Transmitting radio signals with intent, or lack of duty-of-care to cause interference, disruption or min-information even more so.

That being said, there are transmissions that are not illegal. Transmissions in the ISM bands are license-free in most jurisdictions, once certain power levels are adhered to. This is bearing the section on causing interference, above, in mind.

Amateur Radio Operators, depending on jurisdiction, license category, adherence to terms-of-license, local laws etc. have access to far more spectrum, and far higher power levels. Again, without causing interference through intent or lack of care.

13 Receiving

In general, in most jurisdictions, receiving radio signals is not illegal. Your specific case should be verified with a local legal professional.

13.1 Usage cautions

⚠️ The LNA/PA of HackRF is sensitive and can get damaged! ⚠️

14 Prevention

14.1 SMA port

- DO NOT swap the antenna while power is on.
- If you use an antenna that has an exposed metal part, DO NOT touch the antenna with you hands/body (Static electricity can damage the LNA, or it is possible that the spec change of it, like bouncing, while you touching it, could lead issues).
- DO NOT transmit without an antenna.

14.2 Monitoring range

- You shouldn't receive near high power transmitters, even if it's not in the range that you are monitoring.

15 Diagnostic

- Try to listen to local radio. Try it with AMP both ON and OFF.
- Try to capture and replay a non-rolling-code. Try it with AMP both on and off.

Check more details about tx and rx issues.

16 Repair

How to replace the broken component

16.1 First steps

17 Identify your device

There are many different versions of the PortaPack. It is important that you identify the correct version as this can affect how you install the firmware and device initial start-up. It is recommended you read the description of versions here. The H2 and H2+ versions are a slightly improved version, but the people making these copies never published the circuit diagrams or Code, hence breaking the license terms.

The main difference are the controls and the screen size, along with the codec and processor chip. The H2 or H2+ has extra circuitry to manage the battery charging and powering on/off. In this document, there is a brief description about the details. Check a more technical comparison here.

18 Up to date information

There is a Discord server "HackRF PortaPack Mayhem". This has all the current talk and information.

19 Power on/off

The original H1 powers instantly when you plug a power supply into the USB port. To turn it off, just unplug it. Similar to the issues with some USB cables while upgrading the firmware, the quality of your cable might affect the performance.

To power on/off the H2, you need to hold the middle button (knob or pushbutton) for few seconds. See more details here.

For most H2+ models, click the knob to power on, double-click the knob to power off.

20 Extra functionality (H2 and H2+)

20.1 Charging

This version can charge the internal lipo battery via the USB. There is a led indicator that turns off when the charging is done, but it might flicker. On some models (H2+) there are 4 leds below the knob that represent the state of the battery charge: 25%, 50%, 75%, 100%. When charging, one will flash dependent on the current charge state of the battery. See more details here.

20.2 Charging indications on screen (H4M)

This version can charge the internal lipo battery via the USB. There is an on screen indicator of the level of charge. The icons can be a level on a battery, or a percentage.

20.3 Battery life

An internal battery between 1000 and 2500 mAh should last for several hours of use, depending on use of apps. The standby consumption is very low, around 52 µA, so you do not need to worry about having to remove/disconnect the battery in normal circumstances.

20.4 About firmwares

If you bought a standalone HackRF, it probably came with the GSG firmware flashed onto it. This enables usage over USB from a computer.

When buying pre-assembled HackRF + PortaPack bundles, they typically come with some version of the Mayhem firmware. One important thing to understand is that the firmware is always flashed onto the HackRF board; the PortaPack has no flash nor CPU. It is basically an interface module that enables standalone usage without a computer. It provides an LCD screen, buttons, an audio codec IC, an SD card slot, a coin cell battery to preserve settings and time between uses, and in some versions, a high-precision clock signal and a battery.

Although the Mayhem firmware allows you to directly use many functions in the field, standalone without a computer, it also provides a "HackRF mode," which enables the user to start a version of the original GSG firmware and use your HackRF via USB, controlled by a computer. However, if you separate the two boards, you won't be able to use the menu GUI and enable "HackRF mode." So, if you want to use your HackRF board alone (detached), you'll need to flash it with the GSG firmware.

In case you bought a PortaPack separately or want to upgrade your firmware, check out our Update Firmware page! Update firmware

20.5 Redirections on this manual/project

We are using Grabify to get statistics from our links. This may include affiliate links or involve multiple redirections.

Some ad-blocks and privacy lists can report/block these links. Please consider adding our links to your whitelist and sending a request to companies blocking this service, as it does not make sense—any service could be used to monitor clicks.

20.6 Firmware update procedure

20.7 Common workflow

> Note: The TLDR guide is for experts. Please try to read the full guide before asking in the community, if you are having trouble understanding the TLDR guide.

1. Update your SD card content
2. Connect USB cable
3. Use **ONE OF** the following methods:
   - (Method1) Enter HackRF mode, flash with hackrf_spiflash
   - (Method2) Stay in Mayhem mode, use hackrf.app website
   - (Method3) Copy firmware all in one tar package into SD card, and offline update with Flash Utility app in portapack
     - (Method3-1) Use the filename that ends with `.bin` => only update flash
     - (Method3-2) Use the filename that ends with `.ppfw.tar` => update both flash and external SD card apps
     - The `.bin` is useful when you need to make a first flash or a recovery flash because you lost mayhem. If possible, you always want to update using the `.ppfw.tar`

```
citations:
- method 2 and 3 would automatically update external apps too;
- you don't have to update the entire sdcard content everytime, however it is recommended
to do so, since every update mostly have new sdcard content;
- YOU WILL NEED A WORKING DEVICE WITH MAYHEM FIRMWARE TO USE METHOD 2 and 3.
```

20.7.1 Detailed video guide

If you prefer watching a video guide, check this.

21 Detailed Firmware Update procedures

21.1 Using firmware file from github release page and the Flash Utility

- Get the latest firmware from the GitHub release (latest by date) page
- Copy it to your SDCARD, put it back in your PortaPack
- Flash it via the Flash Utility in "Utilities" menu on your PortaPack

Note

- If you don't see the Flash Utility app, try using other methods to update to latest version
- The Flash Utility can also be used to program new firmware from a bin file stored on a MicroSD card

21.2 Using firmware file from github release page and the classic hackrf_spiflash tool

Note ⚠️ Using that method, you NEED to update the SDCARD content accordingly to the release you choose, else the external apps will not work !

21.2.1 Windows

1. Connect the device via USB.
2. Switch to HackRF mode via the on-screen option. (In the PortaPack)
3. Double click `mayhem_flasher.bat` and follow the instructions.
4. Reboot the device.

21.2.2 Linux

1. Connect the device via USB
2. Switch to HackRF mode via the on-screen option. (In the PortaPack)
3. Upload the firmware with `hackrf_spiflash -w new_firmware_file.bin` (eg. portapack-h1_h2-mayhem.bin for mayhem firmware or hackrf_one_usb.bin for stock HackRF firmware)
4. Reboot the device

NOTE

- To Ubuntu and Mint user: Ubuntu based distro never maintains their repo for HackRF host (cli client) package. You'll face a lot of weird problems if your HackRF is R9. To resolve these, please compile the HackRF package yourself, or use another distro.

21.2.3 MacOS

1. If necessary, install the HackRF tools: `brew install hackrf`
2. Connect the device via USB
3. Switch to HackRF mode via the on-screen option (in the PortaPack)
4. Upload the firmware with `hackrf_spiflash -w new_firmware_file.bin`
5. Reboot the device

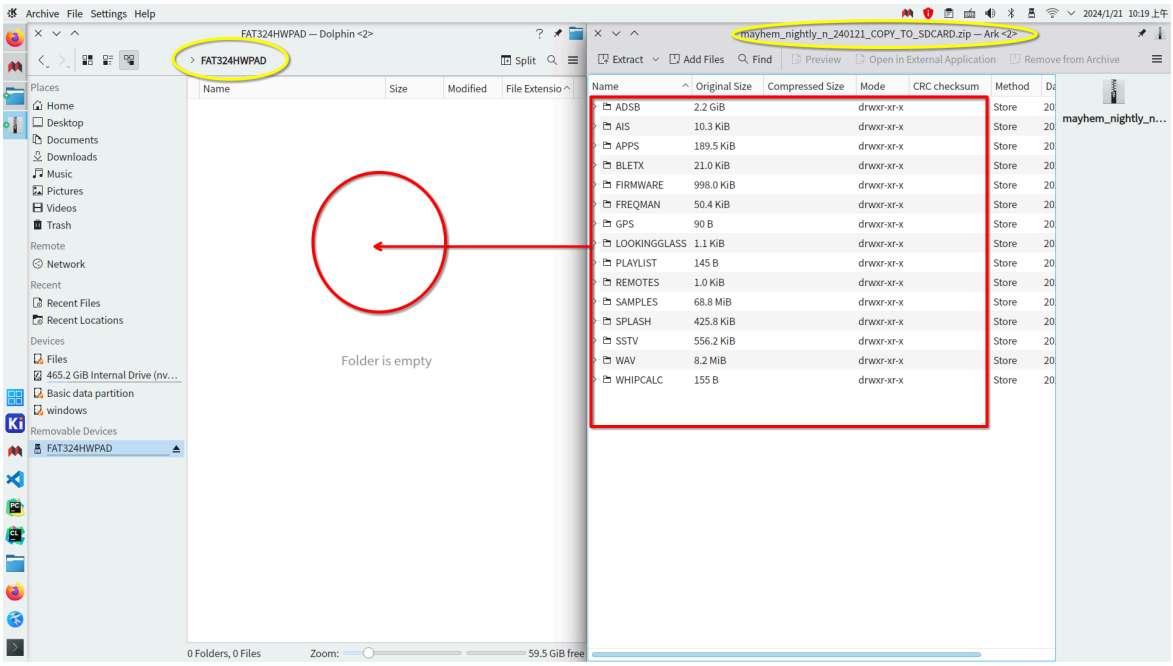
21.2.4 Update SDCARD files

You need to provide a MicroSD with enough space, 16g is good enough unless you have a bunch of record file. This is necessary for certain functionality, like the world map, GPS simulator, external apps, and others.

Get the latest files from the GitHub release (latest by date) page. You need to uncompress (using 7-zip) the files from `mayhem_vX.Y.Z_COPY_TO_SDCARD.7z` to a FAT32 formatted MicroSD card.

NOTE How to put into SD card?
Extract the root directory of the 7z archive into the root directory of your SD card.

- Don't put the 7z archive file directly into your SD card.
- Don't extract into any sub-folder of SD card.
- Don't use unbranded SD card.



21.2.5 Details about the SD card contents

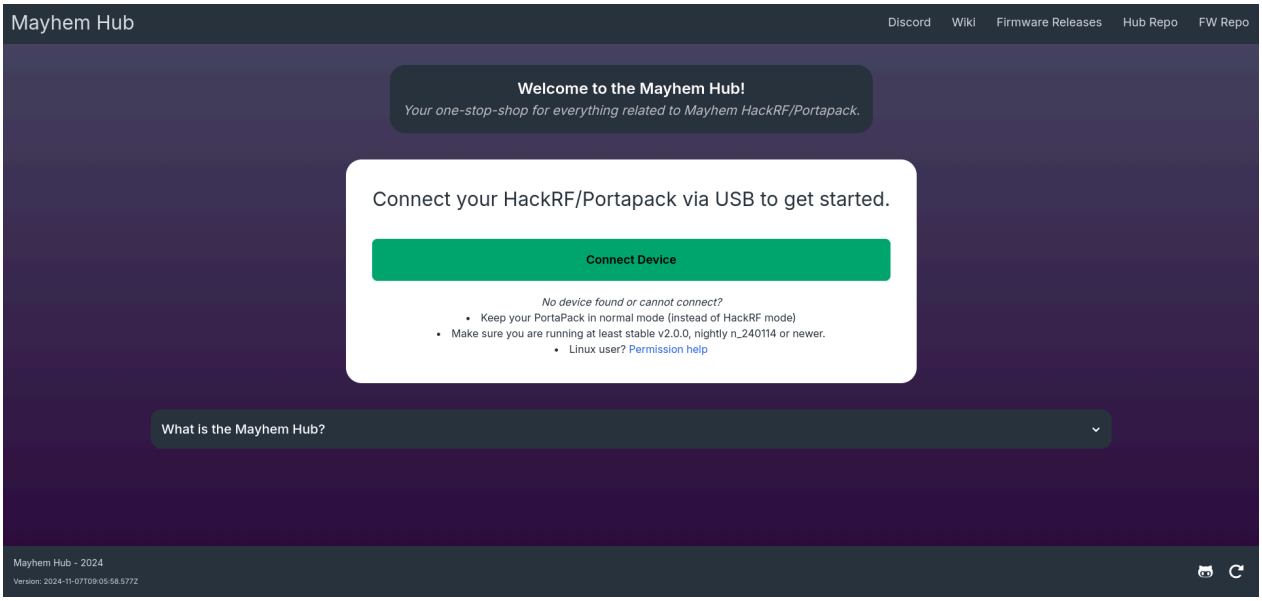
Please go and read the SD card Content article.

21.3 Using the hackrf.app website

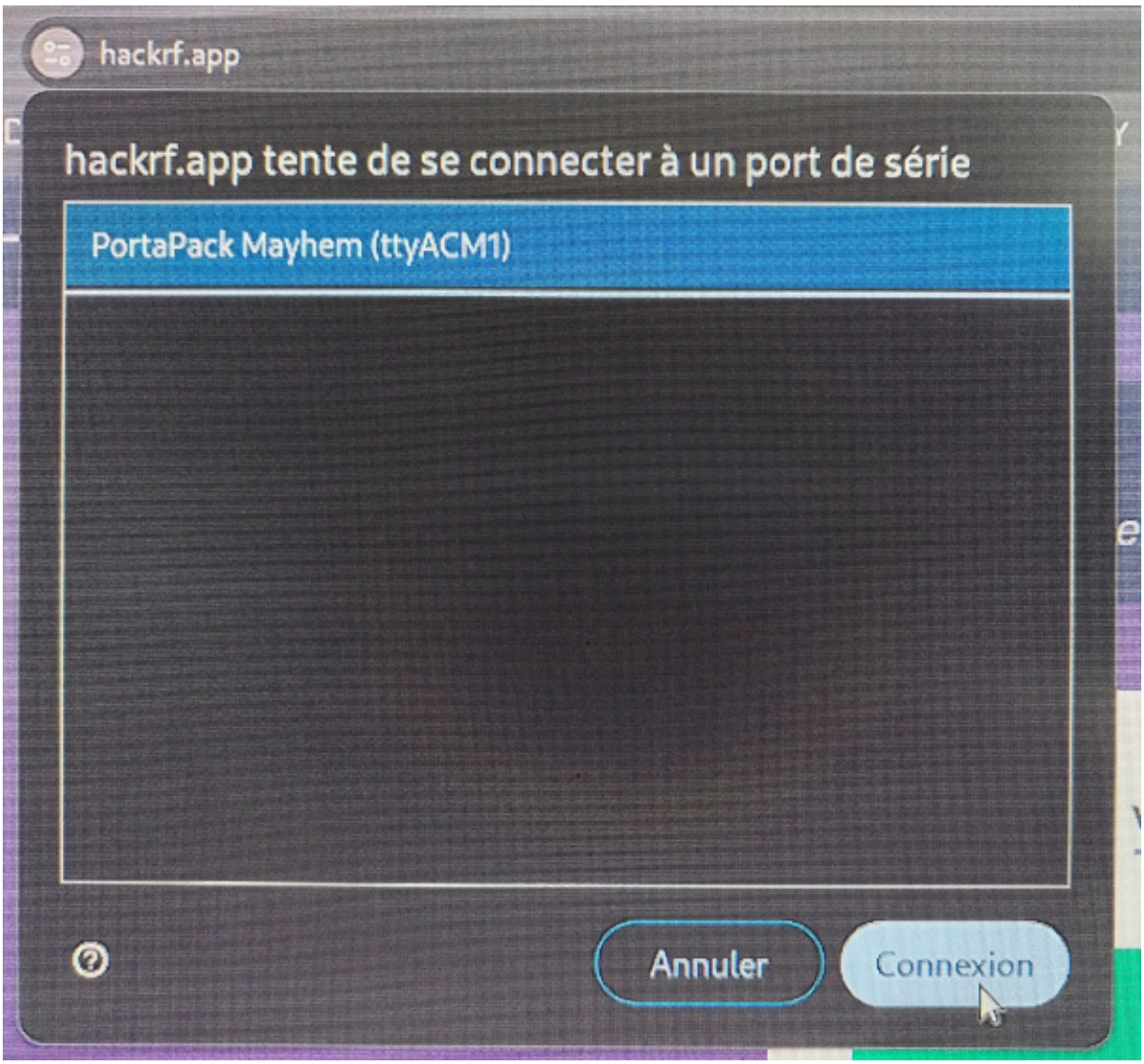
- You need to use a browser compatible with Web USB Serial
- The update using the website is only updating the firmware and external apps on the sdcard. If you want/need a full sdcard content update, please follow first the Update SDCARD files article, then update using the website

21.3.1 Procedure

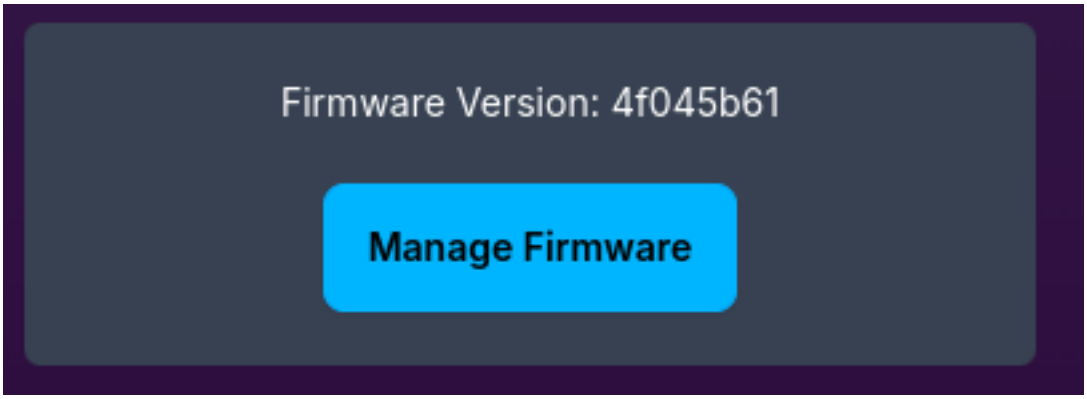
- Connect portapack using an USB cable with DATA enabled
- Navigate to https://hackrf.app/
- If never done, click on "Connect Device"



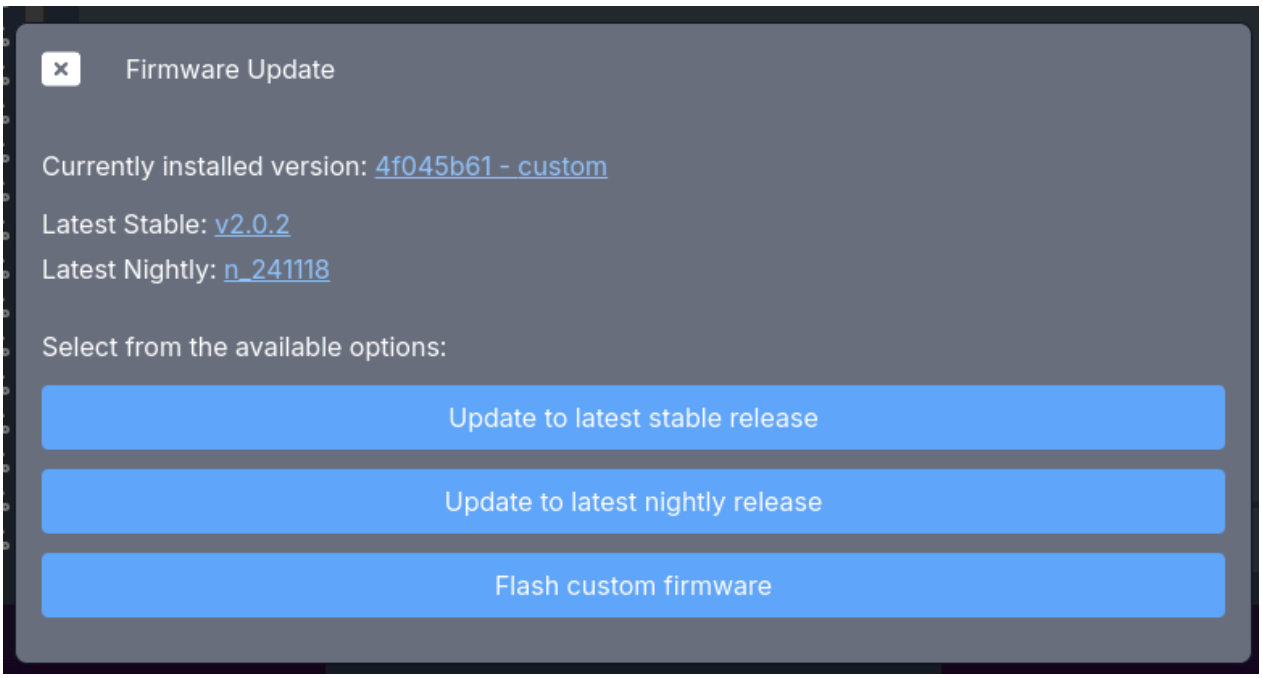
- On the little pop up, select your Portapack, click 'Connect'



- You should see “HackRF Connected” on top of the page, and the Live Screen will load the first screenshot
- Scroll down to “Manage Firmware”, and click it



- Choose the firmware version to install (stable, nightly, or from a custom ppfw file)



- Wait for the file to be uploaded, and for your device to automatically reboot
- Done !

22 Troubleshooting

- Please check this guide
- External apps (.ppma files in the APPS folder) will only function if their version matches the firmware version
- Please check the FAQ
- If your fimware version is lower or equal to v2.0.0 or nightly n_240114, you can only update firmware in HackRF mode with classic procedure
- Updating using the hackrf.app website is only working for stable version bigger or equal to v2.0.1, nightly version newer than n_240114
- When downgrading, please make sure SD card content versions match.
- Mayhem firmware contains HackRF firmware. You DO NOT need to flash HackRF firmware.
- In theory, it is impossible to brick the device, since you can always try the DFU recovery procedure. However, the updating might become fiddly in certain conditions. So you probably need basic computer usage knowledge (running/installing software, read documents, knowing USB ports, adjust system settings, reading English, etc.) if you don't, you probably should ask your friends or someone who knows these things to do it for you.

22.1 Flash Mayhem onto bare HackRF

22.2 Using firmware file from github release page and the classic hackrf_spiflash tool

Note ⚠️ This guide works for bare HackRF instead of HackRF with a PortaPack.

- Get the latest firmware from the GitHub release (latest by date) page

22.2.1 Windows

1. Connect the device via USB.
2. Double click `mayhem_flasher.bat` and follow the instructions.
3. Reboot the device.

22.2.2 Linux

1. Connect the device via USB

2. Upload the firmware with `hackrf_spiflash -w new_firmware_file.bin` (eg. portapack-h1_h2-mayhem.bin for mayhem firmware or hackrf_one_usb.bin for stock HackRF firmware)
3. Reboot the device

NOTE

- To Ubuntu and Mint user: Ubuntu based distro never maintains their repo for HackRF host (cli client) package. You'll face a lot of weird problems if your HackRF is R9. To resolve these, please compile the HackRF package yourself, or use another distro.

22.2.3 MacOS

1. If necessary, install the HackRF tools: `brew install hackrf`
2. Connect the device via USB
3. Upload the firmware with `hackrf_spiflash -w new_firmware_file.bin`
4. Reboot the device

22.3 Updating the Xilinx CPLD on hackrf board

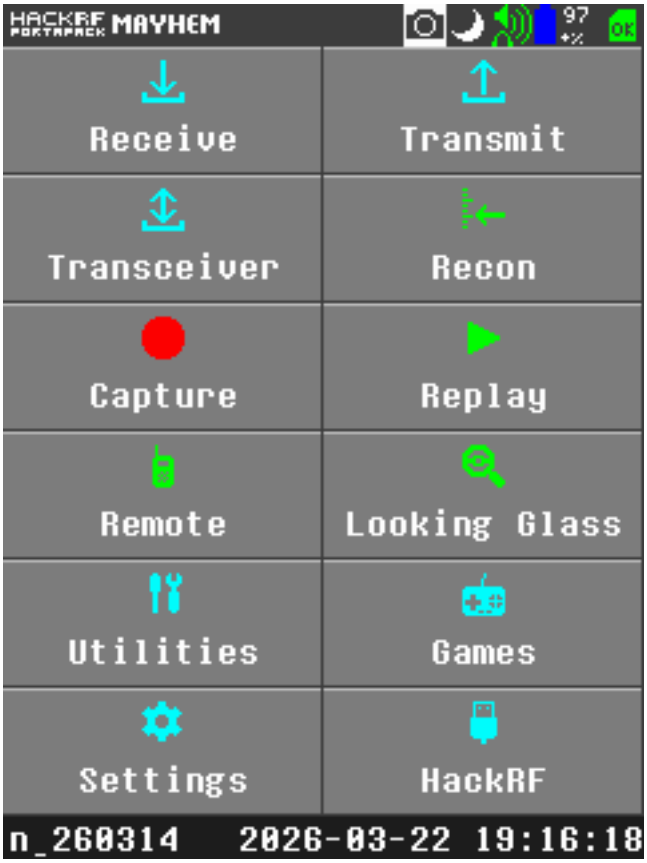
Since HackRF Version 2021.03.1, the CPLD will be upgraded within the SPI and DFU firmware upgrade process. For older versions you have to, and now you can, update the CPLD manually:

- Clone the original HackRF project to your computer:
  - `git clone https://github.com/greatscottgadgets/hackrf`
- If not present, install the HackRF tools.
  - See here how, at the example `hackrf_spiflash`. The Process installs all necessary tools.
- Connect your PortaPack into your Computer, put it into HackRF mode and use the following command to update the CPLD:
  - `hackrf_cpldjtag -x hackrf/firmware/cpld/sgpio_if/default.xsvf`

LED1/2/3 blinking means CPLD program success. LED3/RED steady means error. Wait for the message 'Write finished' and power OFF or Disconnect your PortaPack from your machine.

22.4 User interface

23 User Interface



The PortaPack display is 240×320 pixels. The screen is divided into three horizontal bands that are always present:

```
┌──────────────────────────┐
│                          │
│      App content area    │
│                          │
├──────────────────────────┤
```

← 16 px Title bar

← 288 px

← 16 px Information bar

23.1 Title bar

The top 16 pixels form the [[Title-Bar|title bar]]. It contains:

- Navigation controls (back arrow, app name) on the left
- Configurable status icons and quick-action buttons on the right

23.2 App content area

The middle 288 pixels are used by the currently running app. On the [[main-menu|main menu]] this shows the 2-column button grid. Inside apps it shows the app's own controls and display.

23.3 Information bar

The bottom 16 pixels always show:

- **Left:** Firmware version string (e.g. `n_250606`). If a firmware checksum error is detected at boot, this shows `FLASH ERR` in red instead.
- **Right:** Live UTC clock, updated every second. Date display can be enabled via **Settings → User Interface**.

23.4 Splash screen

A splash screen is shown on each boot before the main menu appears. Tap the screen or press any button to skip it immediately.

By default a random image is selected from the `SPLASH/` folder on the SD card. If a file named `splash.bmp` exists in the SD card root, that file is used instead and the random selection is skipped.

Splash display can be enabled or disabled under **Settings → Display**. See [[Create-a-custom-splash-screen]] for image format requirements.

23.5 Sub-pages

- [[Title-Bar]] — all title bar icons, their functions and active colours
- [[Main-Menu]] — main menu layout and available entries
- [[Main-Controls]] — common controls: frequency field, step size, bandwidth, gain
- [[Text-Entry]] — the on-screen keyboard used for text input
- [[Screenshots]] — how to capture and save screenshots
- [[Create-a-custom-splash-screen]] — boot splash screen image and format requirements
- [[Sleep-mode]] — display sleep and backlight auto-timeout
- [[Stealth-Mode]] — automatic backlight-off during transmission
- [[SD-Card-Content]] — SD card folder structure used by the firmware

23.6 Splash screen

24 Splash screen

For converting back and forth between the image and source you can use `xxd`. In a Windows environment you can use this command in the Linux Subsystem installing one of the versions available (i.e. Ubuntu).

24.0.1 From .bmp to .hpp

```
xxd -i input_image.bmp output.hpp
```

24.0.2 From .hpp to .bmp

```
xxd -r -p input.hpp output_image.bmp
```

24.1 Python converter for Splash Screen

There is also a python helper in firmware/tools/bmp_tools

25 Firmware icons

The special file `firmware/application/bitmap.hpp` contains all firmware icons.
This file should not be edited directly! Instead, new icons should be copied in firmware/graphics and then run `python [firmware/tools/bitmap_tools/pp_png2hpp.py](https://github.com/portapack-mayhem/mayhem-firmware/blob/next/firmware/tools/bitmap_tools/pp_png2hpp.py) <DIRECTORY> bitmap.hpp` to create `bitmap.hpp`.

For operating that Python3 script, `Pillow` library is required: `pip install pillow`

25.1 make_bitmap.py - Convert a folder contains one or more icon.png to one bitmap.hpp

The `make_bitmap.py` is the traditional helper, well tested. The folder with the icons is given as argument and generates the `bitmap.hpp`. This file needs to be copied to `mayhem-firmware/firmware/application/`. Starting from the `tools/ folder, the command is:

```
python3 bitmap_tools/make_bitmap.py ../graphics/ && mv bitmap.hpp ../application/bitmap.hpp
```

The icon size needs to be a multiple of 8. The generated icon is black/white.

25.2 bitmap_arr_reverse_decode.py - Convert bitmap array to icon.png

The `bitmap_arr_reverse_decode.py` takes an array from the bitmap.hpp and convert it back to a png.

25.3 pp_png2hpp.py - Convert both ways

The `pp_png2hpp.py` is based on the previous scripts, as all in one solution.

With the `--hpp bitmap.hpp` file and the `--graphics /folder_to/png_icons/` arguments, this script will generate a `bitmap.hpp`.
Add the `--reverse` argument to generate png icons from a given `bitmap.hpp`.

The reverse function got a parser, for automatic get the filename and size of the image.

25.3.0.1 Example

From the `tools/` folder:

```
$ ./bitmap_tools/pp_png2hpp.py ../application/bitmap.hpp ../graphics
Converting from png to hpp
From path ../graphics/ to file ../application/bitmap.hpp
Find your bitmap.hpp at ../application/bitmap.hpp
```

To reverse the icons from the .hpp to png icons use the parameter `--reverse`. Default is only the mayhem logo, named `titlebar_image`:

```
$ ./bitmap_tools/pp_png2hpp.py ../application/bitmap.hpp /home/lupus/work/hackrf/tmp/ --
reverse
Reverse: Converting from hpp to png
Converting icon titlebar_image
```

Add the parameter `--icon <icon name>` to convert another icon or `all` to convert all available icons at once:

```
$ ./bitmap_tools/pp_png2hpp.py ../application/bitmap.hpp /home/lupus/work/hackrf/tmp/ --
reverse --icon all
```

Note: It is not a recovery, the icons will be new generated. The new icons are black/white, even if the original icons where transparent.

25.4 Online editor

The current icons were designed on Piskelapp. To create or update icons, the current sources (16x16 or 8x8) can be used.

25.5 Title bar

26 Title Bar

The title bar appears at the top of every screen and contains navigation controls, quick-action buttons, and status indicators. It is 16 pixels tall and is always visible.

26.1 Navigation

The left side of the title bar contains two fixed elements:

- **Back arrow** — returns to the previous screen or main menu. Also functions as a back button when tapped.
- **App name or title** — displays the current application name. Tapping it has the same effect as the back arrow.

26.2 Status icons

The right side contains a configurable set of icons, added left to right in this order. Most can be individually hidden via **Settings → User Interface**.

- Icon: Camera; What it shows: —; When tapped: Takes a [[Screenshots; Active colour: screenshots]]
- Icon: Sleep (moon); What it shows: —; When tapped: Activates [[Sleep-mode; Active colour: sleep mode]]
- Icon: Stealth; What it shows: Stealth mode on/off; When tapped: Toggles [[Stealth-Mode; Active colour: stealth mode]]
- Icon: Converter (up/down arrow); What it shows: Up- or down-converter selected; When tapped: Toggles converter on/off; Active colour: Red when active

- Icon: Bias Tee; What it shows: DC power on antenna port on/off; When tapped: Toggles antenna bias with confirmation; Active colour: Yellow when active
- Icon: Clock (int/ext icon); What it shows: Current clock reference source (internal or external); When tapped: Toggles clock output (CLKOUT) on/off; Active colour: Green when CLKOUT is enabled
- Icon: Mute (speaker/headphone); What it shows: Audio mute on/off; When tapped: Toggles audio mute; Active colour: Green when muted
- Icon: Speaker; What it shows: Speaker output on/off; When tapped: Toggles speaker output; Active colour: Green when enabled
- Icon: Battery icon; What it shows: Battery level and charging status; When tapped: Opens battery details; Active colour: —
- Icon: Battery %; What it shows: Numeric battery percentage; When tapped: Opens battery details; Active colour: —
- Icon: SD card; What it shows: SD card present/absent; When tapped: Opens SD card info screen; Active colour: Green when card is detected

NOTE The **Speaker** icon only appears on PortaPack models with an AK4951 audio codec where the speaker can be independently controlled. Do not enable it if the speaker already works without it enabled — on some OpenSourceSDRLab H2 models this can cause a conflict.

WARNING The **Bias Tee** supplies DC voltage on the SMA antenna port. Only enable it if your antenna or LNA is designed to accept DC power via the coax. Applying DC to unprepared equipment can cause damage.

26.3 Converter icon detail

The converter icon switches between an up-arrow (`icon_upconvert`) and a down-arrow (`icon_downconvert`) based on the up/down converter setting in **Settings → Converter**. The icon turns **red** when the converter is enabled, regardless of direction.

26.4 Clock icon detail

The clock icon switches between `icon_clk_int` (internal reference) and `icon_clk_ext` (external reference) based on the currently active clock source. The icon turns **green** when the clock output (CLKOUT on the expansion port) is enabled. Tapping it toggles CLKOUT on/off — it does not switch between internal and external clock sources.

26.5 Navigating the title bar with hardware buttons

On H1/H2 models, pressing **Up** while the focus is on the main menu area moves the highlight into the title bar. Icons can then be selected using the directional buttons. On H4M, the touchscreen is the primary method.

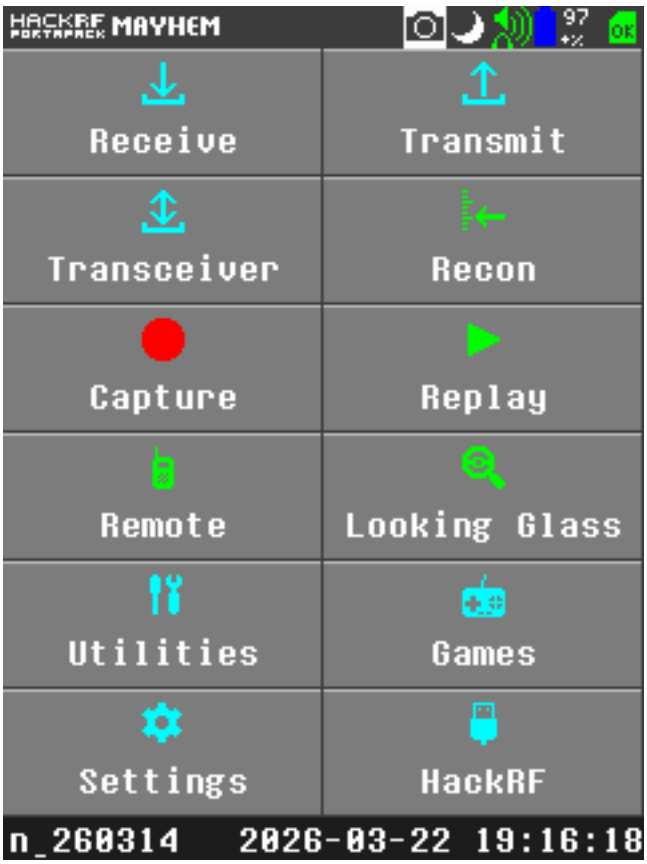
26.6 Hiding icons

All status icons can be individually shown or hidden via **Settings → User Interface**. This is useful for decluttering the title bar on small screens or hiding features that are not needed.

26.7 Main menu

27 Main Menu

The main menu is the first screen shown after boot. It is a 2-column button grid that provides access to all major areas of the firmware.



27.1 Menu items

- Entry: Receive; Color: Cyan; Description: Receiver applications — audio, ADS-B, AIS, APRS, BLE, POCSAG, and more
- Entry: Transmit; Color: Cyan; Description: Transmitter applications — APRS TX, BLE TX, OOK, RDS, TouchTunes, and more
- Entry: Transceiver; Color: Cyan; Description: Bidirectional apps — Microphone TX/RX and KISS TNC
- Entry: Recon; Color: Green; Description: Automated frequency scanning and signal reconnaissance
- Entry: Capture; Color: Red; Description: Raw IQ capture to SD card
- Entry: Replay; Color: Green; Description: Playback of previously captured IQ files
- Entry: Looking Glass; Color: Green; Description: Wideband spectrum sweep visualizer
- Entry: Utilities; Color: Cyan; Description: Tools — File Manager, Freq. Manager, IQ Trim, Notepad, Flash Utility, Debug, and more
- Entry: Games; Color: Cyan; Description: Built-in games
- Entry: Settings; Color: Cyan; Description: Device configuration — radio, display, UI, P.Memory, and more
- Entry: HackRF; Color: Cyan; Description: Switch to standard HackRF mode (USB SDR)

> **NOTE HackRF** always appears as the last item in the menu. It is not a navigation target but a mode switch — selecting it opens a confirmation dialog. See [[HackRF]] for details.

27.2 External apps in the main menu

External apps (`.ppma` or `.ppmp` files in the `APPS/` folder on the SD card) can be configured to appear directly in the main menu rather than inside a submenu. Apps registered at the `HOME` location are inserted between **Settings** and **HackRF** in the order they appear in the `APPS/` directory.

See [[External-apps-informations]] for how to configure an external app's menu location.

27.3 App Search

Tapping the **info bar** (the status bar at the top of the screen) opens an alphanumeric keyboard. Type part of an app name (up to 20 characters) and confirm to search across all built-in and external apps. A results list is shown with matching entries — select one to launch it directly. If no match is found a modal displays "No matching app found." Pressing back or submitting an empty query cancels the search.

27.4 Navigation

From the main menu, pressing the **back button** (or the left hardware button on H1/H2, or navigating back on H4M) has no effect — the main menu is the root of the navigation stack.

The **title bar** at the top of every screen provides access to quick settings (converter, clock output, speaker, stealth mode) and shows the current UTC time, battery status, and SD card status. See [[Title-Bar]] for details.

27.5 Main Controls

28 Main Controls

This page describes the common controls that appear across most receiver and transmitter apps.

28.1 Frequency

When the frequency field is selected, the frequency can be changed in three ways:

- **Encoder dial:** Adjusts the frequency by the currently set step size.
- **Short press of Select:** Opens the direct frequency entry screen, where a frequency can be typed in.
- **Long press of Select:** Toggles *digit mode*. In digit mode, the cursor highlights a single digit of the frequency. Left/Right move the cursor between digit positions; Up/Down (or the encoder) adjust the value of the selected digit. Long press again to exit digit mode.

The nine digit positions represent: 1 GHz, 100 MHz, 10 MHz, 1 MHz, · (decimal), 100 kHz, 10 kHz, 1 kHz, 100 Hz.

28.2 Step size

The step size used when turning the encoder dial can be changed as a secondary option when the frequency field is focused. The available steps are:

`10Hz` · `50Hz` · `0.1kHz` · `1kHz` · `5kHz (SA AM)` · `6.25kHz (NFM)` · `8.33kHz (AIR)` · `9kHz (EU AM)` · `10kHz (US AM)` · `12.5kHz (NFM)` · `15kHz (HFM)` · `25kHz (N1)` · `30kHz (OIRT)` · `50kHz (FM1)` · `100kHz (FM2)` · `250kHz (N2)` · `500kHz (WFM)` · `1MHz`

28.3 Bandwidth

The IF filter bandwidth selects which FIR filter is applied to the received signal. Available options depend on the modulation mode:

AM: `DSB 9k` · `DSB 6k` · `USB+3k` · `LSB-3k` · `CW`

- `DSB 9k` and `DSB 6k` are standard double-sideband AM (9K00A3E and 6K00A3E).
- `USB+3k` and `LSB-3k` are single-sideband (2K80J3E), upper and lower respectively.
- `CW` uses a 200 Hz filter centred at 700 Hz.

NFM: `8k5` · `11k` · `12k5` · `16k`

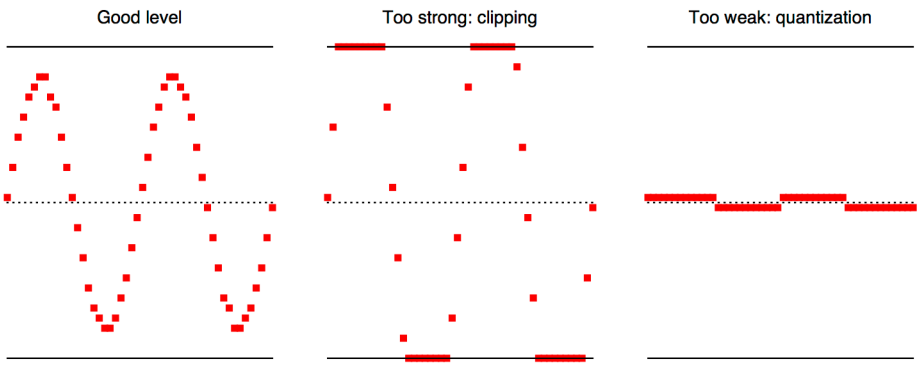
- `16k` (16KF3E) is for 25 kHz channel spacing.
- `12k5` is standard for 12.5 kHz channels; overmodulated systems may benefit from `11k`.
- `8k5` (8K5F3E) is for narrow 12.5 kHz spacing where tighter filtering is needed.

WFM: `80k` · `180k` · `200k`

- `200k` covers a full stereo FM broadcast signal.
- `80k` is a narrower option useful for weaker stations or reduced bandwidth.

28.4 Gain controls

The HackRF has three independent RX gain stages and two TX gain stages. The image below shows their position in the Audio receiver app — the layout is the same across all receiver apps.



28.4.1 RX gain

- Label in UI: AMP; Stage: RF amplifier (at antenna port); Range: 0 dB or +14 dB; Step: on/off
- Label in UI: LNA; Stage: IF or low-noise amplifier; Range: 0 – 40 dB; Step: 8 dB
- Label in UI: VGA; Stage: Baseband variable gain amplifier; Range: 0 – 62 dB; Step: 2 dB

The RF amp is shown as `0/1` in most RX apps. In the Audio app, it appears as `Amp` on the line below LNA/VGA when one of those fields is selected.

A good starting point for most signals is: **AMP off, LNA 16, VGA 16**. Increase LNA and VGA roughly equally until the signal is clear. Enable AMP only for very weak signals — it can degrade SNR if the noise floor is already high, or if a strong signal is present.

> **NOTE** Setting gain too high causes ADC saturation, which appears as broadband noise or spurious signals across the waterfall. The **Satu%** field in the [[DFU-overlay]] shows the current saturation percentage in real time and helps dial in the optimal setting.

28.4.2 TX gain

- Label in UI: AMP; Stage: RF amplifier (at antenna port); Range: 0 dB or +14 dB; Step: on/off
- Label in UI: Gain; Stage: TX VGA; Range: 0 – 47 dB; Step: 1 dB

The TX amp is shown as `0` or `14` in TX apps.

28.5 SD Card content and modification

29 SD Card Content in Portapack Mayhem

The **SD card** is an essential component of the **Portapack Mayhem** firmware, extending the capabilities of the device by providing additional storage for files, configurations, and user-generated data. While the Portapack itself contains built-in functionality, many features rely on an SD card to function optimally.

29.1 Key Uses of the SD Card

- **Records** – Custom **IQ recordings** and pre-generated waveforms for replay attacks or signal analysis.
- **External Apps** – Applications stored in the `APPS` directory of the SD card.
- **Configuration Files** – User preferences, profiles, and settings that persist across sessions.
- **Firmware Updates** – Bootloader and firmware upgrade files stored and loaded from the SD card.
- **Captured Data** – Logs, frequency sweeps, and captured transmissions saved for later analysis.

29.2 File System & Organization

- The SD card should be formatted in **FAT32** or **exFAT**.
- Specific folders and naming conventions are required for certain features.
- Maintaining a well-structured file hierarchy ensures seamless operation.
- Proper management of SD card content unlocks the full potential of the Portapack Mayhem firmware, enabling advanced signal processing, automation, and enhanced usability in the field.
- The SD card provides a flexible storage resource that can be tailored to the user's needs.

30 Downloading and Installing SD Card Files for Portapack Mayhem

To fully utilize the Portapack Mayhem firmware, you need to prepare your SD card with the necessary files.

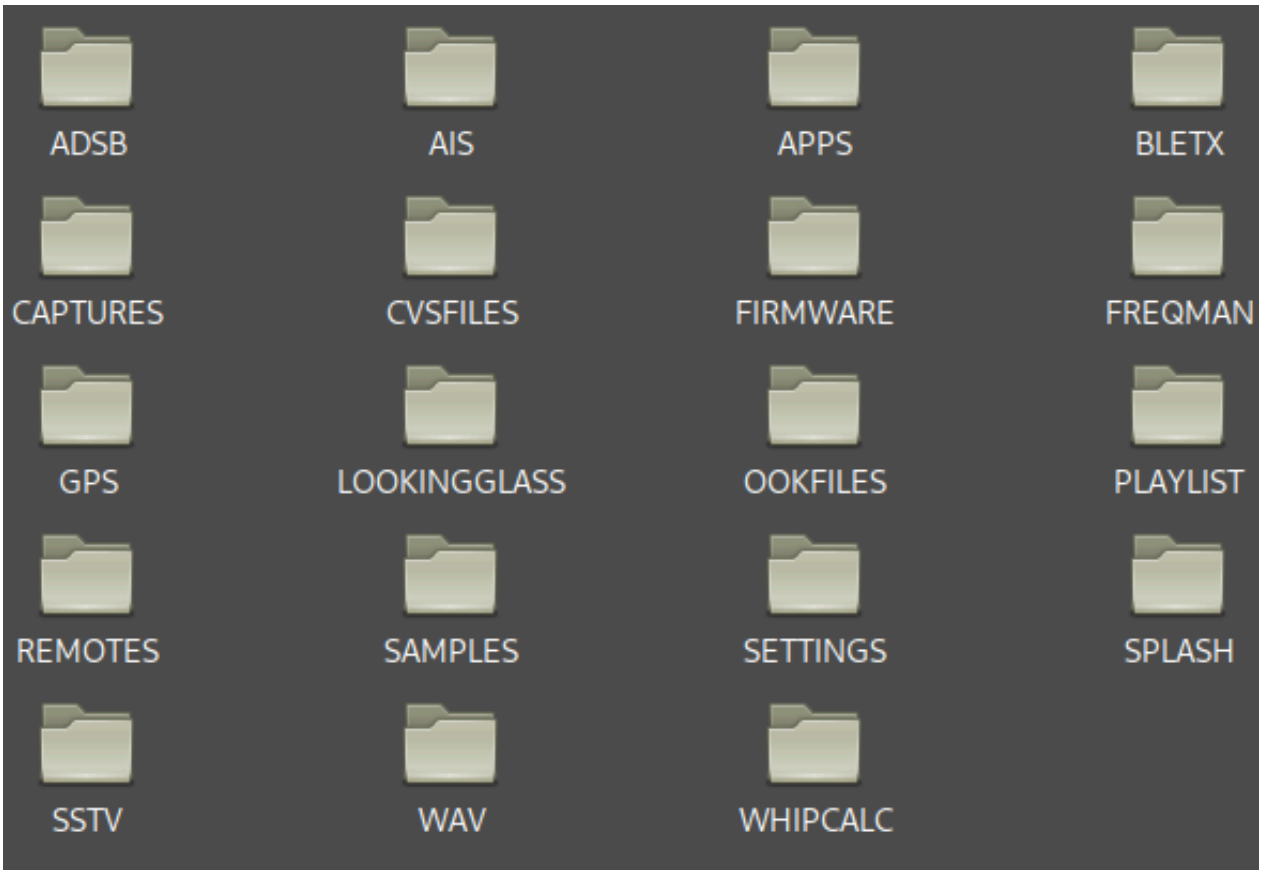
You can refer to the quick procedure below to know where to download the required content and how to install it properly, or have a look at the detailed SD card content update in the Update firmware article.

The standard SD card image is provided as part of the firmware download in the assets list of each release.

30.1 Procedure

1. Navigate to the releases page.
2. Select the version you want (nightly or stable).
3. Under assets, find the two files ending with `...COPY_TO_SDCARD.zip` and `...COPY_TO_SDCARD-no-world-map.zip`.
4. Download the appropriate file:
   - If you have already installed the SD card content once, you do not need the larger version containing the ADS-B world map.
5. Extract the contents of the ZIP **to the root of the SD card**.

The data structure **at the root of your SD card** must match the following example:



IMPORTANT The ZIP file **must** be extracted.

IMPORTANT The directories shown in the screenshot **must be at the root of your SD card**.

31 SD Card Details

31.1 SD Card Contents

The following folders on the SD card store various app-specific data. Some of these folders are included in the SD card image, while others may be created later by the firmware or the user:

- **ADSB** – Contains databases for airlines and ICAO. The world map is also stored here. See the documentation for generating a custom map.
- **AIS** – Stores the AIS database.
- **APPS** – Stores external applications (`.ppma` files) offloaded from the firmware flash ROM.
- **APRS** – Logs from the APRS RX app.
- **AUDIO** – WAV audio recordings from the Audio app.
- **BLERX** – Captures from the BLE RX app.
- **BLETX** – Packet files for the BLE TX app (`.TXT` format).
- **BMP** – BMP images saved by the NOAA APT app. Created automatically by the app on first use.
- **CAPTURES** – IQ capture files (`.C16`, `.C8`, `.CU8`, and metadata `.TXT`) from the Capture app.
- **CVSFILES** – Files for the CVS Spam app.
- **DEBUG** – Debug dump files created from the Debug Dump app.
- **EPIRB** – Files for the EPIRB RX and TX apps. EPIRB RX reads `FREQ.TXT` for the frequency selector. EPIRB TX reads `BEACONS.TXT` for beacon playback in file mode.
- **FIRMWARE** – Contains `.bin` files used to update the PortaPack firmware.
- **FREQMAN** – Stores frequency database files for loading in various apps.
- **GPS** – Stores information for generating GPS simulation files.
- **HOPPER** – Files for the Hopper app.
- **KEELOQKEYS** – Holds the `MFCODES` key file for the KeeLoq TX and SubGhzD apps. `MFCODES` is a plain text file with one entry per line in the format `mf_name;key_hex;type`. Without a matching entry, KeeLoq and Superrollo signals cannot be decrypted or retransmitted.
- **KEELOQREMOTES** – Holds `.KEELOQ` remote files created by SubGhzD when saving a decoded KeeLoq or Superrollo signal. These files are loaded by the KeeLoq TX app for retransmission.
- **LOGS** – Logs from various apps including ADSB RX, ERT, TPMS, POCSAG, Radiosonde, SubGhzD (`SUBGHZDLOG_*.CSV`), and SubCar (`SubCarLOG_*.CSV`).
- **LOOKINGGLASS** – Holds preset frequency ranges for the Looking Glass app. All `.TXT` files in this folder are loaded automatically.
- **MACADDRESS** – Holds MAC address files.
- **OOKFILES** – Signal files for the OOK Editor app.
- **PLAYLIST** – Stores `.PPL` playlist files for the Replay app.
- **REMOTES** – Stores remote files for the Remote app.
- **SAMPLES** – Contains example capture files.
- **SCREENSHOTS** – Stores screenshot files.
- **SECPLUS** – Holds `.SECPLUS` remote files for the Security+ TX app.
- **SETTINGS** – Stores application settings (`.ini` files). If an app malfunctions, deleting its settings file resets it. A blacklist file for disabling apps may also be stored here.
- **SPECTRUM** – Stores `.BMP` images for the Spectrum Painter app.
- **SPLASH** – Stores custom splash screen `.BMP` files (active splash screens must be in the root directory as `splash.bmp`).
- **SSTV** – Contains images for the SSTV app.
- **SUBGHZ** – Holds `.sub` signal files for the FlipperTX app.
- **TPMS** – Holds log files from the TPMS RX app.
- **WATERFALLS** – Holds preset files for the Waterfall Designer app.
- **WAV** – Stores sound files for the Soundboard app.
- **WHIPCALC** – Stores whip length calculation data. Additional whips can be added with specific details.

SD card contents can be managed using the File Manager application, the "SD over USB" app, or by removing the SD card and accessing it via a computer.

WARNING Improper reinsertion may cause the SD card to fall into the device, requiring disassembly to retrieve it.

31.2 Technical Details

- Detailed technical specifications are available here.

WARNING The File Manager app has a memory limitation: an "out of memory" error may occur if a folder contains too many files.

31.3 Text Entry

32 Text Entry

The text entry screen (title: **Text entry**) is used whenever an app needs free-text input — for example entering a callsign, filename, or label. It is opened automatically by apps that require it; it cannot be accessed directly from the menu.

32.1 Layout

The keyboard is a 5-column grid of 29 character buttons. Below the grid are four action controls:

- Control: Shift (icon); Function: Cycle shift state: off → on (once) → locked → off
- Control: <DEL; Function: Delete the character before (to the left of) the cursor
- Control: Mode button (e.g. 123); Function: Switch to the next character set (UPPER or lower-case letters, digits and marks)
- Control: OK; Function: Confirm entry and send the string back to the calling app

Above the grid, the current text is shown with a cursor. Below the grid are two helper fields:

- **Raw:** — a number field (1–255) for inserting a character directly by its ASCII code. Use it for special characters that are not on any keyboard page: rotate the encoder to set the value, check the resulting character in the **AKA:** field, then press the center button to insert it.
- **AKA:** — shows the character corresponding to the current Raw value.

32.2 Character sets

The keyboard cycles through three sets using the Mode button. The button always shows the *next* set's name:

32.2.1 `abc` — Letters

- Normal: `a b c d e f g h i j k l m n o p q r s t u v w x y z , .`
- Shifted (once or locked): same in uppercase `A B C … Z , .`

32.2.2 `123` — Digits and punctuation

- Normal: `1 2 3 4 5 6 7 8 9 0 ( ) ' ` " + - * / = < > _ \ ! ? , .`
- Shifted (locked only): `! @ # $ % ^ & * ( ) [ ] ' ` " { } | : ; < > - _ ~ ? , .`

NOTE In `123` mode, the Shift button skips the one-time shift state and goes directly to ShiftLock. Single-character capitalisation is only available in `abc` mode.

32.2.3 `hex` — Hexadecimal

- `1 2 3 4 5 6 7 8 9 0 A B C D E F` (space)
- Shift has no effect in this mode.

32.3 Shift states

The Shift button cycles through three states:

- State: Off; Shift button colour: Dark (default); Effect: Normal characters
- State: On (once); Shift button colour: Darker; Effect: Next character typed is uppercase/shifted; then resets to off automatically
- State: Locked; Shift button colour: Blue; Effect: All characters are uppercase/shifted until unlocked

32.4 Navigation

- **Encoder dial:** moves the highlight through all 29 character buttons in order.
- **Touchscreen:** tap any button directly.
- **Hardware buttons (D-pad):** navigate between buttons.

The cursor position in the text preview can be set by tapping a position in the text field if supported by the calling app.

32.5 Powering the PortaPack

The PortaPack is powered via the USB connector on the HackRF One. The connector type and battery support depend on the model:

- Model: H1; USB connector: Micro-USB; Internal battery: No
- Model: H2; USB connector: Micro-USB; Internal battery: Optional LiPo
- Model: H4M; USB connector: USB-C; Internal battery: Optional LiPo

32.6 H1

The H1 has no internal battery and must be powered via USB at all times.

32.7 H2

The H2 can be fitted with an internal LiPo (Lithium Polymer) battery, charged via the Micro-USB port.

The maximum battery size that fits in a typical metal enclosure is 2500 mAh; 2100 mAh is a better fit. Based on testing, the H2/HackRF combination consumes between 250 mA (1.25 W) at the idle menu screen and 550 mA peak (2.75 W) with audio at full volume while receiving. When charging from USB with the device in use, peak current draw reaches approximately 1.5 A, dropping as charging progresses. The battery charges at a peak of ~1000 mA, settling to 600–700 mA, and takes 3–4 hours depending on battery capacity and cable quality — longer cables charge slower.

With a 2500 mAh battery, expect 4–5 hours of runtime depending on usage and battery condition.

32.8 H4M

The H4M uses a **USB-C** connector and has a dedicated **power switch**. It can also be fitted with an internal LiPo battery.

> **NOTE** Power consumption and charge time measurements for the H4M have not yet been documented here. Contributions welcome.

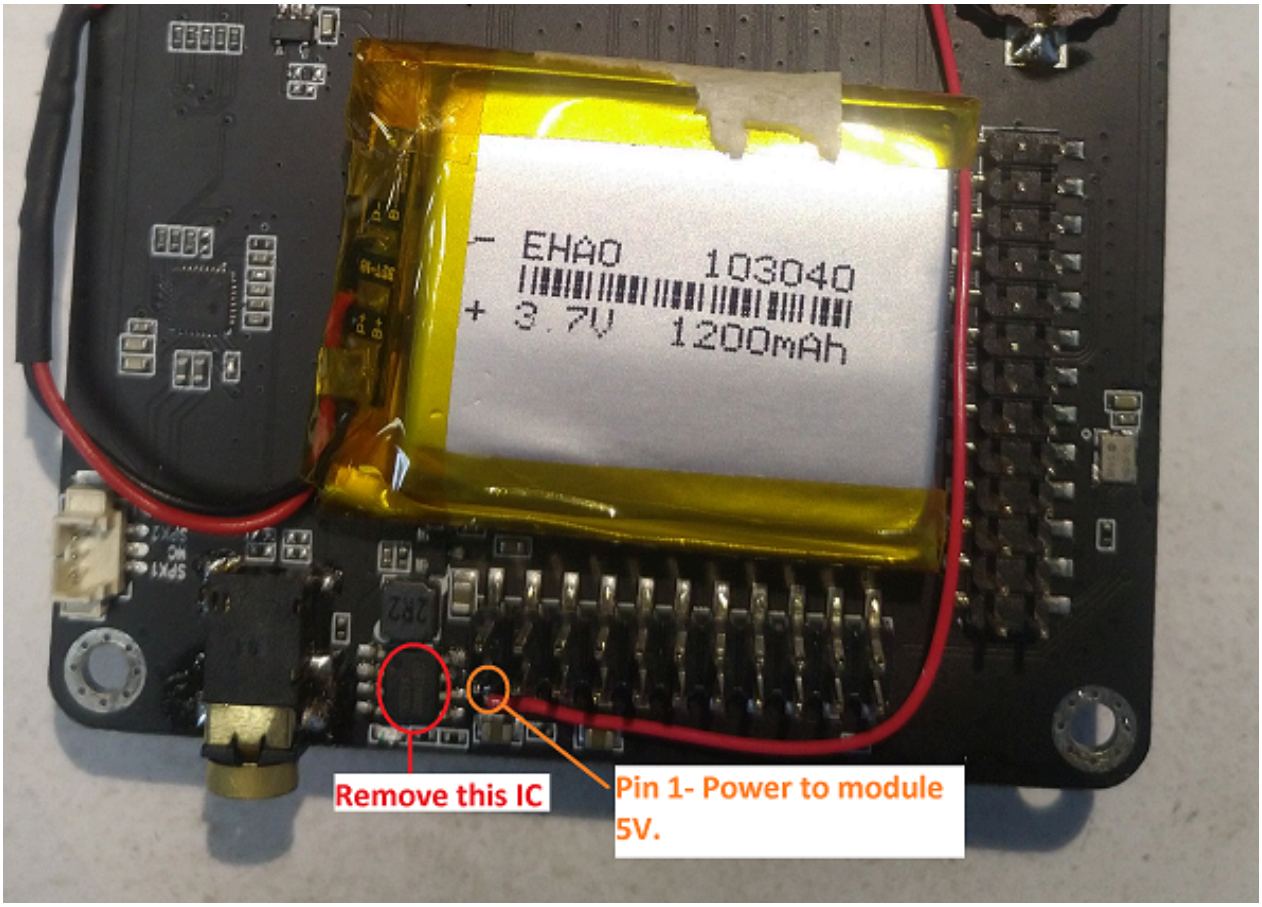
32.9 Battery charger modification (H2)

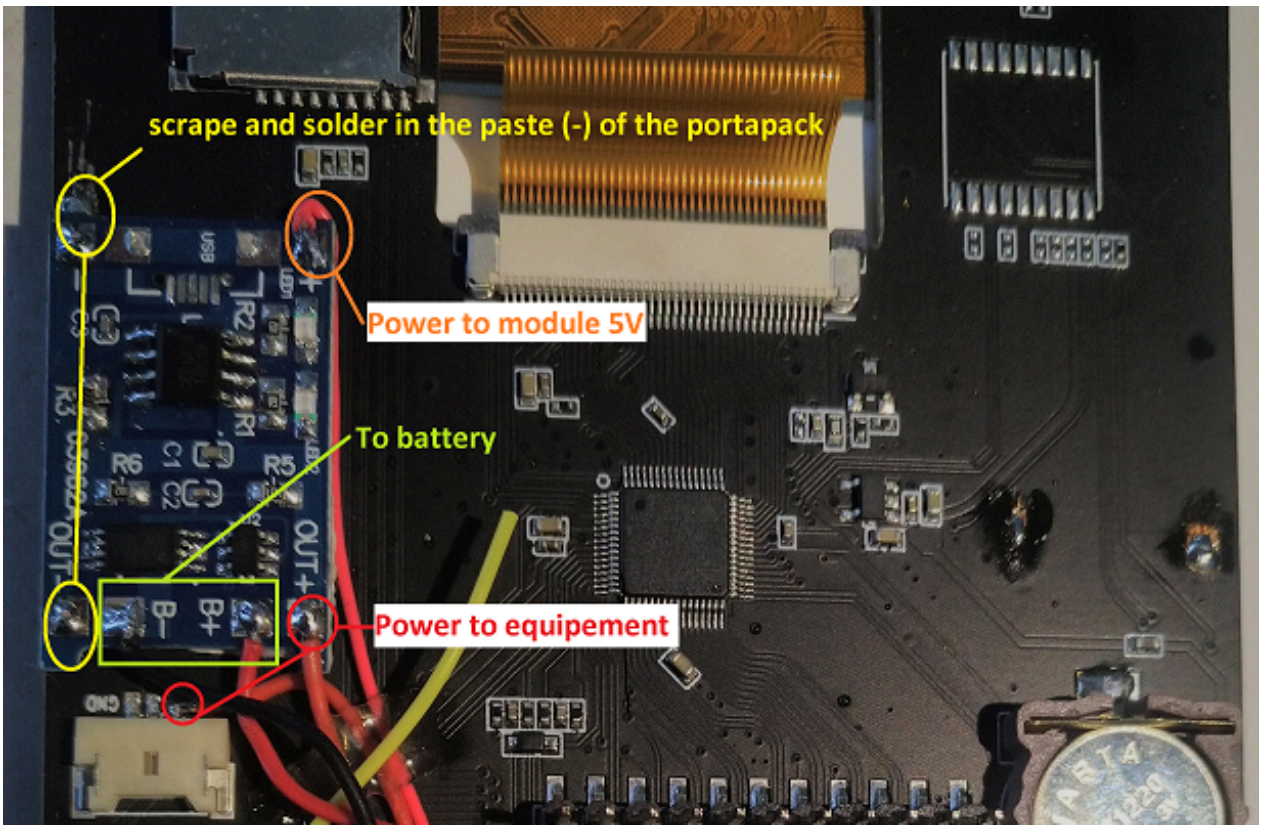
When connected to a computer or power bank, some H2 units generate interference and harmonic signals caused by the onboard battery charging circuit. This only occurs while the battery is charging, and not all units are affected.

If you have confirmed the charging circuit is the source of the interference, one solution is to replace the charging IC with a TP4056 module. The TP4056 offers battery overcharge protection (4.2 V), discharge protection (2.9 V), and adjustable charge current (via a resistor on the module), and eliminates the interference seen with the original IC.

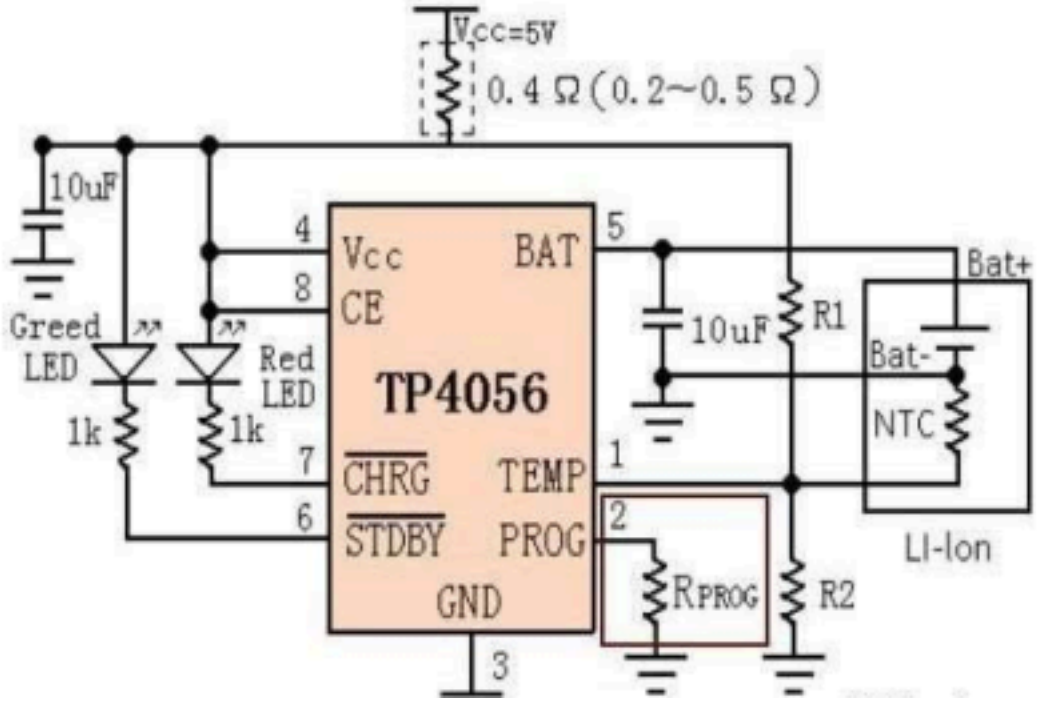
See the following pictures for more information, and take a look at the related issue

TP4056 Lithium Battery Charging Processor
DW01A Lithium Battery
Protection Circuit
input: 4.5 - 8.0V
max. 1.2A
charging:
4.2V, max 1A
output:
Vbatt, max. 3A
Charging Indicator
Charging End Indicator
8205A Dual Power MOSFET
Battery Disconnect Switch


EHAO 103040
3.7V 1200mAh
Remove this IC
Pin 1- Power to module
5V.


scrape and solder in the paste (-) of the portapack
Power to module 5V
To battery
Power to equipement
OUT+
OUT-
B-
B+
R1
R2
R3
R5
R6
C1
C2
USB

R_{PROG} (k)	I_{BAT} (mA)
10	130
5	250
4	300
3	400
2	580
1.66	690
1.5	780
1.33	900
1.2	1000



32.10 Troubleshooting

33 Troubleshooting

Something not working? Start with the **FAQ** and the **SolveBoard** quick problem → solution table — they cover most common issues. If that doesn't help, pick the matching topic below.

33.1 🔌 Boot & power

- **Won't boot** — blank, white or black screen, or the device won't start
- **Config Menu** — reach the low-level config or recovery menu
- **Dead Coin Cell Battery** — clock and date keep resetting: replacing the RTC coin cell
- **H2+ RESET and DFU fix** — repairing the reset or DFU push-buttons on H2+ boards

33.2 💾 Firmware & recovery

- **Firmware upgrade** — problems while updating the firmware
- **Diagnose firmware update in Windows** — Windows-specific update diagnosis
- **DFU overlay** — recover the device using the DFU overlay
- **Factory Defaults** — restore the default settings
- **Full reset** — wipe everything back to a clean state

33.3 📡 Receive & transmit

- **Receive Quality Issues** — weak or no reception: gain and antenna advice
- **No TX/RX** — no transmit or receive at all
- **TX Carrier Only** — only an unmodulated carrier is transmitted

33.4 💳 SD card

- **How to format the SD card** — correct FAT32 formatting
- **SD card not recognized by PC** — SD-over-USB not showing up on the computer

33.5 🔗 USB connectivity

- **WebUSB serial not working (Linux)** — Linux permissions or udev rules for the serial console
- **USB-C port intermittent connection** — connection drops when the cable moves

33.6 🧩 Apps

- **What if I don't like some of the apps** — hide or blacklist apps you don't want

33.7 ❓ Still stuck?

Check the **FAQ** or report an issue on GitHub.

33.8 Won't boot

34 My device wont start up

34.1 White screen

If you device won't boot and leaves you on white screen, then you will need to power off the device and then holding the **UP** button while you power it on. This could take up to 10 seconds.

If this does not work, then try the same thing but this time holding the **DOWN** button (Remembering to wait up to 10 seconds).

What this is doing is loading a different LCD driver to get your devices screen to work.

Once that is done you should be able to reboot normally without any issues.

34.2 Black screen

If you device won't boot and leaves you on a black screen, then you will need to power off the device and then hold down the **LEFT** button while you power it on. This could take up to 10 seconds.

35 Description

- UP key = LCD driver 1
- DOWN key = LCD driver 2
- OK/SELECT key = Reset/Automatic detection
- LEFT key = LCD driver 2 QFP100 chip
- RIGHT key = LCD driver 1 QFP100 chip

If you are having trouble understanding these procedures, here is a step-by-step instruction:

1. power off your device
2. press and holding one of the buttons listed above. □**DO NOT RELEASE IT YET**□

3. power on
4. Wait for at least 10 seconds.
5. if : you see the screen displaying any valid content*, release the button you have chosen.
   else if : you still not seeing any valid content* displayed, choose another button and start again from step 1.
6. power off, wait 5s, power on again **without** holding any buttons listed above, check if it boot successfully.
   if : you see the screen display any valid content* : we are done.
   else if : it boots successfully last round, but fail again this time : check the coin battery.

p.s. ^valid content : either a splash screen or any of the interface

NOTE H2+ usually require you to hold the **UP key** on the first boot to configure them.

*This is valid from nightly version **n_220412** and stable release version: **v1.5.1***

35.1 H2+ (and H1) not powering up, just black LCD (after flashing new Mayhem FW, and it is ignoring previous above key buttons init description):

If your device was working correctly before updating the firmware, and now you see a black screen after it's been flashed and it is ignoring the keyboard of the above instructions and looks like it's bricked (or also if the Mayhem menus appear but running any app causes an M0 guru fault indicating that the ROM may be partially flashed), then you have several options to try below depending the situation. Make sure the PortaPack is charged using a separate 5V USB power adapter (not connected to the PC). After charging, connect the USB cable from your PC to the device and follow the steps below:

35.1.1 (1) If your device is still detecting correctly the USB plug to your USB PC

(showing in the front of the HackRF board a green LED when plug in USB cable), then you are lucky; it means that your device is in "hackrf mode", with good USB communication, **then just follow the below process (I)**

35.1.2 (2) If you have an H1 device (without integrated battery) that does not detect any USB communication

(usually no need to disassemble), try to set up it to DFU mode (by holding in the DFU button while turning on power or attaching the USB cable). In DFU mode, the device will show up as an "NXP LPC" USB device. While in DFU mode, execute the following command:

```
dfu-util --device 1fc9:000c --alt 0 --download hackrf_one_usb.dfu
```

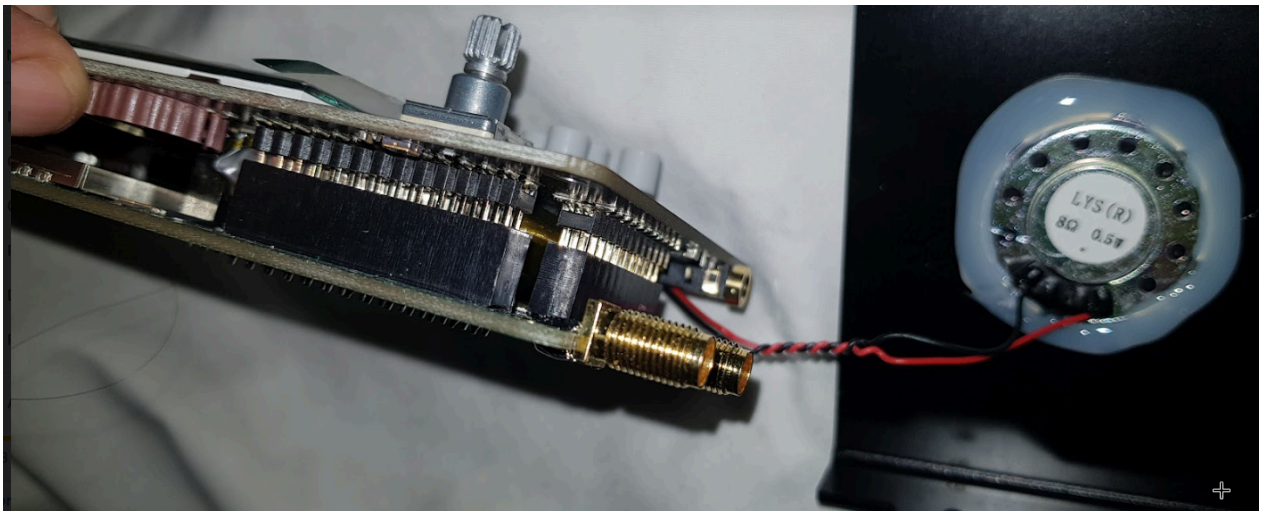
(Alternatively, run the `mayhem_flasher.bat` file and select flash DFU mode, which performs the command above.)

You will see from that point that the green USB LED from the Hackrf becomes active when the USB cable is connected, and the linux command lsusb should show a "Great Scott Gadgets HackRF One" device. If hackrf mode has been entered successfully, **then just follow the below process (I)**. If not, see more DFU information: Update firmware troubleshooting.

35.1.3 (3) If you have an H2+ (with integrated battery) device that does not detect any USB communication,

first try the steps under (2) above. If that does not work, then disassembling the HackRF from the PortaPack will be necessary to get the device into DFU mode. In that case, Dissassemble carefully both boards from the metal case box. Separate with maximum care to not bend or force so much the PCB's and the LCD Panel , both boards (Hackrf - Portapack) , It is a matter

of patience , step by step from both connectors , trying to keep coplanarity of both boards ,



(A) Once disassembled, pick up your hackrf board,

and connect it to USB. If it's not bricked, you will have a green LED when plugging USB (indicating HackRF mode is enabled). if it is bricked you will need to enter DFU mode (by holding in the DFU button while turning on power or attaching the USB cable). In DFU mode, the device will show up as an "NXP LPC" USB device. Execute the following command to get the device into HackRF mode:

```
dfu-util --device 1fc9:000c --alt 0 --download hackrf_one_usb.dfu
```

(Alternatively, run the `mayhem_flasher.bat` file file and select flash DFU mode, which performs the command above.)

You will see from that point that the green USB LED from the Hackrf becomes active when the USB cable is connected, and the linux command lsusb should show a "Great Scott Gadgets HackRF One" device. If hackrf mode has been entered successfully, **then just follow the below process (I)**. If not, see more DFU information: Update firmware troubleshooting.

35.2 Special Step process (I) to recover it (usually only needs to be done just once)

Assuming that you are here, with already in correct Hackrf mode (with green LED when connecting USB cable to the USB). If you have already done the steps below in the past, you can skip to just flashing the firmware to the latest version using HackRF mode; see Update firmware.

In case that you have H2+ Portapack with big CPLD QFP100: flash it with special fw jumbo77 1.43 first

In case that you have H1/H2+ Portapack with standard small CPLD QFP64: flash it with official Mayhem fw 1.43 first

If you came from case (3) above, assemble both boards again.

Then , (a) Confirm correct boot up of fw version 1.43 (special jumbo 1.4.3 for big CPLD QFP100, or official 1.4.3 for std CPLD QFP64)

(b) From that 1.43 fw ,with correct boot, put the device in "hackRF mode" and flash it again, but now with latest Mayhem firmware.

(c) At first power up, keep pressing the appropriate button for your unit for more than 2 secs, until getting correct LCD display, and it should work all correctly !

35.3 Explanation

Firmware starting at version 1.5.4 and onward contain the Pull Request 662 that uses the persistent memory to test and store the hardware and LCD config settings . That memory uses the same back up voltage than the Real Time Clock calendar, both needs a healthy cell battery button voltage. Sometimes, in a re flashing process , although we got good battery cell button voltage, the unit seems to be badly initializing those persistent bytes and we got strange black screen. Doing those above steps probably reset the persistent memory. That's just a guess.

35.3.1 Additional notes :

I used to have many frequent "black LCD boot brick" when exchanging binaries compiled with different gcc-arm… version , from 9.4 to 10.3 or 12 . But thanks to @u–foka's PR fix pmem -> make backup_ram_t data members volatile #1135, all those problems are gone , and now I do not have any persistent memory boot problems , so I do not need to go back to any old version 1.4.3 anymore.

35.4 H4 users

If you got any addon plugged in, try to remove it, and power off, wait a bit, and power on your device. If the addon board or module has problems, it can extremely slow down the boot, and the working. (usually happens when the I2C line goes wrong, and each operation have to wait for the timeout)

35.5 HackRF does not boot with sd-card inserted.

When turning on the HackRF with the sd-card inserted, it does not fully boot into the Mayhem firmware and is stuck on the following screen:

When removing the sd-card, the HackRF does fully boot into the Mayhem firmware.

Try a persistent memory reset (SETTINGS, P.Mem Mgmt, reset, reboot device) and check if SETTINGS/ SD Card is not having “enable high speed IO enabled”. This will fix the problem.

35.6 Config Menu

The Config Menu is a minimal pre-boot configuration interface that runs before the main firmware loads. It uses only the hardware buttons and LEDs — **the LCD screen is not used**.

35.7 Entering the Config Menu

The Config Menu activates automatically after **3 consecutive failed boot attempts** (crashes or resets before the firmware fully loads). To enter it intentionally, press the RESET button 3 times in quick succession — each time before the firmware finishes loading.

> **NOTE** Some PortaPack models do not have a working RESET button. On those, unplugging and replugging the USB cable counts as a reset, and doing so 3 times quickly enough will trigger the Config Menu.

> **NOTE** The Config Menu can be disabled entirely via **Settings → Config Mode**. If disabled, it will never activate automatically or manually.

When entered successfully, all three LEDs (TX, RX, USB) will blink rapidly (~8 times per second). At this point, press the DFU button once to display the currently active settings.

35.8 Available settings

Settings are displayed via the TX and RX LEDs. The USB LED is only used during the initial waiting phase and goes dark once the DFU button is pressed.

35.8.1 LCD driver (TX LED)

- TX LED: Off; Setting: Option 1: Automatic detection (default)

- TX LED: On; Setting: Option 2: LCD driver 1
- TX LED: Blinking fast; Setting: Option 3: LCD driver 2
- TX LED: Blinking slow; Setting: Option 4: LCD driver 2 QFP100 chip
- TX LED: Inverse blink slow; Setting: Option 5: LCD driver 1 QFP100 chip

NOTE The LCD driver can also be set at power-on using the directional buttons. See [[Won't-boot]] for details.

35.8.2 Disable external TCXO (RX LED)

- RX LED: Off; Setting: External TCXO enabled or autodetect (default)
- RX LED: On; Setting: External TCXO disabled

NOTE This setting can also be changed via **Settings → Radio**.

NOTE On HackRF R9 hardware, if the Config Menu is entered automatically due to boot failures, the external TCXO is temporarily disabled for the duration of the Config Menu session. This is because the TCXO is the most likely cause of the crash. The original setting is restored when you exit.

35.9 Switching between settings

Each press of the DFU button advances to the next combination of settings. Both the LCD driver and TCXO settings are cycled together — after **10 presses** you return to the original active combination.

WARNING Do not press the DFU button too quickly. Hold it down for at least **200ms** or the press may be ignored.

Each new selection is saved immediately to persistent memory. The device can be restarted at any time and the selected settings will be active.

35.10 Unwanted Config Menu activation

In some cases a PortaPack may enter Config Menu mode unexpectedly — for example when the power button is pressed rapidly multiple times, or if electrical noise occurs when a USB cable is attached (causing false resets). If this happens frequently (dark screen with blinking LEDs), the Config Menu can be permanently disabled via **Settings → Config Mode**, which sets a flag in persistent memory to skip the Config Menu entirely on all future boots.

35.11 Firmware upgrade

One of the main sources of problems while updating the firmware is the quality of the USB cable. Try with several ones just to be sure before trying any other solution.

35.12 Black or white screen?

If you get a black or white screen after updating, please check the wiki here: Won't boot

35.13 An application is not starting anymore?

Maybe an update in the application settings broke the settings file for the app. Just go in the SETTINGS folder at the root of the SD card and delete the related settings file. As an example lets take the audio app: if we follow the previously mentioned method, then we just go and delete the rx_audio.ini file in SETTINGS directory. This will force a new settings file to be generated with all new needed parameters at the next attempt to launch the app.

If an app is missing entirely, or greyed out, make sure that apps are placed in the APPS folder of the SD card and that their version matches that of the firmware. Many apps have been moved from flash ROM to the SD card to make room for new apps.

35.14 Update failed on Ubuntu/Mint/Ubuntu based distro?

For Ubuntu and Mint user: Ubuntu based distro never maintains their repo for hackrf package. You'll face a lot of weird problems if your hackrf is R9.
To resolve these, please compile hackrf package yourself, or use other distro.

35.15 DFU

This is a special mode to update the firmware in case of problems. To enable this, you should reset your device holding the RESET and DFU buttons at the same time, while doing this, release RESET, and then release DFU. The leds should be ON and the screen wont show anything.

Sometimes it's tricky to entering DFU mode, here are some ways to entering it if you have no luck with the method above:
W1. Press and holding DFU button, then plug the USB cable, then release the DFU button.
W2. Press and holding DFU button, then single press the knob, then release the DFU button, then plug the USB cable.
W3. Plug in USB. Press and hold DFU button and unplug USB. Release DFU button and plug USB back in.

35.15.1 Windows

If you are in Windows, from the release package double click `mayhem_flasher.bat` and follow the instructions to flash DFU. Do not disconnect or reset your PortaPack after that procedure, run mayhem_flasher.bat again and flash the mayhem firmware from the selection options.

35.15.2 MacOS

DFU Utils CLI tools for MacOS available through MacPorts or Homebrew

1. If necessary, install the DFU tools: `brew install dfu-util`
2. Connect the device via USB
3. Switch to DFU mode as per the section above: *DFU*
4. Upload the firmware with `dfu-util --device 1fc9:000c --download hackrf_one_usb.dfu`
5. Do not disconnect or reset your PortaPack after that procedure, and run the following code: `hackrf_spiflash -w your_firmware_file.bin`

35.15.3 Linux

DFU Utils CLI tools for Linux available in standard repositories

1. If necessary, install the DFU tools (example for Debian/Ubuntu variants): `sudo apt install dfu-util`
2. Connect the device via USB
3. Switch to DFU mode as per the section above: *DFU*
4. Upload the firmware with `dfu-util --device 1fc9:000c --download hackrf_one_usb.dfu`
5. Do not disconnect or reset your PortaPack after that procedure and run the following code: `hackrf_spiflash -w your_firmware_file.bin`

35.15.4 Alternative environment

You may be able to try in a virtual environment, completely isolated from your current OS:

1. Download http://eu2-dist.gnuradio.org/ and burn the ISO and boot from it, select the LIVE Image
2. Once in the environment, Open a Command Prompt and type: `sudo apt-get install dfu-util`
3. Connect your device to your computer
4. Press RESET and DFU buttons at the same time, and while doing this, release RESET, and then release DFU
5. Open a terminal and type `dfu-util --device 1fc9:000c --download hackrf_one_usb.dfu`
6. After that, without disconnecting it, you can upload the firmware with `hackrf_spiflash -w new_firmware_file.bin`

35.16 Notes

If you have an existing HackRF One and you've acquired a Portapack separately (from third party suppliers), do the following:

1. Make sure you flash the HackRF One first before attempting anything with the HackRF One + Portapack combo. The code to run Portpack comes from the HackRF One firmware as flashed.
2. If you don't flash the HackRF One and just plugged in the Portapack and powered up the setup, the screen on the Portapack will flash and nothing else will be seen. This does not mean that the Portapack is faulty.
3. After flashing the HackRF One, you can plug in the Portapack and when the setup is powered up. you will see the screen show the UI for the version of the firmware that was flashed into the HackRF One.

35.17 Video

The following video explains all possible outcomes while trying to update the firmware and possible mitigations: https://www.youtube.com/watch?v=_zx4ZvurgOs

There is another very good video about the general overview, also including the update procedure: https://www.youtube.com/watch?v=kjFB58Y1TAo

36 Still stuck?

If you still experience problems after reading this and checking the video, submit an issue.

Also, there is a lot more details on the HackRF wiki.

36.1 Diagnose firmware update in Windows

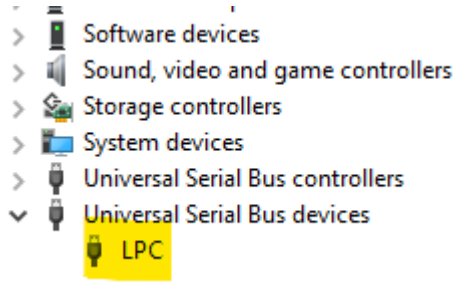
If you cant get firmware to update/install on your device do these steps.

Try boot into DFU mode by:

1. Plug device into computer
2. Press and HOLD the DFU button, While still holding the DFU button, press and release the reset button (WHILE STILL HOLDING THE DFU BUTTON).

   Note : in some particular H2 (ex H2R4 with battery), reset button does not make a proper reset , it just freeze the screen , and do not allow you to enter in DFU . In that case, unplug it from the computer, switch off the unit with two short pushes to the big encoder knob button (power off). And then keep pressing DFU button , a short push button to the encoder knob button (power on) leads you into a proper DFU Mode. And now you can plug it through USB cable to the computer, and continue with process.

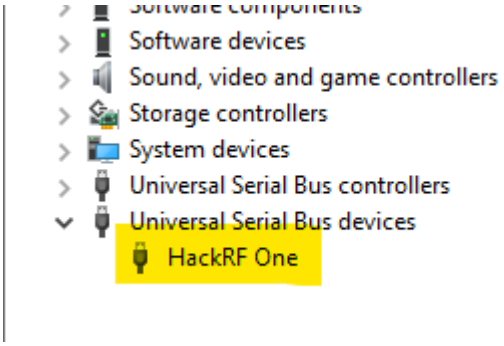
3. Check Windows device manager and see if you have LPC showing up



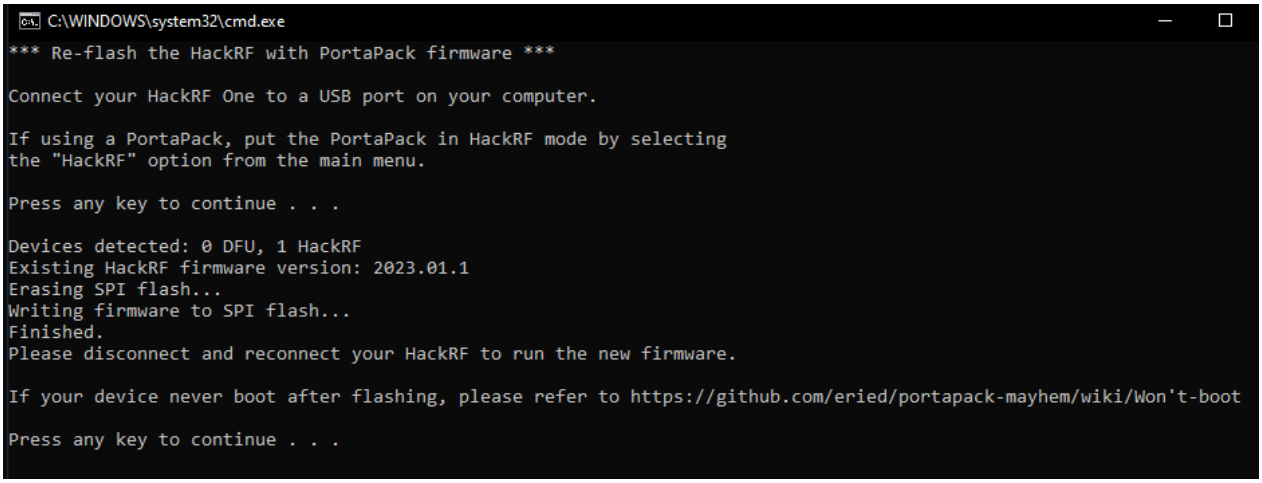
4. If it shows up then make sure you download the latest version of **mayhem_vx.x.x_FIRMWARE.zip** from GitHub here https://github.com/portapack-mayhem/mayhem-firmware/releases/latest. You can then run “mayhem_flasher.bat”. Now double check the file name as they look quite similar.
5. Once it opens, press enter on your keyboard and you should see the following:

```
C:\WINDOWS\system32\cmd.exe
*** Run HackRF firmware in RAM via LPC DFU ***

This is used to "unbrick" your HackRF, if you are no longer able to use
HackRF tools to flash or operate your HackRF.

Connect your HackRF One to a USB port on your computer.

Hold down both the DFU and RESET buttons on the HackRF.
Then release the RESET button (closest to the edge).
Then release the DFU button.

Press any key to continue . . .

dfu-util 0.9

Copyright 2005-2009 Weston Schmidt, Harald Welte and OpenMoko Inc.
Copyright 2010-2016 Tormod Volden and Stefan Schmidt
This program is Free Software and has ABSOLUTELY NO WARRANTY
Please report bugs to http://sourceforge.net/p/dfu-util/tickets/

DFU suffix CRC does not match
A valid DFU suffix will be required in a future dfu-util release!!!
Opening DFU capable USB device...
ID 1fc9:000c
Run-time device DFU version 0100
Claiming USB DFU Interface...
Setting Alternate Setting #0 ...
Determining device status: state = dfuIDLE, status = 0
dfuIDLE, continuing
DFU mode device DFU version 0100
Device returned transfer size 2048
Copying data from PC to DFU device
Download        [=========================] 100%        42564 bytes
Download done.
unable to read DFU status after completion
can't detach
Resetting USB to switch back to runtime mode

Press any key to continue . . .
```

6. Once that is done, you need to check device manager and see if you can now see a new device showing up



7. If this device is showing up, then you can now run “mayhem_flasher.bat” and follow the on screen instructions. You should now see the following:



8. Once that is done, you should be able to press the reset button and your device will boot up. If not, then please refer to the wiki here: [Won't boot](https://github.com/portapack-mayhem/mayhem-firmware/wiki/Won't-boot

36.2 Unknown device

If you see **Unknown device** then you will need to install your device drivers.

36.2.1 1. Download Zadig for USB driver installation

Go to the Zadig website and download the latest version of Zadig. Zadig is a tool that allows you to install the necessary drivers for USB devices like the HackRF.

36.2.2 2. Connect your HackRF device

Plug in your HackRF device into a USB port on your computer.

36.2.3 3. Run Zadig

Open Zadig as an administrator (right-click and choose "Run as administrator"). In Zadig, go to Options > List All Devices.

36.2.4 4. Select the HackRF device

From the dropdown list, select HackRF One or HackRF. If you can't find it, look for USB devices and match the HackRF device by its USB ID (usually it's something like 1d50:6089).

36.2.5 5. Install WinUSB driver

Once you have selected the HackRF device, set the driver to WinUSB (this is usually the default driver). Click Install Driver or Replace Driver if there is already a driver installed. Wait for the installation to complete.

36.2.6 6. Verify Installation

After the driver installation, you can confirm that the HackRF device is working by using tools like SDR# (SDRSharp) or GNU Radio. Download SDR# from AirSpy to test your HackRF and ensure it's detected properly.

36.2.7 7. Install HackRF Tools (Optional)

If you want to use the HackRF command-line utilities, you can download the official HackRF tools from the Great Scott Gadgets GitHub. Extract the tools and add the folder to your system's PATH to use the commands like hackrf_info, hackrf_transfer, etc. After following these steps, your HackRF device should be ready to use on Windows 11.

36.3 Receive Quality Issues

37 Receive Quality Issues

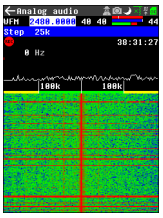
37.1 Not receiving anything or Noisy waterfall

37.1.1 Antenna

Is the correct antenna connected, and plugged in securely? An incorrect or missing antenna is the most common cause of poor reception. See [[Antennas]] for guidance on antenna selection and matching.

37.1.2 Gain settings

Gain controls appear in the same layout across all receiver apps: **LNA** on the left, **VGA** on the right.



LNA (Low Noise Amplifier): Adjusts the front-end gain before the signal is mixed down. Valid values are `0, 8, 16, 24, 32, 40` dB (steps of 8). The LNA attempts to improve signal-to-noise ratio; values of 24–32 are a good starting point for most signals.

VGA (Variable Gain Amplifier): Adjusts the baseband gain after mixing. Valid values are `0` to `62` dB in **steps of 2** (even values only). It amplifies the entire received signal including noise; values between 8–20 are typical for most use cases.

> **NOTE** Setting gain too high causes the ADC to clip, which introduces noise and distortion across the entire waterfall — the same effect as audio overload. If the waterfall looks uniformly noisy or "washed out", try reducing both LNA and VGA.

37.1.3 RX amp

The hardware RX pre-amplifier (AMP button) adds approximately 14 dB of additional gain ahead of the LNA. It is useful for very weak signals but is easy to overdrive and can be damaged if a strong signal is present on the antenna while active. Use it with caution.

37.1.4 Frequency correction

If you can see a signal on the waterfall but it appears offset from where it should be, check **Settings → Radio → Freq. Correct** (correction in PPB). Most clone HackRF/PortaPack units have a small frequency offset. The [[Tuner]] app can help you measure this offset against a known-accurate reference.

37.1.5 Software filters

Mayhem's receiver apps include selectable IF bandwidth filters (the bandwidth field next to the modulation selector). Choosing a bandwidth that matches your signal reduces adjacent-channel interference and noise. Using too wide a bandwidth lets in more noise than necessary.

External hardware filters placed between the antenna and the SMA port can also help significantly:

- **Band-pass filters** pass only a specific frequency range and reject everything outside it. Useful when you want to isolate a specific band.
- **Low-pass filters** reject high-frequency signals and harmonics above a cutoff frequency.
- **High-pass filters** reject low-frequency signals and DC components below a cutoff.
- **Notch filters** (band-stop) reject a specific narrow frequency — useful for eliminating strong local interference such as an FM broadcast station.

37.2 Noise sources

37.2.1 Nearby strong transmitters

High-powered transmitters nearby — FM broadcast stations, trunking repeaters, pager networks — can overload the HackRF's RF front end and create broadband noise across the entire waterfall. An FM-band block filter (88–108 MHz) is a common and inexpensive fix if FM overload is suspected.

37.2.2 Local interference sources

Many common devices generate wideband noise: computer monitors and screens, USB hubs, switching power supplies, poorly shielded cables, and faulty wiring. To isolate the cause, try unplugging devices one at a time, or simply step outside — outdoor reception is often noticeably cleaner.

37.2.3 Power banks

Many USB power banks generate significant RF noise. If you are using one, try a different power bank or a direct mains adapter.

37.2.4 H2 internal battery charger

Some H2 model PortaPacks have been reported to generate noise when the USB cable is connected, caused by the internal battery charger circuit. This is a hardware issue specific to certain H2 units. If you suspect this, try operating on battery only (no USB connected) and compare the waterfall.

> **NOTE** This issue applies to the H2 model specifically. It is not known to affect H1, H2+, or H4M models.

37.2.5 Cables and grounding

Use well-shielded coax cables. Poorly shielded cables can act as antennas for local noise. Ensure the device and any connected equipment share a common ground where appropriate.

37.3 Intermittent signal loss or noise

37.3.1 Loose connections

Wiggle the antenna and any coax connectors while watching the waterfall. A dramatic change in the noise floor or signal level indicates a loose or failing connection.

37.3.2 Nearby intermittent transmitters

Pager networks in particular transmit intermittently and at high power, which can cause brief overloads. Nearby trunking repeaters can have the same effect.

37.3.3 H2 internal battery charger (intermittent)

The same H2 charger noise issue described above can also occur intermittently, appearing only when the charging circuit is active. See the section above for details.

37.4 Broken RF chain

If none of the above steps help and the device receives nothing at all, the RF front end may be damaged — typically the TX amplifier (a component easily damaged by transmitting into an open or mismatched antenna). The best course of action is to open an issue on the HackRF GitHub with a description of the problem. If you have a genuine Great Scott Gadgets HackRF One, a replacement may be possible.

37.5 TX Carrier Only

If your portapack is only transmitting a carrier wave, then this could be an indication of an outdated CPLD image. See Updating the Xilinx CPLD on hackrf board for how to update the CPLD.

37.6 Dead Coin Cell Battery

The coin cell battery is used to preserve settings stored in persistent memory (P.Mem), as well as to power the RTC clock. The smaller coin cell battery used on the PortaPack H2 does not last long and replacing the coin cell can be a chore, depending on the case style.

To partially work around a dead or missing coin cell battery, persistent memory settings can be configured to save settings to the SD card (see Settings -> P.Memory Mgmt). This allows all settings to be preserved, but does not resolve the issue with not keeping time.

Those with an H2 model and soldering skills might want to consider adding a larger CR2032 battery holder such as the one below (attached with double-stick foam tape), to lengthen the time that the coin battery lasts by about 6X:

37.7 Factory Defaults

38 Factory Defaults

38.1 Why reset to factory defaults

After a firmware update, some settings may be stored in an incompatible format, causing apps to malfunction or the device to behave unexpectedly. Resetting to factory defaults clears these settings and returns the device to a known-good state.

There are two separate stores that may need to be reset:

- **Persistent memory (P.Mem):** Stores device-wide settings such as frequency, corrections, UI options, and radio configuration. Held in battery-backed RAM (requires a working coin cell battery).
- **SETTINGS directory on SD card:** Stores per-app settings saved by the "App Load and Save" feature.

38.2 Resetting persistent memory (P.Mem)

38.2.1 Via the Settings menu

Navigate to **Settings → P.Memory Mgmt**. The following options are available:

- Button or Control: Use SD card for P.Mem; Function: Store P.Mem on SD card instead of coin-cell-backed RAM (useful if coin cell is dead or missing)
- Button or Control: Save P.Mem to SD card; Function: Back up current P.Mem contents to SD card
- Button or Control: Load P.Mem from SD Card; Function: Restore P.Mem from a previously saved SD card backup
- Button or Control: Reset P.Mem to defaults; Function: Reset all persistent memory settings to factory defaults

Pressing **Reset P.Mem to defaults** opens a confirmation dialog:

> *"This will reset the P.Mem to default settings."*

Confirm with **Yes** to apply the reset. The status bar refreshes immediately.

⚠️ ! if you are using "Use SD card for P.Mem" option because your coin battery is dead or absent, don't forget to hit "Save P.Mem to SD card" after clicking on "Reset P.Mem to defaults", else you are going to have old settings at reboot ! ⚠️

38.2.2 Via button combination at boot

If the device is in a state where it cannot be navigated normally, persistent memory can be bypassed at startup. Hold the **Left** and **Right** buttons simultaneously while powering on or pressing reset. The device will boot as if P.Mem is empty, loading factory defaults without modifying the stored contents.

> **NOTE** This bypass is temporary — P.Mem is not permanently cleared by this method. To permanently reset, use the Settings menu as described above.

38.2.3 Automatic reset

P.Mem is reset to defaults automatically in two cases:

- After a firmware upgrade or downgrade that changes the P.Mem data structure version
- When the coin cell battery is discharged (P.Mem loses power and becomes invalid)

38.3 Resetting SD card settings

38.3.1 SETTINGS directory

Delete all files under the `SETTINGS/` directory on the SD card. This resets all per-app settings saved via "App Load and Save", as well as the P.Memory Mgmt configuration (e.g. the SD pmem flag file). It does **not** reset the persistent memory itself.

> **NOTE** Updating the SD card content to the latest release is a good opportunity to do this cleanup. The firmware does not automatically remove old or incompatible settings files.

38.4 Full reset procedure

For a complete reset (both P.Mem and SD card settings):

1. Remove the SD card and connect it to a computer.
2. Delete all files under `SETTINGS/`.
3. Optionally update the SD card content to the latest release.
4. Reinsert the SD card.
5. Boot the device without the SD card first (hold the boot key if required), go to **Settings → P.Memory Mgmt**, and press **Reset P.Mem to defaults**.
6. Shut down, reinsert the SD card, and boot normally.

The device will now behave as if freshly set up.

> **NOTE** The "Debug Dump" app (under **Settings → Debug → Debug Dump**) can generate a file listing all current persistent memory settings — useful for diagnosing issues before resetting.

38.5 SD card not recognized by PC with the SD-card over USB selected

If the SD card is not recognized by your PC after starting [[SD-Over-USB|SD Over USB]], the cause is usually a wrong or missing USB driver on Windows, or simply not waiting long enough for the drive to appear.

38.6 First: wait up to 20 seconds

After pressing **Run** in the SD Over USB app, it can take **up to 20 seconds** for the drive to appear on the PC. Wait before assuming something is wrong.

38.7 Windows: wrong driver assigned

Windows sometimes assigns the HackRF's USB serial driver instead of the USB Mass Storage driver when the device switches modes. To fix this:

1. Open **Device Manager** (right-click the Start button → Device Manager).
2. Under **Universal Serial Bus devices** or **Other devices**, find the HackRF One entry.
3. Right-click it and select **Update driver**.
4. Select **Browse my computer for drivers**.
5. Select **Let me pick from a list of available drivers on my computer**.

6. Select **USB Mass Storage Device** from the list.
7. Click **Next** and confirm.

The SD card should now appear as a removable drive in File Explorer.

> **NOTE** You may need to repeat this step after rebooting or reconnecting the device, as Windows may revert to the previous driver assignment.

38.8 Linux and macOS

On Linux and macOS, USB Mass Storage is handled natively by the kernel and no driver installation is needed. If the drive does not appear, check that the SD card is properly inserted in the PortaPack and is not corrupted. If the SD card has filesystem errors, the host OS may fail to mount it — try formatting the card as FAT32 and restoring the SD card contents from a fresh release package.

38.9 General checks

- Make sure the SD card is inserted in the PortaPack before starting the app.
- Use a USB cable that supports data transfer, not a charge-only cable.
- Try a different USB port, preferably directly on the PC rather than through a hub.
- If the drive appears but shows errors, the SD card filesystem may be corrupted — see [[How-to-Format-SDCard]].

38.10 DFU overlay

While Mayhem firmware is running, a short press of the DFU button cycles through two diagnostic overlays and then hides them again:

- DFU press: 1st; Result: Show Performance overlay
- DFU press: 2nd; Result: Show Radio Settings overlay
- DFU press: 3rd; Result: Hide overlay

The overlay is drawn on top of the current screen without leaving the active app. Values update approximately once per second.

> **NOTE** The fields shown in both overlays are defined directly in the firmware source (`ui_dfu_menu.hpp`) and depend on the shared memory structure between the two processors (`portapack_shared_memory.hpp`). They may change between firmware versions if either file is modified. The table below reflects the current source code.

38.11 Overlay 1 — Performance

Displays CPU and memory diagnostics for both processors.

- Field: M0 core; Description: M0 free core memory (bytes)
- Field: M0 heap; Description: M0 free heap memory (bytes)
- Field: M0 frags; Description: M0 heap fragment count
- Field: M0 stack; Description: M0 free stack space (bytes)
- Field: M0 cpu %; Description: M0 CPU utilisation (%)
- Field: M4 heap; Description: M4 heap usage (bytes)
- Field: M4 stack; Description: M4 stack usage (bytes)
- Field: M4 cpu %; Description: M4 CPU utilisation (%)
- Field: M4 miss; Description: M4 baseband buffer miss count
- Field: Uptime; Description: Device uptime (seconds)

38.12 Overlay 2 — Radio Settings

Displays the current live radio configuration for both receiver and transmitter models.

- Field: RX Freq; Description: Receiver target frequency (Hz)
- Field: RX BW; Description: Receiver baseband bandwidth (Hz)
- Field: RX SampR; Description: Receiver sampling rate (Hz)
- Field: RX Satu%; Description: M4 saturation percentage
- Field: Modulatn; Description: Modulation index

- Field: AM cfg; Description: AM configuration index
- Field: NBFM cfg; Description: Narrowband FM configuration index
- Field: WFM cfg; Description: Wideband FM configuration index
- Field: TX Freq; Description: Transmitter target frequency (Hz)
- Field: TX BW; Description: Transmitter baseband bandwidth (Hz)
- Field: TX SampR; Description: Transmitter sampling rate (Hz)

38.13 Using the DFU button to flash firmware

The DFU button serves a second, separate purpose: entering DFU (Device Firmware Upgrade) mode for flashing the HackRF base firmware from a PC. This only works **before** the Mayhem firmware has fully loaded — pressing DFU while the firmware is already running will only toggle the overlay described above.

To enter DFU mode for flashing, try one of these methods while the device is not yet booted into Mayhem:

- Hold **DFU**, plug in the USB cable, then release **DFU**.
- Hold **DFU** and **RESET** simultaneously, release **RESET** first, then release **DFU**.
- Hold **DFU**, power on the device (press the knob on H1/H2, or use the power switch on H4M), release **DFU**, then plug in USB.

When in DFU mode, the LEDs will be on and the screen will remain blank.

See [[Full-reset]] and [[Update-firmware-troubleshooting]] for full flashing instructions.

38.14 Full reset

A full reset reflashes both the HackRF firmware and the Mayhem firmware from scratch, and reinstalls all SD card contents. Use this as a last resort when other troubleshooting steps have failed.

WARNING This process erases all settings and SD card contents. Back up any important files first.

38.15 Steps

1. Back up any important files from your SD card to a computer.
2. Download the latest release from the Releases page. The `mayhem_vX.X.X.ppfw.tar` file contains both the firmware and all SD card contents in a single archive.
3. Remove the SD card and the coin cell battery from the PortaPack. Keep them somewhere safe and ensure they do not touch each other.
4. Enter DFU mode on the HackRF. Try each method in order until one works — the LEDs should light up and the screen will remain blank:
   - Hold RESET and DFU simultaneously, release RESET first, then release DFU.
   - Hold DFU, plug in the USB cable, then release DFU.
   - Hold DFU, power on the device (press the knob on H1/H2, or use the power switch on H4M), release DFU, then plug in USB.
   - Plug in USB, hold DFU and unplug USB, release DFU, plug USB back in.
5. Flash the HackRF base firmware using DFU:
   - **Windows:** Double-click `mayhem_flasher.bat` from the release package and follow the prompts. Do not disconnect or reset the device afterwards.
   - **Linux or macOS:**
     1. Install `dfu-util` via your package manager (e.g. `sudo apt install dfu-util`, `brew install dfu-util`, or `pacman -Sy dfu-util`).
     2. Extract the release package, open a terminal in that directory, and run:

        ```
        dfu-util --device 1fc9:000c --download hackrf_one_usb.dfu
        ```

   3. Do not disconnect or reset the device afterwards.
6. Flash the Mayhem firmware — **without disconnecting the device from step 5**:
   - **Windows:** Run `mayhem_flasher.bat` again and follow the prompts.
   - **Linux or macOS:**

6. Install the `hackrf` host tools (e.g. `sudo apt install hackrf`, `brew install hackrf`, or `pacman -Sy hackrf`).

   Note

   Ubuntu's stable repository may ship an outdated `hackrf` package. If you encounter errors, consider compiling the hackrf host tools from source or using a rolling-release distribution.

7. Extract the firmware package and run (adjust the filename to match the version you downloaded):

   ```
   hackrf_spiflash -w portapack-h1_h2-mayhem.bin
   ```

7. Prepare the SD card:

   7. Format the SD card as FAT32 on your computer. See [[How-to-Format-SDCard]] for instructions.
   8. If you downloaded the `ppfw.tar` file, extract it and copy all contents onto the SD card. If you downloaded the separate `COPY_TO_SDCARD.zip`, extract and copy that instead.
   9. Do not insert the SD card yet.

8. If the device still fails to boot at this point, follow the [[Won't-boot]] steps.
9. Power off the device, wait at least 1 minute, then insert the SD card and reinstall the coin cell battery.
10. Power on the device, go to **Settings → P.Memory Mgmt → Reset P.Mem to defaults → YES**, then power off and wait at least 1 minute.
11. If the device still does not boot correctly, repeat the [[Won't-boot]] steps (the boot configuration requires both the SD card and coin cell battery to be present to save correctly).
12. Power on — the full reset is complete.

38.16 Troubleshooting notes

- If the device is not recognized by your computer no matter what you try, try a different computer running Linux, and try at least 5 different USB cables.
- If DFU flashing fails on Linux, check that `dfu-util` is up to date.
- As an optional additional step, you can update the Xilinx CPLD on the HackRF board: [[Updating-the-Xilinx-CPLD-on-hackrf-board]].
- If the problem you were trying to resolve persists after a full reset, open a GitHub issue.

38.17 SolveBoard

39 SolveBoard

A quick problem → solution lookup table. For the full list of troubleshooting articles, see Troubleshooting.

39.0.1 Does not boot or white screen or black screen

Check Won't boot and Config Menu.

39.0.2 Only turn on once after flashing

Flashing with DFU only make the firmware go to RAM. You need to flash to the SPI flash once again(with regular flash methods)

39.0.3 Crash or Guru Meditation.

Try Full reset. If still does not work, try Updating the Xilinx CPLD on hackrf board. If nothing works, report an issue on GitHub.

39.0.4 Freeze when launch specific app (but no crash or Guru Meditation)

It's mostly from power part. Try: -Full charge -Change battery -Plug into a reliable power source -Change battery -Change at least 5 cables -Remove add-on board if plugged in

39.0.5 No splash screen or No time & date or Touchscreen does not work or Status bar icon disappeared or Backlight timeout abnormal or Encoder dial abnormal

Check settings.

39.0.6 It says xxx missing or can't find xxx or xxx read error or xxx is outdated or xxx empty or you can't see xxx.

Make sure sdcard contents matches your firmware version.

39.0.7 Apps are missing or greyed out after upgrading.

Make sure sdcard contents matches your firmware version, particularly the external APPS folder, and make sure you correctly put the sdcard contents.

39.0.8 `hackrf_info` says `Hardware does not appear to have been manufactured by Great Scott Gadgets`

It's because you have a Hackrf clone instead of original from GSG. If you believe it's a mistake, contact the seller or GSG official support.

39.0.9 Unexpected spike on my waterfall or IQ or spectrum.

It's either a DC offset or noise from power. For reducing the noise, try using the internal battery. For DC offset, there's nothing you can do, however it's normal on SDR.

39.0.10 My computer can't see my HackRF

Try at least 5 different cables. If you are on Windows, try to install drivers in release package. Also try your luck with Zadig. If not working, try on a Linux machine (exclude Ubuntu)

39.0.11 My sound board app can't see the files i put in, but my fileman app can

follow Soundboard to make correct file

39.0.12 Screen went black when transmit or replay

Check Stealth Mode

39.0.13 Fail or freeze when recording or Can't see files even if I correctly put sdcard content

Make sure your sdcard is FAT32 format, and make sure it's a branded one with acceptable quality

39.0.14 My GPS-Simulation Doesn't work. / My bluetooth doesn't work.

Most modern devices use cellular to verify GPS. Make sure your device is vulnerable to gps-sim. Go to Settings - Radio, and make sure the "Reference Source" says "External" or "portapack". If not, then check the last checkbox and see if it's getting better.

39.0.15 Charging is very slow

Check the following things:

- Try at least 5 cables.
- Use a good adapter, for example a 5V1A **branded** adapter
- Don't use computer or laptop or similar thing to charge.

39.0.16 How to turn on/off

One of these may turn on and another may turn off:

- Single press the knob
- Double press the knob
- Long press the knob
- Plug in USB cable
- Unplug USB cable

39.0.17 Replay app shows empty list

You didn't press correct button. Check wiki

39.0.18 Issues related to sdcard

- use BRANDED sdcard from reliable seller.
- Use FAT32 format
- correctly put sdcard content which matches your firmware versions

39.0.19 DFU suffix CRC doesn't match or Cannot open DFU device

If you are on Windows, try to install drivers in release package. Also try your luck with Zadig. If not working, try on a Linux machine (exclude Ubuntu)

39.0.20 Screen backlight never turned off automatically even I correctly set the timeout count

Auto set threshold, follow this: https://github.com/portapack-mayhem/mayhem-firmware/wiki/Settings#touchscreen-threshold

39.0.21 Issue that related to dial/knob/encoder

Go to setting, try all the dial sensitivity and rate multiplier

39.0.22 "FLASH ERR" on my screen

It means you didn't correctly flash the firmware, for example you unplugged your device while it was still flashing the firmware. If you are sure you did everything correctly but it still says that, then your SPI flash on your hacked board may be broken.

39.0.23 Something else

Check more *Troubleshooting* topics on the sidebar

39.0.24 I don't like xxx apps

check here

39.1 How to Format SDCard

PortaPack NEEDS FAT32 or exFAT format for SDCard! Here's how you can format them with your PC.

40 Linux

Usually all the Desktop Environment provide it's own GUI partition tool. for example GPartition or KDE Partition Manager. If you wanna use CLI way:

1. Identify the SD card device with `sudo fdisk -l`.
2. Unmount the SD card with `sudo umount /dev/sdX1`.
3. Format the SD card with `sudo mkfs.vfat -F 32 /dev/sdX1`.
   - if `mkfs.vfat: command not found` install dosfstools dependency (eg: `sudo pacman -S dosfstools`)
4. (Optional) Label the SD card with `sudo mkfs.vfat -F 32 -n "SDCARD" /dev/sdX1`.
5. Mount the SD card with `sudo mount /dev/sdX1 /mnt`.

41 macOS

Just use Disk Utility app.

42 Windows

- If your sdcard capacity is smaller or equal than/to 32G, just right-click the sdcard driver in Explorer
- If your sdcard capacity is greater than 32G, Download a free partition manager (for example Disk Genius) and format with it.

43 I DON'T KNOW WHAT ARE Windows/ Linux/macOS

Just follow the Windows chapter.

43.1 What if I don't like some of the apps

Mayhem firmware comes with a large collection of apps. There are three ways to manage apps you don't want to see.

43.2 Option 1: Hide apps with the App Manager (recommended)

Go to **Settings → App Manager** to hide individual apps from the menu. Hidden apps remain on the SD card and can be re-enabled at any time. This is the easiest and fully reversible method.

Alternatively, you can directly edit the blacklist config file — see [[Settings#blacklist]].

> **NOTE** Hidden apps take up no RAM or CPU while not in use. External apps stored on the SD card each occupy less than 10 KB of storage, so there is no meaningful space benefit to deleting them. Games

and other non-radio apps have no impact on core radio functionality — if you don't want to see them, simply hide or delete them.

43.3 Option 2: Delete the app file from the SD card

If you want to remove an app entirely, delete its file from the `APPS/` folder on the SD card. External apps use two file extensions:

- `.ppma` — standard external apps (must match the current firmware version)
- `.ppmp` — standalone external apps (version-independent)

You can access the SD card by physically removing it, or via the [[SD-Over-USB|SD Over USB]] app.

WARNING Deleted app files are not restored by firmware updates. To get a deleted app back, you will need to copy it from a fresh release package.

43.4 Note on built-in apps

Some apps — particularly the built-in ones compiled directly into the firmware — cannot be removed without recompiling the firmware. However, all of them can be hidden via the App Manager.

44 Applications

44.1 Applications

44.2 Receivers

44.3 2-Tone-RX

44.4 Introduction

Receives Motorola or QCII style two-tone paging sequences from FM voice channels and logs detected tone pairs on screen. The app can optionally require a matching CTCSS tone before detection starts, which helps reduce false detections on shared or noisy channels.

This is an external app. The `.ppma` file must match the currently running firmware version.

44.5 Controls

- Frequency: Sets the receive frequency. It can be adjusted with the encoder, entered directly, or loaded from saved frequencies in the normal Mayhem way.
- RF amplifier (0 or 1): Enables or disables the internal RF amplifier.
- LNA gain: Sets the LNA gain.
- VGA gain: Sets the VGA gain.
- Volume: Sets the audio output volume.
- CTCSS: Selects a required CTCSS tone. Set it to `None` to detect pages without CTCSS gating.
- Sq: Sets the squelch level used by the app.
- Start or Stop: Starts or stops the receiver and tone detector.
- Clr: Clears the detected tone pair history shown on screen.
- Bias-T: Enables Bias-T power for an external LNA while the app is running, then restores the previous antenna bias state when the app stops.

The status line shows the current detection phase:

- `T1 ...` means the app is currently collecting the first tone.
- `T1:xxxx.xHz xxxxms T2...` means the first tone has been accepted and the app is collecting the second tone.
- `T1:xxxx.xHz xxxxms T2 ?` can briefly appear when the second tone has a weak or noisy dropout window.

The tone log stores completed tone pairs in the order they were received. New entries are added below older ones. If more entries are collected than fit on screen, the list can be scrolled.

Each line shows:

- Tone A frequency and duration

- Tone B frequency and duration

Example:

```
584.8Hz 1000ms 810.2Hz 3040ms
```

44.6 Notes

- This app is designed for Motorola or QCII style pages, commonly using approximately 1 second for Tone A and 3 seconds for Tone B.
- The app finalizes a page either when the tone detector reports the end of the tone sequence, or when a valid long Tone B transitions directly into dispatch voice audio without the RF carrier dropping.
- For debugging, the app can write detailed trace output to `DEBUG/TWOTONERX.TXT` on the SD card, but the normal UI log only shows completed tone pairs.

44.7 Practical Tips

- Real-world paging channels are often weak or noisy. Best results usually come from using the correct antenna, appropriate gain settings, and any needed external filtering.
- If the monitored system uses CTCSS, selecting the correct tone can reduce false detections significantly.
- External apps in Mayhem are version-sensitive. If the app hard faults or behaves strangely after an update, make sure the firmware and the `2-Tone RX.ppma` were built from the same firmware version.

44.8 ACARS

ACARS (Aircraft Communications Addressing and Reporting System) is a digital data link protocol used to exchange short messages between aircraft and ground stations over VHF radio. This app decodes ACARS frames and displays the extracted fields in a scrolling console.

This is an external app (`.ppma` file on the SD card).

45 Settings

- **Frequency**: Default **131.825 MHz** (primary ACARS VHF frequency in Europe and globally). Adjustable via encoder knob or tap for keypad. Other common frequencies include 129.125 MHz and 130.025 MHz depending on region and airline.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain.
- **LOG**: Checkbox to enable logging to file. When checked, decoded messages are written to `LOGS/ACARS.TXT`. The log file is opened automatically on startup; the checkbox controls whether entries are written.

46 Decoded Message Fields

Each received and CRC-verified ACARS frame is decoded into the following fields, displayed in the console:

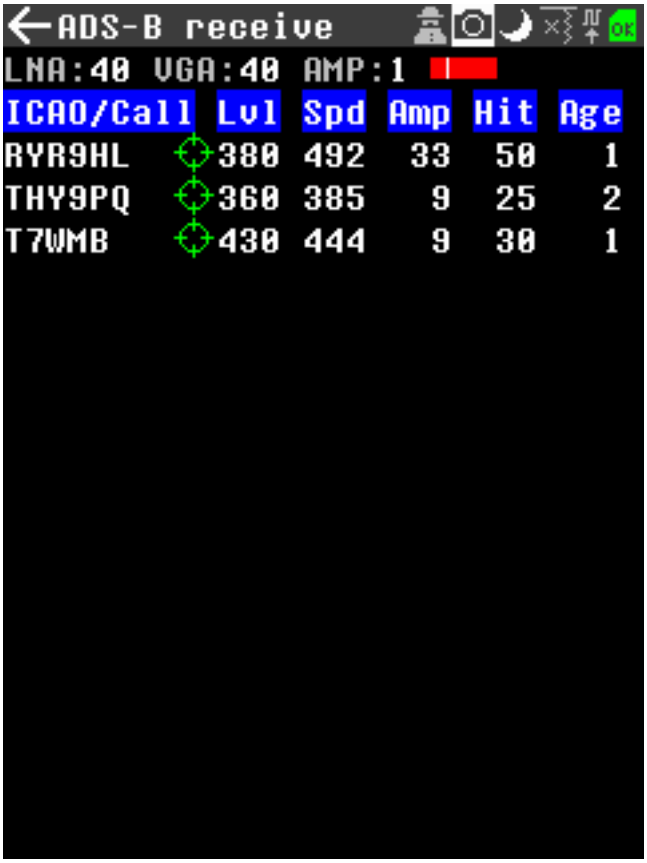
- **CRC**: `OK` if the CRC-16/CCITT checksum over the frame matches the two trailing CRC bytes; `FAIL` otherwise. Frames with CRC failures are still displayed but should be treated as potentially corrupt.
- **Registration**: 7-character aircraft registration (tail number), e.g. `D-ABCD` or `N12345`.
- **Label**: 2-character message type code identifying the category of message (e.g. flight plan update, ATIS, position report).
- **BlockID**: 1-character sequence counter for multi-block messages.
- **MsgNum**: 3-character message number within the current session.
- **FlightID**: 6-character flight identifier, typically the IATA flight number.
- **Message**: Free-text payload of variable length. Content depends on the label type and airline ground system.

A minimum valid frame must be at least 23 bytes (21-byte fixed header + 2 CRC bytes). Frames shorter than this are shown with an error message in the text field instead of decoded content.

47 Logging

When **LOG** is enabled, each decoded message is written as a single timestamped entry to `LOGS/ACARS.TXT`. The file is always opened on launch; entries are only written when the checkbox is active. Settings are saved to `rx_acars.ini`.

47.1 ADS-B



NOTE This is an external app and must be present on the SD card in the `APPS/` folder.

48 Introduction

With this app you can receive real-time ADS-B used for air traffic control.

49 Functionality

Select a received aircraft and more detail will be shown. Select “Show on map” to see the aircraft location. ADS-B operates on 978 to 1090 MHz. Even a short antenna attached directly to the Portapack will receive ADS-B when the receive preamp is enabled (“1” displayed).

49.1 Map

For map functionality a world map must be loaded on the SD card. The map image and other SD card content is available from the portapack-mayhem github download page. The map can be zoomed in or out by moving the cursor down to the map region and turning the encoder dial (resolution is still limited by the map file, but zooming in helps to separate overlapping plane icons, and zooming out may help if you have really good reception). The map position can also be changed by modifying the latitude and longitude values.

49.2 Airlines

49.2.1 Log into microSD

The ADSB RX app will log each frame including the following columns:

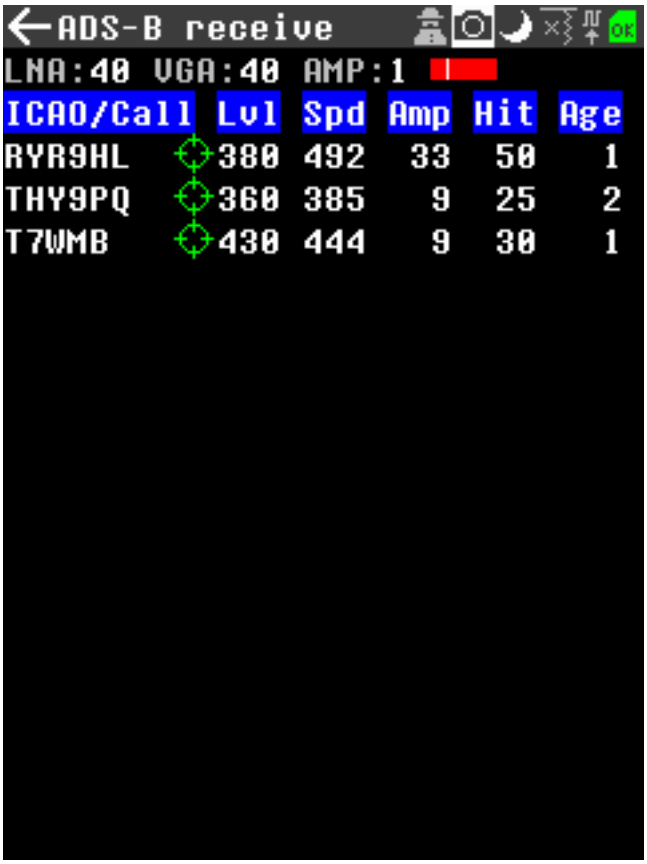
- **(YEAR MONTH DAY HOUR MIN SEC)**
- **(RAW PACKET IN HEX)**
- ICAO: **(ICAO)**
- **(CALLSIGN)**
- Alt: **(ALTITUDE)**
- Lat: **(LATITUDE)**
- Lon: **(LONGITUDE)**

Example:

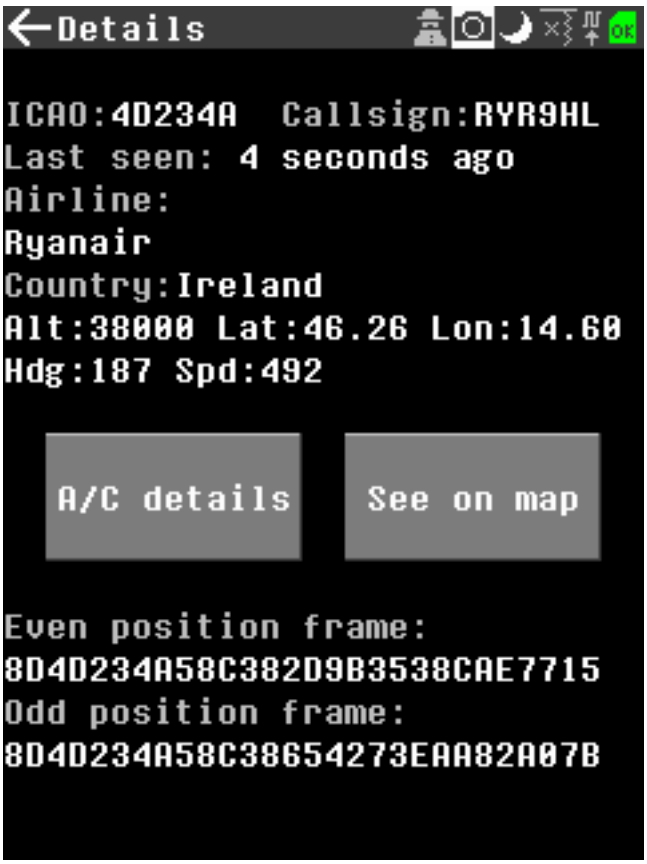
```
20171103100227 8DADBEEFDEADBEEFDEADBEEFDEADBEEF ICAO:nnnnnn callsign Alt:nnnnnn Latnnn.nn Lonnnn.nn
```

49.3 Screenshots

Main ADS-B View. In the columns, the ICAO/Call column will show the Callsign if known, otherwise the ICAO. The Lvl field is the Altitude in 100's of feet. Spd is the speed in mph. Amp is the signal strength, which may be increasing/decreasing depending on whether the plane is getting closer. Hit is the number of packets received from the same plane, and age is the number of seconds since the last packet was received.

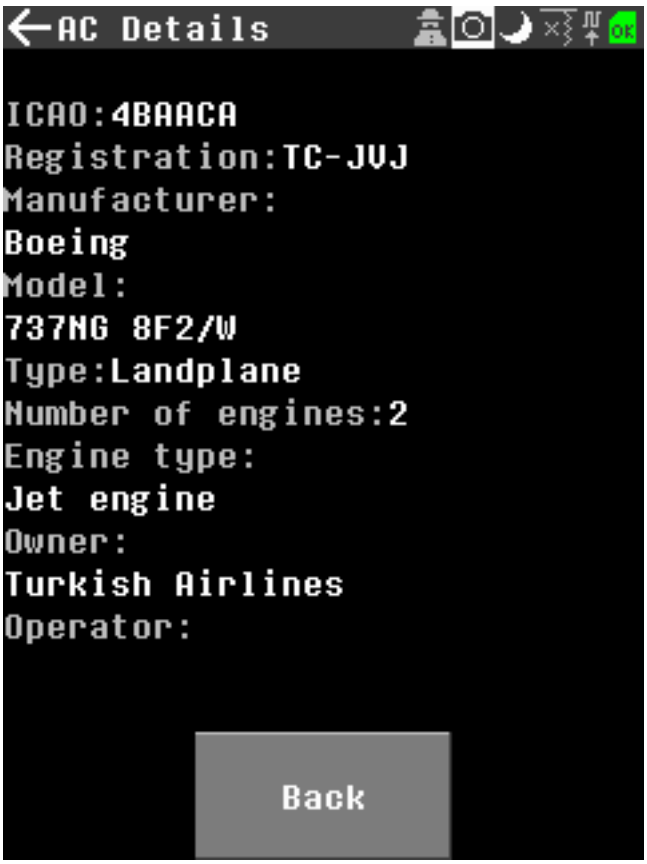


Aircraft View. The altitude is in feet, and the speed is in mph.

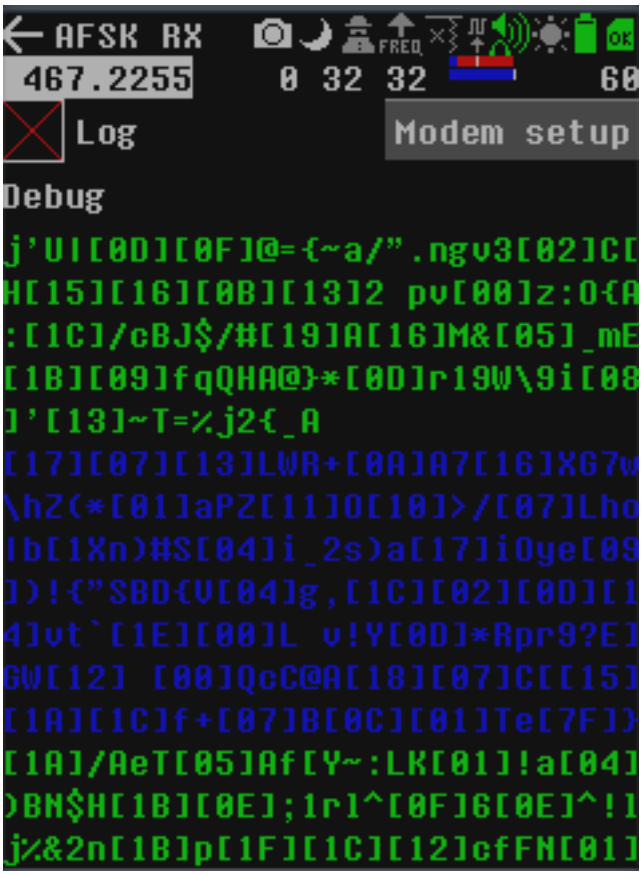


Map View. The map can be zoomed in or out by moving the cursor down to the map area and then turning the encoder dial.

A/C Details


AC Details
ICAO:4BAACA
Registration:TC-JVJ
Manufacturer:
Boeing
Model:
737NG 8F2/W
Type:Landplane
Number of engines:2
Engine type:
Jet engine
Owner:
Turkish Airlines
Operator:
Back

49.4 AFSK


AFSK RX
467.2255
0 32 32
60
Log
Modem setup
Debug
j'UI[0D][0F]@={~a/".ngv3[02]C[
H[15][16][0B][13]2 pv[00]z:O{A
:[1C]/cBJ$/#[19]A[16]M&[05]_mE
[1B][09]fqQHA@}*[0D]r19W\9i[08
]'[13]~T=%j2{_A
[17][07][13]LWR+[0A]A7[16]XG7w
\hZ(*[01]aPZ[11]O[10]>/[07]Lho
Ib[1Xn)#S[04]i_2s)a[17]iOye[09
])!{"SBD{V[04]g,[1C][02][0D][1
4]vt`[1E][00]L v!Y[0D]*Rpr9?E]
GW[12] [00]QcC@A[18][07]C[[15]
[1A][1C]f+[07]B[0C][01]Te[7F]}
[1A]/AeT[05]Af[Y~:LK[01]!a[04]
)BN$H[1B][0E];1rl^[0F]6[0E]^!]
j%&2n[1B]p[1F][1C][12]cfFN[01]

50 Important

This App Does not work as a general-purpose AFSK decoder. The app is hardcoded at startup to the modem settings and default frequency of the LCR/TEDI app (467.225 MHz, Bell 202, 7-bit Even parity, LSB first). This means it produces a constant stream of random characters on any other frequency or signal type. For LCR reception use the dedicated LCR/TEDI app instead.

This is an external app and must be present on the SD card in the `APPS/` folder.

50.1 Context

AFSK stands for **A**udio **F**requency-**S**hift **K**eying, and is a protocol for encoding digital information into audio for transmission over radio frequencies. Its use exists within many technologies, including in a few apps in the mayhem-firmware such as APRS-RX.

This app was (until now) poorly documented and has remained in *beta* for a significant amount of time, relegating it to an external app to free up space in the core firmware. It is believed by most to not work at all, and some to not work as well as it should.

50.1.1 Main Page

The top bar of the main page of the app has configurable items that are common to many other apps, including frequency selector, AMP, LNA, VGA, and Volume. It also has a toggle for outputting data to a log file writing to `AFSK.TXT` in the logs directory, as well as Modem Setup.

Finally, it has a section labelled as Debug. Here the user is presented with an ever-scrolling dump of what may appear to be random characters, but are the applications attempt to decode audio frequencies into textual output using UTF8 1.

Each time the app processes a new message 2 it will change the colour of this output cycling through `Red`, `Blue`, `Green`, and then `Light Blue`. In the log file, these are simply split by new lines.

50.1.2 Modem Setup

This page allows for the selection of a number of presets for processing AFSK

1. Bell202
2. Bell103
3. V21
4. V23 M1
5. V23 M2

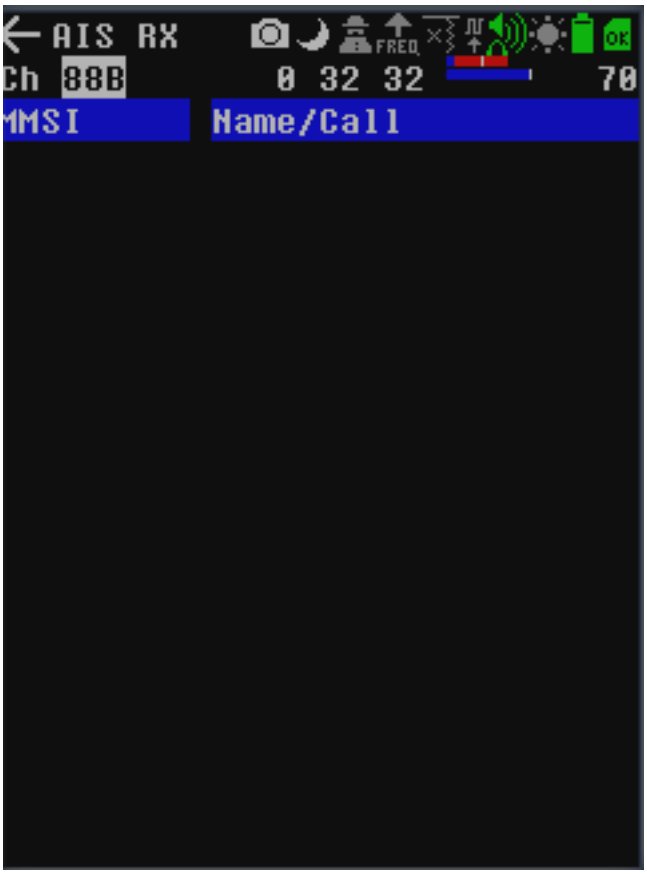
These can be selected by scrolling the selection wheel to the desired option, and then pressing the SET button. It is also possible to manually set the `Baudrate`, `Mark`, `Space`, `Repeat`, and `Serial Format`. Once the user has the desired settings selected, pressing the `Save` button at the bottom of the screen will bring them back to the Main Page. Note: due to the hardcoded startup configuration, modem settings changed here do not take effect correctly.

[1] At least, I think it's UTF-8. The printed characters of UTF-8 line up with the values that the C++ code is reading and declaring as a printed character, but at time of writing I am not familiar enough with C++ to know if it would be UTF-8 or another encoding type that has an overlap with UTF-8 such as windows 1252/latin1/etc.

[2] This is based on the code attempting to decode the audio into a byte(s) value between `DEC 32/HEX 0x20/UTF-8 (SPACE)` and `DEC 126/HEX 0x7e/UTF-8 ~(TILDE)`, and printing out the corresponding UTF-8 character. Any values outside of this range (with exception to below) are placed between square brackets and printed in their HEX form.

[3] If the decoded value is `DEC 127/HEX 0x7f/UTF-8(Control Character)` then this is treated as the end of a message, changing the colour of the text where displayed on the main page, or breaking onto a new line in the log file.

50.2 AIS Boats



NOTE This is an external app and must be present on the SD card in the `APPS/` folder.

51 AIS

Automatic Identification System (AIS) is a tracking system used by water-going vessels. It is described in ITU Recommendation M.1371..

The PortaPack AIS receiver decodes information coming from vessels and base stations, on one of two selectable VHF frequencies, 161.975 MHz (channel 87B) or 162.025 MHz (channel 88B). Others frequencies can be added to the SD Card Under AIS.

The Key Items on the App that can be selected with the cursor and changed with the encoder knob are:

- **Title bar:** The usual Items may be changed and displayed.
- **Ch:** The selection of the channel number, eitehr 87B or 88B.
- **Gain:** Setting are shown in order of Amp 0=0db or 1=14dB, LNA(IF) (0-40) and VGA (Baseband Gain) (0-62).

The App page shows two columns of received data, the Maritime Mobile Service Identity and the vessel name or call sign, if it was received. By selecting the received data line this opens a detailed view, giving the most recent details received from the vessel. It should be noted that dependant on the activity of the vessel that the information will be sent every few minutes to every few seconds if traveling at speed.

51.1 APRS RX

52 Automatic Packet Reporting System

NOTE This is an external app and must be present on the SD card in the `APPS/` folder.

The APRS RX app receives and displays messages transmitted using the APRS (Automatic Packet Reporting System) protocol. It shows a live list of received stations including callsign, position (if available), hit count, and last received time.

52.1 Pre-configured Regions

Select a region to automatically tune to the correct APRS frequency for that area:

- Code: MAN; Frequency: user-defined; Region: Manual — frequency set manually
- Code: NA; Frequency: 144.390 MHz; Region: North America
- Code: NZ; Frequency: 144.575 MHz; Region: New Zealand
- Code: JAP; Frequency: 144.640 MHz; Region: Japan
- Code: PHI; Frequency: 144.740 MHz; Region: Philippines
- Code: EUR; Frequency: 144.800 MHz; Region: Europe or UK
- Code: THA; Frequency: 144.900 MHz; Region: Thailand
- Code: AUS; Frequency: 145.175 MHz; Region: Australia
- Code: BR; Frequency: 145.570 MHz; Region: Brazil
- Code: ISS; Frequency: 145.825 MHz; Region: International Space Station

When a region is selected, the frequency field updates automatically. If you change the frequency manually afterwards, the region selector switches to **MAN**.

Frequencies not in the preset list can be tuned manually (MAN), for example **432.500 MHz** (Europe, UHF).

52.2 Controls

52.2.1 Frequency

- Use the rotary encoder over the frequency field to adjust in steps
- Press the middle button over the frequency field to enter a frequency manually via keypad
- Load a frequency from SD card using the frequency manager

52.2.2 Radio Settings

From left to right:

- **AMP** — RF amplifier: `0` = off, `1` = on (+14 dB). Enable if signal is weak.
- **LNA** — IF gain (0–40 dB in 8 dB steps)

- **VGA** — Baseband gain (0–62 dB in 2 dB steps)

See Description of gain settings for more details.

52.2.3 Volume

Audio volume is shown after the signal strength bars. Use the rotary encoder when focused on it.

52.3 Station List

The main screen shows a list of recently received APRS stations with the following columns:

- Column: Source; Description: Callsign of the transmitting station (with SSID if present)
- Column: Location; Description: GPS coordinates if the packet contained a position report
- Column: Hits; Description: Number of times this station has been received
- Column: Time; Description: Time of last received packet

Tap or select an entry to open the **Detail View**.

52.4 Detail View

Selecting a station from the list opens a detail screen showing the full decoded APRS packet information including the raw info field content. Two buttons are available:

- **Done** — return to the station list
- **See Map** — opens the map view centered on the station's position. This button is only visible if the station transmitted a valid GPS position.

52.5 Logging

Received packets are automatically logged to the SD card at:

`LOGS/APRS.TXT`

Each line contains the raw decoded APRS stream data. Logging is active as long as the app is running and a writable SD card is present.

52.6 Important Notes

- APRS RX on the HackRF is relatively insensitive. Enable **AMP** if you are not receiving stations that are clearly audible on a dedicated receiver.

⚠️ Be careful not to connect strong signals directly — the HackRF has no input protection.

- Some stations transmit with a bandwidth wider than the default 8.5 kHz. If you are missing packets, increase the reception bandwidth by editing the settings file on the SD card:
  1. Navigate to `SETTINGS/rx_aprs.ini` using the File Manager app
  2. Select the file, press down to select the pen icon, press OK
  3. Find the line starting with `nbfm_config_index` and change the value:
     - `0` = 8.5 kHz (default)
     - `1` = 11 kHz
     - `2` = 16 kHz

52.7 Protocol

APRS is transported over **AX.25** using a **Bell 202** modem at **1200 bit/s** (AFSK): a **1200 Hz** tone for *mark* (binary 1) and **2200 Hz** for *space* (binary 0).

52.8 SSID (Secondary Station ID)

The SSID is a numeric suffix added to a callsign to tell apart multiple stations under one call. APRS allows 16 SSIDs (0–15); the commonly recommended meanings:

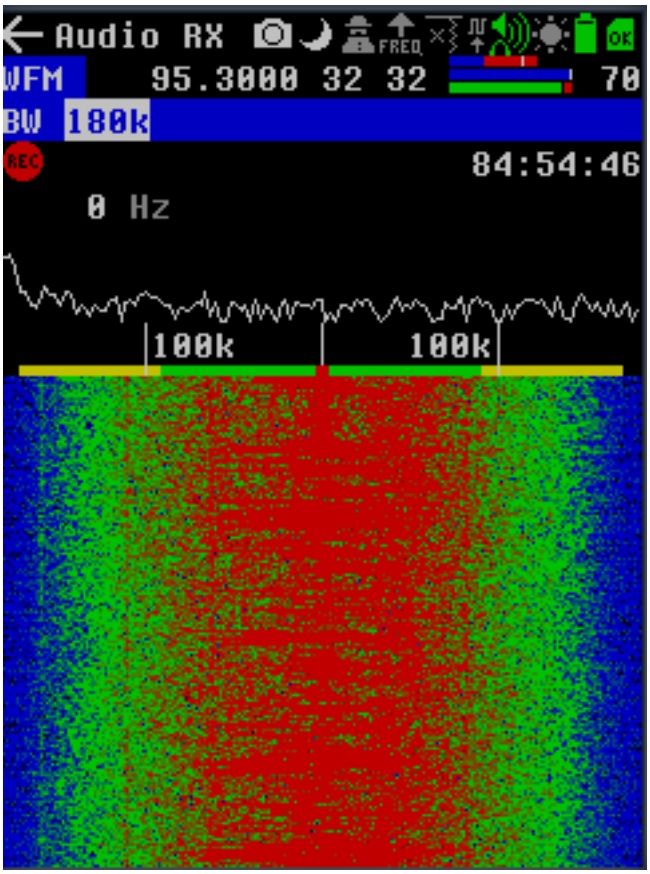
- SSID: 0; Typical use: Primary station, usually fixed and message-capable
- SSID: 1–4; Typical use: Generic additional station (digi, mobile, wx, etc.)
- SSID: 5; Typical use: Other networks (D-STAR, smartphones, etc.)
- SSID: 6; Typical use: Special activity, satellite ops, etc.
- SSID: 7; Typical use: Handheld transceivers or human-portable
- SSID: 8; Typical use: Boats, sailboats, RVs, or second main mobile
- SSID: 9; Typical use: Mobiles or trackers
- SSID: 10; Typical use: Internet, iGates, Echolink, Winlink, AVRS, APRN, etc.

- SSID: 11; Typical use: Balloons, aircraft, spacecraft, etc.
- SSID: 12; Typical use: APRStt, DTMF, RFID devices, one-way trackers, etc.
- SSID: 13; Typical use: Weather stations
- SSID: 14; Typical use: Truckers or full-time drivers
- SSID: 15; Typical use: Generic additional station (digi, mobile, wx, etc.)

52.9 Further Reading

- APRS TX — transmit APRS packets
- APRS Protocol Reference Manual
- aprs.fi — Live APRS map

52.10 Audio



53 Introduction

The Audio App is the main way that signals can be heard and seen in detail. Three types of decoders are provided for audio modulated signals and a spectrum view of the signals. The user interface has the ability to view and change:

53.1 Controls

- **SPEC:** Display a Spectrum of the received signal and allow viewing of 10MHz of RF Spectrum, centered on a configurable frequency, with 5MHz above the frequency and 5MHz below.
- **AM:** Demodulate and Record RF Signals modulated using the Amplitude Modulation scheme. It can demodulate Double-Sideband AM (ITU Designation: A3E) and both Lower-Sideband and Upper-Sideband Single-Sideband AM (ITU Classification: R2E, H3E, J3E) signals.
- **NFM:** The Narrow Band Frequency Modulation decoding ITU Classification: FM3
- **WFM:** The Wide FM Receiver is a Sub-Application of the Audio Receiver Application. Its purpose is to Demodulate and Record RF Signals modulated using the Frequency Modulation scheme. It can demodulate mono and stereo Wide FM signals of 200KHz bandwidth. Such signals are commonly used for VHF FM Broadcast services.

- **AMFM:** Here in that Audio App we can tune , and RECORD the APT signal from a HF-Weather Fax map Broadcasting Station into audio_format_APT_file.wav ,mainly for investigation , debugging , and check signal levels (signal strenght , saturation , filters check ,...). But we have also in Mayhem fw , another external receiver Wefax App (based on that AMFM demodulation engine) , more convenient to the user, because it combines both things : (that APT demodulation and the additional display weather map to the LCD (in low resolution) able to record the decoded weather map into a bmp file (in full received resolution. -much better than the displayed in the low resolution screen) We are combining in that demodulation , two consecutive demodulations. First RF USB audio baseband demodulation , and later demodulating the FM subcarrier audio baseband tones.That modulation , it is commonly used in Wefax (HF Weather Fax transmissions) . It demodulates APT signal of the HF weather fax maps. , and sends the demodulated low scan map APT signal in parallel to the speaker (just for our audio tuning reference) and it can be also RECORDED to the SD card , through audio.wav file (SR =12KHZ, audio mono track). Inside audio baseband spectrum , there is a sub-carrier audio tone , modulated in FM with the low scan APT image map signal . (Audio Baseband with f_subcarrier fsc (1900Hz) FSK mod +-400Hz ; Shift White pixel +400 Hz = 2.300 Hz , Shift Black pixel −400 Hz = 1.500 Hz). For better usability , in both Applications (this Audio App AMFM mode, and Wefax App) we are internally applying a proper tune-offset . Therefore you just need to tune the official published Weather Fax frequency transmitter. (example 3.855 Khz ,7.880 Khz , 8.040 Khz ,...).
- **FMAM:** Here in that Audio App we can tune , and RECORD the APT signal 137 Mhz , from the discontinued NOAA Analog saltellite weather image service. Fax image Broadcasting Station into audio_format_APT_file.wav ,mainly for investigation , debugging , and check signal levels (signal strenght , saturation , filters check ,...). But we have also in Mayhem fw , another external receiver NOAA APT App

Hackrf has poor sensitivity to receive directly those HF Wefax signals, (mainly broadcasted in HF band : 3 Mhz to 16 Mhz) Normally we should add to the Hackrf receiver a front up-converter with better HF filtered pre-amplifiers, and a reasonable good enough HF antenna. WEFAX's format is a derivative of the Automatic Picture Transmission (APT) originally developed for transmission from the polar satellites of the USA. WEFAX has 60, 90, 100, 120, 180 and 240 LPM (Lines per minute) speeds, and two IOC (Index of Cooperation) modes, IOC 576 and IOC 288. Most weather forecasts are sent in IOC 576 , 120 LPM . Our initial target is WEFAX-IOC576 standard. https://www.sigidwiki.com/wiki/WEFAX. Here you can see the actual Worldwide Weather fax HF Stations with his frequencies , and exact GMT map transmission time . https://weatherfax.com/stations/

The Key Items on the App that can be seen or selected with the cursor and changed with the encoder knob are:

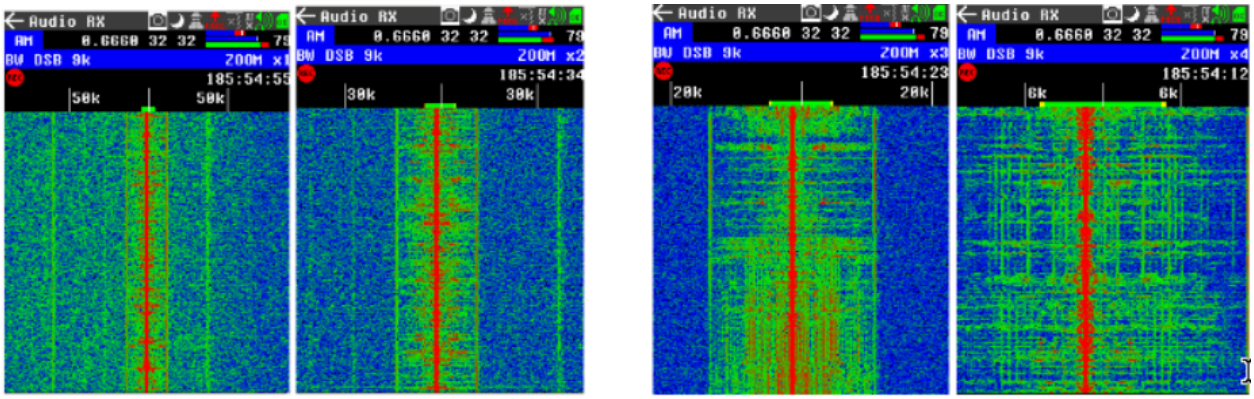
- **Title bar:** The usual Items may be changed and displayed.
- **Mode:** On the line below title bar is the demodulation mode AM, NFM, WFM, SPEC. When either of these are selected it will bring up a secondary set of relevant items on the line below. These are discussed in secondary items below.
- **Frequency:** The Centre frequency of the demodulation band.

> Note 1 : From Frequency field, you can move down to the below Step field , to adjust your best suitable Freq-step, according to your needs.
>
> Note 2 : If you have load/save App Settings from SD card enabled, it will continue to use what it was last set at. And you can always edit the file in the SD card, /SETTINGS/rx_audio .ini and edit the step size there, according to your needs in that App.
>
> Note 3 : Additionally , to be more user friendly , from version 1.7.4+ holding in the Select button on the Frequency field for a second until a digit turns blue, then you can use Left/Right select which digit you'd like to adjust, and then you can use the Encoder Dial to adjust any digit up/down by 1 to tune more precisely. (Press Select again to exit this tuning mode.

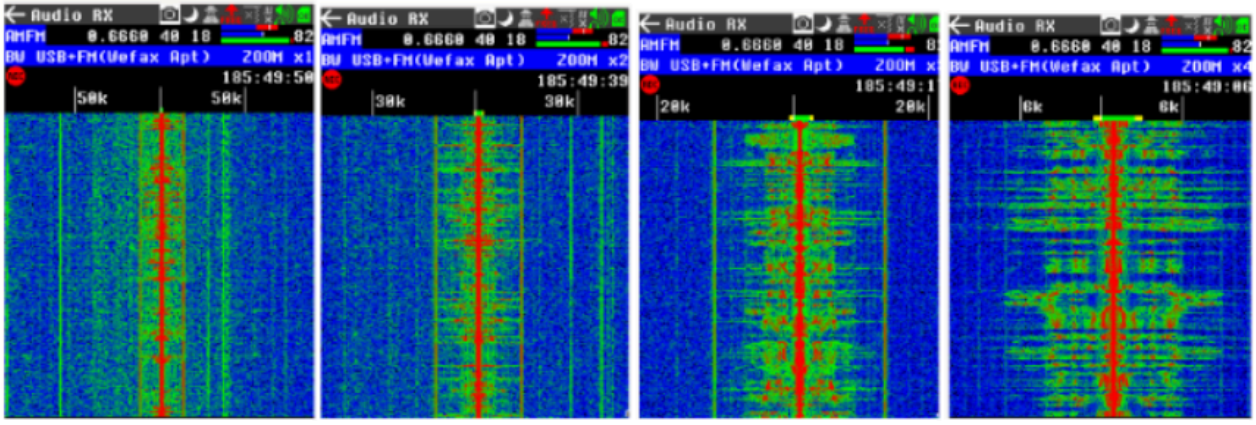
- **Gain:** Settings are shown in order of LNA(IF) (0-40) and VGA (Baseband Gain) (0-62). When either of these are selected in the secondary line the AMP setting is shown and can be as either set to 0=0db or 1=14dB.
- **Signal Display:** The three coloured displays are top to bottom RSSI(Red/Blue) with an average marker in the line. Next is the Baseband signal and last the Audio level.
- **Volume:** The Last item on this line is the audio volume control (O-99) that is used with either headphone or speaker if fitted.
- **Secondary Information:** This line provides associated information for the following items these are:
  - **AM:** Bandwidth settings of DSB 9k, DSB 6k, USB+3k, LSB-3k, CW. The Spectrum view can go from +/-90khz till +-11k25 depending on the selected zoom (x1,x2,x3,x4). It has also applied the Sliding frequency feature . There is also an **SQ** field, shown as `SQ  /99`, for an optional channel-power squelch (range 0-99, **default 0 = off**). This is separate from the NFM squelch. When set above 0, the audio is muted while the channel power stays below the threshold (about −80 dBFS at 1, rising to about −20 dBFS at 99), which silences an idle AM channel between transmissions.
  - **NFM:** Bandwidth Settings of 16k,11k,8k5. Note there is no setting for the more common 6k5 used in European Spectrum plans. Next item is SQ: which is shown in the format of 40/99 allow the noise squelch point to be set Between 0-99. Typically, around 40-50 is a good threshold. It has also applied the Sliding frequency feature .
  - **Gain:** The RF Amp settings. The Spectrum view is +/-20k.
  - **USB+FM:** It is usually used to demodulate Wefax (HF Weather fax transmissions) , demodulating the signal in two steps, first demodulating the tuned RF signal in USB , extracting audio tones. And then demodulating again those FM modulated subcarrier audio tones, to extract APT low scan weather fax map signal. (that signal can be recorded into audio.wav file). For a correct demodulation ,currently we need to tune that published frequency , (in our application , you do not need to apply any usual - 1.900Khz offset . The -tuning offset , is already applied internally). So we have to just tune the (published) Wefax Station frequency directly .
  - **FM-AM:** It was used to demodulate NOAA analog APT Weather Image satellite in 137 Mhz band, demodulating the signal in two steps, first demodulating the tuned RF signal in WFM (38KHZ BW aprox) , extracting AM carrier modulated with APT (2khz BW aprox) . And then demodulating again this AM modulated subcarrier to extract APT low scan Image weather satellite signal. (that signal can be recorded into audio.wav file).
- **Record button:** press to record a file, press one more time to stop the record.
- **Marker Frequency** Just below REC Icon we have a marker Frequency field input (0 Hz) , that we can adjust from 0..48kHz range with the knob rotary encoder, with 200 Hz step jumps. It will move a short vertical red line pointer cursor across the Multiplex FM demodulated baseband FFT spectrum graphic. It could be useful to see and confirm the exact frequency of those spectrum peaks, like the below picture , showing 19Khz pilot carrier tone from a usual Stereo FM Broadcasting.
- **Waveform display** if you get down the marker frequency and put focus on the wafeform display, you can disable it with a single click
- **Frequency shift** if you get down the Waveform display or just touch the screen, you'll enter in the frequency shift mode. A flashing red dot will indicate the next position. Use rotary encoder or click on the screen, and press enter to set the new frequency.

Note : All AM modes (DSB 9K, DSB 6K, USB, LSB, CW) and AMFM mode has the option to expand the waterfall, selecting **ZOOM x1** / **ZOOM x2** / **ZOOM x3** / **ZOOM x4** , but in addition, AM modes has also implemented Slider Frequency feature.

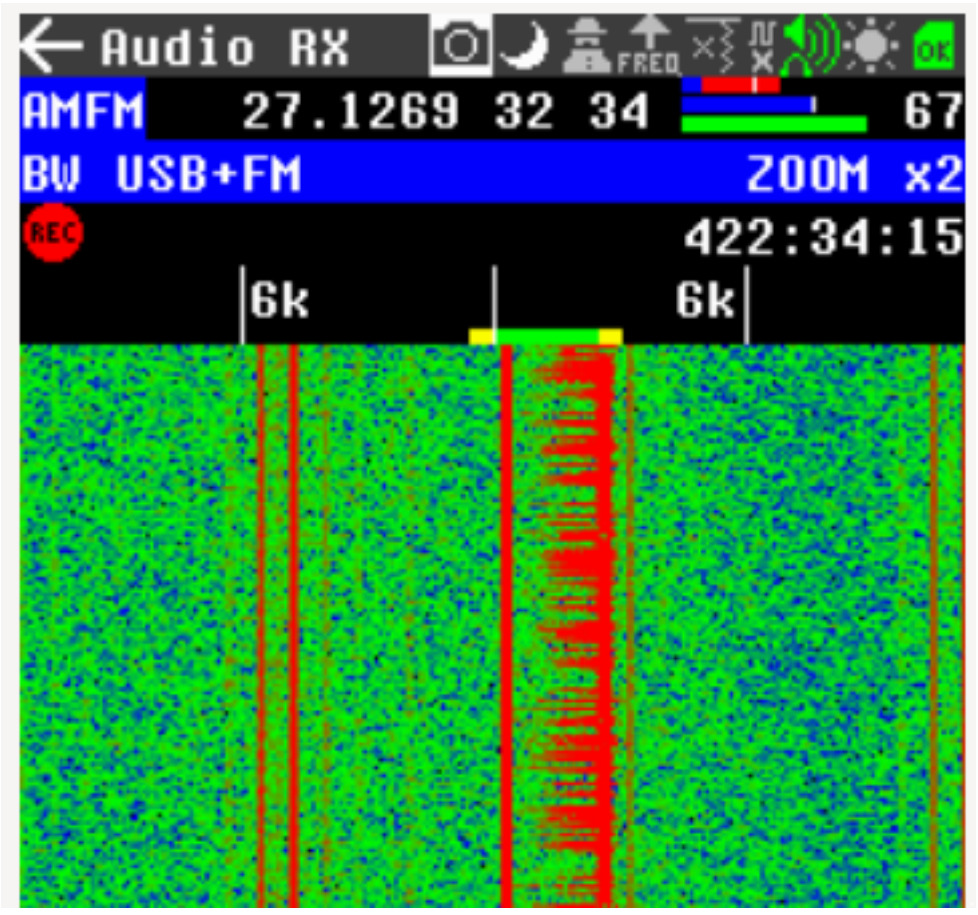
AM MODES



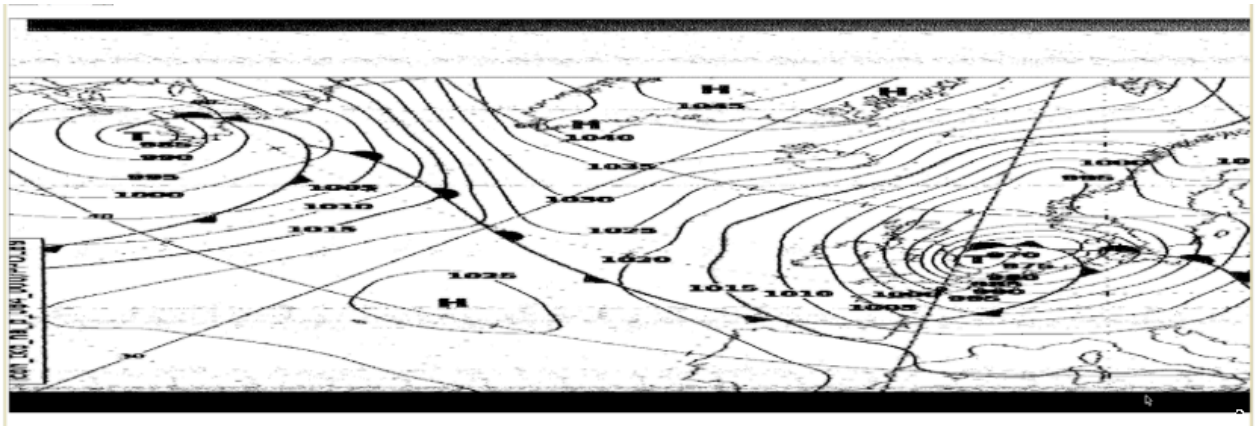
AMFM (Wefax)



Here in that Audio App we can tune , and RECORD the APT signal into file.wav ,mainly for investigation and check signal levels (signal strenght , saturation , filters check , debugging).



(as example, that weather map, was decoded directly from a weather HF Broadcasting station, by the other Wefax App)



53.1.1 WFM

There are three option filters in that Secondary settings:

1. 200k ,the original filter for commercial FM stations with soft transition,
2. 180k with sharp transition, for also commercial FM broadcast station, specially useful to improve demodulated S/N in around 6 to 8 dB in weak signals.
3. 40k with also sharp transition, for supporting NOAA APT weather satellite reception in 137 MHz . (that filter is too narrow for WFM with 75khz delta deviation and can produce audio distortion in the demodulated sound). The Spectrum view is +/-100k with a marker That may be changed (though seem the incorrect value).

53.1.2 SPEC

The spectrum Secondary Items allows the view of the RF spectrum with different setting for maximum bandwidth shown:

1. 20M with markers at +/- 5M
2. 10M with markers at +/- 3M

3. 5M with markers at +/- 2M
4. 2M with markers at +/- 500k
5. 1M with markers at +/- 300k
6. 500k with markers at +/-200k

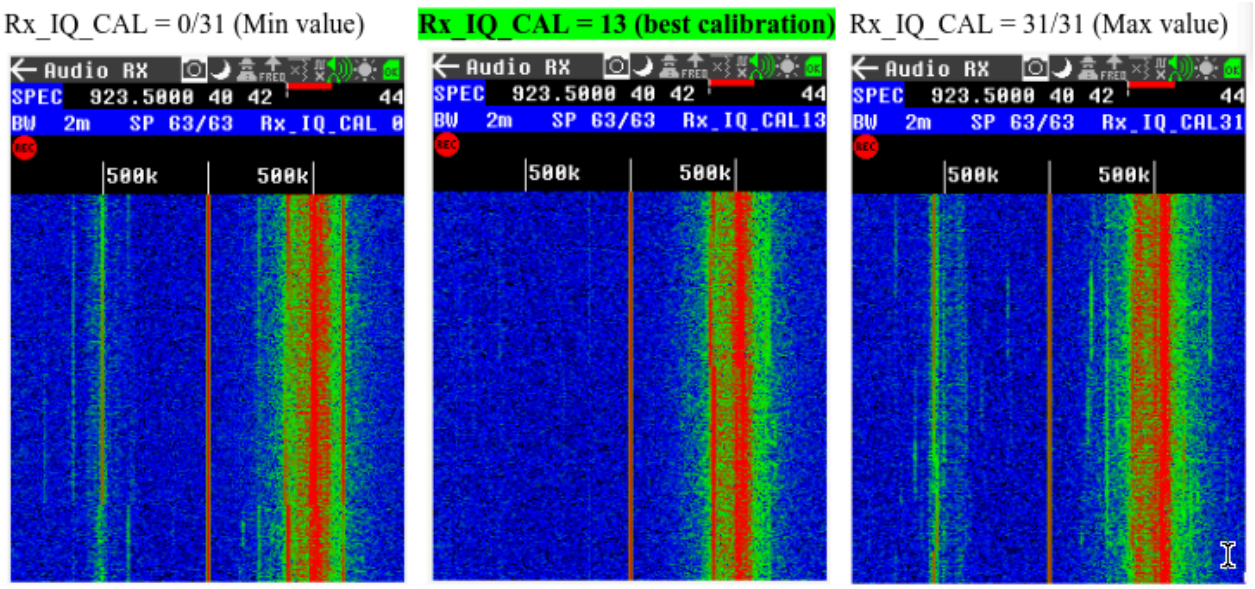
The next item is the setting of the bin sizes used for the waterfall (0-63) with "0" being the minimum information being the fastest display and "63" the maximum information collected the slowest display. Adjust to give a balance of speed and information seen.

53.1.2.1 Calibration

And next right to that field ,while you are still in SPEC mode, you will see the Rx_IQ_CAL field. It actually can improve (8 to 10 dB's from worse CAL point to the best one) the receiver Image Reject Ratio (IRR) only in the Receiver Applications that are using Zero IF-frequency tuning ,like this SPECTRUM mode. This calibrated value will be stored in the SD card , /settings/rx_audio.ini .

To calibrate it , you will need to try to find some clean isolated continuous air carrier signal (please make sure that it is a real air signal, not internal harmonic residual interference clock beats) , and try to tune it , (for example, to the right of the positive frequencies). Select properly the SPEC BW to try to isolate that air carrier signal (in below example , I selected BW = 2 Mhz). Then, to make slightly more visible the bare Image mirror frequency -that appears in the other symetric left part screen.-, you will need to start set up CAL to 0 (min). Then , you will need to adjust AMP , LNA and GAIN to receive strong enough signal to the right part (but at the same time , not so strong to not produce intermodulations , otherwise re-adjust again decreasing AMP, LNA, GAIN) , and not so weak , till just getting the carrier +freq. clean (desired carrier signal) , and its bare visible Image mirror Signal (in the other symetrical -xx mirror freq . (unwanted Image signal to be minimized) . Then, when you got similar situation as left below picture, you can start adjusting Rx IQ phase CAL till minimizing the received Image "mirror" signal (as in the middle below picture).

That below example is calibrating H2+ old r6 version (max2837) that has 31 steps ,(Note , using Hackrf r9 (max2839 ,you will have 64 steps)



53.1.3 NFM

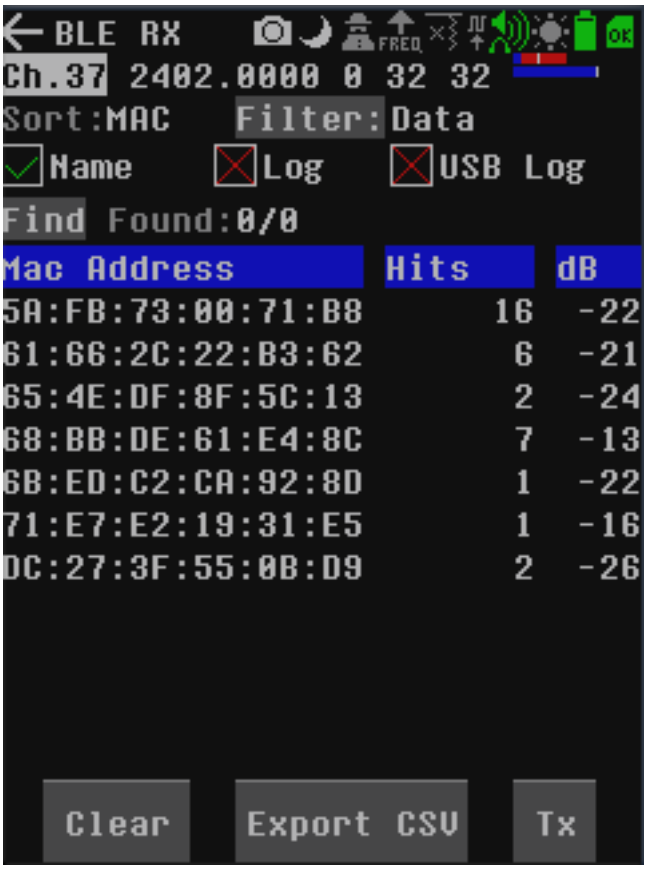
- **CTCSS:** This Continuous Tone Coded Squelch System is a display at the end of the secondary information line. It is used by many systems and standardised by EIA/TIA, with a description [here.](https://en.wikipedia.org/wiki/Continuous_Tone-Coded_Squelch_System) In the NFM mode in Audio app, the tone field is increased from 11 to 14 characters. Now , the CTCSS tone frequency and CTCSS code numbers are displayed at once (if a matching freq was found in the tone_key table). See the Annex to this document below. It should be noted that most of the time the display is jumping around and only clearly displays the received tone when there is a gap in the voice and the Signal is of good quality.
- **Record:** The record button if selected will show the record file name, % of the SD Card used, and at the end of the line is the total recording time available left on the SD card and this decrements when

recording. The file name of the recorded 16-bit mono WAV file includes the date, time, and radio tuning frequency. The WAV file sampling rate depends on the receiver mode; 48KHz for WFM, 24KHz for NFM, or 12KHz for AM.

53.2 CTCSS Tone List

```
None      0.0
0 XZ      67.000
1 WZ      69.400
2 XA      71.900
3 WA      74.400
4 XB      77.000
5 WB      79.700
6 YZ      82.500
7 YA      85.400
8 YB      88.500
9 ZZ      91.500
10 ZA      94.800
11 1ZB      97.400
12 21Z      100.000
13 1A      103.500
14 1B      107.200
15 2Z      110.900
16 2Z      114.800
17 2B      118.800
18 3Z      123.000
19 3A      127.300
20 3B      131.800
21 4Z      136.500
22 4A      141.300
23 4B      146.200
24 5Z      151.400
25 5A      156.700
40 --      159.800
26 5B      162.200
41 --      165.500
27 6Z      167.900
42 --      171.300
28 6A      173.800
43 --      177.300
29 6B      179.900
44 --      183.500
30 7Z      186.200
45 --      189.900
31 7A      192.800
46 --      196.600
47 --      199.500
32 M1      203.500
48 8Z      206.500
33 M2      210.700
34 M3      218.100
35 M4      225.700
49 9Z      229.100
36 --      233.600
37 --      241.800
38 --      250.300
50 0Z      254.100
Axient 28kHz      28000.0
Senn. 32.768k      32768.0
Senn. 32.000k      32000.0
Sony 32.382k      32382.0
Shure 19kHz      19000.0
```

53.3 BLE RX



54 Introduction

As of 10/31/23, the previous BLE implementation has been updated, and improved to correct some of the issues in the previous BTLE app. This BLE app has several features which I will highlight in a brief overview.

55 Main Menu:

This is the main view which provides the user with incoming packet entries captured by the BLE Scanning.

1. The BLE app upon entry will scan for **only** BLE advertisement packets, and report them on the screen. The **Channel** knob can be used to select which advertisement channel to listen on. There is an Auto feature which will switch channels incrementally in 25ms steps.
2. Once found the user can then select an individual MAC Address entry to pull up a more detailed view of most recently captured data packet from this MAC Address.
3. The **Sort Knob** will sort the list of MAC indices by either Ascending MAC Address, device name, number of hits, dB, or by most recently updated entry.
4. The **Filter** button allows the user to filter based on the hex data of each packet. It also allows for filtering based on the ASCII name of the device (if found). More on that below.
5. The **Name** toggle allows the user to toggle off and on name display, if there is a name associated with the device. Not all devices have a name string. This name string is being parse from the Shortened and Complete Local Name packet type. See BLE Spec for information on packet types.
6. The **Log** toggle allows BLE app to write all filtered packet data into SD card under `/BLERX/Logs`. Each log entry contains timestamp, packet type, length of the packet, MAC address, and packet data.
7. The **USB Log** toggle allows BLE app to write all filtered packet data to serial over USB. Messages can be viewed by any serial terminal client (eg: PuTTY) by opening the port (eg: '/dev/ttyACM0') with a baud rate of 9600.
8. The **Clear** button allows the user to clear entries as they fill up the screen.

9. The **Export CSV** file allows the user to export the current list of packet entries into a csv style file. Upon resaving the file, the file will update the existing entries with new data, as well as append new entries to the existing file.
10. The **Tx** button allows the user to switch to the BLE Tx app. See BLETX for more information.

Note: The dBm displayed on the list of devices is an estimated receive power, relative to the Tx power of the device. Estimation takes into account the LNA/VGA/AMP when determining the dBm value.

56 Packet Detail Menu:

This menu is a more single detailed view on a specific entry selected by the Main Menu. More specific packet information is laid out here such as:

1. The **MAC Address**
2. The **PDU Type** which displays the last received PDU packet type as noted here:

- Type: 00; Name: ADV_IND; Description: Scannable advertising indication for all devices
- Type: 01; Name: ADV_DIRECT_IND; Description: Directed advertising to a specific device indicating that only that device can connect
- Type: 02; Name: ADV_NONCONN_IND; Description: Advertising indication but not accepting connections or scans
- Type: 03; Name: SCAN_REQ; Description: Sent by a device to receive more info from scannable advertisers
- Type: 04; Name: SCAN_RSP; Description: Response to 03 containing more info
- Type: 05; Name: CONNECT_REQ; Description: Sent to an advertiser to initiate a connection
- Type: 06; Name: ADV_SCAN_IND; Description: Scannable advertising indication but not accepting connections
- Type: 07; Name: ADV_EXT_IND; Description: BT 5.0, points to additional data on secondary channels
- Type: 09; Name: AUX_ADV_IND; Description: BT 5.0, scannable advertising indication for all devices on secondary channels
- Type: 0A; Name: AUX_SCAN_REQ; Description: BT 5.0, sent by a device to receive more info from scannable advertisers on secondary channels
- Type: 0B; Name: AUX_SCAN_RSP; Description: BT 5.0, response to 0A containing more info
- Type: 0C; Name: AUX_CONNECT_REQ; Description: BT 5.0, sent to an advertiser to initiate a connection
- Type: 0D; Name: AUX_CHAIN_IND; Description: BT 5.0, chains advertising packets together when they get too big
- Type: 0E; Name: AUX_CONNECT_RSP; Description: BT 5.0, response to 0C
- Type: 10-26; Name: Link Layer; Description: Link layer setup and operation
- Type: 26-FF; Name: Data; Description: General data and proprietary

3. The last packets received for each data type, their length, and their contents.
4. The **Send** button immediately moves the user to the BLE TX app and allows the user to transmit the selected packet data.
5. The **Save** button allows the user to save packets in the format used by the BLE TX app.
6. The **Done** button brings the user back to the Main Menu.

57 Video:

Below short example on using the BLE Rx App.

https://vimeo.com/886881206?share=copy

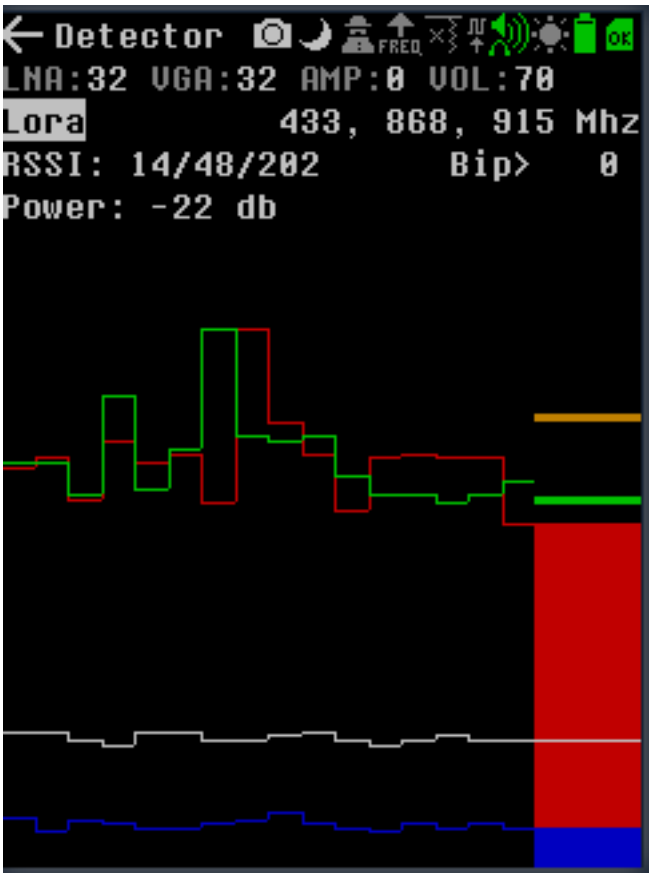
58 References:

BLE Tx App: Bluetooth Low Energy Transmitter

BLE Packet Types: https://www.bluetooth.com/wp-content/uploads/Files/Specification/Assigned_Numbers.pdf?id=3

Reference Code Used in Porting Protocol: https://github.com/JiaoXianjun/BTLE

58.1 Detector



59 Introduction

The Detector app is based on the Level app and allows quick signal detection across preset frequency lists for LoRa, remote controls, and TETRA uplink. It scans through all frequencies in the selected preset automatically, and produces an audible beep whenever a signal exceeds the configured threshold.

This is an external app (`.ppma` file on the SD card).

60 Controls

60.1 First line — Gain and Volume

- **LNA** — IF gain (0–40 dB). Adjust with rotary encoder when focused.
- **VGA** — Baseband gain (0–62 dB). Adjust with rotary encoder when focused.
- **AMP** — RF amplifier: `0` = off, `1` = on (+14 dB).
- **VOL** — Beep volume (0–99).

60.2 Second line — Preset selection

Select the signal type to monitor:

- Preset: TETRA UP; Frequency range: 380–390 MHz; Description: TETRA uplink band (emergency services, Europe)
- Preset: Lora; Frequency range: 433, 868, 915 MHz; Description: LoRaWAN EU433, EU868 and US915 bands
- Preset: Remotes; Frequency range: 315, 433.92 MHz; Description: Common remote control and key fob frequencies

The current frequency being scanned is displayed on the right side of this line.

60.3 Third line — Beep threshold and signal display

- **Bip** — Detection threshold in dBm (-100 to +20). When the received signal power exceeds this value, a beep is triggered. Lower values = more sensitive. The beep pitch scales with signal strength: a weak signal at threshold produces a 400 Hz tone, a maximum strength signal (+20 dBm) produces a 2600 Hz tone. Beep duration is 150 ms.
- **RSSI: min/avg/max** — Raw RSSI values (min/average/max) from the RSSI graph over recent history.
- **Power: -XX db** — Current received signal power in dBm.

61 How scanning works

The app cycles through all frequencies in the selected preset on every display frame (~16 ms per frequency step). Each scan window has a bandwidth of **750 kHz**. If the signal level on any frequency exceeds the threshold during its scan window, a beep is produced.

62 Signal display

The main area shows two displays:

- **RSSI graph** — scrolling history of min/avg/max RSSI values and signal power
- **RSSI bar** — current instantaneous RSSI level on the right side

63 Preset frequencies

63.1 Remotes

- Center frequency: 315.000 MHz; Coverage: 314.625 – 315.375 MHz
- Center frequency: 433.920 MHz; Coverage: 433.545 – 434.295 MHz

63.2 LoRa

- Center frequency: 433.375 MHz; Coverage: 433.000 – 433.750 MHz; Band: EU433
- Center frequency: 434.125 MHz; Coverage: 433.750 – 434.500 MHz; Band: EU433
- Center frequency: 434.875 MHz; Coverage: 434.500 – 435.250 MHz; Band: EU433
- Center frequency: 867.375 MHz; Coverage: 867.000 – 867.750 MHz; Band: EU868
- Center frequency: 868.125 MHz; Coverage: 867.750 – 868.500 MHz; Band: EU868
- Center frequency: 868.875 MHz; Coverage: 868.500 – 869.250 MHz; Band: EU868
- Center frequency: 914.250 MHz; Coverage: 913.875 – 914.625 MHz; Band: US915
- Center frequency: 915.000 MHz; Coverage: 914.625 – 915.375 MHz; Band: US915
- Center frequency: 915.750 MHz; Coverage: 915.375 – 916.125 MHz; Band: US915

63.3 TETRA Uplink

- Center frequency: 380.375 MHz; Coverage: 380.000 – 380.750 MHz
- Center frequency: 381.125 MHz; Coverage: 380.750 – 381.500 MHz
- Center frequency: 381.875 MHz; Coverage: 381.500 – 382.250 MHz
- Center frequency: 382.625 MHz; Coverage: 382.250 – 383.000 MHz
- Center frequency: 383.375 MHz; Coverage: 383.000 – 383.750 MHz
- Center frequency: 384.125 MHz; Coverage: 383.750 – 384.500 MHz
- Center frequency: 384.875 MHz; Coverage: 384.500 – 385.250 MHz
- Center frequency: 385.625 MHz; Coverage: 385.250 – 386.000 MHz
- Center frequency: 386.375 MHz; Coverage: 386.000 – 386.750 MHz
- Center frequency: 387.125 MHz; Coverage: 386.750 – 387.500 MHz
- Center frequency: 387.875 MHz; Coverage: 387.500 – 388.250 MHz
- Center frequency: 388.625 MHz; Coverage: 388.250 – 389.000 MHz
- Center frequency: 389.375 MHz; Coverage: 389.000 – 389.750 MHz
- Center frequency: 390.125 MHz; Coverage: 389.750 – 390.500 MHz

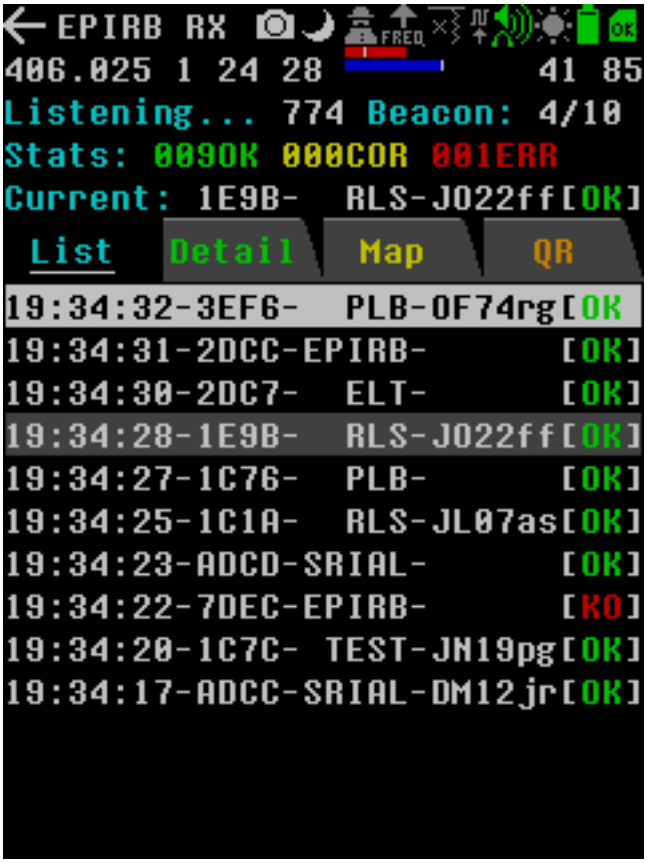
64 Settings

The beep threshold is saved automatically to `SETTINGS/rx_detector.ini` on the SD card when the app is closed (requires Load App Settings to be enabled in the Settings menu).

64.1 EPIRB RX

65 🎯 Overview

The **EPIRB RX** application receives and decodes **ELT/EPIRB/PLB** COSPAS/SARSAT emergency beacon frames.



This is an external app (`.ppma` file on the SD card).

65.1 🆘 Beacons

The **EPIRB RX** application decodes 406 MHz distress beacon transmissions from EPIRBs (Emergency Position-Indicating Radio Beacons), PLBs (Personal Locator Beacons), and ELTs (Emergency Locator Transmitters).

COSPAS/SARSAT emergency beacons are used in boats (**EPIRB**), planes (**ELT**) or by individuals (**PLB**) to support Search And Rescue (**SAR**) operations by helping to locate accident sites.

65.2 🦺 Error Correction

Received beacons are decoded using BCH error detection and single-bit correction. Each beacon state is color-coded:

- **Green** = valid packet
- **Yellow** = corrected (1-bit error fixed)
- **Red** = unrecoverable error

66 🎁 Features

RX parameters can be configured using standard RF settings: RF amplifier, LNA gain, and VGA gain.

Frequencies can be selected from a predefined list stored in the EPIRB/FREQ.TXT file.

The **EPIRB RX** application also outputs audio of the monitored frequency, with adjustable squelch and volume settings.

The application status header displays:

- A time counter showing seconds since startup, or a 50-second countdown after each frame
- The total number of received beacon frames
- Statistics for valid, corrected, and invalid frames
- The currently selected beacon information

The application uses a 4-tab layout:

- List Tab: displays a summary of recently received beacons
- Detail tab: shows detailed information about the selected beacon
- Map tab: displays the beacon's location
- QR tab: shows a QR code linking to the beacon location or details from SARSAT Beacon Hex Decoder service

The **EPIRB RX** application decoding algorithm is very sensitive to frequency alignment. Due to size limitation of Mayhem applications, the **EPIRB RX** application does not provide a spectrum waterfall view. Hearing a beacon frame does not necessarily mean you are tuned to the correct frequency, as COSPAS/SARSAT channels are very close to each other. If decoding fails, it is recommended to use the Audio application's waterfall view to verify frequency alignment.

67 📶 Protocol

Beacon frame data is encoded using **Manchester BPSK modulation** at **400 bits per second**, with a phase of **±63°**, in accordance with COSPAS/SARSAT specifications.

Each frame is preceded by a 160 ms carrier.

68 📻 Baseband

A dedicated proc_epirb **baseband processor** has been implemented using a state machine to:

- Detect the 160 ms carrier
- Detect the first phase transition and start Manchester decoding at each phase change
- Read packets using a dedicated PacketBuilder

An **Automatic Frequency Correction (AFC)** loop continuously tracks the carrier offset in the idle state, allowing the decoder to acquire signals up to **±5 kHz** off the nominal frequency before carrier lock. This eliminates the need for manual frequency fine-tuning in most cases and makes the decoder tolerant of oscillator drift and Doppler shift.

69 ❓ User Guide

69.0.1 📡 Receiver section

This section is used to adjust receiver parameters for optimal decoding.

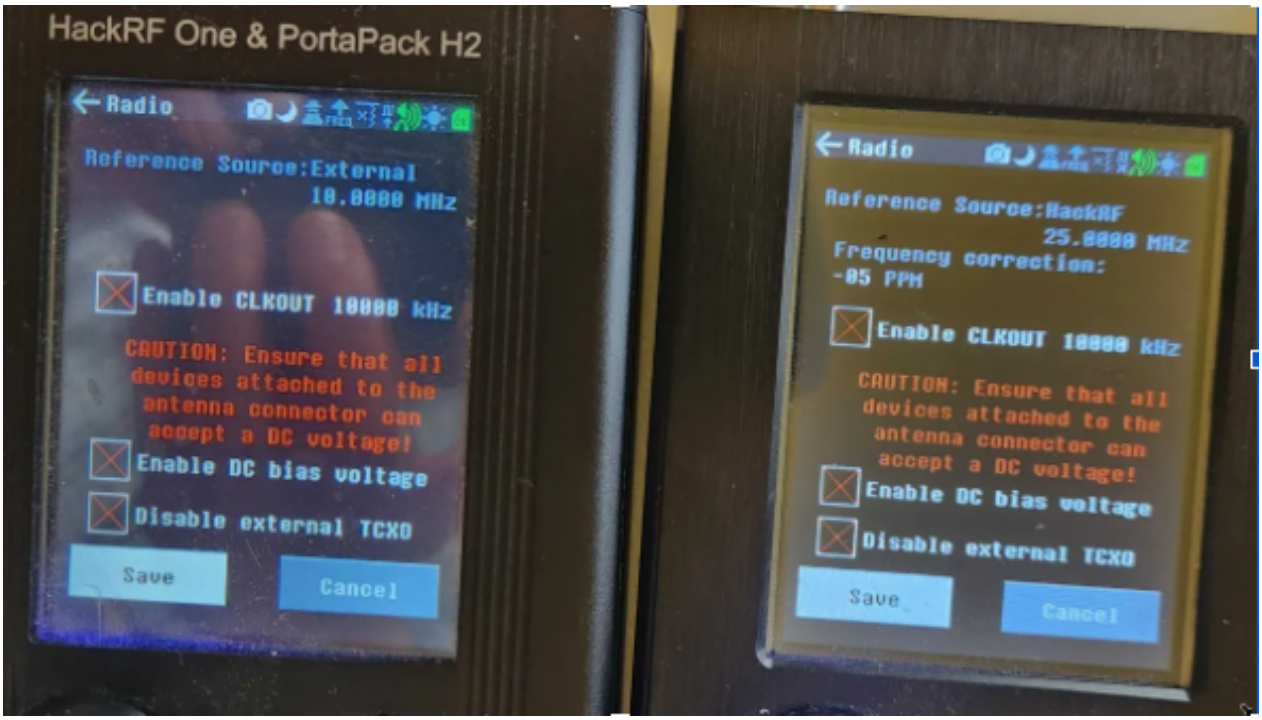
69.0.2 i Application status

This section displays the current application status.

69.0.3 📓 List Tab

The first tab displays a list of recently received beacons.

Note : In case that you are not decoding any received audible EPIRB burst signal , and you have selected a correct tuning frequency, please make sure if your system is using an external portapack built-in , 10 Mhz TCXO reference clock or not. You can see it , just going to menu , Settings or Radio you can see in the top of the picture if your system is using external 10 Mhz TCXO or internal 25Mhz Hackrf xtall.



In case that your system is using the internal (non TCXO) 25Mhz Hackrf xtall, you can confirm that possible freq. offset using some good external PLL digital receiver (HF, of FM ,VHF, or UHF) and use any TX App (like example MIC App) and check and eventually compensate the potential +/- frequency drift through the menu Settings or Radio or Frequency correction , example −05 PPM, as above right picture (save that PPM compensation and from now on it will be automatically applied to both TX and RX App's .

69.0.4 🔍 Detail Tab
This tab provides detailed information about the selected beacon.

69.0.5 🗺️ Map Tab
Displays the selected beacon on a map. Other beacons within the visible map area are also displayed.

69.0.6 🔗 QR Tab
Display QR codes that can be scanned to access location or details from SARSAT Beacon Hex Decoder service

70 ⚙️ Settings

70.0.1 📝 SETTINGS/rx_epirb.ini file
In addition to receiver settings, that are available from the UI, other settings can accessed by editing `SETTINGS/rx_epirb.ini` file when **App Settings** is enabled in the global Settings menu.

This settings include:

- countdown: duration (in seconds) of the countdown that starts after receiving a frame

70.0.2 📝 EPIRB/FREQ.TXT file
The list of frequencies available from the frequency selector comes from EPIRB/FREQ.TXT file.

You can edit this file to add custom frequencies.

Default frequencies:

- **406.025 MHz** (406025000)
- **406.028 MHz** (406028000)

- **406.037 MHz** (406037000)
- **406.040 MHz** (406040000)
- **406.049 MHz** (406049000)
- **406.052 MHz** (406052000)
- **406.061 MHz** (406061000)
- **406.064 MHz** (406064000)
- **433.025 MHz** (433025000)
- **144.875 MHz** (144875000)
- **406.022 MHz** (406022000)

70.1 ERT Meter

71 Introduction

Encoder receiver transmitter (ERT) is a packet radio protocol developed by Itron for automatic meter reading of water, gas and electricity. The system uses OOK short range radio which is transmitted in the unlicensed 902-928 MHz ISM band, so that meters can be read from a passing vehicle. This is mainly used in USA. The data is not encrypted, and does not have functions to enable full control. The system is not as advanced as a smart meter and lacks the control or security in the data transferred. Supported message protocols are IDM, SCM, and SCM+.

This is an external app (`.ppma` file on the SD card).

72 Controls

The Key Items on the App that can be selected with the cursor and changed with the encoder knob are:

- **Title bar:** The usual Items may be changed and displayed.
- **Frequency:** The default frequency of 911.6 MHz can be modified in newer firmware (the ISM band is commonly ~902 MHz to 928 MHz).
- **Gain:** Setting are shown in order of AMP 0=0db or 1=14dB, LNA(IF) (0-40) and VGA (Baseband Gain) (0-62).

73 Indications

Fields displayed in the App are as follows:

- **ID:** The meter ID in decimal format. Should match the number or barcode printed on the physical meter.
- **Ty:** The packet type — one of `IDM`, `SCM`, or `SCM+`.
- **Consumpt:** The meter consumption reading value.
- **Tamp:** Tamper flags. For SCM type meters the tamper flags are shown as two single-digit values (physical/encoder tamper flags respectively). For SCM+ or IDM type meters the tamper flags are shown as a 4-digit hexadecimal value. Note: for IDM packets, bits 32–47 of the tamper flags are not displayed due to screen width limitations.
- **Ct:** The count of the number of readings received with the same meter ID. Displayed as `++` if the count exceeds the field width.

The PortaPack ERT receiver monitors approximately 2.5 MHz centered around 911.6 MHz. It does not implement channel filters, so sensitivity is reduced in exchange for monitoring more simultaneous "channels".

74 Logging

If a FAT-formatted micro SD card is present when this mode is entered, the receiver will log received packets to a file named `LOGS/ERT.TXT`.

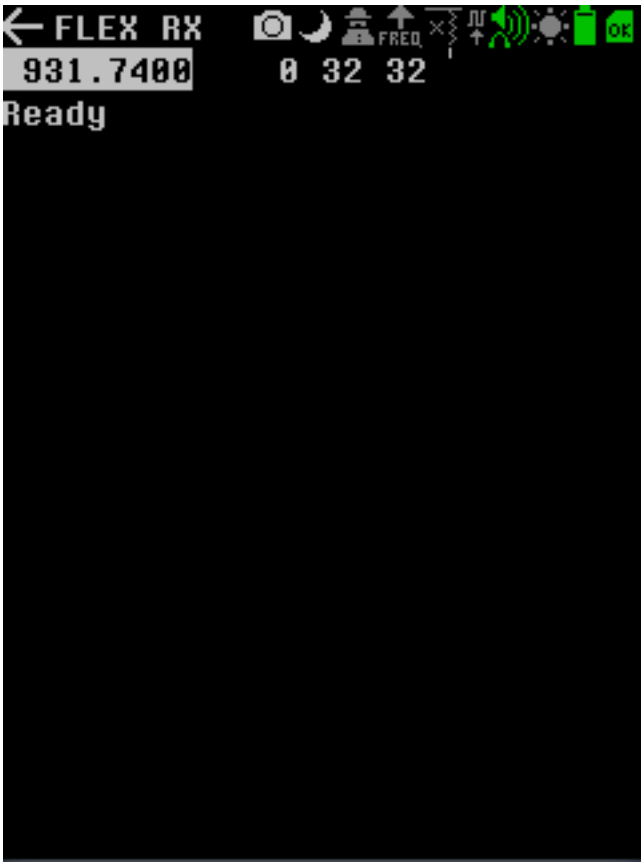
Each log entry is one line with the following format:

```
YYYYMMDDHHMMSS FREQUENCY TYPE data/errors ID:meterID
```

- `YYYYMMDDHHMMSS` — timestamp in sortable format
- `FREQUENCY` — the tuned frequency in Hz at the time of reception
- `TYPE` — packet type: `IDM`, `SCM`, or `SCM+`
- `data/errors` — Manchester-decoded data bits, a `/`, and a per-bit Manchester coding error indicator (`1` = error on that bit)
- `ID:meterID` — the decoded meter ID in decimal

Received data that looks like it might be a packet (based on the sync bits) but has an invalid checksum will not be shown on screen, but will still be written to the log file for debugging purposes. Adjusting the gain values may help reduce these invalid packets.

74.1 Flex RX



75 Introduction

FLEX is a one-way paging protocol developed by Motorola in the 1990s, used by commercial paging networks to deliver messages to pagers. It supports multiple baud rates and modulation schemes, allowing higher capacity than its predecessor POCSAG. FLEX paging networks were widely deployed across North America, Europe, and Asia, and some systems remain in operation today.

The Flex RX app receives and decodes FLEX pager messages, displaying the capcode (pager address) and message content on screen.

This is an external app (`.ppma` file on the SD card).

76 Controls

The key items that can be selected with the cursor and changed with the encoder knob are:

- **Title bar:** The usual items may be changed and displayed.
- **Frequency:** The default frequency is 931.74 MHz. Adjust to match the local paging network channel.
- **AMP or LNA or VGA:** Gain controls. Start with AMP off, LNA 24, VGA 24 and adjust based on received signal strength shown in the RSSI indicator.
- **Config:** Opens the configuration screen (logging, capcode filter, display font) — see Config screen below.

76.1 Common frequencies

FLEX paging frequencies vary by country and operator. Some examples:

- Region: USA (primary); Frequency: 931.8125 MHz
- Region: USA (secondary); Frequency: 929.7125 MHz, 931.7375 MHz
- Region: USA (various); Frequency: 929.0375 – 932.0 MHz band
- Region: UK or Europe; Frequency: 153.025 MHz, 466 MHz band
- Region: Australia; Frequency: 148.0125 MHz

Note: The app default (931.74 MHz) approximates the US 931.7375 MHz channel. Local frequencies depend on the operator and region. Scanning across known paging bands with Looking Glass or Scanner may help identify active channels.

77 Protocol details

FLEX uses FM modulation with automatic detection of baud rate and modulation level from the sync codeword. The following modes are supported by the Mayhem decoder:

- Mode: 1600/2; Baud rate: 1600 baud; Modulation: 2-FSK; Phases: A
- Mode: 1600/4; Baud rate: 1600 baud; Modulation: 4-FSK; Phases: A, B
- Mode: 3200/2; Baud rate: 3200 baud; Modulation: 2-FSK; Phases: A, C
- Mode: 3200/4; Baud rate: 3200 baud; Modulation: 4-FSK; Phases: A, B, C, D

The app demodulates at a sample rate of 3.072 MHz (decimated to 24 kHz for symbol detection) with a channel bandwidth of 1.75 MHz.

78 Display

Received messages are shown in a scrolling console. Each decoded message entry shows:

- **Capcode** — the numeric pager address (up to 10 digits)
- **Message text** — decoded content (alphanumeric, numeric, or "Tone Only")

Up to 20 messages are stored in the display buffer. When the buffer is full, the oldest messages are removed to make room for new ones. A blank line separates each message entry.

78.1 Message types

The following message types can be decoded:

- Type: Alphanumeric; Description: Free-text message (7-bit ASCII)
- Type: Numeric; Description: Digits only, using BCD encoding
- Type: Tone Only; Description: Alert to the pager, no message text
- Type: Secure; Description: Encrypted alphanumeric (content not decoded)
- Type: Binary; Description: Raw binary data (not decoded)

Numbered Numeric and Special Numeric variants are also supported.

79 Config screen

A **Config** button at the top of the screen opens a settings page, much like POCSAG RX. The choices are kept in the app settings and restored the next time the app is opened:

- **Enable Log** — writes each decoded page to `LOGS/FLEX.TXT`, timestamped.
- **Log Raw Data** — in addition, logs every packet with all of its fields (the serial output format), including system information (BIW). This is the complete capture and is not affected by the capcode filter.
- **Use Small Font** — shows the message list in the compact 5×8 font, fitting more lines on screen.
- **Hide Tone Only** — tone-only pages carry no message text and are hidden by default. Uncheck to list them as well.
- **Filter Mode** — `None`, `Drop` or `Keep`, used together with a **Capcode** value (64-bit, so long addresses fit):
  - `None` — no filtering (default).
  - `Drop` — hide the pages addressed to the given capcode.
  - `Keep` — show only the pages addressed to the given capcode.

Note: The capcode filter affects only the pages shown on screen and the normal log (**Enable Log**). **Log Raw Data** and the USB serial output always keep the complete, unfiltered capture.

80 Status bar

The top two rows of the screen show network metadata decoded from BIW words. An RSSI meter at the top of the screen shows live signal strength.

Row 1: `C<cycle>/F<frame> <bitrate> <polarity>  HH:MM:SS UTC+N`

Time is reconstructed from BIW TIME as `Frame-0 time + frame*1.875s + ext_sec*0.9375s`. If the transmitter sends an ExtSec value it is shown as `UTC+N es=N`.

Row 2: Local ID, coverage zone, country code from SSID BIWs. R flag = roaming supported.

81 Settings persistence

The selected frequency is saved automatically and restored the next time the app is opened. The Config-screen choices (logging, raw logging, small font, hide tone-only, and the capcode filter mode and value) are saved as well. Settings are stored in `SETTINGS/rx_flex.ini` on the SD card.

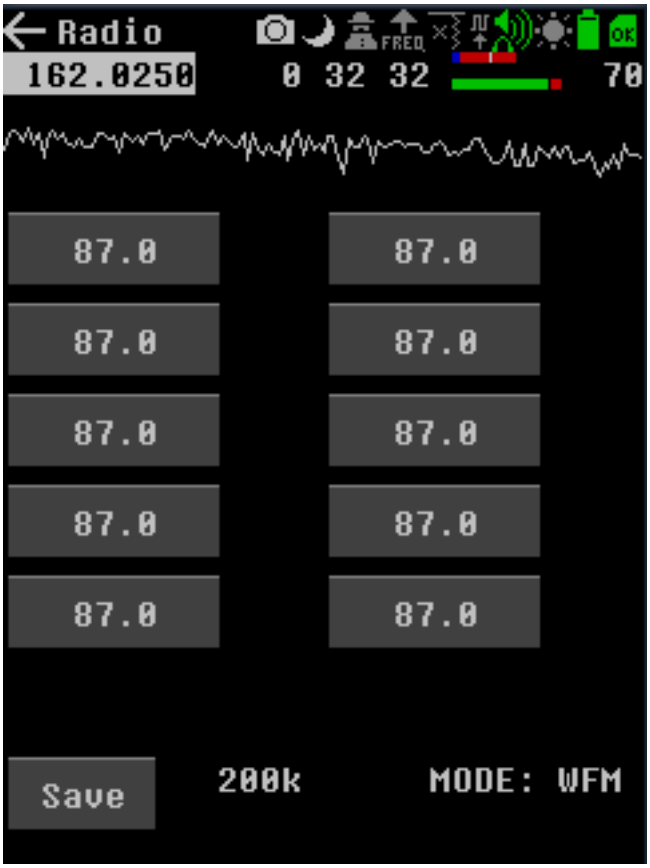
82 Legal notice

Receiving pager transmissions may be restricted in your country. In many jurisdictions, intercepting communications intended for others — even if unencrypted — is regulated under telecommunications or wiretapping law. Check local regulations before use. Transmitting on paging frequencies without authorization is illegal.

83 Further reading

- FLEX protocol — Wikipedia
- FLEX paging frequencies — RadioReference

83.1 Radio



This is an external app (`.ppma` file on the SD card).

84 Introduction

The Radio app is a general-purpose audio receiver with 10 quick-access favourite slots. It supports AM, NFM (Narrow FM), WFM (Wide FM), USB and LSB demodulation modes and is primarily intended for listening to broadcast FM, amateur radio, and other audio transmissions. Switching between presets requires only a single button press or touch, making it convenient for scanning across known frequencies.

85 Controls

Top row (always visible):

- **Frequency:** The tuned frequency. Use the rotary encoder to step in 25 kHz increments (suitable for FM broadcast), or click to open the numeric keypad. The default frequency is 87.0 MHz.
- **AMP:** RF preamplifier — 0 = off, 1 = on (+14 dB). Avoid enabling AMP on strong local signals as it will cause overload.
- **LNA:** Low Noise Amplifier gain (0–40 dB). Use the rotary encoder to adjust.
- **VGA:** Variable Gain Amplifier or baseband gain (0–62 dB). Use the rotary encoder to adjust.
- **Volume:** Audio output level. Use the rotary encoder to adjust.

Waveform or Spectrum display:

Below the top row, an audio spectrum waveform is shown during reception. In WFM mode only, clicking on the waveform toggles between the waveform view and a graphic equalizer (GraphEq) display. Clicking the GraphEq cycles through 20 built-in colour themes.

Favourite buttons (middle):

Ten buttons arranged in two columns show the frequency of each saved preset. Clicking a button immediately loads that preset (frequency, modulation, and bandwidth). Buttons show the frequency in MHz format (e.g., `87.0`). Unassigned slots default to 87.0 MHz or WFM.

Bottom row:

- **Save:** Activates save mode. The label "Select slot" appears above the favourite buttons. Click any favourite button to save the current frequency, modulation, and bandwidth to that slot. Save mode is cancelled automatically after saving or by pressing Save again.
- **BW (Bandwidth):** Adjusts the demodulation bandwidth. Available options depend on the selected modulation mode (see table below).
- **MODE:** Selects the demodulation mode. Available modes: `AM`, `NFM`, `WFM`, `USB`, `LSB`.

85.1 Modulation modes

- Mode: WFM; Description: Wideband FM — broadcast FM radio (87–108 MHz); Audio rate: 48 kHz; Default BW: 200 kHz
- Mode: NFM; Description: Narrowband FM — PMR, amateur VHF/UHF; Audio rate: 24 kHz; Default BW: 16 kHz
- Mode: AM; Description: Amplitude Modulation — MW, shortwave; Audio rate: 24 kHz; Default BW: DSB
- Mode: USB; Description: Upper Sideband — HF amateur bands; Audio rate: 24 kHz; Default BW: (AM baseband)
- Mode: LSB; Description: Lower Sideband — HF amateur bands; Audio rate: 24 kHz; Default BW: (AM baseband)

Note: USB and LSB use the AM baseband processor internally with the sideband filter applied. The BW selector changes the AM configuration accordingly.

85.2 Saving a favourite

1. Tune to the desired frequency and select the correct modulation and bandwidth.
2. Press **Save**. The text "Select slot" appears.
3. Press any of the 10 favourite buttons to store the current settings.

The slot is immediately updated and the new frequency label appears on the button. All slots are saved persistently to the SD card.

86 Settings persistence

All 10 favourite slots (frequency, modulation, bandwidth) and the current colour theme for the GraphEq are saved automatically and restored on next launch. Settings are stored in `SETTINGS/rx_fmradio.ini` on the SD card.

86.1 What this app does not support

86.1.1 DAB and DAB+

DAB (Digital Audio Broadcasting) and DAB+ cannot be received or decoded by this app or any other app in Mayhem. DAB/DAB+ is a digital standard that uses OFDM modulation across a 1.536 MHz channel, with audio encoded in MPEG-4 HE-AAC (DAB+) or MPEG-1 Layer II (DAB). Decoding requires significant DSP processing for OFDM demodulation, channel correction, Viterbi decoding, and audio codec processing — well beyond what the HackRF's M4 baseband processor can handle standalone.

DAB/DAB+ is therefore only receivable with a dedicated DAB receiver or a PC-based SDR solution (e.g. Welle.io or a hardware DAB stick). The PortaPack with Mayhem cannot substitute for these.

NOTE FM broadcast (WFM) worldwide: The Radio app receives FM broadcast normally wherever it is available:

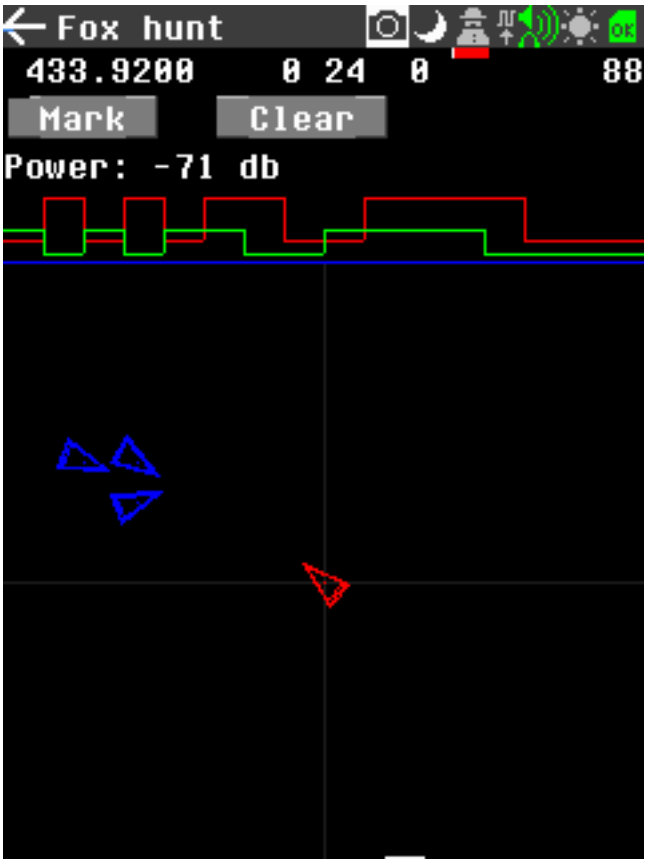
- **Europe or most of the world:** 87.5–108 MHz
- **USA or Canada:** 87.9–107.9 MHz (same band, fully compatible)
- **Japan:** 76–95 MHz — note the different lower band edge; tune accordingly
- **Regions with DAB+ rollout (Germany, UK, Norway, Switzerland, Australia etc.):** FM and DAB+ often run in parallel. FM remains receivable with WFM; DAB+ is not.
- **Regions where FM is still dominant (most of Africa, South/Central America, Southeast Asia):** WFM works as expected with no DAB interference to consider.

DAB/DAB+ via PC: When the PortaPack is switched to [[HackRF]] and connected to a PC via USB, DAB/DAB+ reception becomes possible using PC-based SDR software such as **Welle.io** (Windows/ Linux/macOS, supports HackRF natively). The HackRF has more than enough bandwidth for a full DAB multiplex (1.536 MHz channel). In this case the PC handles all demodulation and decoding — the PortaPack itself does nothing beyond acting as an RF frontend.

86.1.2 Other unsupported digital audio standards

Similarly, **HD Radio** (USA), **DRM** (Digital Radio Mondiale, HF), and **ISDB-Tsb** (Japan/Brazil) are not supported for the same reasons — all require complex digital demodulation and audio decoding that exceeds the hardware's processing capacity.

86.2 Fox-Hunt



The Fox Hunt app is designed to help with transmitter hunting (also known as fox hunting). Its use is not limited to the game — it can be used to locate any AM signal source.

The receiver demodulates in **AM mode only** and outputs audio through the speaker. Fox hunt transmitters typically send Morse code, which can be heard directly. A real-time RSSI graph and power level display help judge signal strength changes as you move toward or away from the source.

The app integrates with external GPS and orientation (compass) modules to show your current position and heading on a built-in map, and allows you to drop markers at bearing positions to triangulate the transmitter's location.

This is an external app (`.ppma` file on the SD card).

87 Controls

Top row:

- **Frequency:** The frequency to monitor. Use the rotary encoder to step in **100 Hz increments** (very fine tuning, useful for centering on a weak AM carrier), or click to open the numeric keypad.
- **AMP:** RF preamplifier — 0 = off, 1 = on (+14 dB). Use with caution near strong signals.
- **LNA:** Low Noise Amplifier gain (0–40 dB).
- **VGA:** Variable Gain Amplifier or baseband gain (0–62 dB).
- **Volume:** Audio output level.

Below frequency row:

- **Power: X db** — Displays the current peak channel power in dB. Updates in real time.
- **Mark** — Drops a marker at your current GPS position and compass bearing. Use this when the signal peaks in a particular direction, then move to a new position and mark again.
- **Clear** — Removes all previously placed markers from the map.

RSSI graph:

A scrolling bar graph (64 columns) showing the history of minimum, average, and maximum RSSI values alongside the peak power. This gives a visual trend of signal strength over time, helping identify whether you are approaching or moving away from the transmitter.

Map:

The lower portion of the screen shows a geographical map. The map auto-follows your GPS position — it is not manually scrollable. If an orientation module is connected, a directional arrow shows your current heading. Dropped markers (from the **Mark** button) appear on the map and remain until **Clear** is pressed.

Zooming in the map helps significantly when navigating to a nearby transmitter — use the encoder or touch to zoom.

88 External modules

The app benefits from two optional external modules connected via the PortaPack's I²C/GPIO expansion port:

- **GPS module** — Provides real-time position updates. The map centres on your location and tracks movement automatically.
- **Orientation or compass module** — Provides a bearing angle. The map displays a heading arrow. Each marker stores the bearing at the moment it was dropped, which is useful for triangulation (two or more bearings from different positions narrow down the transmitter's location).

Without these modules the map is static and markers cannot be placed.

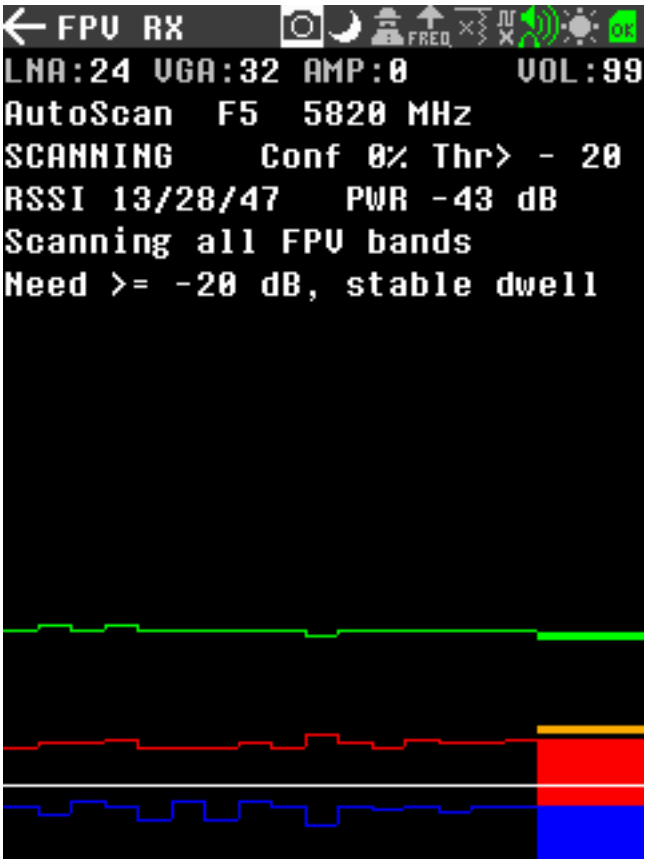
89 Typical usage

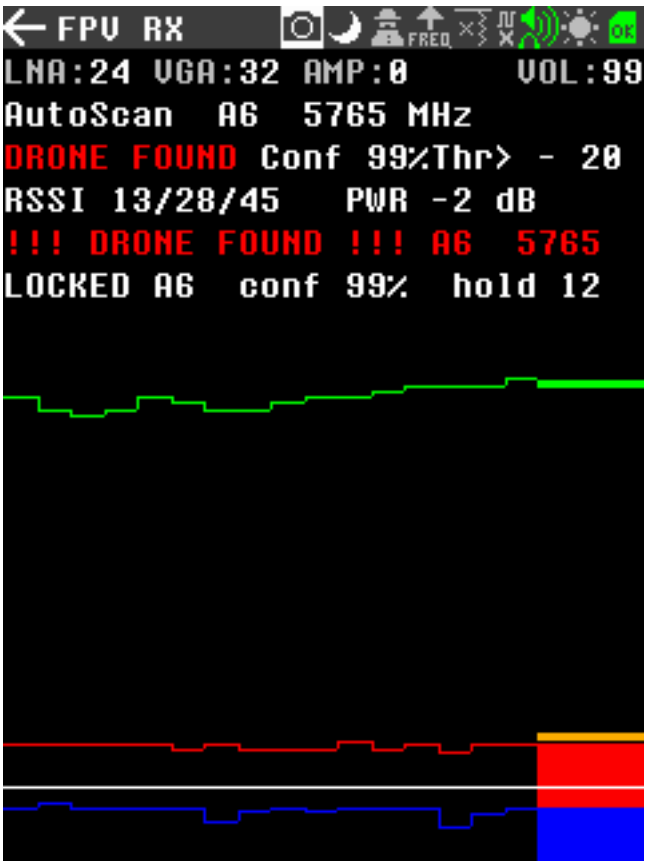
1. Tune to the known fox hunt frequency.
2. Adjust LNA and VGA so the signal is audible but not overloaded.
3. Move to an open area and rotate — the RSSI graph and Power indicator will peak when pointing toward the source.
4. Press **Mark** to record your bearing and position.
5. Move to a second location at least 50–100 m away, repeat, and press **Mark** again.
6. The intersection of the two bearing lines (visible on the map) indicates the transmitter's approximate position.

90 Settings persistence

The last used frequency and gain settings are saved automatically and restored on next launch. Settings are stored in `SETTINGS/rx_foxhunt.ini` on the SD card.

90.1 FPV-Detect





91 Introduction

FPV Detect is a PortaPack application for detecting analog FPV (first-person view) video signals and identifying their channel across the 5.8 GHz band. It covers all standard FPV bands — A, B, E, F, and R — used by most analog drone video transmitters (VTX) and goggles. The app can scan a single selected band or run **AutoScan** across all 40 channels simultaneously.

Its use is not limited to drone racing — it can be used to locate any analog FPV carrier in the 5.6–5.9 GHz range.

This is an external app (`.ppma` file on the SD card).

92 Controls

Top row:

- **LNA:** Low Noise Amplifier gain (0–40 dB).
- **VGA:** Variable Gain Amplifier or baseband gain (0–62 dB).
- **AMP:** RF preamplifier — 0 = off, 1 = on (+14 dB). Use with caution near strong signals.
- **VOL:** Audio output volume.

Second row:

- **Band selector:** Choose which band to scan — `Band A`, `Band B`, `Band E`, `Band F`, `Band R`, or `AutoScan` (default). AutoScan steps through all 5 bands × 8 channels = 40 channels in order.
- **Frequency display:** Shows the current channel being monitored in the format `B3  5771 MHz` (band letter, channel number, frequency).

Third row:

- **State badge:** Current detection state — `SCANNING`, `VERIFY FPV`, or `DRONE FOUND` (red when locked).
- **Conf X%:** Detection confidence percentage (0–99%). Increases during verification and lock.
- **Thr> / threshold field:** Detection threshold in dBm, adjustable from −100 to +20. Default: **−38 dB**. Only signals above this level trigger detection. Increase to reduce false positives in noisy environments; decrease if signals are weak.

Fourth row:

- **RSSI min/avg/max:** Scrolling RSSI bar graph (256 columns) showing min, average, and maximum received levels.
- **PWR X dB:** Current peak channel power in dB.

Status lines:

- **Status text:** Human-readable summary — band/channel/frequency when verifying or locked.
- **Detail text:** During scanning: shows the threshold requirement. During verification: shows average dB, peak dB, hit/sample counts. During lock: shows locked channel, confidence, and hold counter.

The lower part of the screen shows the scrolling RSSI history graph and a vertical peak-hold RSSI indicator on the right.

93 Detection algorithm

The app uses a three-state machine. The key insight is that a real FPV carrier is a **narrow peak** — it stands out clearly from adjacent channels. Broadband interference (Wi-Fi, noise) does not, so it gets filtered out during the neighbor check in the Candidate state.

```
stateDiagram-v2
    [*] --> Scanning

    Scanning --> Candidate : power ≥ threshold (default −38 dB)\nbeep 1150 Hz or 60 ms
    Candidate --> Locked : ≥4/5 hits AND conf ≥72%\nAND center ≥ neighbors+3 dB\nbeep 1850
Hz or 180 ms
```

```
    Candidate --> Scanning : verification failed\n(too few hits or weak neighbor margin)

    Locked --> Scanning : hold counter reaches 0\n(power stays below threshold-6 dB)\nbeep
650 Hz or 90 ms

    state Scanning {
        [*] --> stepping
        stepping --> stepping : 1 frame dwell per channel\nAutoscan: 40 ch total\nSingle
band: 8 ch
    }

    state Candidate {
        [*] --> sampling
        sampling --> neighbor_check : after 2nd sample
        neighbor_check --> sampling : retune left+right neighbors\nto measure their power
        sampling --> lock_decision : 5 samples collected
    }

    state Locked {
        [*] --> holding
        holding --> holding : power ≥ threshold-6 dB\nhold counter stays at 12\nconf++ (up
to 99%)
        holding --> draining : power < threshold-6 dB\nhold counter-- per frame
    }
```

Scanning — the receiver steps through channels one by one, dwelling for 1 frame per channel. When power exceeds the detection threshold, it enters Candidate.

Candidate — the app stays on the suspected channel and collects up to 5 power samples. After the 2nd sample it briefly re-tunes to the adjacent channels (left neighbor, then right neighbor) to measure their current power. A confidence score is computed from: average power above threshold, peak margin, hit/miss ratio, and neighbor margin (center power minus the strongest adjacent channel). Lock requires all three: ≥4 hits out of 5 samples, confidence ≥72%, and center channel at least 3 dB above neighbors. A fast-lock path fires earlier if peak exceeds threshold by ≥8 dB and confidence ≥80%.

Locked — the app stays on the confirmed channel, shows `DRONE FOUND` in red, and emits a long beep. A hold counter (max 12) prevents brief signal fades from unlocking immediately — each frame below the unlock threshold (= detection threshold − 6 dB) decrements the counter; unlock only occurs when it reaches zero.

94 FPV channel frequencies

All 40 standard channels covered by the app:

- Ch: 1; Band A (MHz): 5865; Band B (MHz): 5733; Band E (MHz): 5705; Band F (MHz): 5740; Band R (MHz): 5658
- Ch: 2; Band A (MHz): 5845; Band B (MHz): 5752; Band E (MHz): 5685; Band F (MHz): 5760; Band R (MHz): 5695
- Ch: 3; Band A (MHz): 5825; Band B (MHz): 5771; Band E (MHz): 5665; Band F (MHz): 5780; Band R (MHz): 5732
- Ch: 4; Band A (MHz): 5805; Band B (MHz): 5790; Band E (MHz): 5645; Band F (MHz): 5800; Band R (MHz): 5769
- Ch: 5; Band A (MHz): 5785; Band B (MHz): 5809; Band E (MHz): 5885; Band F (MHz): 5820; Band R (MHz): 5806
- Ch: 6; Band A (MHz): 5765; Band B (MHz): 5828; Band E (MHz): 5905; Band F (MHz): 5840; Band R (MHz): 5843
- Ch: 7; Band A (MHz): 5745; Band B (MHz): 5847; Band E (MHz): 5925; Band F (MHz): 5860; Band R (MHz): 5880
- Ch: 8; Band A (MHz): 5725; Band B (MHz): 5866; Band E (MHz): 5945; Band F (MHz): 5880; Band R (MHz): 5917

The receive bandwidth is 750 kHz per channel (IQ capture mode, no audio demodulation during scanning).

Note: Band E channels 5–8 (5885–5945 MHz) and Band R channel 8 (5917 MHz) fall outside the standard 5.725–5.875 GHz ISM band. These are legal to receive but may not be legal to transmit on in all regions.

95 ⚠️ False positives

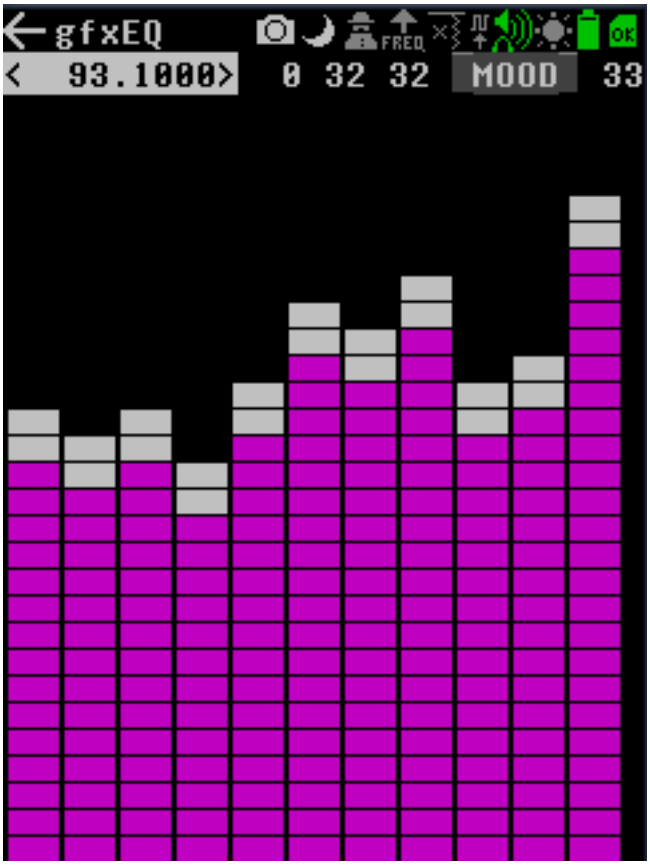
FPV Detect can trigger false positives, particularly **indoors or in urban environments** where the 5.8 GHz band is crowded with non-FPV sources. Detection is based on signal power, not video content analysis — any sufficiently strong narrow emission near an FPV channel frequency can cause a lock. Common interference sources include Wi-Fi 5/6 routers (5.725–5.825 GHz overlap), ISM band devices (wireless cameras, baby monitors), wireless HDMI senders, and harmonic emissions from other transmitters.

If false locks occur frequently: raise the detection threshold, select a specific band instead of AutoScan, or move to an area with less RF congestion. When locked, the audio output lets you verify: an actual FPV analog video carrier produces a characteristic buzz, while interference typically sounds like noise or silence.

96 Settings persistence

The detection threshold value is saved automatically and restored on next launch. Settings are stored in `SETTINGS/rx_fpv.ini` on the SD card.e.

96.1 gfxEQ



97 Introduction

gfxEQ is a Wideband FM (WFM) audio receiver with a full-screen graphic equalizer display. It receives FM broadcast radio and visualises the demodulated audio spectrum as 11 animated bar segments covering the audible frequency range. The display fills almost the entire screen. 20 colour themes are available and can be cycled without interrupting reception.

The gfxEQ widget is also embedded inside the Radio app, where it appears when you click the waveform display in WFM mode.

This is an external app (`.ppma` file on the SD card).

98 Controls

- **Frequency button** (leftmost, labelled `<93.10>` by default): click to open the numeric keypad for direct frequency entry. Use the rotary encoder while focused to step in **25 kHz increments** — suitable for tuning FM broadcast channels. The default frequency is **93.1 MHz**.
- **AMP:** RF preamplifier — 0 = off, 1 = on (+14 dB). Avoid enabling AMP near strong local broadcast transmitters.
- **LNA:** Low Noise Amplifier gain (0–40 dB). Default 32.
- **VGA:** Variable Gain Amplifier or baseband gain (0–62 dB). Default 32.
- **MOOD:** Cycles through 20 built-in colour themes. Each press advances to the next theme. The selected theme is saved and restored on next launch.
- **Volume:** Audio output level. Use the rotary encoder. If the audio is inaudible, try increasing to around 70.

The graphic equalizer occupies the rest of the screen below the controls and updates continuously from the WFM audio spectrum data.

99 Frequency bands

The display shows 11 bars, each representing a fixed audio frequency range:

- Bar: 1; Range (Hz): 0–375; Character: Deep bass, low rumble
- Bar: 2; Range (Hz): 375–750; Character: Upper bass punch
- Bar: 3; Range (Hz): 750–1500; Character: Lower midrange fullness
- Bar: 4; Range (Hz): 1500–2250; Character: Midrange clarity
- Bar: 5; Range (Hz): 2250–3375; Character: Upper midrange bite
- Bar: 6; Range (Hz): 3375–4875; Character: Presence and edge
- Bar: 7; Range (Hz): 4875–6750; Character: Lower brilliance
- Bar: 8; Range (Hz): 6750–9375; Character: Brilliance and air
- Bar: 9; Range (Hz): 9375–13125; Character: High treble sparkle
- Bar: 10; Range (Hz): 13125–16875; Character: Upper treble airiness
- Bar: 11; Range (Hz): 16875–24375; Character: Extreme treble or noise floor

100 Technical details

The app uses WFM demodulation with a **180 kHz** channel bandwidth at a 3.072 MHz sampling rate. Audio is output at 48 kHz. The modulation and bandwidth are fixed — this app is specifically intended for FM broadcast reception (87–108 MHz and above).

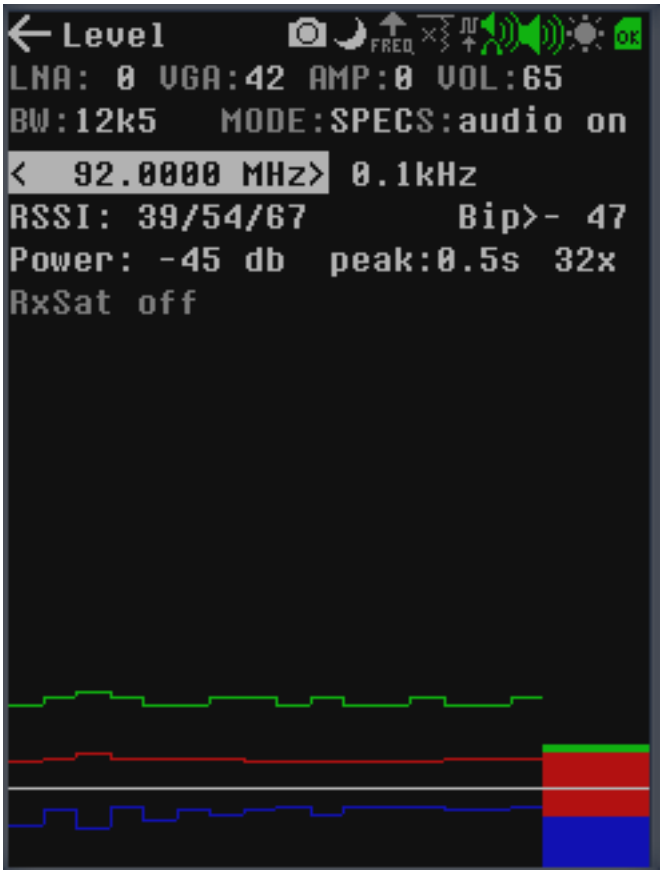
The frequency can also be set via the USB Serial Console using the standard `setfreq` command.

101 Settings persistence

The selected frequency and colour theme are saved automatically and restored on next launch. Settings are stored in `SETTINGS/rx_gfx_eq.ini` on the SD card.

101.1 Level

102 Level App



This is an external app (`.ppma` file on the SD card).

103 Introduction

The Level app provides a comprehensive single-screen signal monitoring view combining RSSI meters, a power level readout, receiver saturation indicator, a scrolling history graph, and optionally audio output and an audio squelch beep. It is designed for signal level measurement and monitoring across AM, NFM, WFM, and spectrum (capture) modes.

104 Controls

Row 1 — gain:

- **LNA:** Low Noise Amplifier gain (0–40 dB).
- **VGA:** Variable Gain Amplifier or baseband gain (0–62 dB).
- **AMP:** RF preamplifier — 0 = off, 1 = on (+14 dB).
- **VOL:** Audio output volume (only relevant when audio is on).

Row 2 — reception parameters:

- **BW:** Bandwidth selector. Options depend on the selected modulation mode (AM: DSB/USB/LSB/CW variants; NFM: various kHz widths; WFM: 200k/180k/80k; SPEC: various sampling rates from 12.5k to 20M).
- **MODE:** Demodulation mode — AM, NFM, WFM, or SPEC. SPEC (spectrum/capture) mode uses the IQ capture baseband and shows signal power without audio demodulation. Cycling past WFM wraps to SPEC, cycling past SPEC wraps back to AM.

- **S (Step):** Frequency tuning step size for the encoder. A short list of standard step sizes from 100 Hz upward.

Row 3 — frequency:

- **Frequency button** (labelled "): click to open the numeric keypad for direct entry. Use the rotary encoder while focused to step in the selected step size.
- The frequency persists from the last used receiver state.

Row 3 (right) — audio:

- **AUDIO:** `audio off` / `audio on` — enables or disables audio output. Audio is not available in SPEC mode even if set to on; in SPEC mode the beep function takes over instead when a signal is above the squelch level.

Row 4 — squelch and display:

- **Bip> / beep squelch field:** Threshold in dBm (−100 to +20) above which an audio beep is emitted. The beep pitch scales with signal strength: −100 dBm maps to 400 Hz, +20 dBm maps to 2600 Hz, duration 150 ms. The beep fires on every statistics update while the signal exceeds this threshold.
- **RSSI: min/avg/max** — current RSSI values from the scrolling graph.
- **peak:** Sets peak hold time for the right vertical RSSI bar — none, 0.25 s, 0.5 s, 1 s, 3 s, 5 s, or 10 s. Default: 0.5 s.
- **History columns:** Number of columns in the scrolling RSSI/power graph — 16, 32, 64, 128, or 240. Default: 32.

Row 5:

- **Power: X db** — peak channel power in dB (range −100 to +20).
- **RxSat: X%** — receiver saturation percentage (see below). Displayed with a colour gradient. Shows `RxSat off` in WFM+audio mode or SPEC mode where saturation measurement is not available.
- **CTCSS tone** — in NFM mode, the detected CTCSS/DCS sub-tone or code is shown to the right of RxSat if one is present on the received signal.

105 Displays

105.1 Right vertical level meter

A narrow vertical bar on the right edge of the screen:

- **Blue fill** — RSSI minimum
- **White horizontal bar** — RSSI average
- **Red fill** — RSSI maximum
- **Green marker** — peak hold (when peak hold is enabled and a peak has been captured)

105.2 Scrolling history graph

The main graph area below row 5 shows four overlaid traces over a scrollable history:

- **Blue line** — RSSI minimum (clipped to RSSI range 31–170, scaled to graph height)
- **White line** — RSSI average (same range)
- **Red line** — RSSI maximum (same range)
- **Green line** — channel power in dB (clipped to −100 to +20 dB, scaled to graph height)

The number of visible history columns is set by the **history columns** selector.

105.3 RxSat colour coding

The RxSat indicator uses a colour gradient to show how hard the receiver front-end is working:

- Range: 0–80%; Colour: Blue → green; Meaning: Normal — increasing as signal strength grows
- Range: 80%; Colour: Green; Meaning: Ideal operating point
- Range: 80–100%; Colour: Green → red; Meaning: Overloading — reduce LNA/VGA gain

RxSat is calculated from the M4 baseband performance counter. It is **not available** in SPEC mode or in WFM mode with audio enabled.

106 Modulation modes

- Mode: AM; Audio rate: 12 kHz; Notes: DSB default. BW selector changes AM configuration (DSB/USB/LSB/CW).
- Mode: NFM; Audio rate: 24 kHz; Notes: 16 kHz bandwidth default. CTCSS/DCS detection active.
- Mode: WFM; Audio rate: 48 kHz; Notes: 200 kHz bandwidth default. RxSat not shown when audio is on.
- Mode: SPEC; Audio rate: 24 kHz; Notes: IQ capture mode. BW sets the baseband sampling rate. Beep squelch active. No audio.

107 Settings persistence

The beep squelch threshold, audio on/off state, modulation mode, and bandwidth selection are saved automatically and restored on next launch. Settings are stored in `SETTINGS/rx_level.ini` on the SD card.

The frequency is not saved in the app's own settings — it follows the global receiver frequency. scaled along the height of the RSSIGraph widget

107.1 Morse RX

108 Introduction

Morse RX is a Morse code receiver that automatically detects and decodes incoming CW signals. Tune to the signal frequency and wait for clear audio — the app handles tone frequency detection, WPM estimation, and character decoding without manual configuration.

The decoder uses automatic tone and speed detection, which may take a few moments to lock onto a signal. Decoding quality depends heavily on signal clarity.

109 Controls

Top row:

- **Frequency:** Click to open the numeric keypad, or use the rotary encoder for fine tuning.
- **AMP:** RF preamplifier — 0 = off, 1 = on (+14 dB). Avoid enabling near strong signals.
- **LNA:** Low Noise Amplifier gain (0–40 dB).
- **VGA:** Variable Gain Amplifier or baseband gain (0–62 dB).
- **Volume:** Audio output level.

Second row:

- **Squelch (0–99):** Sets the squelch level. Signals below this threshold are ignored. Adjust to suppress background noise without cutting off the desired signal.
- **Mode:** Selects the demodulation mode — `AM/CW`, `NFM`, `AM/DSB`, `AM/USB`, or `AM/LSB`. Use `AM/CW` for standard HF CW, `NFM` for VHF/UHF Morse beacons.
- **Speed: XX wpm:** Displays the estimated receive speed in words per minute. This is calculated automatically from the detected dit/dah timing. If decoding goes wrong the value will visibly adapt as the algorithm self-corrects.

Third row:

- **Log checkbox:** Enables logging to a daily file in the `LOGS/` directory on the SD card. Each log session is stamped with a timestamp header and the selected mode.
- **Tone: XXXX Hz:** Displays the detected audio tone frequency of the incoming Morse signal. The colour indicates whether the tone is in a good range for decoding:
  - **Green** — 580–1220 Hz: optimal range
  - **Yellow** — 400–580 Hz or 1220–1400 Hz: acceptable but at the edge
  - **Red** — below 400 Hz or above 1400 Hz: outside the reliable decoding range; results will be poor

Fourth row:

- **Last seq.:** Shows the most recently detected dit/dah pattern before it was decoded into a character, useful for debugging weak or noisy signals.

Decoded text console:

The large area in the lower half of the screen shows the decoded characters. The colour of each character indicates the decoder's confidence:

- **Green** — confidence ≥ 90%: high confidence
- **Yellow** — confidence 80–90%: moderate confidence
- **Red** — confidence < 80%: low confidence, likely incorrect
- **White** — spaces and unmatched sequences

Bottom:

- **CLR:** Clears the decoded text console.

110 Tips for best results

The decoder requires a clean, steady audio tone in the 400–1400 Hz range. Before starting:

- Adjust frequency so the signal produces a clear audio tone — listen with the speaker enabled.
- Use a narrowband mode (`AM/CW` or `NFM`) and set gain so the signal is strong but not clipping (a `clipping` warning appears on screen if the input is saturated).
- Keep the scan range small and avoid strong nearby interferers.
- If the tone frequency display shows red and jumps around, there is too much noise or the wrong mode is selected.

111 Settings persistence

The selected demodulation mode is saved automatically and restored on next launch. Settings are stored in `SETTINGS/rx_morse_radio.ini` on the SD card.

111.1 NOAA

112 Update regarding decommissioning of all NOAA POES satellites

All three NOAA POES satellites have been decommissioned and are no longer available for receiving.

This is an external app (`.ppma` file on the SD card).

112.1 Decommissioning Timeline

NOAA-18: Decommissioned on June 6, 2025 at 17:40 UTC due to an unrecoverable S-Band transmitter failure.

NOAA-19: Decommissioned on August 13, 2025 at 16:55 UTC due to a battery failure on August 9, 2025 that threatened safe spacecraft operations. This occurred earlier than the planned August 19 date.

NOAA-15: Decommissioned on August 19, 2025 at 20:37 UTC (postponed from the original August 12 date).

113 Update regarding decommissioning of NOAA-18

NOAA-18 was decommissioned on June 6, 2025 and is no longer available for receiving. This is due to a failing S-Band transmitter used in the control of the satellite. At the time of this update both NOAA-15 and NOAA-19 continue to remain operational

https://www.nesdis.noaa.gov/about/documents-reports/notice-of-changes/2025-notice-of-changes/decommissioning-of-noaa-18-scheduled-june-6-2025-1733-1749-utc

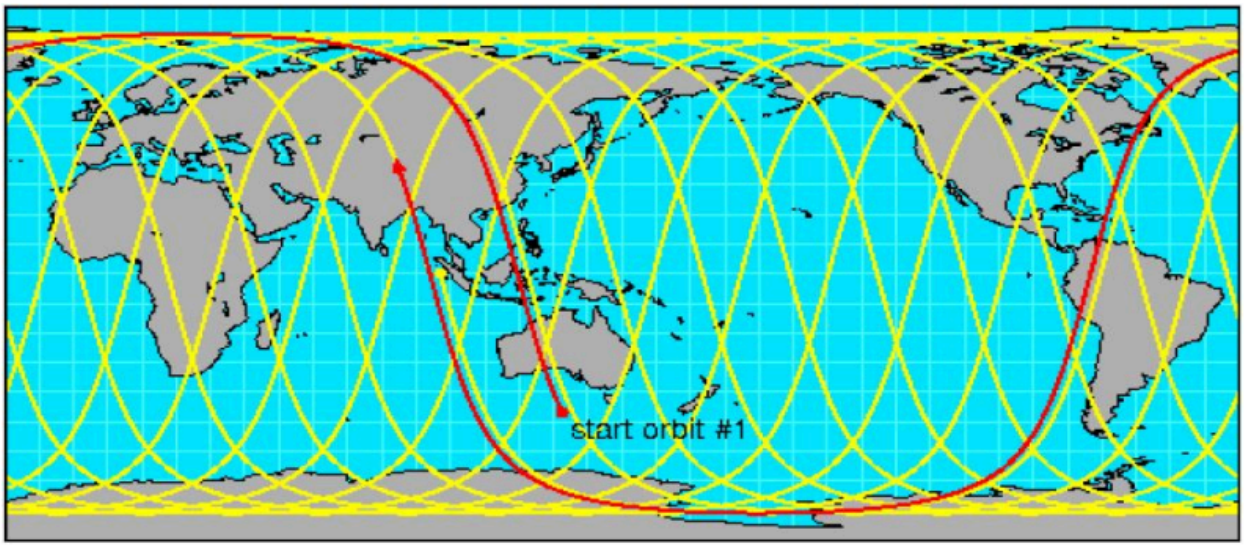
114 Introduction

That Rx NOAA App, allows us to receive any of the 3 current operating NOAA weather satellites that are transmitting analogue APT maps in the 137 MHz band . Currently, they are using those frequencies, (but in future they may change them, then , better time to time, double check it)

Frequencies

Satellite	Frequency	Mode	Bandwidth	Polarisation
NOAA 15	137.62 MHz	FM	50 kHz	RHCP
NOAA 18	137.9125 MHz	FM	50 kHz	RHCP
NOAA 19	137.1 MHz	FM	50 kHz	RHCP

NOAA Polar Operational Environmental Satellites (POES) have an orbital period of approximately 102 minutes, approximately 520 miles (800 kms) above Earth, meaning they circle the Earth in about that time.The Earth's rotation allows the satellite to see a different view with each orbit, and each satellite provides two complete views of weather around the world each day. This allows them to complete about 14 orbits per day, passing over the same location twice daily, once during daylight and once at night.Example of the 14 orbits , in 24h (Image from NOAA):

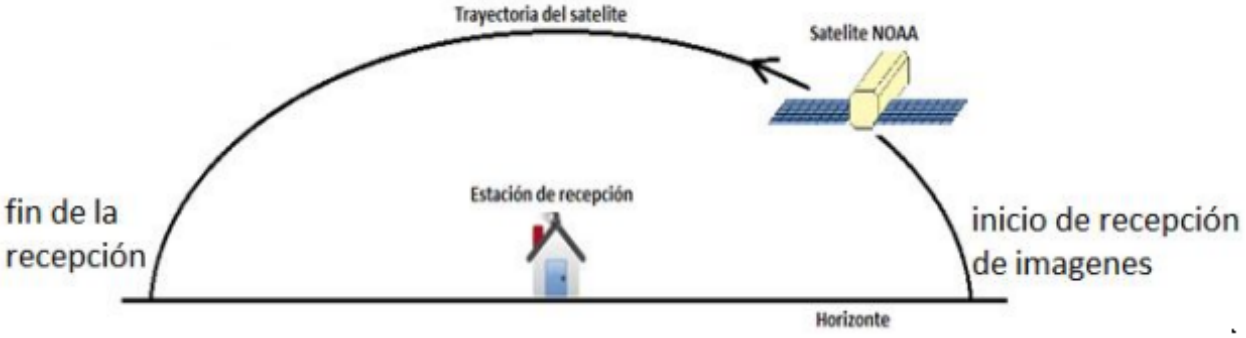


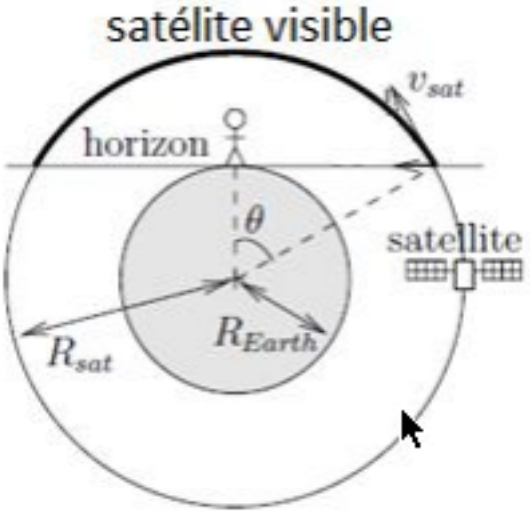
The APT signal is a 256-level amplitude-modulated 2400 Hz subcarrier, which is then frequency modulated onto the 137 MHz-band RF carrier. The satellite is broadcasting it with an effective radiated power of approximately 5 watts (37 dBm) at 800 kms from the ground earth . These signals are frequency modulated (FM) and carry low-resolution weather imagery.

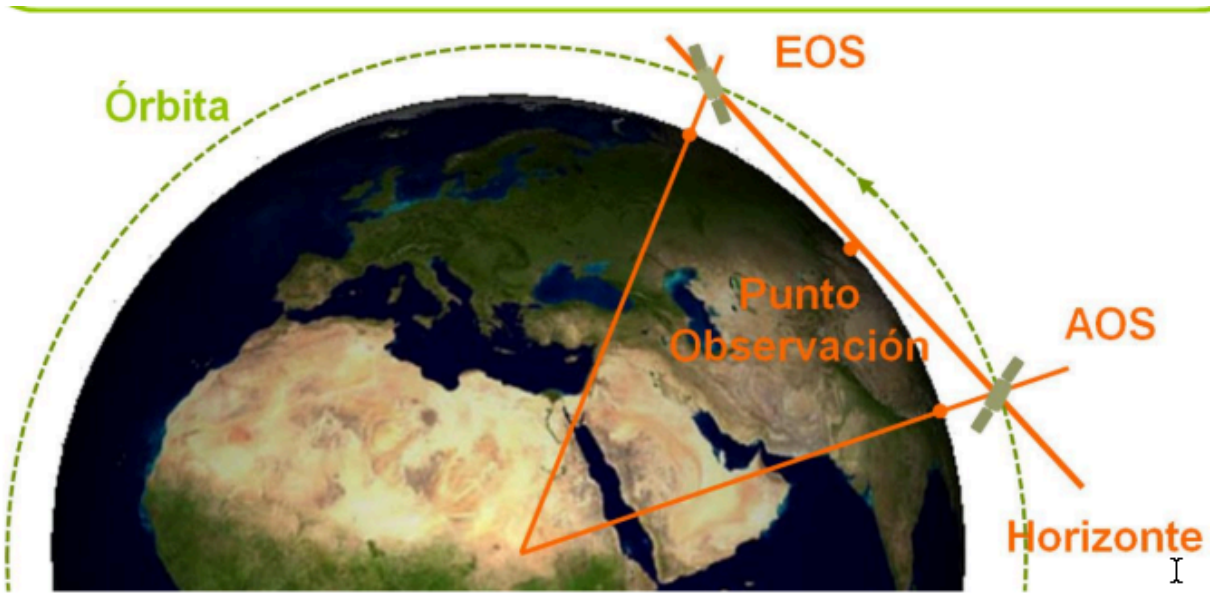
The analogue APT weather map transmission was invented in the 60′s

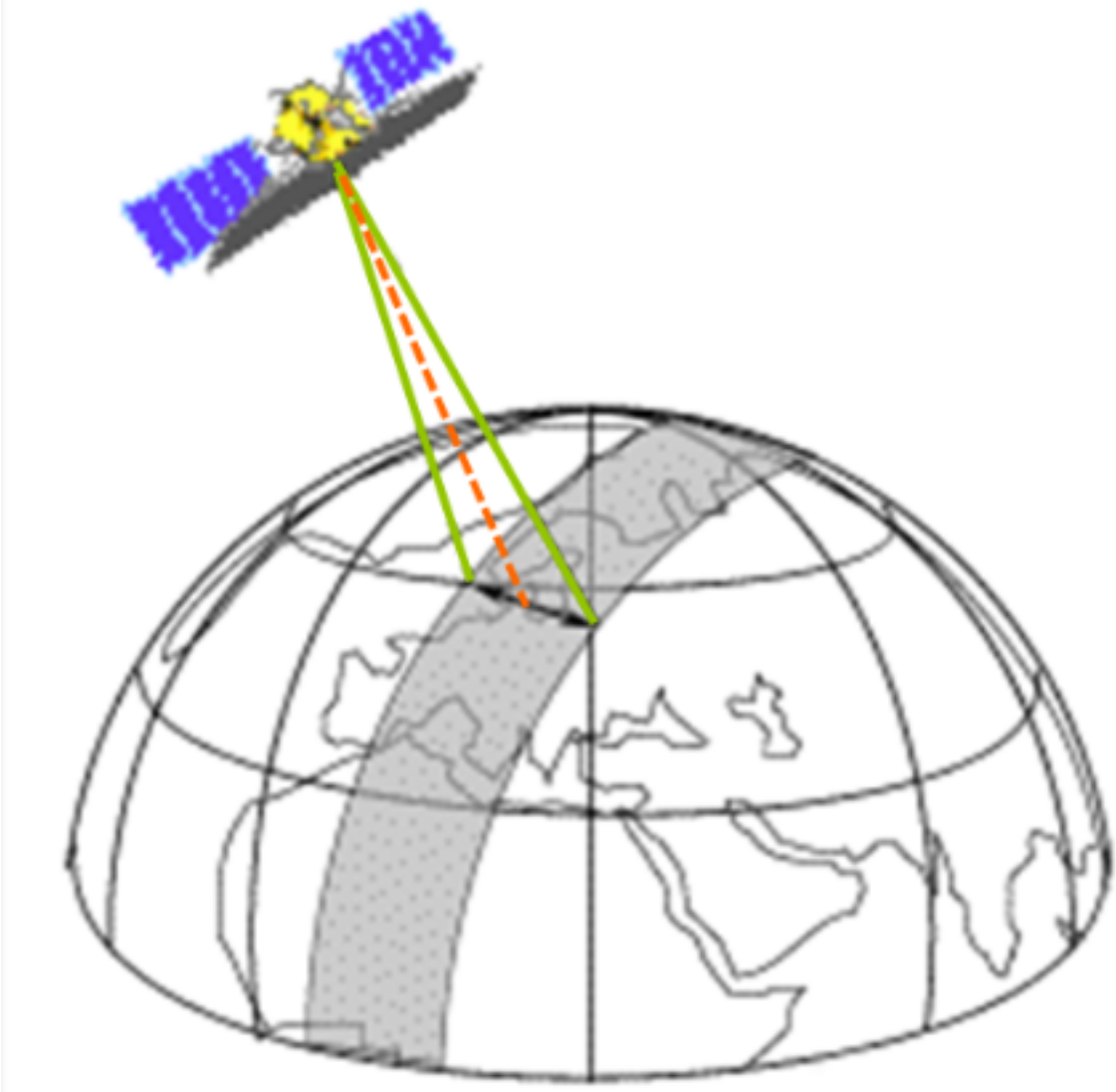
It has a resolución of 4 Kms/pixel. □ Mínimum , 2 passes or day □ Continuos signal in the visual satellite range time.

The maximum pass duration for NOAA satellites, when receiving signals from them using ground equipment, can be around 10 to 15 minutes. This duration refers to the time the satellite is visible and within signal range at a given location, from its initial rise above the horizon to its descent below the horizon on the other side. (If you have a pass just above of your (zenith = visual satellite max. elevation = 90^{o}) , you can record the max. long pass, till around 1800 lines with max 15 min time aprox)

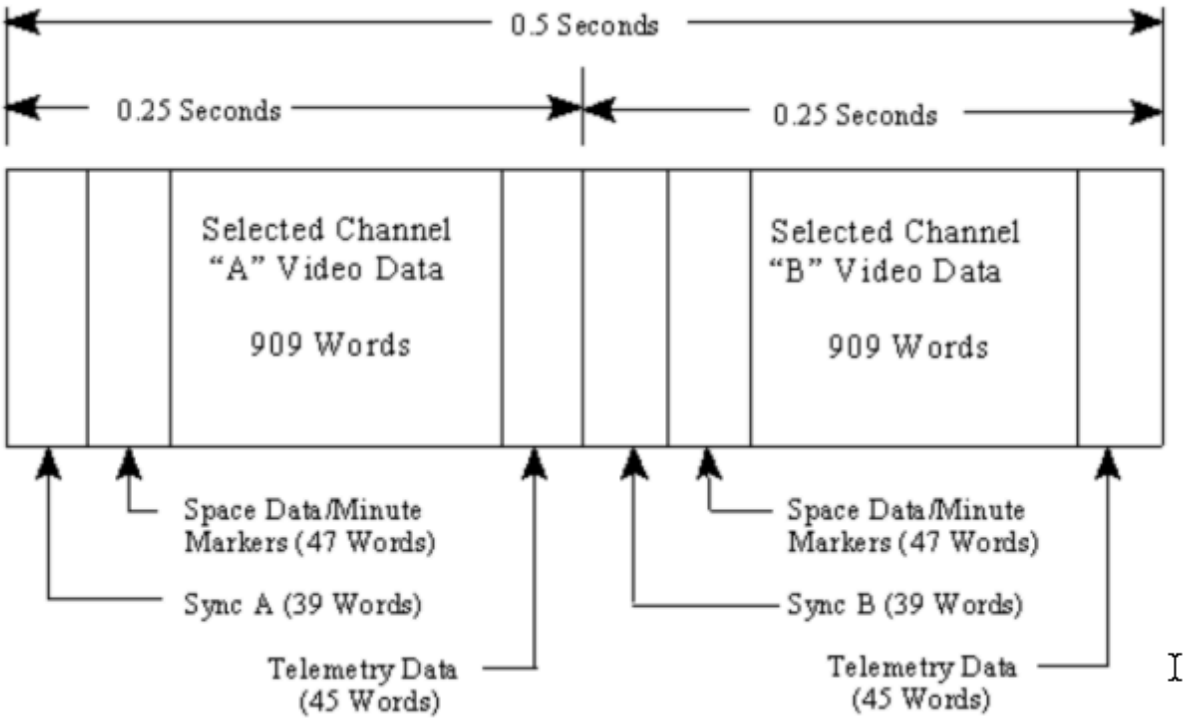

Trayectoria del satelite
Satelite NOAA
Estación de recepción
fin de la recepción
inicio de recepción de imagenes
Horizonte

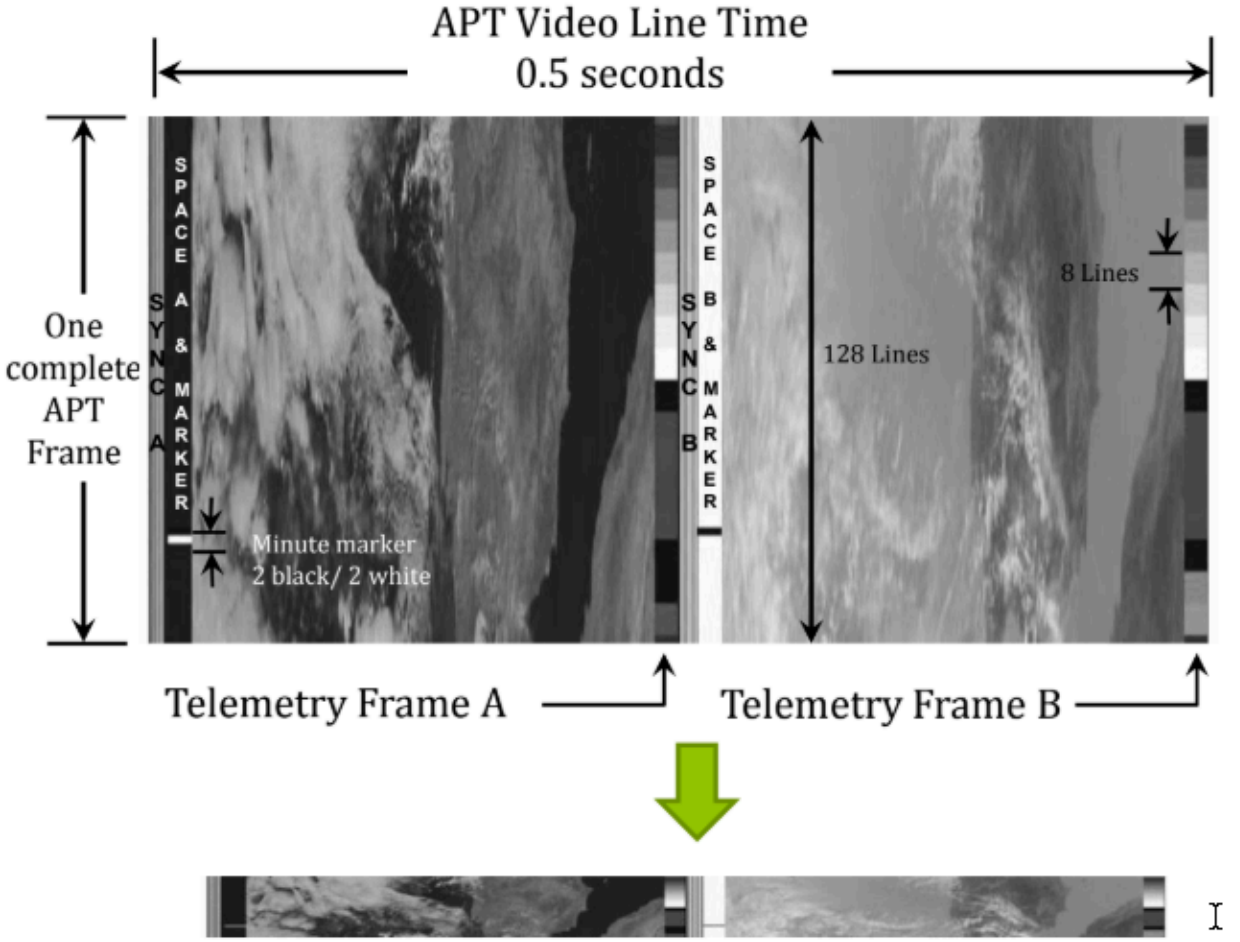

satélite visible
v_{sat}
horizon
θ
satellite
R_{sat}
R_{Earth}

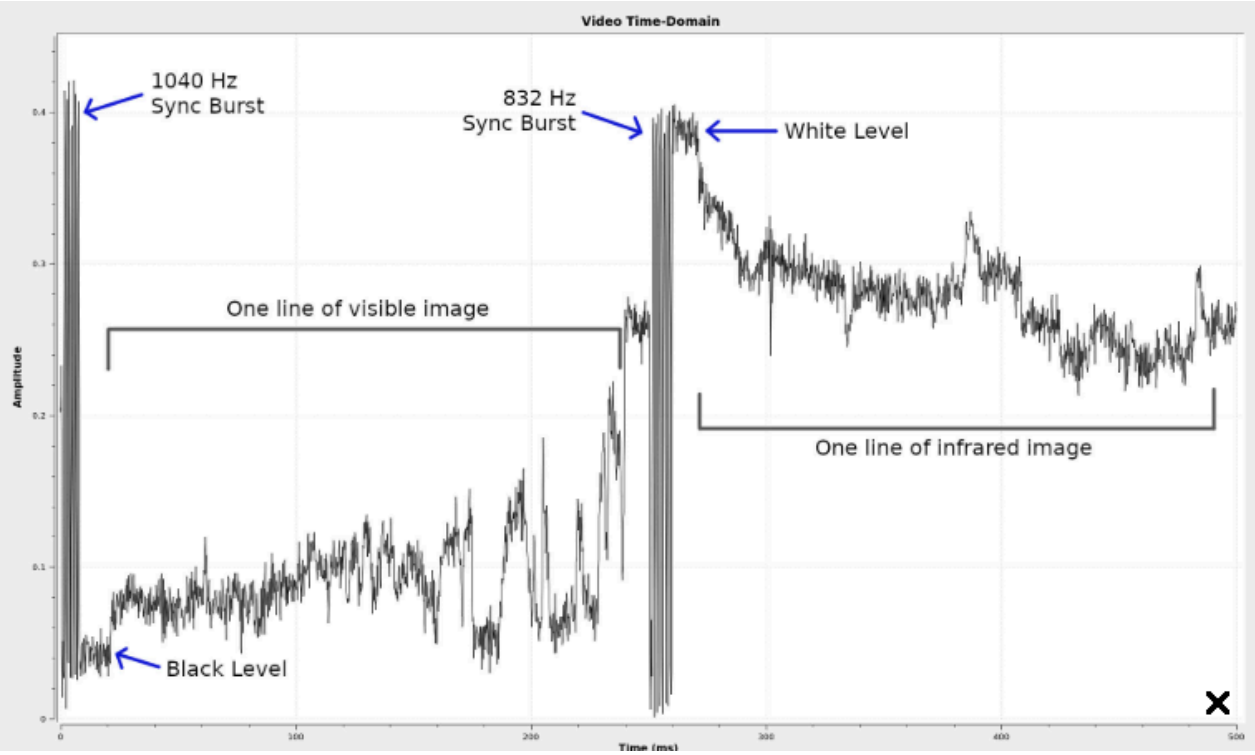




APT signal format : 120 lines or minute , sending two pictures at the same time (visible or infrared) 8 bits/pixel.The signal amplitude represents the brightness of each pixel. Two lines per second, 4160 pixels per second. 2080 pixels per line, each channel has 909 useful pixels per line.


0.5 Seconds
0.25 Seconds
0.25 Seconds
Selected Channel
"A" Video Data
909 Words
Selected Channel
"B" Video Data
909 Words
Space Data/Minute
Markers (47 Words)
Sync A (39 Words)
Telemetry Data
(45 Words)
Space Data/Minute
Markers (47 Words)
Sync B (39 Words)
Telemetry Data
(45 Words)

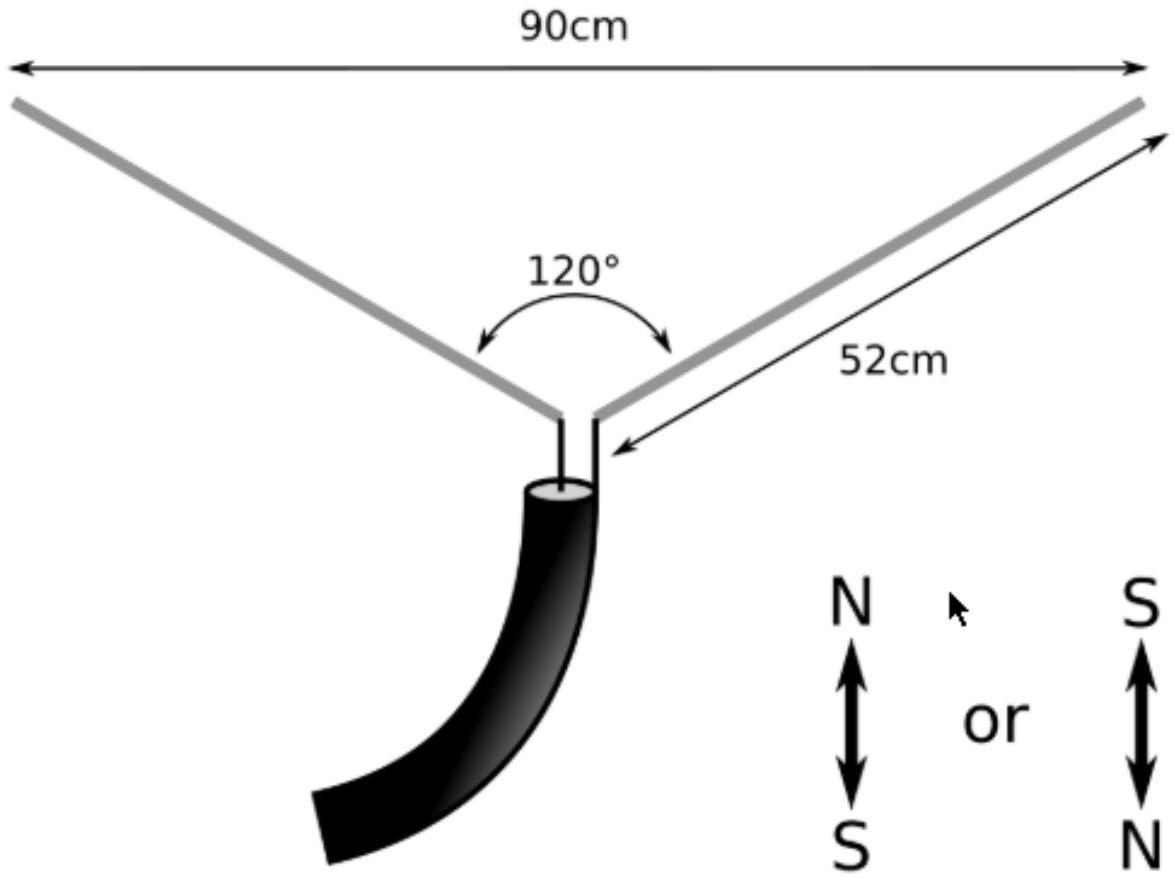

APT Video Line Time
0.5 seconds
One
complete
APT
Frame
SYNC A
SPACE A & MARKER
Minute marker
2 black/ 2 white
SYNC B
SPACE B & MARKER
128 Lines
8 Lines
Telemetry Frame A
Telemetry Frame B



115 Usage

Hackrf, like the rest of SDR receivers, have not enough sensitivity to receive directly those NOAA APT maps with the stock rod antenna. To receive those signals , the best would be to use an omnidirectional QFH 137 Mhz antenna band ,or a proper 137 Mhz dipole + bandpass amplifier (if you have noisy strong adjacent channels, it will help to improve the reception).

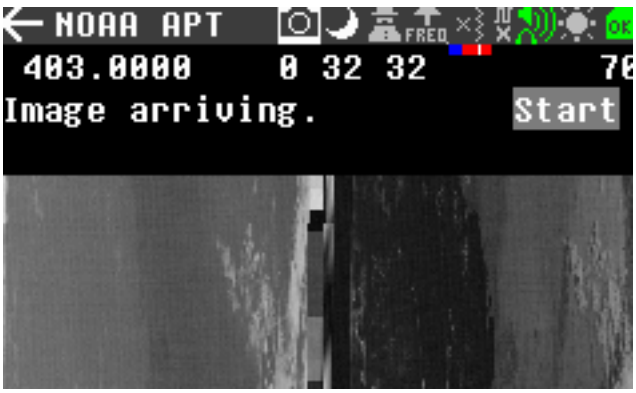
The most simple antenna you can use is a V-dipole antenna, consists of two wires or rods of 52cm, spread apart by 120 degrees. Instead of using a protactor you can use trigonometry, the distance between the tips of each rod should be around 90cm.



You can tune , inputting directly any of the 3 x above current NOAA Satellite frequencies (NOAA 15: 137.62MHz. NOAA 18: 137.9125MHz. NOAA 19: 137.1MHz) ,and check at what exact time , the satellite will be visible in his orbital pass. (there are many programms and web links to check the exact local time pass to your ground station place) . Here you have, some predict and tracking web links examples :

To check NOAA 15, NOAA 18, NOAA 19 real time tracking position or next 10 day prediction passes to your location (based on your IP).

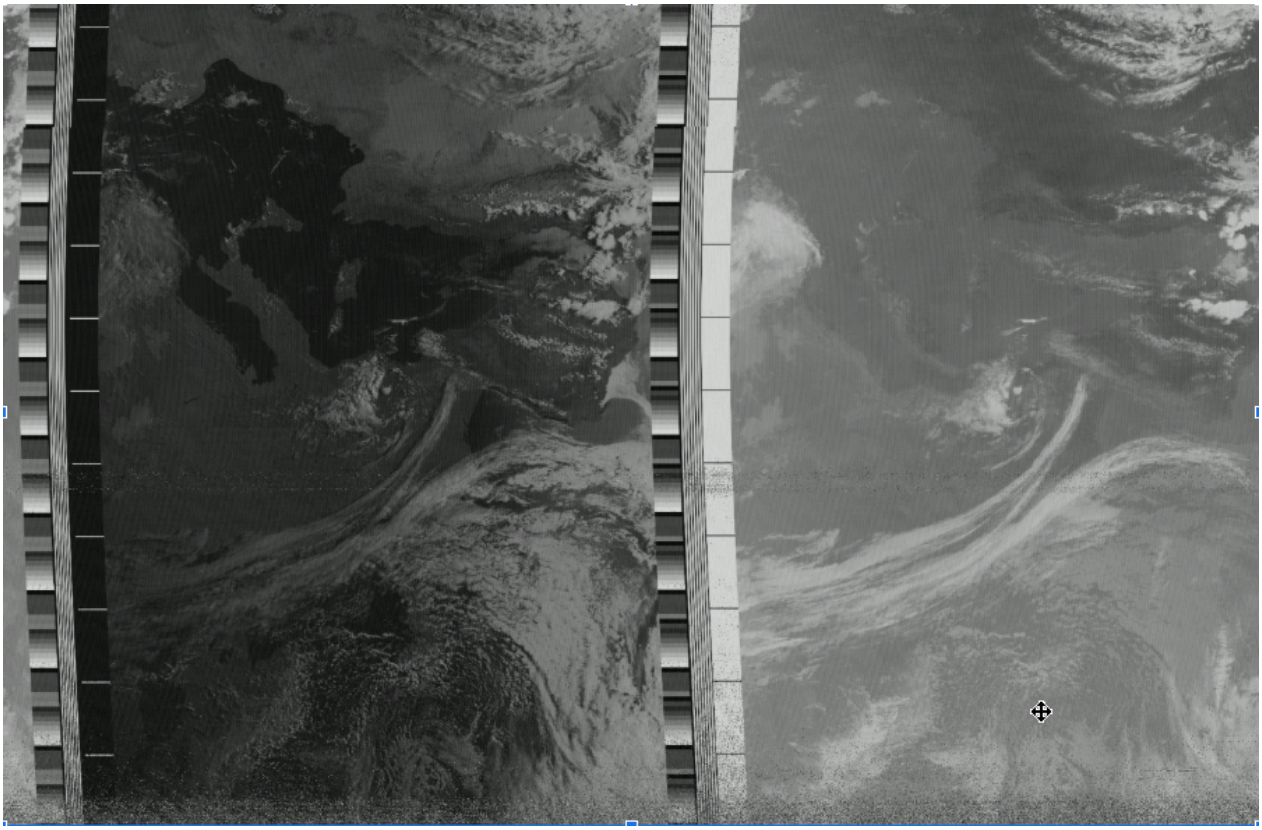
You need to adjust proper LNA , RF GAIN , VGA till getting proper signal (not weak, not saturating) , and you can adjust the audio volume till recognizing the usual beep , beep sound indicating frame A, frame B every 0.25 segs.Once you confirmed all those above points optimizing the gain levels for best LCD screen map decoding. you can press START , and it will start to record the map into SD/BMP folder ,in high resolution. Once you lose the satellite signal , you can press the STOP button , and you can check the recorded image into that SD/BMP folter . The file name will start by “noaa_year-month-date-time.bmp”

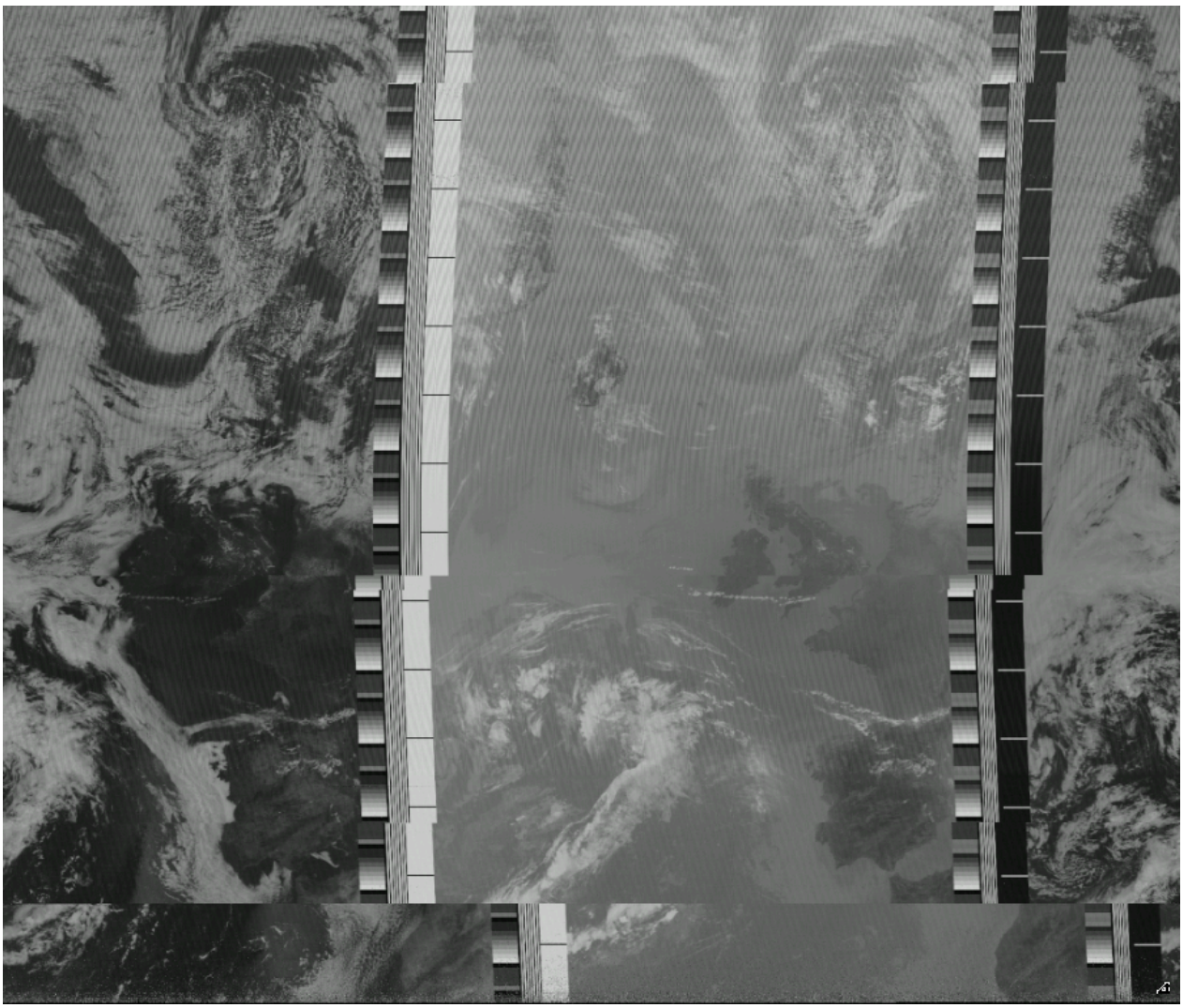


/6B82-5F61/BMP/

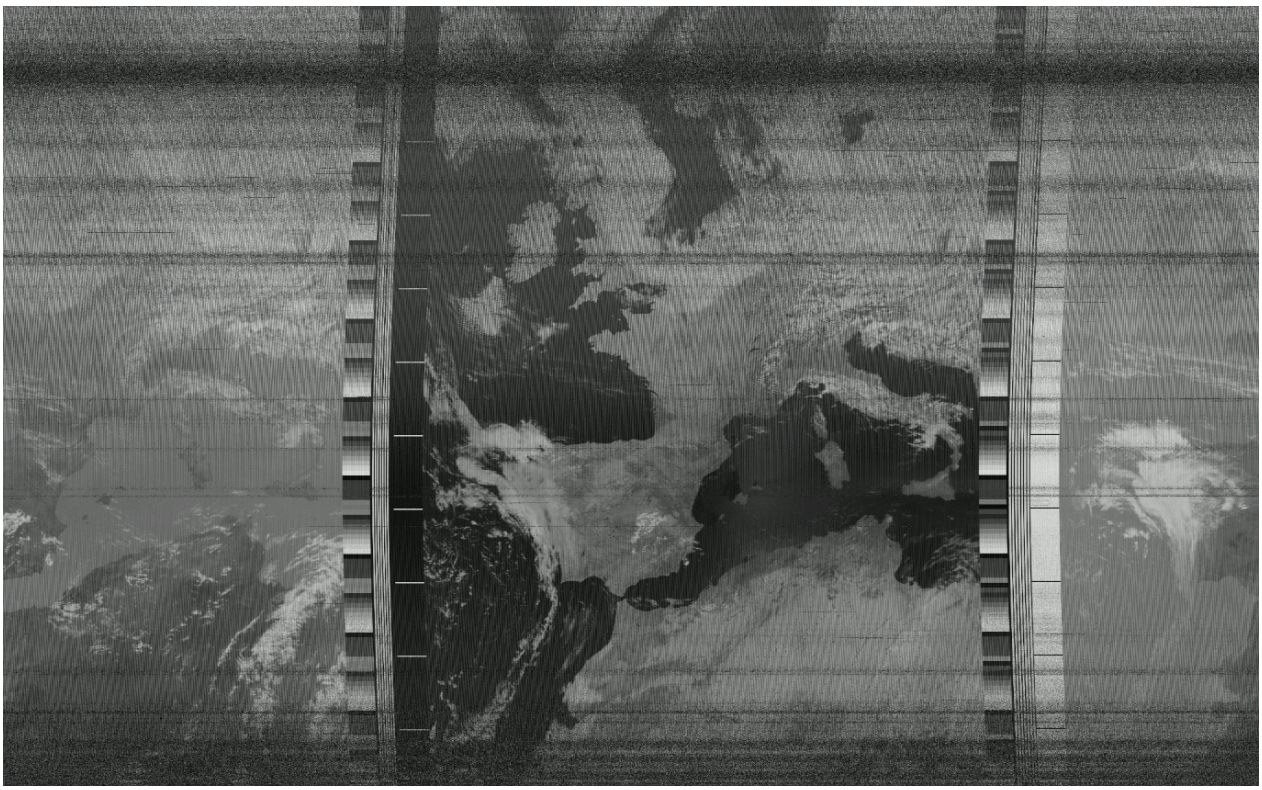
Name	Size	Type
noaa_19811221000203.bmp	10.4 MiB	Windows BMP image
noaa_20250507203913.bmp	7.1 MiB	Windows BMP image
noaa_20250507210801.bmp	7.9 MiB	Windows BMP image
noaa_20250510121043.bmp	10.6 MiB	Windows BMP image
noaa_20250510124523.bmp	5.1 MiB	Windows BMP image
noaa_20250510125556.bmp	10.6 MiB	Windows BMP image
noaa_20250510131142.bmp	207.2 KiB	Windows BMP image

Here some example of local test replay decoded maps from that Mayhem NOAA Application , that we can record in .bmp image format





Previous pictures was from local Replay , when debugging and developping. Here I can share a real captured picture , NOAA 19 pass, with max. elevation of 83º . The satellite came from North (that is why picture appeared with direct proper North on top). I recorded using new binary after #2654 PR ,using QFH 137 Mhz antenna with cheap pre-amplifier. And PP settings :
LNA = 1 , GAIN = 32 , VGA (not relevant 30 or 40) + a direct cheap pre-amplifier to the PP, and short antenna cable . (It was just quick hand made testing) .



115.1 NRF

116 Introduction

The NRF receiver app decodes packets from nRF24L01+ transceivers made by Nordic Semiconductor. These are low-cost, low-power 2.4 GHz devices widely used in wireless keyboards, mice, game controllers, mini-drone

remotes, and general hobbyist projects. They communicate using Nordic's proprietary Enhanced ShockBurst (ESB) protocol.

The app currently supports **250 kbps mode only**. 1 Mbps and 2 Mbps modes are not yet implemented.

This is an external app (`.ppma` file on the SD card).

117 Protocol background

The nRF24L01+ operates in the 2.4 GHz ISM band on 128 channels spaced 1 MHz apart (channel 0 = 2400 MHz, channel 127 = 2527 MHz). Each packet consists of a preamble, a 5-byte address, a 9-bit Packet Control Field (PCF, containing payload length and packet ID), the payload (0–32 bytes), and a 2-byte CRC. The decoder verifies the CRC before displaying a packet — only packets with a valid CRC are shown.

118 Controls

- **Frequency:** The default is **2480 MHz** (channel 80). Use the rotary encoder to step in 100 Hz increments, or click to open the numeric keypad. To scan a specific device, set the frequency to the channel it is using: `2400 + channel_number` MHz. Many devices hop across multiple channels; scanning manually or sweeping with Looking Glass first can help identify active channels.
- **AMP:** RF preamplifier — 0 = off, 1 = on (+14 dB).
- **LNA:** Low Noise Amplifier gain (0–40 dB).
- **VGA:** Variable Gain Amplifier or baseband gain (0–62 dB).
- **Modem setup:** Opens the modem configuration screen. The app auto-configures the modem on launch; this button is available for manual adjustments.

119 Display

Decoded packets are shown in the console area as hex bytes, formatted as:

```
addr: AA BB CC DD EE  data: XX XX XX XX ...
```

Each line represents one received packet that passed the CRC check. The address field is always 5 bytes. The data bytes that follow are the payload content.

120 Limitations

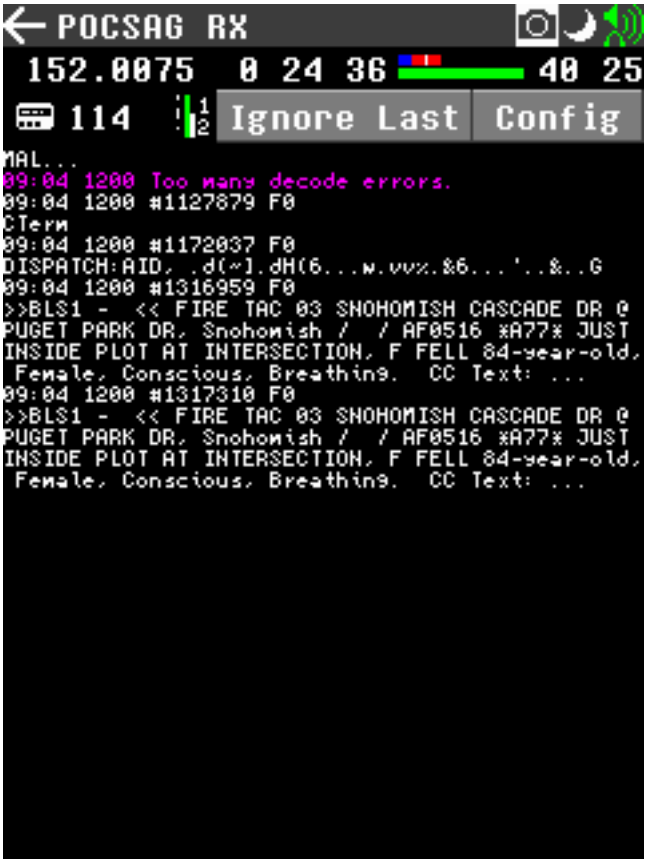
- Only **250 kbps** data rate is supported. Devices transmitting at 1 Mbps or 2 Mbps will not be decoded.
- No channel hopping — the receiver listens on a single fixed frequency. Devices that use frequency hopping must be caught on the right channel.
- The HackRF's 2.4 GHz sensitivity is limited. Passive reception is possible but range is short compared to dedicated 2.4 GHz receivers. Placing the antenna close to the transmitter improves results significantly.

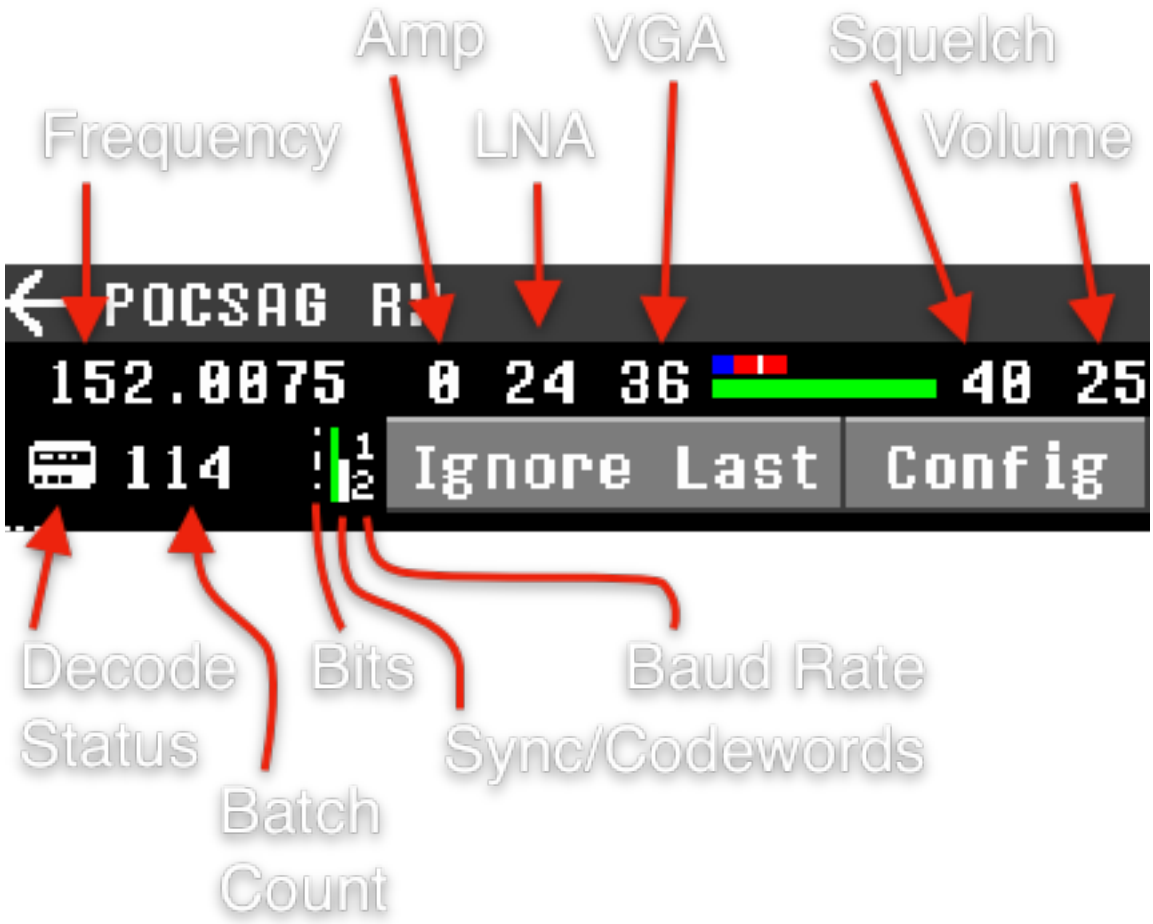
120.1 POCSAG

Receives pager messages using the POCSAG protocol.

This protocol operates in the VHF/UHF bands using FSK modulation.

121 UI Overview





121.0.1 Settings

- **Frequency**: Sets the frequency to receive pager messages on. Can be adjusted with the encoder knob, on-screen numpad, or loaded from frequencies saved on an SD card. 439.9875 MHz is the most popular worldwide frequency used by amateur radio for POCSAG. Amateur radio POCSAG uses 1200 bps.
- **RF amplifier** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0, 8, 16, 24, 32, 40): Sets the LNA gain. See Receive Quality Issues for details.
- **VGA gain** (0 to 62): Sets the VGA gain. See Receive Quality Issues for details.
- **RSSI/Audio**: Top bar indicates signal strength. Bottom bar indicates the audio level. The audio bar can be useful for tuning settings without headphones or a speaker.
- **Squelch** (0–99): Sets the minimum signal level required to start decoding. 0 disables squelch (all signals decoded regardless of strength). Higher values require a stronger signal — use this to avoid decoding weak or noisy signals that would produce garbled output.
- **Volume**: Output volume for the received audio.

121.0.2 Information

- **Decoder Status**: Indicates the status of the decoder state machine. White: Idle, Cyan: Clear, Yellow: Waiting for message start, Green: Waiting for rest of message.
- **Batch Count**: Number of message batches received. A batch has 16 codewords.
- **Bits**: Displays the bits being decoded into codewords.
- **Sync**: Green when the frame decoder has received a sync frame. Messages are not decoded unless a sync frame is found.
- **Codewords**: Shows the number of codewords in the current batch. When the bar fills, the batch is complete and processed.
- **Baud Rate**: The detected rate of the current message. `05`: 512 bps, `12`: 1200 bps, `24`: 2400 bps.

121.0.3 Buttons

- **Ignore Last**: Sets the filter to drop messages sent to the last received address.

- **Config**: Opens the settings page

121.1 Config

- **Enable Log**: Logs decoded messages to `LOGS/POCSAG.TXT` on the SD card.
- **Log Raw Data**: Also logs the batch codewords as hexadecimal. Useful for debugging decoder issues.
- **Use Small Font**: Uses a smaller font in the UI to show more messages on screen at once.
- **Hide Bad Data**: Do not show (or log) codewords that fail checksum validation.
- **Hide Addr Only**: Do not show (or log) codewords that contain an address but no message payload.
- **Filter Mode** / **Filter Address**: Controls address filtering. `None` disables filtering. `Drop` hides messages sent to the specified address. `Keep` shows only messages sent to the specified address and hides all others.
- **Beta**: Enables the newer POCSAG baseband processor. The app must be restarted after changing this option.
- **Save**: Saves the current settings changes.

Note: Settings are only saved and restored if **App Settings** is enabled in the global Settings menu.

121.2 Message Display

Typical message:

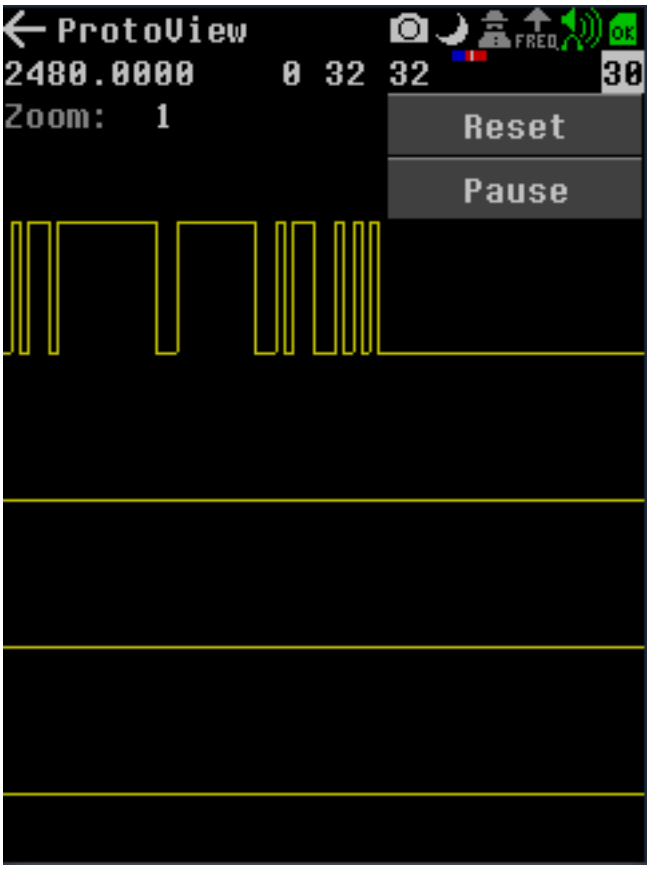
```
12:34 1200 #432123 F2
This is a test message
```

Description (from top-left):

- `12:34` — **Time**: When the message was received (PortaPack clock time).
- `1200` — **Data rate**: The baud rate used to receive the message.
- `#432123` — **Address**: The pager address the message was sent to.
- `F2` — **Function**: A value 0–3 indicating the message type or source category.
- `This is a test message` — **Message**: The decoded message text.

Messages displayed in **magenta** contain a high number of decoding errors and may be unreliable.

121.3 ProtoView



ProtoView is a raw signal visualizer that displays received OOK/ASK pulses as a waveform. It does not attempt to decode protocols — it shows the raw on/off timing of any signal, making it useful for inspecting unknown or custom transmissions that other decoders do not recognize.

This is an external app (`.ppma` file on the SD card).

121.4 Settings

- **Frequency**: The frequency to monitor. Adjustable via the encoder knob, on-screen numpad, or a saved freqman entry. Frequency step is 100 Hz.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain. See Receive Quality Issues.
- **RSSI**: Signal strength indicator in the title bar.
- **Volume**: Audio output volume for monitoring the raw signal.

121.5 Controls

- **Zoom**: Sets the time scale of the waveform display. The value is the number of microseconds (µs) per display pixel. Available steps: 1, 2, 5, 15, 30, 50, 100, 200, 500, 1000. Lower values show finer detail over a shorter time window; higher values compress the signal to show longer bursts.
- **Reset**: Clears the signal buffer and restarts live capture. Also exits pause mode and resets the shift position.
- **Pause or Resume**: Freezes the display to inspect the captured waveform. When paused, the **Shift** field appears (see below). The button label changes to **Resume** while paused.

121.6 Waveform Display

The signal is displayed as four stacked waveform rows, each rendered in yellow. Together they represent up to 600 display columns of pulse data. High signal (above threshold) is shown as a raised level; low signal as a flat baseline.

The display is derived from a buffer of up to 400 pulse-timing entries. Each entry stores the duration in microseconds of one continuous high or low state. The threshold between high and low is determined automatically by the baseband processor and adapts continuously to the received signal level.

Pulses longer than 30,000 µs are treated as gap/idle markers and are not rendered.

121.7 Pause Mode and Shift

When paused, a **Shift** field appears between the Zoom control and the waveform. Turning the encoder knob while the Shift field is focused scrolls the waveform left or right through the captured buffer:

- Positive shift values move the display window to the right (skip earlier samples).
- Negative shift values insert blank padding on the left, moving the signal toward the right of the display.

The shift range is −400 to +400 samples. Use this to align specific pulses or transitions in the center of the screen for easier inspection.

121.8 Signal Processing

The baseband processor samples at 4 MHz and decimates by 8, producing an effective analysis rate of 500 kHz. Signal presence is determined by comparing the instantaneous IQ magnitude against an adaptive threshold, which is recalculated every 90,000 samples and clamped to the range 50–1700. This allows the app to work across a wide range of signal strengths without manual squelch adjustment.

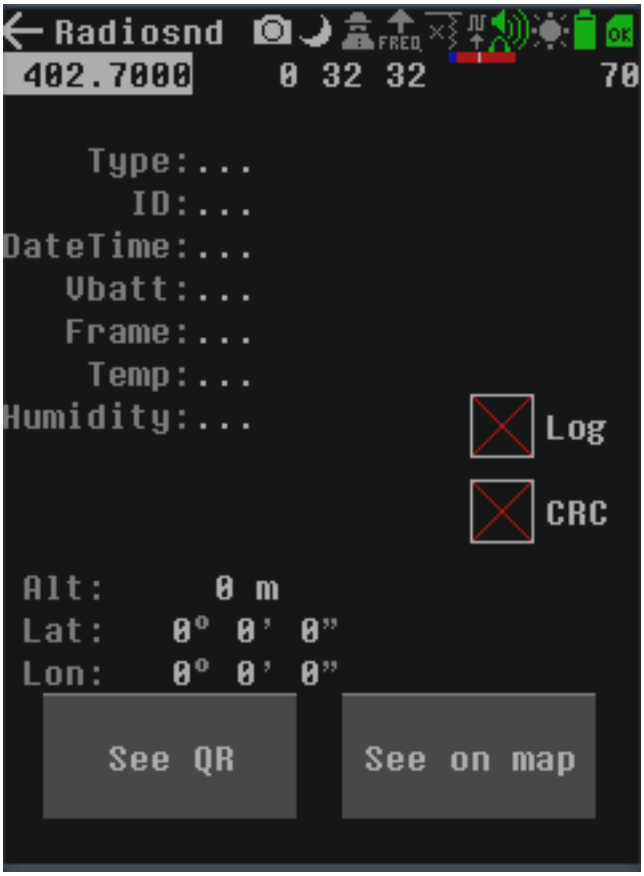
121.9 Settings Persistence

App settings (frequency, gain, zoom) are saved to `rx_protoview.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

121.10 Radiosonde

Receives and decodes telemetry from meteorological radiosondes — weather balloons that transmit GPS position, temperature, humidity, and other atmospheric data as they ascend.

122 UI Overview



122.1 Settings

- **Frequency**: The frequency to monitor. Default is **402.7 MHz**, which is the primary international radiosonde downlink band. Adjustable via the encoder knob or on-screen numpad.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain. See Receive Quality Issues.
- **RSSI**: Signal strength indicator.
- **Volume**: Audio output volume.

122.2 Decoded Fields

The app displays a common set of fields for all sonde types. Fields that are not available for a given sonde type are shown as `....` Vertical speed is calculated by the app from successive GPS altitude readings and packet timestamps.

- Field: Type; Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: ✓; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓
- Field: ID or Serial; Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: —; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓
- Field: DateTime; Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: ✓; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓
- Field: Battery voltage; Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: ✓; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓
- Field: Frame number; Meteomodem M10: —; Meteomodem M20: ✓; Meteomodem M2K2: —; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: —
- Field: Temperature; Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: —; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓
- Field: Humidity; Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: —; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓
- Field: Pressure; Meteomodem M10: —; Meteomodem M20: ✓; Meteomodem M2K2: —; Vaisala RS41-SG: —; Meteo-Radiy MRZ-N1 or MP3-H1: —
- Field: GPS (Lat or Lon or Alt); Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: ✓; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓
- Field: Vertical speed; Meteomodem M10: ✓; Meteomodem M20: ✓; Meteomodem M2K2: ✓; Vaisala RS41-SG: ✓; Meteo-Radiy MRZ-N1 or MP3-H1: ✓

> **Note:** The Meteo-Radiy MRZ-N1 or MP3-H1 (shown as `Meteo-Radiy MRZ`) transmits around 403 MHz and was added in #3388 (issue #3309, October 2026). Until the next release it is only in the nightly builds. It has no running frame counter, so the frame-number field stays empty; the GPS time of day stands in for it internally.

122.3 Checkboxes

- **Log**: Logs received packets to the SD card. A new timestamped file is created each time the app is opened: `LOGS/SONDE_YYYYMMDD_HHMMSS.TXT`. The log contains raw packet bit data and a reception timestamp.
- **CRC**: Enables CRC checking. When active, packets that fail their checksum are discarded rather than displayed. CRC is supported for M10, M20, RS41-SG and MRZ-N1 or MP3-H1; M2K2 does not have a CRC implementation.

122.4 Beep on Packet

The app emits an RSSI-level-dependent audio beep when a packet is received. The beep pitch corresponds to signal strength. This behaviour is controlled by the global **Beep on packets** setting in Settings → Radio, not by a per-app checkbox.

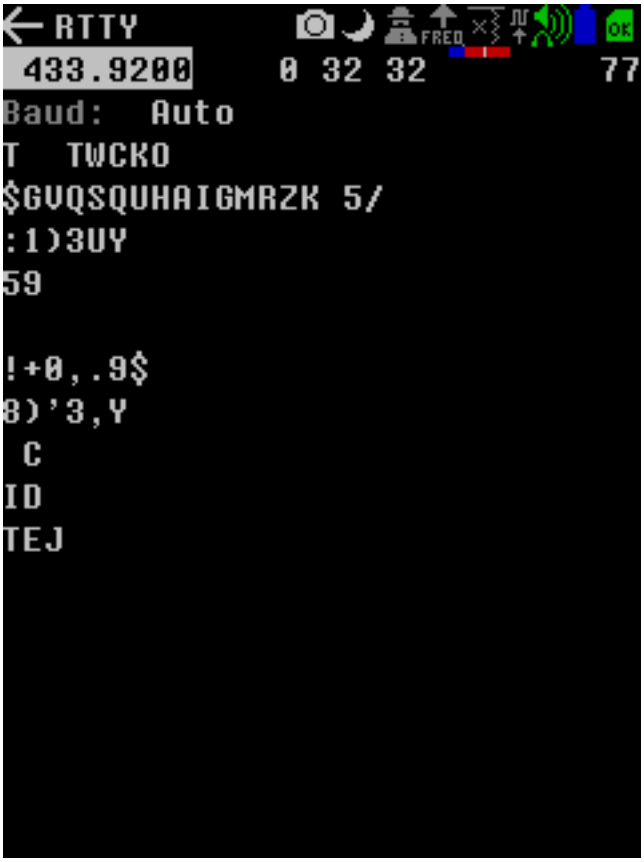
122.5 Map Buttons

- **See QR**: Generates a `geo:latitude,longitude` QR code for the last received GPS fix. Scanning the code on a smartphone opens its map application at those coordinates.
- **See on map**: Opens the built-in world map view centred on the last received GPS fix. Requires `world_map.bin` to be present in the `/ADSB/` folder on the SD card.

122.6 Settings Persistence

The Log and CRC checkbox states are saved to `rx_sonde.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

122.7 RTTY RX



Receives and decodes Radio Teletype (RTTY) transmissions using Baudot/ITA2 5-bit encoding over FSK (frequency-shift keying). The decoder produces live text output and supports automatic baud rate detection.

This is an external app (`.ppma` file on the SD card).

122.8 Settings

- **Frequency**: The center frequency to receive on. Frequency step is 100 Hz. Adjustable via the encoder knob or on-screen numpad.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain. See Receive Quality Issues.
- **RSSI**: Signal strength indicator.
- **Volume**: Audio output volume. The received audio (mark/space tones) is passed to the speaker or headphone output.
- **Baud**: Sets the expected symbol rate. Options: `Auto`, `45`, `45.45`, `50`, `75`, `100`, `110`, `150`, `200` bps. See Auto-Baud below.

122.9 Decoded Text Area

The main body of the screen is a scrolling text console that displays decoded Baudot/ITA2 characters in real time. Both letter and figure shift states are handled automatically. The console does not auto-scroll — new text appends at the current position.

122.10 Shift

The frequency shift between mark and space is fixed at **170 Hz**. This is the standard narrow shift used by most amateur radio RTTY and many commercial RTTY stations. The shift is not user-configurable in this app.

122.11 Auto-Baud

When **Baud** is set to `Auto`, the decoder measures incoming pulse widths and continuously refines its estimate of the symbol rate. It starts with a default of approximately **45.45 baud** (the most common RTTY rate) and updates as more signal is received.

The auto-baud algorithm uses a fast-lock approach: a new baud estimate is accepted after 3 consecutive pulses agree with a candidate width. If no signal is detected for approximately 0.5 seconds (12,000 samples at 24 kHz), the baud estimate resets to the 45.45 baud default.

When a specific baud rate is selected, baud detection is bypassed and the decoder locks immediately to the configured rate. This results in faster and more reliable locking when the transmission rate is known in advance.

122.12 Squelch and Polarity

Squelch is automatic and based on the measured spread between the mark and space signal levels. It opens when spread exceeds a threshold and closes when the signal fades — there is no manual squelch control.

Polarity (which tone is mark and which is space) is also detected and corrected automatically. If only one tone is received continuously for more than ~0.3 seconds, the decoder flips polarity and resets the UART state machine.

122.13 Signal Processing

The baseband runs at 3.072 MHz and passes through three decimation stages to reach a final demodulation rate of 24 kHz (3.072 MHz → 384 kHz → 48 kHz → 24 kHz). FM demodulation is used to discriminate between the mark and space frequencies, followed by envelope tracking, a Schmitt trigger slicer, and a 5-bit asynchronous UART state machine to extract Baudot characters.

122.14 Settings Persistence

The selected baud rate is saved to `rx_rtty.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

122.15 Scanner

The scanner steps through approximately 20 frequencies per second, pausing 50 ms per frequency to allow the receiver to stabilize after retuning.

When a signal above the squelch threshold is found, the scanner locks onto that frequency for up to 500 ms (10 × 50 ms cycles) to confirm the signal is still present before pausing — this avoids stopping on spurious peaks.

This is an external app (`.ppma` file on the SD card).

122.16 Big Numbers Frequency

The large frequency display uses a colour code to indicate scanning state:

- **Grey**: Scanning or idle
- **Yellow**: Analyzing — possible signal detected, confirming
- **Green**: Signal confirmed and locked; audio active if above squelch

122.17 Scan vs Search Modes

The scanner operates in two modes, toggled with the `SRCH/SCAN` button (which shows the mode that will be switched *to* when pressed):

- **Scan mode**: Steps through a list of discrete frequencies loaded from a freqman file.
- **Search mode**: Steps sequentially through all frequencies between a start and end frequency at a chosen step size.

122.18 Frequency List

On startup the app loads from `FREQMAN\SCANNER.TXT` by default. Use the `LOAD` button to select a different freqman file. If app settings are enabled, the last used file is restored on the next startup.

If the loaded file contains a frequency *range* entry (`a=`, `b=`, `s=`), the start, end, and step fields are populated from it. Only one search range per file is supported. You can also set the search range manually by editing the SRCH START and END fields and the STEP selector directly.

Use the Freq manager app to create or edit freqman files.

122.18.1 Manually Selecting Entries or Frequencies

Whenever the `/` button is highlighted, the encoder knob scrolls through the frequency list (Scan mode) or through the search range (Search mode), regardless of whether Pause is currently active. The current item field `X or Y` can also be focused and used with the encoder knob in the same way.

122.19 Radio Settings

- **LNA or VGA or AMP**: Gain settings. AMP: 0 = 0 dB, 1 = 14 dB. LNA: 0–40. VGA: 0–62. These are preserved per app if **App Settings** is enabled.
- **VOL**: Audio output volume.
- **BW**: Signal bandwidth. Available values depend on the selected **MODE**.
- **SQ**: Squelch threshold in dBm (range −90 to +20, default −30). When scanning, the scanner pauses on any frequency whose signal level exceeds this value. When paused for any reason, audio is enabled or muted based on this threshold.
- **MODE**: Modulation mode — AM (DSB 9k, DSB 6k, USB+3k, LSB−3k, CW), NFM, or WFM.
- **STEP**: Frequency step used for sequential Search mode.

122.20 Buttons

- `LOAD`: Loads a freqman file (from the `FREQMAN/` directory) into the scan list and/or search range fields.
- `MCLR`: Clears the in-memory frequency list. Does not modify any file on the SD card. Frequencies can be re-added using `ADD FQ` or `LOAD`.
- `SRCH/SCAN`: Toggles between Search mode and Scan mode.
- `/`: Manually pauses or resumes scanning. When this button is highlighted, the encoder knob can be used to step through frequencies. The squelch threshold still controls audio output while paused.
- `FORWARD/REVERSE`: Changes scanning direction. The button label shows the direction that will be activated when pressed.
- `MIC TX`: Opens the Microphone Transceiver app for 2-way radio operation.
- `AUDIO`: Opens the Audio Receivers app for further signal analysis.

- `DEL FQ`: Removes the currently displayed frequency from the in-memory scan list. Does not modify the file on the SD card.
- `ADD FQ`: Appends the currently displayed frequency to the **currently loaded freqman file** on the SD card. If the frequency already exists in the file, an error is shown. The entry is also added to the in-memory list if it is below the per-file limit.

122.21 Delay Settings

- **Wsa** (Wait while Signal Active, default 5 s, range 0–99): Time to stay on an active frequency before skipping to the next one, even if a signal is still present. Set to 0 to remain on the frequency indefinitely until the encoder knob is used or the signal drops. A non-zero Wsa is also called "Browse" mode.
- **Wsl** (Wait after Signal Loss, default 2 s, range 0–99): Time to remain on a frequency after the signal drops below the squelch threshold before moving on.

122.22 Other Fields

The `X or Y` field shows the current frequency index and the total number of entries in the scan list. If the loaded file is at the maximum freqman entry limit, `(DB MAX!)` is shown. If a description is available for the current entry, it appears above the big frequency display; otherwise the loaded filename is shown, or `SEARCHING...` in Search mode.

122.23 Freqman File Format

See Freqman Manager.

122.24 Settings Persistence

Wsa, Wsl, squelch, search range, and the last-loaded freqman file are saved to `rx_scanner.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

122.25 Example: Using Scanner as a Broadcast FM Radio Tuner

1. Press `LOAD`, select `FM_STANDARD_BAND.TXT`, and confirm. This loads the FM search range, step, mode (WFM), and bandwidth.
2. Optionally adjust VOL, SQ, gain settings, and Wsa.
3. Press " to stop automatic scanning.
4. Navigate focus to the `/` button. When highlighted, use the encoder knob to tune to an FM station.
5. Press `ADD FQ` to save the current frequency to the loaded file. Repeat to store additional stations.
6. In future sessions, load the same file and Scanner will cycle through the saved stations, spending Wsa seconds on each. Press " to stop on a station; use the encoder knob to change stations manually.

122.26 Also see: Recon app

The Recon app is a more fully-featured scanner application for advanced use cases.

122.27 Search

[[img/screenshots/Calls.png]]

The Search app continuously sweeps a defined frequency range using FFT spectrum analysis and locks onto any signal that rises above a configurable power threshold. It is designed for passive signal detection — it does not demodulate audio during scanning, but logs the frequency, time, and duration of each detected signal.

123 UI Overview

123.1 Settings

- **Min or Max**: The frequency range to search. Default is 100 MHz to 400 MHz. Settings are saved per app (see Settings Persistence).
- **LNA** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA** (0–62): Variable-gain amplifier gain. There is no RF amplifier control in this app; the amp is always off.

- **Trig** (5–255, step 5, default 80): Power threshold for signal detection. A signal is detected when its FFT bin power exceeds `Mean + Trig`. Set this higher to ignore weaker signals and reduce false triggers; lower to catch weaker signals.
- **Snap to**: When enabled, the detected frequency is rounded to the nearest step before being displayed and logged. Available snap steps: 25 kHz, 12.5 kHz, 8.33 kHz, 2.5 kHz, 500 Hz. Default is 12.5 kHz (enabled).
- **LOG**: When checked, creates a timestamped CSV log file on the SD card each time it is enabled (see Logging).

123.2 Status Indicators

- **Mean**: The average FFT power across all bins in the current scan, shown as a value from 0–255. The detection threshold is `Mean + Trig`. Setting Trig above the noise floor is the primary tuning task for this app.
- **Slices**: The number of 2.5 MHz slices the scan range has been divided into, shown as `X or 32`. Each slice is one FFT capture. The maximum supported span is 80 MHz (32 slices).
- **Rate**: The detection update rate in Hz, indicating how many full scan sweeps are completed per second.
- **VU meter**: Displays the peak FFT power across all bins. The marker line indicates the current detection threshold (`Mean + Trig`).
- **Timer**: A progress bar showing the detect or release countdown (see Detection Timing).
- **Status text**: Shows the current state — `Listening`, `Locked!`, or `Out of range`.

123.3 Big Frequency Display

Shows the frequency of the currently detected or locked signal. When a signal is found outside the configured Min/Max range, it is shown as `Out of range` and displayed in grey.

123.4 Spectrum Display

A colored gradient spectrum bar is shown across the screen, representing the FFT power across the entire scanned range. This provides a visual overview of band activity.

123.5 Frequency Log Table

The lower portion of the screen is a running log of all signals that were locked during the session, showing frequency, time of detection, and duration (in 100 ms units). Selecting an entry from the log opens the frequency entry screen, where the frequency can be saved to a freqman file on the SD card.

124 Detection Timing

The app operates in 100 ms timing units:

- **Detection**: A signal must remain above the threshold for **500 ms** (5 × 100 ms) before it is locked. This prevents single-frame noise spikes from being logged.
- **Release**: Once locked, a signal is released **600 ms** (6 × 100 ms) after it drops below the threshold. The duration logged is the time from lock to release.

125 Frequency Accuracy

Each FFT bin covers approximately 9.8 kHz (2.5 MHz ÷ 256 bins). The raw frequency resolution is therefore **±4.9 kHz**. Enabling **Snap to** rounds the result to a grid, which can improve readability for known channel plans.

126 Logging

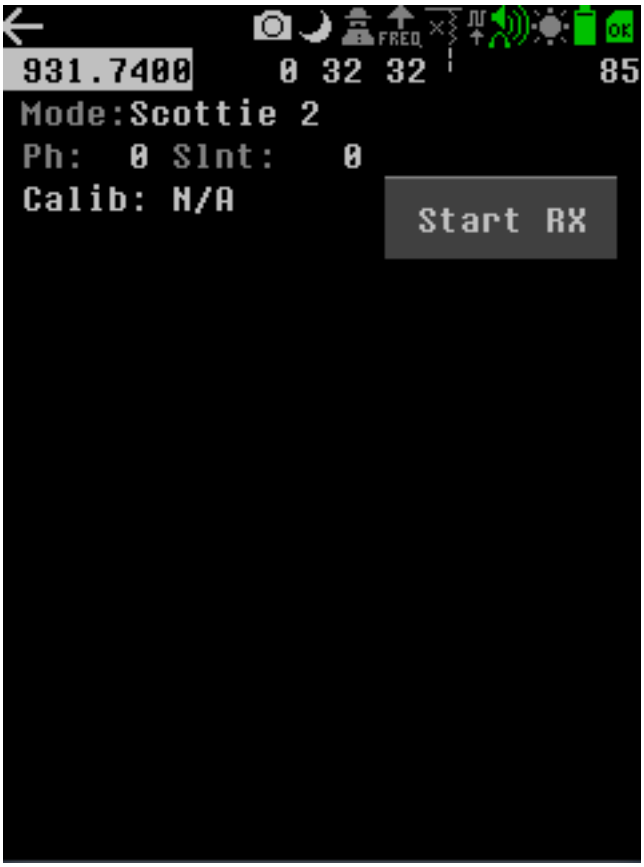
When **LOG** is enabled, a new CSV file is created at the start of the session: `LOGS/SEARCH_YYYYMMDD_HHMMSS.CSV`. The format is semicolon-delimited: `Time;Freq;Duration;`. A new file is created each time LOG is toggled on; existing files are not appended to.

A beep is emitted on signal lock if **Beep on packets** is enabled in the global Settings → Radio menu.

127 Settings Persistence

The power threshold, Min/Max frequencies, and Snap settings are saved to `rx_search.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

127.1 SSTV RX



Receives and decodes Slow Scan Television (SSTV) transmissions, displaying the incoming image in real time and saving it as a BMP file to the SD card.

This is an external app (`.ppma` file on the SD card).

128 Settings

- **Frequency**: The frequency to receive on. Adjustable via the encoder knob or on-screen numpad.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain. See Receive Quality Issues.
- **Volume**: Audio output volume.
- **Mode**: Selects the SSTV transmission mode. Default is **Scottie 2**. The six supported modes are:
- Mode: Scottie 1; VIS Code: 60; Pixels: 320; Lines: 256; ms/pixel: 0.432
- Mode: Scottie 2; VIS Code: 56; Pixels: 320; Lines: 256; ms/pixel: 0.275
- Mode: Scottie DX; VIS Code: 76; Pixels: 320; Lines: 256; ms/pixel: 1.080
- Mode: Martin 1; VIS Code: 44; Pixels: 320; Lines: 256; ms/pixel: 0.458
- Mode: Martin 2; VIS Code: 40; Pixels: 320; Lines: 256; ms/pixel: 0.229
- Mode: SC2-180; VIS Code: 55; Pixels: 320; Lines: 256; ms/pixel: 0.734

- **Ph** (Phase, −50 to +50): Horizontal pixel offset to correct line start alignment. Adjust if the image has a horizontal shift or color misalignment. Applied immediately while receiving. Not saved between sessions.
- **Slnt** (Slant, −100 to +100): Timing drift correction in 0.1% steps. Adjust if the image appears diagonally tilted. Applied immediately while receiving. Not saved between sessions.

129 Calibration Display

A **Calib** text field below the Mode selector shows sync detection status during reception — sync count and suggested phase/slant corrections are displayed here as they are computed from the incoming signal.

130 Image Display

The received image is drawn line by line onto the screen in real time, scaled to fit the available display area below the controls. The internal image resolution is always 320 × 256 pixels regardless of screen scaling.

131 Start or Stop

Press **Start RX** to begin reception. The button label changes to **Stop RX** while active. Each time Start RX is pressed, a new timestamped BMP file is created on the SD card. When Stop RX is pressed (or the app is closed), the current file is finalized and closed.

Images are saved to: `SSTV/RX/SSTV_YYYYMMDD_HHMMSS.BMP`

132 Modulation Note

This app decodes SSTV signals transmitted using **FM modulation** only. SSB (USB/LSB) SSTV, which is the norm on HF amateur radio, is not supported. FM SSTV is used on VHF/UHF (e.g., 144.500 MHz in Europe) and via certain satellite downlinks.

132.1 SubCar

SubCar is a receiver app for decoding automotive remote keyless entry (RKE) signals from car key fobs. It passively listens on a configurable frequency and attempts to match incoming signals against a library of known manufacturer protocols, displaying decoded results in a scrolling list.

This is an external app (`.ppma` file on the SD card).

133 Settings

- **Frequency**: Default is **433.92 MHz**, the standard European RKE band. Adjustable via the encoder knob or on-screen numpad for other regional bands (e.g., 315 MHz in North America).
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain. See Receive Quality Issues.
- **Mode**: Selects the demodulation mode — **AM** (default) or **FM**. Most car key fob protocols operate with AM (OOK) modulation; FM mode may be needed for some regional or newer implementations.
- **Clear**: Clears the decoded signal list from the display (does not affect the log file).
- **Log**: When checked, decoded packets are written to a timestamped CSV file on the SD card: `LOGS/SubCarLOG_YYYYMMDD_HHMMSS.CSV`. The file header is `; Type; Bits; Data;`.

134 Decoded Signal List

Each decoded packet is shown as a row with three columns: **Type** (protocol name), **Bits** (bit length of the decoded frame), and **Age** (seconds since last received). The list is deduplicated by signal content — repeated presses of the same fob button update the age counter rather than adding a new row.

Selecting an entry opens a detail view showing the protocol type, serial number, button code, and additional decoded fields where available for that protocol.

135 Supported Protocols

The following car key fob protocols are currently implemented:

- Protocol: Suzuki; Manufacturer: Suzuki
- Protocol: VW; Manufacturer: Volkswagen (button IDs: Lock, Unlock, Trunk, Panic, combinations)
- Protocol: Subaru; Manufacturer: Subaru
- Protocol: Kia V0–V5; Manufacturer: KIA
- Protocol: Ford V0; Manufacturer: Ford
- Protocol: Fiat V0; Manufacturer: Fiat
- Protocol: BMW V0; Manufacturer: BMW

> Note: Some protocols with full rolling-code encryption (Kia V6, PSA/Peugeot-Citroën) are recognized but not decoded, as the payload is entirely encrypted.

136 Settings Persistence

The modulation mode and Log checkbox state are saved to `rx_subcar.ini` on the SD card when **App Settings** is enabled in the global Settings menu

136.1 SubGhzD

SubGhzD (Sub-GHz Decoder) is a passive receiver app for decoding OOK/AM sub-GHz remote control signals from gate remotes, garage door openers, doorbells, home automation devices, and similar ISM-band transmitters. It identifies the protocol and displays decoded data in a scrolling list.

137 Settings

- **Frequency**: Default is **433.92 MHz**. Adjustable via the encoder knob or on-screen numpad for other ISM bands (e.g., 315 MHz, 868 MHz).
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier. Use with care on strong signals.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain. See Receive Quality Issues.
- **Clear**: Clears the decoded signal list from the screen. Does not affect the log file.
- **Log**: When checked, decoded packets are written to a timestamped CSV file on the SD card: `LOGS/SUBGHZDLOG_YYYYMMDD_HHMMSS.CSV`. The file header is `; Type; Bits; Data;`.

138 Decoded Signal List

Each decoded packet appears as a row with three columns: **Type** (protocol name), **Bits** (frame bit length), and **Age** (seconds since last received). The list deduplicates by signal content — repeated transmissions from the same remote update the age counter rather than adding new rows.

Selecting a row opens a detail view showing the protocol type, serial number, button code, counter value, and any additional decoded fields for that protocol.

For **KeeLoq** and **Superrollo** signals the detail view always shows the Fix code and the Encrypted hop value. The decrypted **Hop** value and the **Save** button are only shown when a matching manufacturer key is found in `KEELOQKEYS/MFCODES`. Without the key, the signal is received and displayed but cannot be decrypted and cannot be saved for retransmission.

When the Save button is available, it saves the decoded remote data to `KEELOQREMOTES/<name>.KEELOQ` on the SD card for use with the KeeLoq TX app.

139 Supported Protocols

45 protocols are currently implemented, covering fixed-code and some rolling-code (KeeLoq) systems across the 300–868 MHz ISM bands:

Princeton, Bett, CAME, CAME Atomo, CAME Twee, Chamberlain Code, Clemsa, Doitrand, Dooya, FAAC, Gate TX, Holtek, Holtek HT12X, Holtek HT6P20B, Honeywell, Honeywell WDB, Hörmann, IDO, Intertechno V3, KeeLoq, Kingates Stylo 4K, Linear, Linear Delta 3, Magellan, Marantec, Marantec24, Mastercode, Megacode, Nero Radio, Nero Sketch, Nice FLO, Nice FLO RS, Phoenix V2, Power Smart, Security+ V1, Security+ V2, SMC5326, Somify Keytis, Somify Telis, Star Line, X10, Legrand, Gangqi, Restaurant Pager, Superrollo (GW60/HCS361).

> Note: Hörmann BiSecure (FM-based) is recognized but not implemented in this app as it requires FM demodulation.

140 Superrollo (GW60/HCS361)

Superrollo GW60 remotes use the HCS361 rolling-code chip with a 67-bit OOK frame at Te=450 µs. SubGhzD decodes these signals and displays them as protocol type **Superrollo**.

Full workflow:

1. Tune to 433.92 MHz and press the Superrollo remote button.
2. The signal appears in the list as **Superrollo**, 67 bits.
3. Open the detail view — Fix code and Encrypted hop are always shown.
4. If a `Superrollo` entry exists in `KEELOQKEYS/MFCODES`, the decrypted Hop value and the **Save** button also appear.
5. Press Save, enter a filename — the decoded remote is saved to `KEELOQREMOTES/<name>.KEELOQ`.
6. Open KeeLoq TX, load the file, and transmit.

Without a valid manufacturer key in MFCODES, steps 4–6 are not available. See KeeLoq TX for the MFCODES file format.

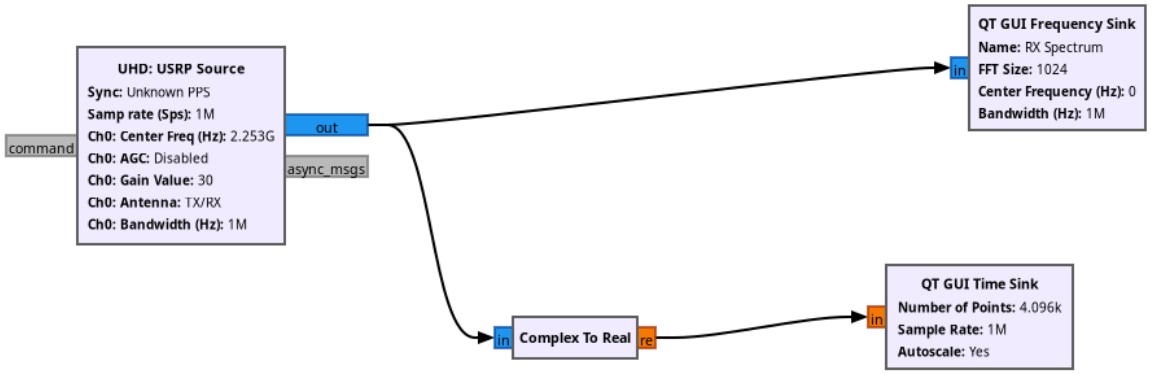
141 Settings Persistence

The Log checkbox state is saved to `rx_subghzd.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

141.1 Time Sink

Time Sink is a time-domain waveform viewer — an oscilloscope for RF. It plots the I (in-phase) component of the received IQ stream as an amplitude-vs-time trace across the screen. Apps like Looking Glass display the frequency domain (spectrum); Time Sink displays the time domain (signal shape over time).

The equivalent GNU Radio block is the **Time Sink** source block:



142 Display

The waveform area covers the lower portion of the screen (below the 3-row header). Each horizontal pixel corresponds to one of **240 data points** sampled from the received baseband buffer. The vertical axis represents amplitude, with the centre line at zero. Only the I (real) component is shown.

143 Settings

143.1 Row 0 — Frequency and Gain

- **Frequency**: Default 433.920 MHz. Use the encoder knob or tap to open the keypad.
- **Frequency step**: Step size for encoder tuning.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain.

143.2 Row 1 — Sample Rate and Decimation

- **SR (Sample Rate)**: Baseband sampling rate. Options: **1.0 MHz, 2.0 MHz, 5.0 MHz, 10.0 MHz, 20.0 MHz**. Default: 2.0 MHz. Higher rates capture faster signal edges; lower rates give a slower, more zoomed-in time base.
- **DEC (Decimation)**: Range 0–128. Controls how many baseband buffers are skipped between display updates. Each buffer is ~2048 samples. At DEC=0, the display updates every buffer; at DEC=128, it updates every 129th buffer. Increase DEC to slow down the trace refresh on slow or periodic signals.

143.3 Row 2 — Persistence and Trigger

- **PST (Persistence)**: Number of past waveform frames to keep visible on screen simultaneously. Options: **1, 2, 4, 8, 16**. At PST=1 only the current trace is shown; higher values overlay multiple recent traces, similar to the persistence mode of a digital oscilloscope. Maximum is 16 frames due to memory constraints in the external app format.
- **TRM (Trigger Mode)**: Controls waveform synchronisation.
  - `Off` — free-running, no synchronisation
  - `Rise` — triggers on a rising edge crossing the trigger level
  - `Fall` — triggers on a falling edge crossing the trigger level

  When triggered, the app finds the crossing point in each frame and aligns the display to that position, stabilising the waveform. A ±2 hysteresis band is applied to prevent false triggers. The trigger locks to the nearest crossing to the previous one for frame-to-frame stability.

- **LVL (Trigger Level)**: Range −127 to +127. Sets the amplitude threshold for trigger detection relative to the signal midpoint (0 = centre). Has no effect when TRM is set to Off.

144 Settings Persistence

All settings (sampling rate, DEC, persistence, trigger mode, trigger level) are saved to `rx_time_sink.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

144.1 TPMS RX

Tire Pressure Monitoring System (TPMS) sensors transmit in licence-free ISM bands — 315 MHz in North America and 433.92 MHz in Europe and most of the rest of the world. This app decodes TPMS transmissions and displays tyre pressure, temperature, sensor ID, and status flags in a live list.

This is an external app (`.ppma` file on the SD card).

145 Settings

- **Frequency**: Three options are available — **314.9 MHz**, **315.0 MHz**, and **433.9 MHz**. Both 314.9 and 315.0 are provided because "315 MHz" sensors may actually transmit at either frequency; the default is 314.9 MHz.
- **Pressure unit**: **kPa**, **PSI**, or **BAR**. Selectable inline; applies to both the list view and detail view.
- **Temperature unit**: **°C** or **°F**. Selectable inline.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain. See Receive Quality Issues.

146 Decoded Signal List

Each row in the list represents one unique sensor (identified by type + ID). The columns are:

- **Tp**: Sensor type number (shown as a 2-digit number in the list; the detail view also shows the name). Currently decoded types:
  - `0` = None
  - `1` = FLM_64
  - `2` = FLM_72
  - `3` = FLM_80
  - `4` = Schrader
  - `5` = GMC_96
  - `9` = Renault
  - `10` = Hyundai-VDO

  Renault and Hyundai-VDO are decoded from the same FSK 19k2 Schrader baseband path as the FLM types — no extra baseband signal path is needed for them.

  The firmware also contains CRC-verified decoders for further sensor types (Toyota, Ford, Citroen, Nissan, Kia, Abarth124, Jansite, Jansite-Solar, Elantra2012, PMV-107J, Renault-0435R, AVE). These are present in the code but are **not yet wired into the active decode path**, so they do not currently appear in the list — most of them require additional baseband signal paths (other modulations/baud rates) that are not implemented yet.
- **ID**: The 8-digit hexadecimal sensor ID. Note: some sensors print their ID in decimal on the housing.
- **Pres**: Tyre pressure in the selected unit (kPa, PSI, or BAR).
- **Temp**: Temperature in the selected unit (°C or °F). Spurious readings can occur due to receive errors, unsupported sensor models, or failing sensors — compare against the vehicle's dashboard reading where possible.
- **Cnt**: Number of packets received from this sensor since the app was opened.
- **Fl**: Sensor flags as a 2-digit hexadecimal value. The meaning of individual flag bits varies by sensor manufacturer and model.

Selecting a row opens a detail view with all fields expanded. A **Save** button in the detail view writes the sensor data to `TPMS/TPMS_.TXT` in a format compatible with the TPMS TX app.

147 Transmission Behaviour

Each sensor transmits at intervals that vary from a few seconds to several minutes, depending on the vehicle and sensor model. Transmission is typically triggered by wheel movement or by a rapid pressure change. Most sensors can also be triggered immediately using a TPMS activation tool operating at 125 kHz.

148 Logging

When an SD card is present on app startup, received packets are automatically logged to `LOGS/TPMS.TXT`. Each log entry contains a timestamp in `YYYYMMDDHHMMSS` format, followed by the receiver frequency, signal type (`FSK 38400 19200 Schrader`, `OOK - 8192 Schrader`, or `OOK - 8400 Schrader`), the raw Manchester-decoded hex data, a `/` separator, and a per-bit Manchester coding error indicator.

149 Settings Persistence

The selected pressure unit and temperature unit are saved to `rx_tpms.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

149.1 Weather

The Weather app decodes wireless transmissions from consumer weather stations and sensors operating in the ISM licence-free bands. It receives both OOK (AM) and FSK signals and presents a live list of decoded sensor readings, updated in real time. Besides temperature and humidity, many Fine Offset or Ecowitt sensors also report wind, rain, pressure, UV, light, soil moisture, PM2.5 and lightning.

This is an external app (`.ppma` file on the SD card).

150 Settings

- **Frequency**: Default 433.920 MHz. Adjustable in 10 kHz steps using the encoder knob, or tap to open the keypad. Common frequencies are listed below.
- **Temperature unit**: **°C** or **°F**. Applies to both the list view and detail view.
- **RF Amp** (0 or 1): Enables/disables the internal RF amplifier. Use with caution on strong nearby signals.
- **LNA gain** (0–40): Low-noise amplifier gain. See Receive Quality Issues.
- **VGA gain** (0–62): Variable-gain amplifier gain.
- **Volume**: Volume for packet beep tones. Only active when **Beep on packets** is enabled in the global Settings.
- **Clear**: Clears the in-memory sensor list. Does not affect any log file.
- **Log**: Checkbox to enable CSV logging. When checked, a new log file is created immediately. The log state is saved to `rx_weather.ini` and restored on next launch — if it was enabled when you last left the app, logging resumes automatically on the next startup.
- **S.Hem**: Checkbox for the southern hemisphere. When checked, the wind direction shown and logged is rotated by 180°. Only the arrays with a flat solar panel that has to point north in the southern hemisphere are rotated: FO-WH24 (HP1000), FO-WH65, FO-WS69 and FO-WS68. Other sensors, e.g. the WS80 and WS90, always show the direction they send. Off by default. See Wind direction in the southern hemisphere.

Gain tip: This app is sensitive to signal strength. If nothing decodes, try starting with AMP 0, LNA 32, VGA 20, and adjust from there. Both over-amplification and under-amplification will prevent decoding.

151 Common Frequencies

- **433.92 MHz** — Europe, Australia, most of the world
- **315 MHz** — North America, Japan
- **868 MHz** — Europe and others
- **915 MHz** — North America, Japan

> Fine Offset or Ecowitt sensors are sold in 433, 868 and 915 MHz versions. The frequency is usually printed on the sensor or its box. Set the app to match the version you have.

152 Sensor List

Each row represents one unique sensor. Entries are kept for up to 64 sensors at a time. Columns:

- **Type**: Sensor protocol name (see Supported Protocols below).
- **Temp**: Temperature in the selected unit, or - if the sensor does not measure temperature (e.g. rain gauges, PM2.5, soil moisture, lightning and leak sensors).
- **Hum**: Relative humidity in percent, or **–** if the sensor does not transmit humidity.
- **Ch**: Channel number reported by the sensor, or **–** if not applicable.
- **Age**: Seconds since the last packet was received from this sensor. Resets to 0 on each new reception.

Selecting a row opens a **detail view** with all fields expanded: Type, Id (hexadecimal), Temperature, Humidity, Channel, Battery status (OK or LOW), and Age. Press **Done** to return to the list.

If the sensor reports more values, they are listed below those fields. Only the values the sensor actually sends are shown:

- Line: Wind; Unit: m/s; Notes: Average wind speed
- Line: Wind dir; Unit: deg; Notes: 0 = North, 90 = East. Rotated by 180° when S.Hem is checked
- Line: Gust; Unit: m/s
- Line: Rain; Unit: mm; Notes: The sensor's running total, not rain per hour or per day
- Line: Pressure; Unit: hPa; Notes: Absolute (station) pressure, not adjusted to sea level
- Line: UV index
- Line: Light; Unit: lux
- Line: Moisture; Unit: %; Notes: Soil moisture
- Line: PM2.5; Unit: ug/m3
- Line: Storm; Unit: km; Notes: Distance to the last lightning strike
- Line: Strikes; Notes: Lightning strike counter
- Line: Leak; Notes: 1 = water leak alarm, 0 = no leak (WH55)

153 Logging

When the **Log** checkbox is enabled, received sensor data is written to `LOGS/WEATHERLOG_<timestamp>.CSV` on the SD card. The file starts with the header line:

```
;Type; id; Temp; Hum; CH; Batt; Wind m/s; Dir; Gust m/s; Rain mm; hPa; UVI; Lux; Moist;
PM2.5; Storm km; Strikes; Leak
```

Each subsequent line contains one decoded sensor reading in the same column order. Columns a sensor does not report are left empty. The wind direction column follows the **S.Hem** setting.

154 Fine Offset or Ecowitt sensors

Fine Offset Electronics makes the sensors sold under the Ecowitt, Ambient Weather, Froggit, Misol and many other brands (e.g. the HP1000 station). In the list they show as `FO-` followed by the model, e.g. `FO-WH24`.

> Fine Offset or Ecowitt support and the **S.Hem** option were added in #3383 (October 2026). Until the next release they are only in the nightly builds.

Most current models use **FSK**; the older ones use OOK. The app handles both automatically, so no extra setting is needed. The decoders are based on rtl_433, and the values are scaled the same way rtl_433 does.

- Shown as: FO-WH24; Sensors: WH24 or HP1000 outdoor array; Values: Temp, hum, wind, dir, gust, rain, UV, light
- Shown as: FO-WH65; Sensors: WH65 outdoor array; Values: Temp, hum, wind, dir, gust, rain, UV, light
- Shown as: FO-WS69; Sensors: WS69 outdoor array with pressure; Values: As WH65, plus pressure
- Shown as: FO-WH25; Sensors: WH25 indoor sensor; Values: Temp, hum, pressure
- Shown as: FO-WH32; Sensors: WH32, WH32B, WN32B; Values: Temp, hum
- Shown as: FO-WS80; Sensors: WS80 ultrasonic array; Values: Temp, hum, wind, dir, gust, UV, light
- Shown as: FO-WS85; Sensors: WS85 ultrasonic wind & rain; Values: Wind, dir, gust, rain
- Shown as: FO-WS90; Sensors: WS90 all-in-one array; Values: Temp, hum, wind, dir, gust, rain, UV, light, pressure
- Shown as: FO-WS68; Sensors: WS68 wind or solar; Values: Wind, dir, gust, UV, light
- Shown as: FO-WH31; Sensors: WH31E, WH31B; Values: Temp, hum, channel
- Shown as: FO-WN34; Sensors: WN34, WN38; Values: Temp
- Shown as: FO-WH40; Sensors: WH40 rain gauge; Values: Rain
- Shown as: FO-WN20; Sensors: WN20 rain gauge; Values: Rain
- Shown as: FO-WH51; Sensors: WH51 soil moisture; Values: Moisture
- Shown as: FO-WH0290; Sensors: WH0290 or WH41 PM2.5; Values: PM2.5
- Shown as: FO-WH57; Sensors: WH57 or WH31L lightning; Values: Storm distance, strikes
- Shown as: FO-WH55; Sensors: WH55 water leak; Values: Channel, battery, leak alarm
- Shown as: FO-WH2; Sensors: WH2, WH2A, Telldus; Values: Temp, hum
- Shown as: FO-WH5; Sensors: WH5, TFA 30.3225; Values: Temp, hum
- Shown as: FO-WH0530; Sensors: WH0530 rain gauge; Values: Temp, rain
- Shown as: FO-WH1080; Sensors: WH1080 or WH3080 (OOK); Values: Temp, hum, wind, dir, gust, rain
- Shown as: FO-WH3080UV; Sensors: WH3080 UV or light packet; Values: UV, light
- Shown as: Ecowitt WH53; Sensors: WH53 temperature probe; Values: Temp, channel

Most sensors transmit every 16 to 80 seconds (e.g. WH24 about every 16 s, WH25 about every 64 s), so it can take a minute before a sensor appears.

154.1 Wind direction in the southern hemisphere

Some outdoor arrays (WH24 or HP1000, WH65, WS69 and WS68) have a flat solar panel that must face the equator, so the array has to be installed with the panel pointing south in the northern hemisphere and north in the southern hemisphere. In the northern hemisphere the array's "N" marking then points north. In the southern hemisphere the array is installed turned around, so the marking points south and the sensor sends a wind direction that is 180° off. The station consoles have a hemisphere setting to correct this.

The sensor always sends the uncorrected direction. If you are in the southern hemisphere and installed one of these arrays with the solar panel pointing north, check **S.Hem** so the wind direction matches your console. **S.Hem** only rotates these four types. Other sensors are left as they are, e.g. the WS80 and WS90, which have a ring-shaped solar panel around the top of the unit and are installed with their "N" marking pointing north everywhere, so both kinds of array show the right direction at the same time.

155 Supported Protocols

49 protocols are decoded:

Acurite592TXR, Acurite5in1, Acurite606TX, Acurite609TX, Acurite986, Ambient, AuriolAhfl, AuriolTH, Bresser3CH, Ecowitt WH53, EmosE601x, FO-WH0290, FO-WH0530, FO-WH1080, FO-WH2, FO-WH24, FO-WH25, FO-WH3080UV, FO-WH31, FO-WH32, FO-WH40, FO-WH5, FO-WH51, FO-WH55, FO-WH57, FO-WH65, FO-WN20, FO-WN34, FO-WS68, FO-WS69, FO-WS80, FO-WS85, FO-WS90, GT-WT02, GT-WT03, InFactory, Kedsum, LaCrosse TX, LaCrosse TX141THBv2, NexusTH, Oregon2, Oregon2B, Oregon3, OregonV1, SolightTE44, ThermoPro TX4, TX 8300, Vauno EN8822, Wendox W6726

156 Limitations

- The app shows the raw values the sensor sends. Calibration offsets set on a station console are not applied, so readings can differ slightly from the console.
- Pressure is the absolute pressure measured at the sensor. Consoles usually also show relative (sea-level) pressure, which is higher above sea level.
- Rain is the sensor's running total. It does not reset each day like the console's daily rain.
- Not supported yet: Fine Offset WH1050, WH52, WH45 or WH46 air quality sensors and the 868 MHz FSK version of the WH1080.

157 Settings Persistence

The temperature unit, log state and **S.Hem** setting are saved to `rx_weather.ini` on the SD card when **App Settings** is enabled in the global Settings menu.

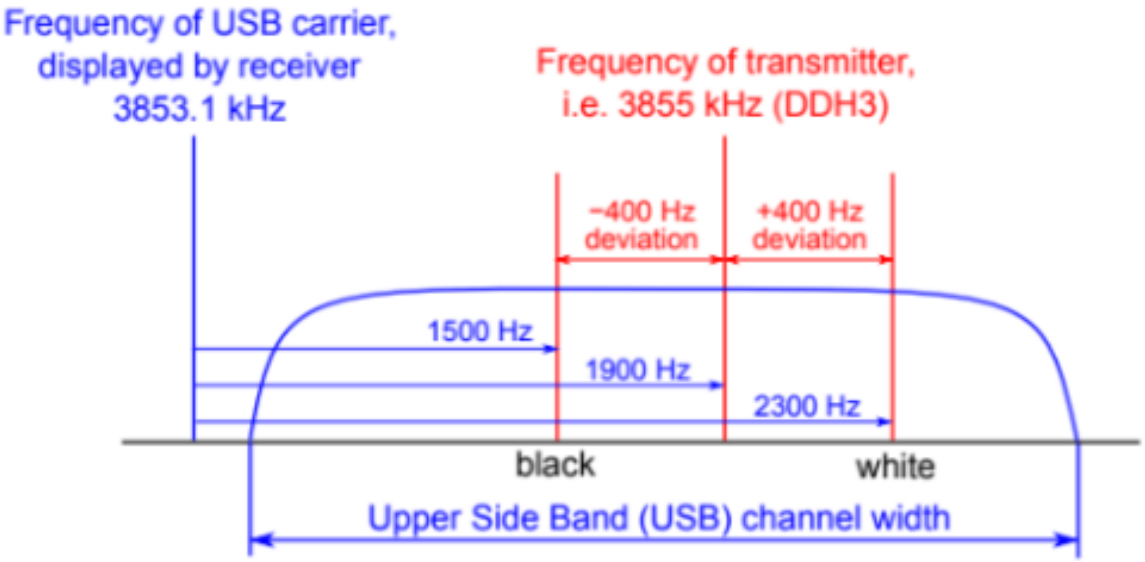
157.1 WeFax

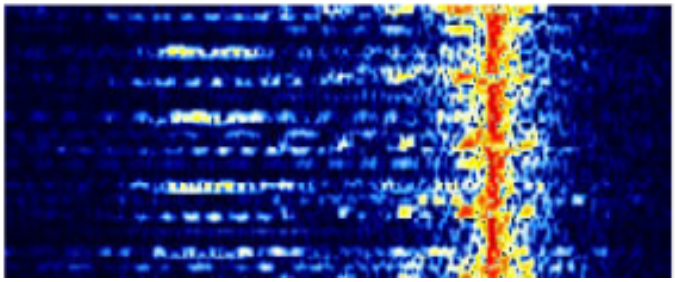
This is an external app (`.ppma` file on the SD card).

158 Introduction

That Wefax App, allows to receive any HF Weather Station Brodcasting (Wefax, https://www.sigidwiki.com/wiki/WEFAX) , and decode its analogue weather fax maps transmissions .

Here is the list of current Wefax stations and its frequencies,and the exact GMT map transmission time https://weatherfax.com/stations/. To save RF spectrum , and optimize the range , all those broadcasting are transmitting in USB . Some example of the spectrum frequencies are the following ones ,





For better usability , we are internally applying that tune-offset . Therefore you just need to tune the official published WeatherFax frequency transmitter. (3.855 Khz in that example, or 7.880 Khz , 8.040 Khz ,...)

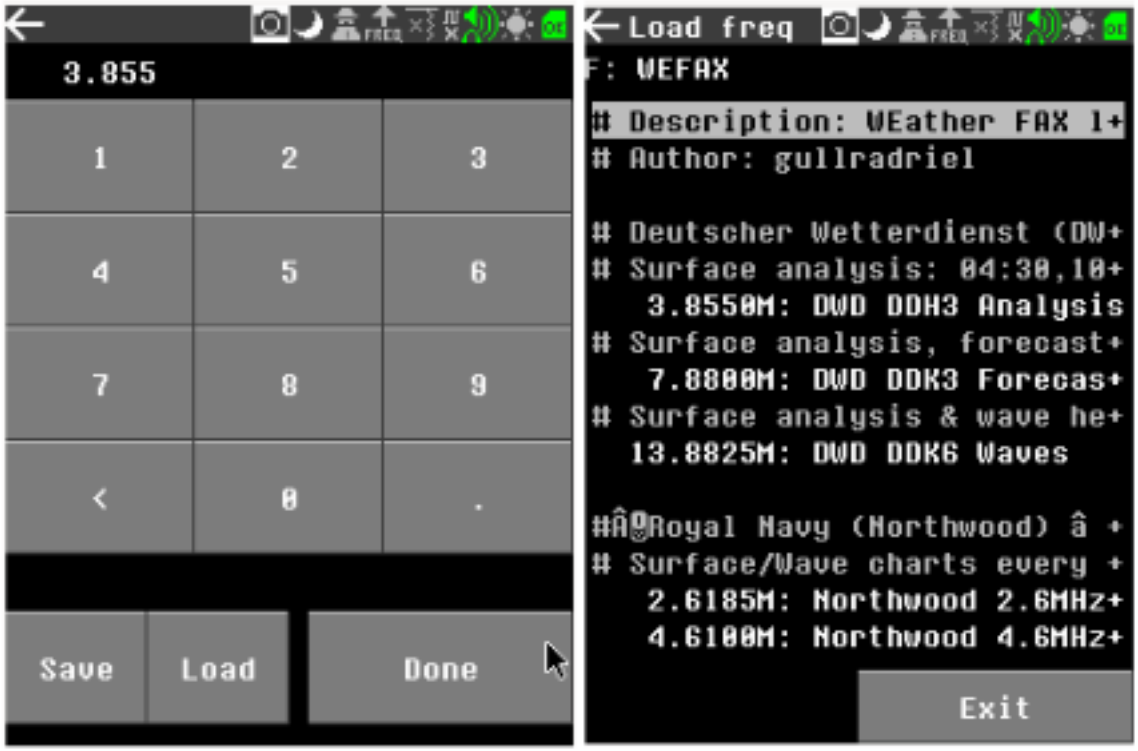
That Wefax App is combining two consecutive demodulations. First RF USB audio baseband demodulation , and later demodulating the FM subcarrier audio baseband tones. (Audio Baseband with centered f_subcarrier fsc (1900Hz), and modulated in FSK +-400Hz ; Shift White pixel +400 Hz = 2.300 Hz, and Shift Black pixel −400 Hz = 1.500 Hz).

That analogue AM-FM modulation , it is commonly used in Wefax (HF Weather Fax transmissions) . It demodulates APT signal of the HF weather fax maps. , and sends the demodulated low scan map APT signal in parallel to the speaker (just for our audio tuning reference) and to the LCD display in low resolution. And it allows to the user , been able to RECORD (with START or STOP) that map in full resolution to the SD/BMP folder.

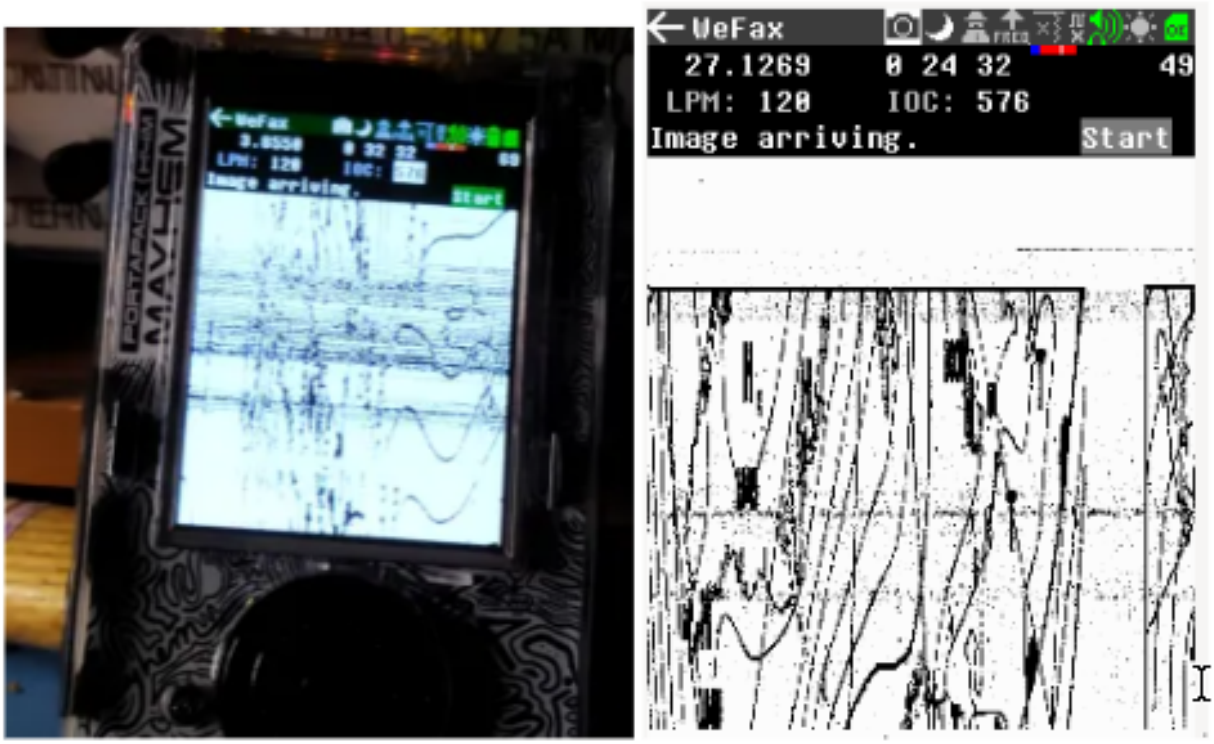
159 Usage

Hackrf has poor sensitivity to receive directly those HF Wefax signals, (mainly broadcasted in HF band : 3 Mhz to 16 Mhz) Normally we should better add a front up-converter with better HF filtered pre-amplifiers, and a reasonable good enough HF antenna. WEFAX's format is a derivative of the [Automatic Picture Transmission (APT)] originally developed for transmission from the polar satellites of the USA. WEFAX has 60, 90, 100, 120, 180 and 240 LPM (Lines per minute) speeds, and two IOC (Index of Cooperation) modes, IOC 576 (1200 line maps) and IOC 288 (600 line maps). Most weather forecasts are sent in IOC 576, 120 lpm (make sure to select those 120 lpm . Our initial target would be , WEFAX-IOC576 standard.

You can tune , inputting directly the Wefax broadcast frequency Station or pressing LOAD WeatherFax presetting stations , in the (tune menu).

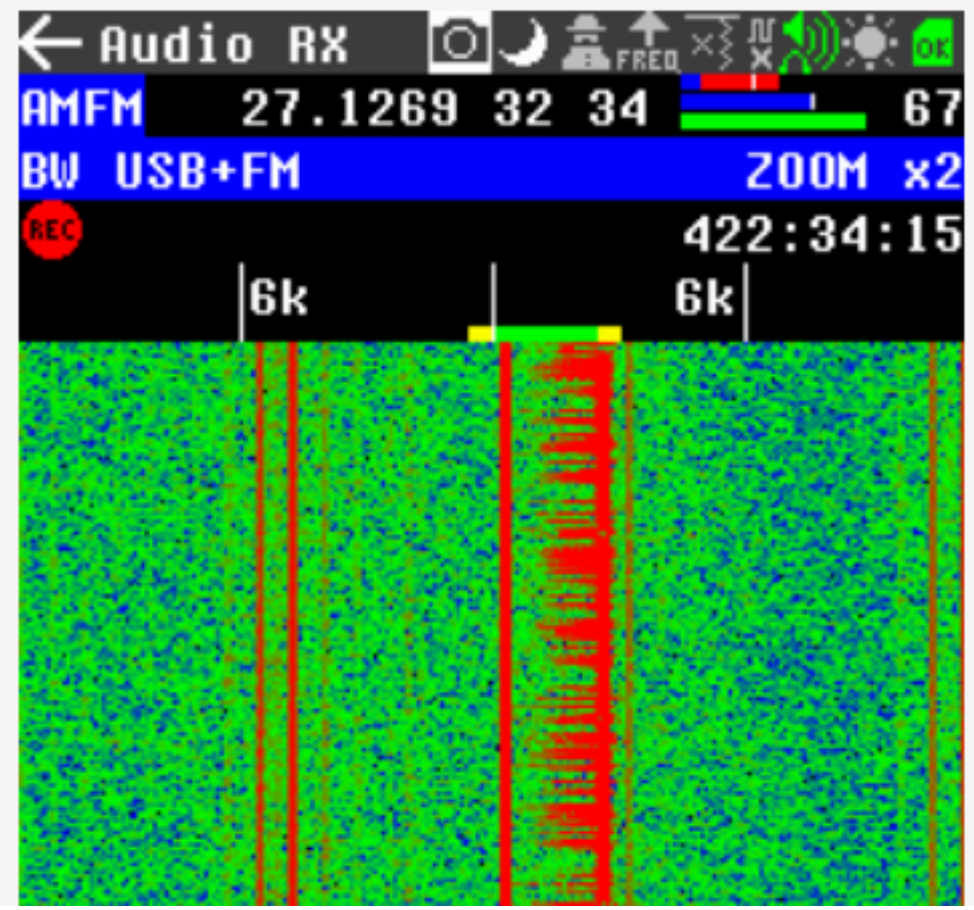


The application is in continuous display mode (trying to plot 2 lines or sec - if you select 120 lpm IOC 576 - the most common-) , and once you input the correct station frequency , you need to adjust the gain reception to the best point (through LNA :0/1, AMP , VGA) , to not saturate and not get too weak signal. If you don't amplify enough it could be darker image, or if you amplify too much, there could be much more whites. So it needs some tweaking too. Looking the screen and hearing the sound you should be able to find a good compromise tuning reception. In good setting conditions , you will start to recognise some part of the map in the screen , (like this below one)

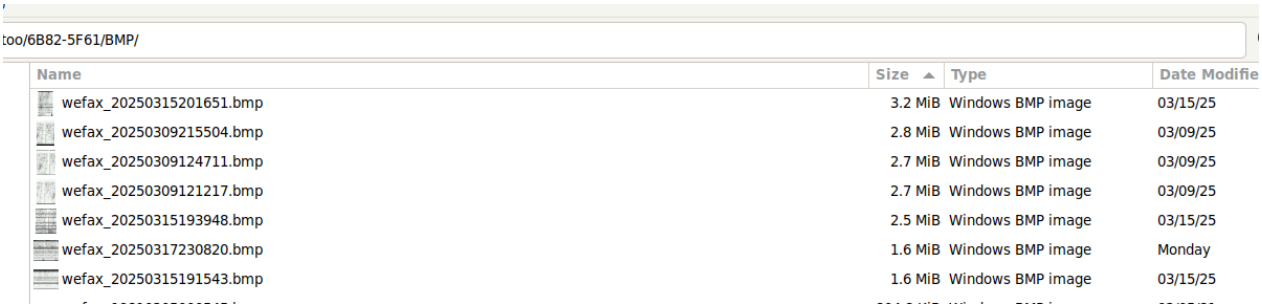


If needed , you can also check the waterfall signal Spectrum (and adjacent channels, interferences,…) using the other Audio App , with AMFM demod. mode in ZOOM x2 (for better view) ,and you can confirm ,

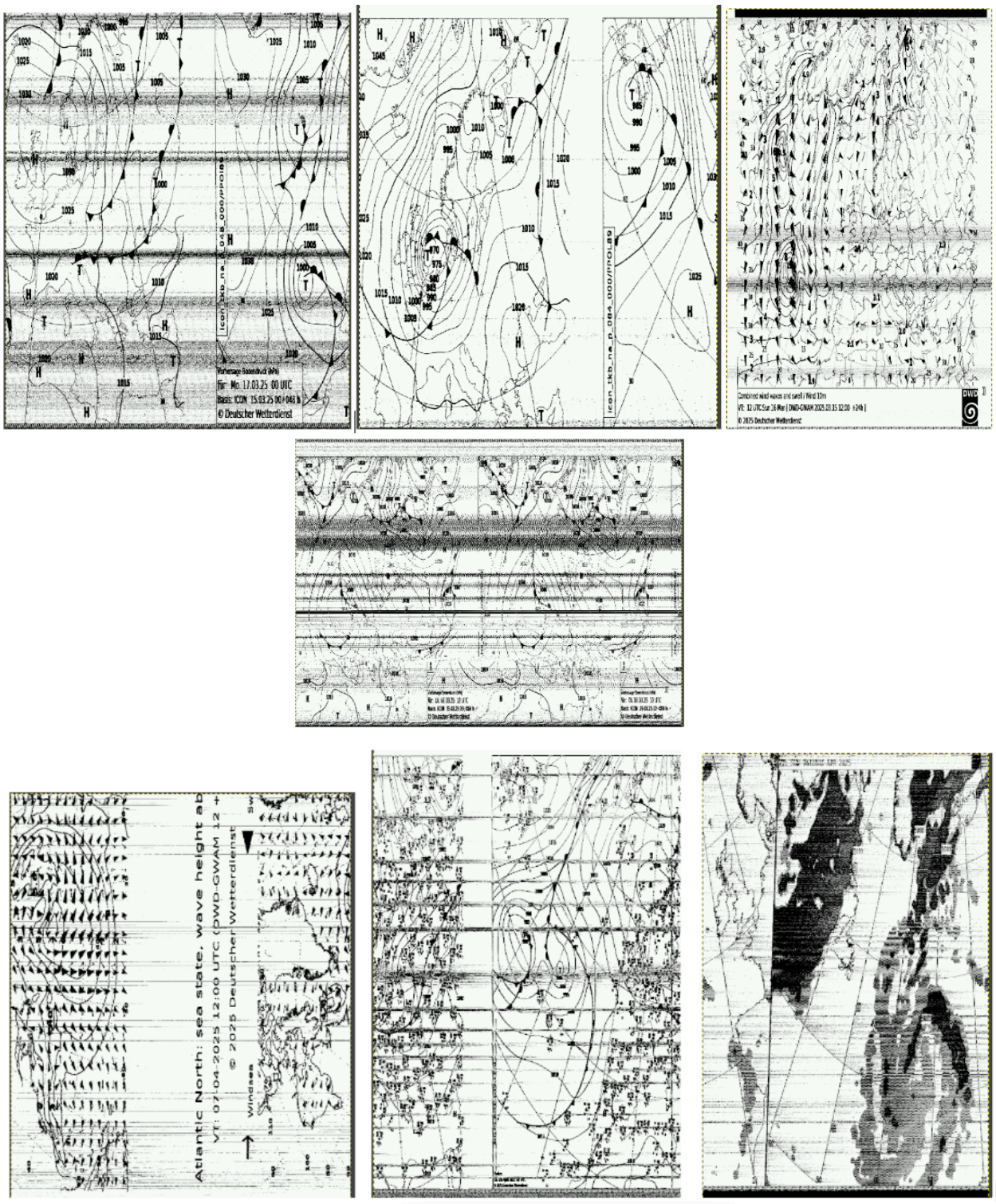
- that the frequency is the correct one (well tuned when top FM +deviation (white map pixels) are aligned to the border of the right yellow mark)
- and the waterfall signal has also some -deviation (black map pixels) left peaks (as in the below picture) .
- that you can confirm , that you have adjusted the best overall gain (proper LNA_AMP or GAIN or VGA) .



Once you confirmed all those above points from Audio App or directly from the Wefax App (optimizing the gain levels for best LCD screen map decoding. you can press START from the Wefax App ,and it will record into SD/BMP , related decoded maps , in high resolution.



Pls. find attached some captured examples,



Receiving & Decoding WeatherFax: https://youtu.be/KJ7u5V44u2o The video starts by showing where the preconfigured WeatherFax frequencies are located in the Portapack menu, making it easy to start receiving weather fax transmissions. You will then see live WeatherFax signal reception, followed by automatic decoding of the signal into weather map images directly on the device. The video also shows where decoded WeatherFax images are stored on the H4M Portapack Mayhem, and includes three real WeatherFax images received with this device.

159.1 Transmitters

159.2 2-Tone-TX

159.3 Introduction

Transmits a Motorola or QCII style two-tone page with configurable Tone A, Tone B, durations, optional gap, and optional CTCSS. It can be used for bench testing, interoperability checks, or generating known paging sequences for decoder testing.

This is an external app. The `.ppma` file must match the currently running firmware version.

159.4 Controls

- CTCSS: Selects the CTCSS tone mixed with the transmitted page. Set it to `None` to send the page tones without CTCSS.
- A: Selects Tone A from the Motorola or QCII frequency table, or `Custom`.
- B: Selects Tone B from the Motorola or QCII frequency table, or `Custom`.
- ADur: Sets Tone A duration in milliseconds.
- BDur: Sets Tone B duration in milliseconds.
- Gap: Sets an optional silent gap between Tone A and Tone B.
- Timing: Selects a timing preset such as `Moto Std`, `Short Alert`, `Fire Std`, `Long Alert`, or `Custom`.
- Slot: Selects one of five preset storage slots.
- Save: Saves the current tone and timing settings into the selected slot.
- Load: Loads the selected preset slot.
- AHz or BHz: Used when Tone A or Tone B is set to `Custom`. This allows direct entry of a custom frequency in whole Hz.
- The TX time line shows the total transmit duration and the selected tone names.
- The status line reports actions such as preset save/load, `Stopped`, or `Done`.
- The progress bar shows the transmission progress while the page is being transmitted.

159.5 Presets

Each preset slot stores:

- CTCSS selection
- Tone A selection
- Tone B selection
- Tone A duration
- Tone B duration
- Gap duration
- Custom A/B frequency values

Preset slots also have a short display name shown on screen.

159.6 Notes

- The app uses the existing Mayhem tone transmit path and does not require a special receive-side detector.
- With CTCSS enabled, the CTCSS tone is mixed simultaneously with the main paging tones.
- When `Moto Std` is selected, the app uses the usual 1000 ms Tone A and 3000 ms Tone B timing.

159.7 Practical Tips

- Use only frequencies and modulation settings that are legal in your region.
- Keep deviation and gain settings conservative when testing with another receiver to avoid distortion or clipped tones.
- If you are testing a decoder, a `0 ms` gap most closely matches standard back-to-back two-tone paging.

159.8 ADS-B(S) TX

This is an external app (`.ppma` file on the SD card).

159.9 WARNING

> **This is a potentially dangerous app. HackRF can generate valid ADS-B messages that, if radiated, can affect how real aircraft and navigation systems perceive each other. It must only be used in a completely closed RF environment with no live transmission into open air.**

- This application is intended solely for **experimental** purposes. It must not be used for any other reason.
- It is your responsibility to comply with all local, state, national, and international laws while using this application. Any illegal activities are strictly prohibited.
- This application is not suitable for use by individuals under the age of 18.
- All experiments conducted with this application are done entirely at your own risk. The developers are not liable for any damages or losses that may result.
- No form of assistance or support is provided for experiments conducted with this application. You are solely responsible for all outcomes and consequences.
- By using this application, you agree to indemnify and hold harmless the developers and all associated parties from any and all claims, damages, losses, liabilities, costs, and expenses arising from your use of this application or violation of these terms.
- This application is provided "as is" without warranty of any kind, express or implied.

ADS-B TX generates and transmits ADS-B Mode S messages at 1090 MHz, simulating an aircraft transponder. The transmitted frames can carry position, callsign, velocity, and squawk code data, which will be visible to ADS-B receivers and applications in range.

The ICAO24 address field is **shared across all four tabs** and is always active. The four tabs each add an independently toggleable message type to the transmission. Settings entered on each tab are **not saved to persistent memory** — only the TX frequency, gain, and RF amp settings are retained between sessions (via `tx_adsb.ini`).

159.10 ICAO24 (All Tabs)

- **ICAO24**: The 6-digit **hexadecimal** address that uniquely identifies the transmitting aircraft (e.g. `4CA1E0`). This 24-bit code, sometimes called the Mode S address, is assigned to each registered aircraft. Enter using the encoder knob on each hex digit.

159.11 Position Tab

Enable **Transmit position** to include airborne position messages (DF17 Type 9–18).

- **Alt**: Altitude in feet. Range: −1000 to 50000 ft.
- **Lat**: Latitude in Degrees° Minutes' Seconds". Degrees: −90 to +90; Minutes: 0–59; Seconds: 0–59. The decimal degree equivalent is displayed to the right.
- **Lon**: Longitude in Degrees° Minutes' Seconds". Degrees: −180 to +180; Minutes: 0–59; Seconds: 0–59. The decimal degree equivalent is displayed to the right.
- **Set from map**: Opens the map view. Tap the desired location on the touchscreen to transfer its coordinates into the Lat/Lon fields. An **OK** button confirms the selection — it is initially hidden behind the coordinate readout but appears when that area is tapped.

159.12 Callsign Tab

Enable **Transmit callsign** to include aircraft identification messages (DF17 Type 1–4).

- **Callsign**: Up to 8 characters. Tap the callsign button to open the text entry pad. Default: `TEST1234`.

159.13 Speed Tab

Enable **Transmit speed** to include airborne velocity messages (DF17 Type 19).

- **Speed**: Ground speed in knots. Range: 0–999 kn, step 5.

- **Bearing**: Ground track angle. Range: 0–359°, step 1. The compass rose in the upper-right corner shows the selected direction visually.
- **Vertical rate**: Climb/descent rate in feet per minute. Range: −4096 to +4096 ft/min, step 64 (per the ADS-B encoding standard). Positive values indicate climbing, negative values descending. The source is simulated as GNSS-derived.

159.14 Squawk Tab

Enable **Transmit squawk** to include Mode A identity replies (DF17 squawk).

- **Squawk**: A 4-digit **octal** transponder code (each digit 0–7), giving 4096 possible codes (0000–7777). Certain codes carry standard meanings: 7700 = general emergency, 7600 = radio failure, 7500 = hijack.

159.15 Common Controls (Bottom Bar)

- **Frequency**: Fixed at **1090.000 MHz** (the international ADS-B frequency). Editable but must not be changed for valid ADS-B simulation.
- **LNA gain** (0–47 dB) and **RF Amp** (0 or 14 dB): Standard TX gain controls.
- **Start or Stop**: Begins or ends the transmission. All enabled tabs transmit their frames in a repeating cycle.

W Wikipedia

W Wikipedia

159.16 Adult Toys

> **Consent warning:** This app can send control signals to compatible Bluetooth devices within radio range — including devices belonging to other people. Using this app to operate, interfere with, or control any device without the explicit consent of its owner is a serious violation of personal boundaries and privacy, and may be illegal under harassment, stalking, or communications interference laws in your jurisdiction. **Only use this app on your own devices.**

Adult Toys TX transmits Bluetooth Low Energy (BLE) control packets compatible with **Love Spouse** brand Bluetooth-connected devices. The protocol is based on the Lovespouse implementation from the Flipper Zero Momentum project. The app operates on BLE advertising channels in the 2.4 GHz band.

This is an external app (`.ppma` file on the SD card).

160 Settings

- **Frequency**: BLE channel frequency (2.4 GHz band). The app cycles through BLE advertising channels 37, 38, and 39 automatically.
- **Start or Stop**: Begins or halts the continuous transmission of control packets.

161 Notes

- The device must be in range and in a connectable/advertisable state to respond.
- Currently only the Love Spouse protocol is implemented. Other brands are not supported.
- Settings are saved to `tx_adult_toys.ini`.

161.1 APRS TX

[Automatic Packet Reporting System] The APRS TX App sends messages on the selected frequency. The App allows the configuration of the APRS stream. The App is marked as Beta, but it does send decodable messages in AFSK in the correct AX.25 format.

NOTE This is an external app and must be present on the SD card in the `APPS/` folder.

161.2 Fields

Source: Your own licensed amateur radio callsign (e.g. W1ABC). The field accepts up to 6 alphanumeric characters. Move the cursor to the right side of the label using the arrow buttons and select each character. Then move to "SSID:" and select a number from 0–15. The most common SSID for a PortaPack is 0.

Dest.: This is **not** the callsign of a receiving station. In APRS, this field must contain a valid APRS destination address. The recommended value for general use is **APRS** (which falls under the generic AP* wildcard accepted by all APRS decoders). For experimental or custom software, **APZNNN** (where NNN is a version number) is the correct choice per the APRS Protocol Reference. Entering a random callsign or leaving this field empty will cause APRS decoders (such as OpenWebRX, Direwolf, or aprs.fi) to ignore or incorrectly parse the packet. Set SSID to 0.

Path: The digipeater path tells the network how far to relay your packet.

- WIDE1-1 — single hop, recommended for most mobile and portable use
- WIDE1-1,WIDE2-1 — two hops, wider area coverage
- Leave empty for direct RF only (no digipeating)

Info field: The message content. Select the "Set" button to enter text (maximum 30 characters). **Important:** The very first character of the Info field must be a valid APRS Data Type Identifier (DTI). Without it, APRS decoders cannot determine the packet type and will not display the content correctly.

Note: The ! DTI is an exception — it may appear anywhere up to the 40th character position. All other DTIs must be the first character.

161.3 APRS Data Type Identifiers

The first character of the Info field defines what kind of data follows. Source: APRS Protocol Reference v1.0, Chapter 5.

- Character: >; Type: Status — simplest, no coordinates needed; Messaging: No; Example Info field: >Hello from PortaPack
- Character: !; Type: Position without timestamp; Messaging: No; Example Info field: ! 4071.27N/07400.59W-Hello from NYC
- Character: =; Type: Position without timestamp; Messaging: Yes; Example Info field: =4071.27N/07400.59W-Hello from NYC
- Character: /; Type: Position with timestamp; Messaging: No; Example Info field: /181654z4071.27N/07400.59W-Hello
- Character: @; Type: Position with timestamp; Messaging: Yes; Example Info field: @181654z4071.27N/07400.59W-Hello
- Character: :; Type: Message to a specific station; Messaging: Yes; Example Info field: :W1XYZ :Hello there
- Character: ;; Type: Object report; Messaging: No; Example Info field: ;OBJECT *181654z4071.27N/07400.59W/
- Character: _; Type: Weather report (no position); Messaging: No; Example Info field: _10092345c000s000g000t068
- Character: T; Type: Telemetry; Messaging: No; Example Info field: T#001,100,200,300,400,500,00000000

"Messaging" means the packet signals that the station is capable of receiving and acknowledging APRS messages. Use = or @ if you want two-way messaging; use ! or or for position-only beacons.

For beginners, start with > (status). It requires no coordinates, no timestamp, and is decoded by all APRS software.

The ?GPS? placeholder in the Info field is automatically replaced by the current GPS coordinates if a GPS module is active or Manual GPS has been set. Example: !?GPS?-Hello from PortaPack

161.4 Timestamp formats

Timestamps appear inside position packets that use the or or @ DTI. The format is always 7 characters.

- Format: DDHHMMz; Example: 181654z; Meaning: Day 18, 16:54 UTC — recommended
- Format: DDHHMM/; Example: 181654/; Meaning: Day 18, 16:54 local time
- Format: HHMMSSh; Example: 165400h; Meaning: 16:54:00 UTC (hours/minutes/seconds)

The APRS Protocol Reference recommends always using zulu (z) format for on-air transmissions.

Note: The MDHM format (Month/Day/Hours/Minutes) seen in some older examples is only valid for positionless weather station reports and must not be used in position packets.

161.5 Position coordinate format

Latitude: DDmm.hhN or DDmm.hhS (8 characters) Longitude: DDDmm.hhE or DDDmm.hhW (9 characters — always 3 digits for degrees, with leading zero if needed)

Example: 4071.27N/07400.59W = New York City area

- 40 degrees 71.27 minutes North
- 074 degrees 00.59 minutes West (note the leading zero)

The character between latitude and longitude is the Symbol Table Identifier (/ = primary table, \ = alternate table). The character immediately after the longitude is the Symbol Code. Common symbol codes: - = house, > = car, [= jogger, / = dot/waypoint. See the APRS Symbol Tables for the full list.

161.6 Altitude

When a real position is present (from a GPS module or the manual position picker), the station's altitude is transmitted as the standard APRS comment extension **`/A=aaaaaa`**, appended after the position. Altitude is entered in **metres** in the position picker and converted to **feet** for the six-digit `/A=` field. An altitude of zero is a valid value and is sent as `/A=000000` rather than being dropped. The last altitude is saved to `tx_aprs.ini`.

161.7 Frequency

APRS operates on dedicated frequencies per region. Transmitting on the wrong frequency means nobody will receive your packets.

- Frequency: 144.390 MHz; Region: North America (USA, Canada, Mexico), Colombia, Chile, Indonesia, Thailand
- Frequency: 144.800 MHz; Region: Europe, Africa, Russia, Middle East
- Frequency: 144.575 MHz; Region: New Zealand
- Frequency: 144.640 MHz; Region: Japan, Eastern Asia
- Frequency: 145.175 MHz; Region: Australia
- Frequency: 145.825 MHz; Region: ISS (International Space Station) — worldwide, only when ISS is in range (~10 min passes)
- Frequency: 432.500 MHz; Region: Europe (UHF)

The default frequency in the app is 144.390 MHz. Change it using the on-screen keypad at the bottom of the screen.

ISS note: Packets sent to 145.825 MHz during an ISS pass are relayed by the onboard APRS digipeater and can be received worldwide. Use ARISS as the path instead of WIDE1-1. Check current ISS pass times for your location before attempting.

⚠️ Transmitting to the ISS requires the same valid amateur radio license as any other APRS operation. Using the ISS digipeater with a fictional callsign is a violation of both national regulations and ITU Radio

Regulations Article 25. The ARISS organization monitors activity on 145.825 MHz. See the ARISS website for schedules and guidelines.

161.8 Gain settings

At the bottom of the screen:

- **Frequency step:** select step size with cursor, adjust with rotary encoder (e.g. 12k5)
- **Deviation:** select with cursor, adjust with rotary encoder (e.g. 5 kHz)
- **Gain IF:** range 0–47. Color indicates combined output power (IF + AMP): 0–17 green, 18–38 yellow, 39–47 orange, 48–61 red
- **Amp:** 0 dB or 14 dB (on/off)

Transmission starts with the "Start" button and stops automatically after the packet is sent.

161.9 Complete working examples

Some example message formats are as follows. Many more are in the APRS Protocol Reference Manual.

Simplest packet — status message:

- Source: W1ABC, SSID: 0
- Dest.: APRS, SSID: 0
- Path: WIDE1-1
- Info field: >Hello from PortaPack
- Frequency: 144.390 MHz (North America)

Position without timestamp (no messaging):

- Info field: !4071.27N/07400.59W-Hello from NYC
- The - after the longitude is the Symbol Code for a house

Position without timestamp (with messaging enabled):

- Info field: =4071.27N/07400.59W-Hello from NYC

Position with UTC timestamp (no messaging):

- Info field: /181654z4071.27N/07400.59W-Hello from NYC
- Format: /DDHHMMzDDmm.hhN/DDDmm.hhW[symbol][comment]

Position with UTC timestamp (with messaging):

- Info field: @181654z4071.27N/07400.59W-Hello from NYC

Position with GPS placeholder:

- Info field: !?GPS?-Hello from PortaPack
- Requires GPS module active or Manual GPS set

Position with altitude:

- Info field: !4071.27N/07400.59W-Hello from NYC /A=105
- Altitude /A= is always in feet per APRS Protocol Reference

Putting it all together — position, UTC timestamp, altitude and comment:

- Info field: @181654z4071.27N/07400.59W-Hello from NYC /A=105
- April 18th at 16:54 UTC, New York City area, 105 feet altitude

161.10 Testing your transmission

A reliable way to verify your packets are correctly decoded is to use a second SDR receiver (e.g. RTL-SDR) with SDR++ and the QTMM AFSK1200 Decoder software, both free. You can also use an online WebSDR or a local OpenWebRX instance. Note that aprs.fi and similar internet-connected receivers will only show your packet if an iGate is within RF range of your transmission.

161.11 Legal notice — amateur radio license required

APRS operates on amateur radio frequencies. Transmitting on these frequencies requires a valid amateur radio license in virtually every country.

- Country or Region: USA; Governing body: FCC; Regulation: Part 97; License required: Technician class or higher
- Country or Region: Europe (ITU Region 1); Governing body: National authority (e.g. BNetzA, Ofcom, ANFR); Regulation: ITU Radio Regulations Art. 25; License required: National amateur license
- Country or Region: Australia; Governing body: ACMA; Regulation: Radiocommunications Act 1992; License required: Foundation class or higher
- Country or Region: Japan; Governing body: MIC; Regulation: Radio Law of Japan; License required: Operator license + equipment type approval
- Country or Region: Canada; Governing body: ISED; Regulation: RBR-4; License required: Amateur radio operator certificate
- Country or Region: Worldwide; Governing body: ITU; Regulation: ITU Radio Regulations; License required: National amateur license

The Source callsign must be your own licensed callsign. Using a fictional, unassigned, or third-party callsign is illegal in all jurisdictions and may result in fines, license revocation, or criminal prosecution depending on local law.

The HackRF One is a test and development device, not a certified amateur radio transceiver. In the USA (FCC Part 97), no equipment certification is required for amateur radio — the licensed operator bears full responsibility for the quality and legality of the transmission. In most European countries, the transmitting equipment must additionally meet national technical requirements (spurious emissions, bandwidth). Ensure your transmission complies with the regulations of your country before transmitting on any amateur frequency.

161.12 Protocol

APRS is transported over **AX.25** using a **Bell 202** modem at **1200 bit/s** (AFSK): a **1200 Hz** tone for *mark* (binary 1) and **2200 Hz** for *space* (binary 0).

161.13 SSID (Secondary Station ID)

The SSID is a numeric suffix added to a callsign to tell apart multiple stations under one call. APRS allows 16 SSIDs (0–15); the commonly recommended meanings:

- SSID: 0; Typical use: Primary station, usually fixed and message-capable
- SSID: 1–4; Typical use: Generic additional station (digi, mobile, wx, etc.)
- SSID: 5; Typical use: Other networks (D-STAR, smartphones, etc.)
- SSID: 6; Typical use: Special activity, satellite ops, etc.
- SSID: 7; Typical use: Handheld transceivers or human-portable
- SSID: 8; Typical use: Boats, sailboats, RVs, or second main mobile
- SSID: 9; Typical use: Mobiles or trackers
- SSID: 10; Typical use: Internet, iGates, Echolink, Winlink, AVRS, APRN, etc.
- SSID: 11; Typical use: Balloons, aircraft, spacecraft, etc.
- SSID: 12; Typical use: APRStt, DTMF, RFID devices, one-way trackers, etc.
- SSID: 13; Typical use: Weather stations
- SSID: 14; Typical use: Truckers or full-time drivers
- SSID: 15; Typical use: Generic additional station (digi, mobile, wx, etc.)

161.14 Further reading

- APRS Protocol Reference Manual
- FCC Part 97 — Amateur Radio Service
- ARRL guide to Part 97
- ARISS — Amateur Radio on the International Space Station
- ITU Radio Regulations
- aprs.fi — Live APRS map
- APRS Symbol Tables

161.15 BHT Xy/EP

BHT TX transmits remote streetlight management messages using two protocols — **Xylos** and **EPAR** — developed by BH-Technologies for public lighting control in France and some other European cities. The

two protocols are selected via tabs and differ in their encoding and address structure. Settings are saved to `tx_bht.ini`.

> **Note:** Transmission on the frequencies used by these systems requires authorisation. Use only in a closed RF environment or with explicit permission from the network operator.

This is an external app (`.ppma` file on the SD card).

161.16 Xylos Tab

Xylos encodes commands as a sequence of 20 CCIR tones (100 ms per tone, 400 ms silence). Each field in the address hierarchy narrows which receivers respond.

- **Header A or Header B**: Two 2-digit numeric codes (0–99 each). These form the protocol header identifying the network.
- **City code**: Numeric code identifying the municipality. Range: 0–99.
- **Family**: Receiver family number. Range: 0–9.
- **Subfamily**: Receiver sub-group. Range: 0–9. The **All** checkbox sends a wildcard that matches all subfamilies in the selected family.
- **Receiver ID**: Individual receiver address. Range: 0–99. The **All** checkbox sends a wildcard that addresses all receivers in the selected subfamily.
- **Relay 1–4**: Four independent relay states. Each relay can be set to **Ignore** (no change), **Off**, or **On**, shown as bulb icons.

161.17 EPAR Tab

EPAR (also called EP) uses OOK encoding at approximately 580 baud. Each transmission is repeated 26 times.

- **City code**: Municipality code. Range: 0–255.
- **Group**: Receiver group on the installation. Options: **A**, **B**, **C**, or **TP** (test point).
- **Relay 1–2**: Two relay states, each **Off** or **On**.

161.18 Common Controls

- **Scan**: When checked, the app steps through the address range automatically after each transmission, incrementing the receiver address until the full range is covered. A progress bar tracks scan progress.
- **Flashing**: After each transmission, the app waits the number of seconds set in the speed field, flips all relay states to the opposite, then retransmits. This creates a flashing/toggling effect on the controlled lights.
- **Speed**: Interval in seconds between flashing toggles. Range: 1–99 s. Default: 1 s.
- **Frequency**: Editable via the keypad. The Xylos protocol operates in the 31–32 MHz band; EPAR deployments vary by installation.
- **LNA gain** (0–47 dB) and **RF Amp** (0 or 14 dB): Standard TX gain controls.
- **TX or Stop**: Starts or stops the transmission.

161.19 BLE TX

BLE TX transmits Bluetooth Low Energy advertisement packets loaded from a .TXT file. It supports two modes: single transmit (sends each packet once) and loop mode (repeats continuously for the number of repeats specified in the file). A file must be loaded before transmitting — unless arriving directly from the BLE RX app, which transfers the captured packet automatically.

161.20 Settings

Open file: Select a .TXT packet file from the SD card. Files are stored in the BLETX/ directory by default.

?? Mac: When checked, a randomised MAC address is used instead of the one defined in the file.

Loop: When checked, the app transmits the full packet file repeatedly until stopped. The progress bar and Packets Left indicator show how many packets remain in the current pass.

Speed: Controls the transmit interval between packets, based on display frame timer ticks (≈16 ms each):

- 1 — 2 ticks (~32 ms per packet)
- 2 — 4 ticks (~64 ms per packet)
- 3 — 6 ticks (~96 ms per packet)

- 4 — 8 ticks (~128 ms per packet)
- 5 — 12 ticks (~192 ms per packet)

Channel: BLE channel selection:

- **Ch.37, Ch.38, Ch.39** — individual BLE advertising channels
- **Ch.00–Ch.36** — individual BLE data channels
- **Auto** — cycles through the three advertising channels (Ch.37, Ch.38, Ch.39)
- **All** — cycles through all 40 BLE channels (Ch.00–Ch.39)

Advertisement PDU Type: The BLE packet type to transmit — set in the UI, not read from the file: DISCOVERY, ADV_IND, ADV_DIRECT, ADV_NONCONN, ADV_SCAN_IND, SCAN_REQ, SCAN_RSP, CONNECT_REQ. To make a packet visible on a BLE scanner app, use ADV_NONCONN.

Marked Data sequence: When bytes are marked in the data editor, controls how they change between transmissions — Ascend, Descend, or Random.

Long-press on packet data: Hold a value in the data packet display to edit it in real time using the encoder knob and buttons.

Clear Marked: Removes all marked bytes from the data editor.

Save Packet: Saves the current packet list to a new file in BLETX/ named BLETX_XXXX.TXT.

Switch to Rx: Switches directly to the BLE RX app.

161.21 File Format

Each line in the .TXT file defines one packet with three space-separated fields. All fields must be separated by a single space.

- **MAC:** Exactly 12 hex characters, no separators (e.g. 010203040506)
- **AdvertisementData:** Up to 62 hex characters of raw advertisement payload (max 31 bytes per BLE spec)
- **RepeatCount:** Number of times to transmit this packet (integer)

Up to 32 packets per file are supported. Each packet goes on its own line.

Example — two packets, 500 repeats each:

```
010203040506 190953445220426c7565746f6f7468204c6f7720456e65726779 500
010203040507 190953445220426c7565746f6f7468204c6f7720456e65726779 500
```

The advertisement data in this example decodes to the ASCII string SDR Bluetooth Low Energy.

Files captured by the BLE RX app are already in this format and can be loaded directly.

161.22 Settings Persistence

Settings are saved to tx_ble.ini.

161.23 BLESpam

> **WARNING** The method this app uses is fixed in Android and iOs too. Only works with old os versions. Since PP can't handle two way communication over BLE within a little time frame, it is not possible to fix it. You can use an ESP based solution for recent OS versions.

> **WARNING** Using this app to disrupt or crash other people's devices without their consent may be illegal in your jurisdiction. The **iOs crash** mode explicitly crashes nearby unpatched iOS devices. Use responsibly and only on devices you own or have permission to test.

BLESpam continuously transmits spoofed Bluetooth Low Energy advertisement packets to trigger pairing popups, notifications, or instability on nearby devices. The MAC address and BLE channel (37, 38, or 39) are randomised automatically with each new packet. Each packet is transmitted 4 times before the app moves to the next one.

The protocol implementations are ported from the Flipper Zero Xtreme-Apps BLE spam module.

This is an external app (`.ppma` file on the SD card).

162 Attack Mode

Select the target platform or behaviour from the dropdown:

- **Android** — Sends Google FastPair advertisement packets. Triggers a pairing popup on nearby Android phones. Android applies a per-device cooldown of approximately 15 minutes before showing the popup again.
- **iOs** — Sends Apple Continuity packets (Proximity Pair or Nearby Action type, chosen randomly). Triggers device suggestion notifications on nearby iOS devices. Effect depends on iOS version.
- **iOs crash** — Sends a specially crafted Apple Continuity packet. Causes unpatched iOS devices to crash after approximately one minute. A power cycle recovers the device.
- **Windows** — Sends Swift Pair packets. Triggers a device popup on nearby Windows machines. Behaviour depends on patch level and supported hardware.
- **Samsung** — Sends Samsung EasySetup packets (Galaxy Buds or Galaxy Watch models, chosen randomly). Triggers a pairing popup on nearby Samsung devices.
- **NameSpam** — Broadcasts one of 9 fixed device names as a generic HID device advertisement: `PortaHack`, `PwnBt`, `iSpam`, `GenericFoodVagon`, `SignalSnoop`, `ByteBandit`, `RadioRogue`, `RadioRebel`, `ByteBlast`. The name is chosen randomly each cycle.
- **NameRandom** — Broadcasts a randomly generated 9-character alphanumeric string as the device name. A new name is generated each cycle.
- **NameCustom** — Broadcasts names read line by line from `/BLESPAM/NAME.txt` on the SD card (one name per line, max 19 characters each). The file is cycled continuously. Falls back to **NameSpam** behaviour if the file is missing or unreadable.
- **All-Safe** — Cycles randomly through all modes except **iOs crash**.
- **All** — Cycles randomly through all modes including **iOs crash**.

163 Controls

- **Start or Stop**: Begins or halts transmission.
- **Rnd device** (checkbox, default: on): When enabled, the device model or payload is randomised with each packet change. When disabled, the same payload is retransmitted repeatedly (MAC and channel are still randomised).

164 Display

The console area shows the raw advertisement data hex of the current packet, updated approximately every 25 packet changes.

165 Settings Persistence

Settings are saved to `tx_blespam.ini`.

165.1 Burger Pager

> **NOTE** This is an external app and must be present on the SD card in the `APPS/` folder.

Burger Pager transmits FSK paging signals compatible with restaurant coaster pager systems. It allows sending a page to a specific pager unit by address, or scanning through a range of addresses automatically.

The default frequency is **433.920 MHz**. Actual pager system frequencies vary by hardware — adjust the frequency field to match the target system before transmitting.

166 Settings

- **Data** (16 hex digits): The full 8-byte payload to transmit. The lower 16 bits (last 4 hex digits) represent the pager address. Default value: `44013B30303034BC`.
- **Scan** (checkbox): When enabled, the app increments the pager address by 1 after each successful transmission and immediately retransmits, stepping through addresses continuously. When disabled, the frame is transmitted once and stops.

167 Frame Format

Each transmission consists of a fixed 19-byte FSK frame:

- Bytes: 0–7; Content: Preamble: 55 55 55 55 55 55 55 55
- Bytes: 8–9; Content: Sync word: 2D D4
- Bytes: 10; Content: Data length: 08
- Bytes: 11–18; Content: Data payload (8 bytes, from the Data field)

Modulation is FSK with a deviation of 5 kHz. Each frame is transmitted 32 times per press.

168 Settings Persistence

Settings are saved to `tx_burger.ini`.

168.1 CVS Spam

WARNING Transmitting signals that trigger retail store assistance systems without authorisation may be illegal in your jurisdiction. Use only for research on systems you own or have explicit permission to test.

CVS Spam replays pre-recorded IQ capture files to trigger wireless customer assistance call buttons, as found in retail environments (CVS Pharmacy and similar). It operates as a raw IQ replay transmitter — it does not generate signals in software, but plays back `.c16` or `.cu8` capture files from the SD card.

The transmit frequency is fixed at **433.920 MHz** and cannot be changed from the UI.

App credit: RocketGod (@rocketgod-git). Original capture files: @jimilinuxguy.

This is an external app (`.ppma` file on the SD card).

169 SD Card Setup

Place `.c16` or `.cu8` IQ capture files in the `CVSFILES/` directory on the SD card. The app lists all files found there. Up to 50 files are shown per page.

If the directory is empty or no supported files are found, the app displays `Empty directory!`.

170 Controls

- **File list**: Scroll through available capture files. The currently highlighted file is the one that will be transmitted with **CALL**.
- **< / >**: Navigate between pages when more than 50 files are present.
- **CALL**: Transmits the currently selected file once, then stops. Selecting the same file while transmitting stops the transmission immediately.
- **CHAOS**: Starts continuous random playback — after each file finishes, a new file is chosen using an LFSR pseudo-random sequence and transmitted after a 100 ms pause. Pressing **CHAOS** again while active stops playback.
- **STOP**: Stops any active transmission immediately.

171 Transmission Parameters

These values are set by the app and cannot be adjusted from the UI:

- Parameter: Frequency; Value: 433.920 MHz
- Parameter: Sample rate; Value: 250,000 Hz
- Parameter: RF Amp; Value: Enabled
- Parameter: TX Gain; Value: 47
- Parameter: Bandwidth; Value: 1,750,000 Hz

172 Progress Bar

The red progress bar updates continuously during playback to show approximate progress through the current file.

172.1 EPIRB

173 🎯 Overview

EPIRB TX application simulates an **ELT/EPIRB/PLB** COSPAS/SARSAT emergency beacon.

Demo vidéo

This is an external app (`.ppma` file on the SD card).

173.1 📡 Beacons

COSPAS/SARSAT emergency beacons are used in boats (**EPIRB**), planes (**ELT**) or by individuals (**PLB**) to facilitate Search And Rescue (**SAR**) missions by helping to locate accident sites. This application is intended for testing COSPAS SARSAT 406 decoding devices used by **HAM radio Civil Security SAR teams** around the world.

There are two generation of 406 MHz beacons:

- First Generation Beacons (FGB) use a **Manchester encoded BPSK signal**,
- Second Generation Beacons (SGB) use **DSSS-OQPSK modulation**.

173.2 ⚠️ Warning ⚠️

This application only transmits **"Self-Test"** beacon frames. It **MUST NOT** be used to send real (non test) beacon data, as it could initiate a **Search And Rescue alert**.

174 🎁 Features

The application features two transmission modes:

- A **406 MHz signal** (Manchester encoded BPSK for FGB or DSSS-OQPSK modulation for SGB) containing beacon data,
- A **127.5 MHz AM distress audio signal**.

Beacon data can be selected from two different sources:

- A set of **test beacons** stored in the `BEACONS.TXT` file in `EPIRB` directory of the SD card,
- A **manual beacon editor** integrated in the application, allowing selection of beacon type and location.

Beacon frames can be sent periodically (using the `Resend` checkbox) every **50 seconds** (adjustable). If `AM signal` checkbox is enabled, the transmission will automatically switch to **121.5 MHz AM emergency**

signal at the end of the BPSK/OQPSK frame. If the `Send on change` checkbox is enabled, a new frame will be sent each time a beacon parameter is modified.

175 📶 Protocol

First Generation Beacon frame data is encoded using **Manchester BPSK modulation** at **400 bits per second**, with a phase of **±63°**, in accordance with COSPAS/SARSAT specifications. Second Generation Beacon frame data is encoded using *OQPSK** at **300 bits per second** with DSSS spreading.

176 📻 Baseband

A dedicated proc_epirb_tx **baseband processor** is used for:

- BPSK/OQPSK data transmission,
- AM emergency signal generation.

The **121.5 MHz AM emergency signal** is an audio signal using a **3Hz frequency sweep from 300 Hz to 1600 Hz** reproducing the typical ELT distress sound.

177 ❓ User Guide

177.0.1 📝 Beacon Editor Mode

177.0.2 🗺️ Map View

177.0.3 📄 Beacon File Mode

177.0.4 📡 Transmission block

177.0.5 📻 Transmitter View

177.1 Flex TX

178 Flex TX

178.1 Overview

This is an external app (`.ppma` file on the SD card).

178.2 Introduction

This application transmits pager messages using the FLEX protocol. FLEX is a one-way paging protocol developed by Motorola that supports higher throughput than POCSAG through multi-level modulation and multiple phases.

The transmitter generates a complete FLEX frame including sync patterns, FIW (Frame Information Word), BIW (Block Information Words) with optional network metadata, address encoding, and message payload. It supports short and long addresses, multiple message types, and configurable network parameters.

It is intended for testing, experimentation, and educational use with licensed equipment or on frequencies for which you hold appropriate authorization.

178.3 Controls

178.3.1 Capcode

The destination pager address (capcode). Enter using the on-screen digit field (0-4297068542).

An info line below the capcode field shows the address type and computed frame/phase:

- Range: 1 - 1,933,312; Type: Short address
- Range: 2,058,240 - 2,062,335; Type: Network address
- Range: 2,041,856 - 2,058,239; Type: Info service
- Range: 2,062,336 - 2,062,351; Type: Temporary group #0-15
- Range: 2,062,352 - 2,062,367; Type: Operator message
- Range: 2,101,249 - 4,297,068,542; Type: Long address (2-word encoding)

The address type, frame number, and phase (A/B/C/D) are decoded from the capcode and shown for information only. In a real network, the infrastructure drives phase assignment and collapse values - these computed values may not reflect what a live network would use.

Only short and long addresses are valid for normal message transmission. Other address types (network, info service, temporary group, operator) may require different vector types or special payload formats. The app does not prevent sending to these addresses, but pagers will ignore messages with an invalid payload format for the address type.

178.3.2 Speed

Currently supports **1600/2FSK** only (1600 baud, 2-level FSK, phase A). This is the baseline FLEX mode supported by all FLEX pagers.

- Mode: 1600/2FSK; Baud Rate: 1600 baud; Modulation: 2-FSK; Phases: A; Bit Rate: 1600 bps; TX Support: Yes
- Mode: 3200/2FSK; Baud Rate: 3200 baud; Modulation: 2-FSK; Phases: A, C; Bit Rate: 3200 bps; TX Support: No
- Mode: 3200/4FSK; Baud Rate: 1600 baud; Modulation: 4-FSK; Phases: A, C; Bit Rate: 3200 bps; TX Support: No
- Mode: 6400/4FSK; Baud Rate: 3200 baud; Modulation: 4-FSK; Phases: A, B, C, D; Bit Rate: 6400 bps; TX Support: No

178.3.3 Type

Message type to transmit:

- Type: Alphanumeric; Description: Free-text ASCII message (up to 240 characters)
- Type: Numeric; Description: BCD-encoded digits using a numeric vector word, with message words for the payload. Valid BCD characters: 0-9 U -] [space and . (reserved, rendered as period for clarity)
- Type: Short/tone; Description: Short numeric vector with an empty payload - the pager typically decodes this as a tone-only alert. This is not a true “address-only” page (which would be an address word with no vector at all); it is a short numeric vector where all digit positions are filled with spaces.
- Type: Short numeric; Description: Short numeric vector with BCD digits encoded directly in the vector word(s). Short addresses carry up to 3 digits; long addresses carry up to 8 digits (3 in the vector word + 5 in a second word). Every pager that supports numeric vectors also supports short numeric vectors.

> **Note:** True tone-only pages (address without any vector word) are not implemented here and are not supported by most alphanumeric pagers in practice.

178.3.4 Message

The text payload. Press **Set message** to enter text using the on-screen keyboard. Maximum length is 240 characters.

For **Numeric** and **Short numeric** types, only valid BCD characters are accepted: `0-9 U - ] space` and `.` (reserved BCD symbol, rendered as period). Invalid characters will trigger an error.

178.3.5 Set Params

Opens the FLEX Parameters configuration screen (see below).

178.4 FLEX Parameters

The **Set params** button opens a configuration screen for the BIW (Block Information Word) fields included in the transmitted frame. These fields carry network metadata that real FLEX transmitters broadcast alongside messages.

178.4.1 Date

When enabled, includes a BIW date word in the frame. Fields: Year (1994-2025), Month, Day.

> **Note:** The FLEX protocol encodes years as a 5-bit offset from 1994, giving a valid range of 1994-2025. For years beyond 2025, the app can automatically find a calendar-equivalent year (same leap year status and day-of-week alignment) within the valid range. Some newer pagers treat year 0 as 2026 (an epoch rollover similar to how GPS or Unix time handles wraparound), but this behavior is pager-dependent.

Date and time fields are pre-populated from the PortaPack's RTC clock when the app opens, but you can change them or disable them with the checkboxes before transmitting.

178.4.2 Time

When enabled, includes a BIW time word. Fields: Hour, Minute, Second. The second value is quantized to 7.5-second steps per the FLEX specification.

The app uses standard mode: BIW TIME encodes the time at Frame 0 of the current cycle (`minute = cycle*4, second = 0`, hour from the RTC). The pager adds `frame*1.875s` internally to recover the real current time.

178.4.3 TZ or DST or ExtSec

When enabled, includes a timezone BIW sysinfo word. Select from a list of UTC offsets. The DST checkbox indicates whether daylight saving time is active (pre-populated from the PortaPack's DST setting).

ExtSec (0-7, default 0): extended seconds trim in 0.9375s steps. Used by real infrastructure to compensate for clock/frame-grid misalignment. Leave at 0 for normal use.

178.4.4 LocID or CovZone

When enabled, includes a SSID1 BIW word with:

- **Local ID** (0-511): Identifies the local transmitter site
- **Coverage Zone** (0-31): Geographic coverage area identifier

178.4.5 Country Code

When enabled, includes a SSID2 BIW word with the country code (0-1023).

178.4.6 Roam

When enabled, sets the roaming flag in the FIW, which tells pagers that this network supports roaming and they may join even if it is not their home network. Enabling roaming automatically enables both LocID and CountryCode fields.

> **Note:** LocID, CovZone, CountryCode, and Roam should all be left disabled unless the target pager's home country and network are known. This transmitter does not implement a proper carousel to broadcast all the BIW words required for correct multi-network operation. By default, pagers treat whatever frequency they are configured on as their home network, so these options are for experimentation only.

178.4.7 Msg#

Message sequence number (0-63). Incremented automatically after each successful transmission. Initialized from the RTC on app launch. Pagers can use this to detect duplicate messages as part of the FLEX extended network feature, but not all pagers implement it - some simply deduplicate based on message content.

178.4.8 ERS

When enabled, transmits ERS (Emergency Re-Sync) signal cycles before the message frame. In a FLEX network with collapse > 0, pagers periodically go to sleep to save battery and only wake up during their assigned frames. The ERS signal must be at least one collapse cycle in length so that any sleeping pager has a chance to wake up and see the re-sync alert before the message frame arrives.

- **Count** (1-10): Number of ERS burst groups to transmit.

178.5 Transmission

178.5.1 Transmit Controls (Bottom Bar)

- **Frequency**: Transmission frequency. Default is 931.74 MHz.
- **Gain** (0-47 dB): TX IF gain.
- **AMP** (0 or 1): RF amplifier (+14 dB when enabled).

Press **START** to begin transmitting. The progress bar shows transmission progress through the multi-step sequence:

1. **ERS bursts** (if enabled) – one step per ERS count
2. **Frame 1/2 (new)** – small ERS preamble + message frame with retrieve=1 (new message)
3. **Frame 2/2 (dup)** – small ERS preamble + message frame with retrieve=0 (retransmit, don't alert if already seen)

Each frame transmission is sent as a single package: a short ERS preamble (8 cycles, 96 bytes) followed by the frame data (379 bytes), totaling about 475 bytes.

The shared memory buffer used to pass data to the baseband processor is 512 bytes, so there is only room for one frame at a time. A real FLEX network transmits a continuous time-synchronized signal - 15 cycles per hour, 128 frames per cycle (each 1.875 seconds), with frame 0 of cycle 0 aligned to the top of each hour. The PortaPack cannot maintain such a signal because the baseband buffer is too small to hold more than one frame, and there is no GPS time reference or real-time frame scheduler. Instead, it transmits individual frames as one-shot bursts, with the carrier dropping between steps. The FIW carries the real FLEX cycle and frame number derived from the RTC at the moment of transmission, so a standard pager reconstructs the correct current time as `BIW TIME + frame*1.875s`. Because of this, it sends two copies of the message frame with small ERS preamble: the first message with the retrieve bit set (new message, pager alerts) and the second without it (retransmit, pager stores but does not alert again if it already received the first). This improves the chance of successful delivery. Transmission stops automatically when complete.

178.6 USB Serial

The Flex TX app can be triggered remotely via USB serial using the `FlexTosend` message type. When a serial message is received, the capcode, type, and message fields are updated in the UI and transmission starts automatically.

178.7 Settings Persistence

Capcode and BIW parameter selections are saved automatically and restored on next launch. Settings are stored in `SETTINGS/tx_flex.ini` on the SD card.

178.8 Legal Notice

FLEX operates on licensed frequencies. Transmitting on commercial paging frequencies without authorization is illegal in all jurisdictions. The HackRF One is a test and development device. The licensed operator is responsible for ensuring compliance with local regulations.

178.9 Further Reading

- [Flex RX - receive and decode FLEX pager messages
- FLEX protocol - Wikipedia
- POCSAG TX - transmit using the POCSAG protocol

178.10 FlipperTX

FlipperTX transmits signals from Flipper Zero `.sub` capture files directly from the PortaPack. It reads the file's encoded protocol timing data and replays it over the air at the configured frequency using the modulation preset defined in the file itself.

This is an external app (`.ppma` file on the SD card).

179 SD Card Setup

Place `.sub` files in the `SUBGHZ/` directory at the root of the SD card. The Browse button opens a file selector starting in this directory, filtered to `.sub` files.

180 Supported File Formats

FlipperTX only supports a subset of the Flipper Zero `.sub` file format:

Protocols:

- Protocol: RAW; Support: ✅ Supported
- Protocol: BinRAW; Support: ✅ Supported
- Protocol: All others; Support: ❌ Not supported

Presets (modulation):

- Preset: OOK; Modulation: On-Off Keying; Support: ✅ Supported
- Preset: 2FSK; Modulation: 2-level FSK; Support: ✅ Supported
- Preset: Custom or unknown; Modulation: —; Support: ❌ Not supported

If the file uses an unsupported protocol or preset, the filename field shows an error message and transmission is blocked:

- `File: err, bad file` — file could not be parsed

- `File: err, not supp. proto` — protocol not RAW or BinRAW
- `File: err, not supp. preset` — preset not OOK or 2FSK

181 Controls

- **Frequency**: Transmit frequency. Defaults to **433.920 MHz**. Must be set manually to match the frequency in the `.sub` file — the app does not read frequency from file metadata automatically.
- **Gain or AMP**: TX gain and RF amplifier toggle.
- **Browse**: Opens the file selector in `SUBGHZ/`. After a valid file is selected, the full path is shown in the filename field.
- **Start or Stop**: Begins or aborts transmission. The file is transmitted once and stops automatically at end of file. There is no loop mode.

182 Transmission Parameters

These values are fixed and cannot be changed from the UI:

- Parameter: Sample rate; Value: 2,280,000 Hz
- Parameter: Bandwidth; Value: 1,750,000 Hz
- Parameter: FM deviation (2FSK only); Value: 60,000 Hz

183 Settings Persistence

Settings are saved to `tx_flippertx.ini`.

183.1 GPS Sim

> **WARNING** This application is intended solely for experimental and educational purposes. Broadcasting GPS signals may be illegal in your jurisdiction and can interfere with navigation systems. You are solely responsible for ensuring compliance with all applicable laws before use.

GPS Sim replays pre-generated GPS baseband signal files (`.C8` format) to simulate GPS satellite signals. It can be used to feed a fixed or moving position to GPS receivers for testing and research.

This is an external app (`.ppma` file on the SD card).

184 SD Card Setup

Place `.C8` data files and their corresponding `.TXT` metadata files in the `GPS/` directory at the root of the SD card. Both files must share the same base name (e.g. `gpssim.C8` and `gpssim.TXT`).

When a `.C8` file is opened, the app automatically reads the matching `.TXT` file and sets the transmit frequency and sample rate accordingly.

185 Metadata File Format

Create a `.TXT` file with the same name as the `.C8` file containing:

```
center_frequency=1575420000
sample_rate=SAMPLE_RATE
```

`center_frequency` is the GPS L1 carrier frequency and should always be `1575420000`. If the metadata file is missing, or if either value is zero, the app falls back to its defaults (1575.420 MHz, 2,600,000 Hz sample rate).

Known working sample rate values:

- `2600000`
- `2500000`
- `1250000`

186 Generating C8 Files

Use the `gps-sdr-sim` tool to generate `.C8` files. It requires a RINEX navigation file (GPS ephemeris data) as input.

Fixed position:

```
gps-sdr-sim -e BRDC_FILE.n -l LAT,LON,HEIGHT -b 8 -s 2600000 -o gpssim.C8
```

Moving position from NMEA file:

```
gps-sdr-sim -e BRDC_FILE.n -g NMEA_FILE.txt -b 8 -o gpssim.C8
```

RINEX navigation files (`.n` / `.20n` etc.) can be downloaded from public GNSS data archives such as NASA CDDIS or IGS. Use the file dated closest to your intended simulation date.

Example metadata file for the above:

```
center_frequency=1575420000
sample_rate=2600000
```

187 Controls

- **Open file**: Opens a file browser in `GPS/`, filtered to `.C8` files. After selection, the filename, sample rate, and estimated duration are shown.
- **Frequency or Gain or AMP**: Transmit frequency, gain, and RF amplifier. Automatically set from the `.TXT` file when a capture is loaded.
- **Loop** (checkbox, default: on): When enabled, the file restarts automatically at end of file and transmits continuously.
- ▸ / ■: Start or stop transmission.

The progress bar shows playback position through the file. A live waterfall is displayed during transmission.

188 Default Transmission Parameters

- Parameter: Frequency; Default value: 1,575.420 MHz (GPS L1)
- Parameter: Bandwidth; Default value: 15,000,000 Hz
- Parameter: Sample rate; Default value: 2,600,000 Hz (overridden by .TXT file)

189 Improving Reception on Target Devices

Modern devices use multiple positioning sources and may reject spoofed GPS. To improve the chance of a lock:

- Enable airplane mode (disables Wi-Fi, Bluetooth, and cellular assisted positioning)
- Test indoors where real GPS signal is weak
- Reboot the device after enabling airplane mode

190 Known Device Results

> **NOTE** Newer devices increasingly use sensor fusion and network-based positioning. Results may vary by firmware version.

- Type: Tablet; Make: Samsung; Model: Note 7; Locks on spoofed GPS?: No; GPS acts “jammed”?: Yes
- Type: Phone; Make: Samsung; Model: S8; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No; Notes: Wi-Fi must be off
- Type: Phone; Make: Huawei; Model: P30 lite; Locks on spoofed GPS?: No; GPS acts “jammed”?: Yes
- Type: Phone; Make: Huawei; Model: Mate 10 Pro; Locks on spoofed GPS?: No; GPS acts “jammed”?: No; Notes: 6–7 satellites: no fix

- Type: Smartwatch; Make: Huawei; Model: Amazfit Bip; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No
- Type: Phone; Make: Apple; Model: iPhone 6; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No; Notes: Works with Wi-Fi on
- Type: Phone; Make: ZTE; Model: Axon 7; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No; Notes: Works with Wi-Fi on
- Type: Phone; Make: Xiaomi; Model: Mi5; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No; Notes: Works with Wi-Fi on
- Type: Phone; Make: Samsung; Model: S20+ 4G; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No; Notes: Wi-Fi must be off
- Type: Smartwatch; Make: Samsung; Model: Gear S3 WiFi; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No; Notes: Location over GPS only
- Type: Phone; Make: OnePlus; Model: 7 Pro; Locks on spoofed GPS?: No; GPS acts “jammed”?: No; Notes: No effect observed
- Type: Phone; Make: OnePlus; Model: 5; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No; Notes: 3D fix
- Type: Phone; Make: Samsung; Model: S21 Ultra; Locks on spoofed GPS?: Yes; GPS acts “jammed”?: No

191 Settings Persistence

Settings are saved to `tx_gps.ini`.

191.1 Hopper

> **WARNING** This application transmits RF signals that may cause interference. Use only for authorised testing on equipment you own or have explicit permission to test. Jamming is illegal in most jurisdictions.

Hopper transmits on a user-defined list of frequencies, cycling through all of them using the jammer baseband. All listed channels are active simultaneously within the baseband; the “hop” interval controls how long each transmission burst lasts before moving to the next channel. It is designed for testing multi-channel interference resilience on systems you own.

This is an external app (`.ppma` file on the SD card).

192 SD Card Setup

Frequency lists are stored as `.PHOP` files in the `HOPPER/` directory at the root of the SD card. Each line in the file contains one frequency in Hz. Lines starting with `#` are treated as comments and skipped.

Example file (`HOPPER/mylist.PHOP`):

```
# My test frequencies
433920000
315000000
868000000
```

Up to **80 frequencies** can be loaded. Additional entries beyond the limit are ignored.

193 Controls

- 📁 **(Load)**: Opens a file browser in `HOPPER/`, filtered to `.PHOP` files. Replaces the current list.
- 💾 **(Save)**: Prompts for a filename and saves the current frequency list as a `.PHOP` file in `HOPPER/`.
- ➕ **(Add)**: Opens the frequency keypad to add a single frequency to the current list.
- 🗑️ **(Delete)**: Removes the currently highlighted frequency from the list.
- 🧹 **(Clear)**: Clears the entire frequency list after confirmation.
- **START or STOP**: Begins or ends transmission.

194 Settings

- **Type**: Signal type generated on each channel. Options (with defaults in bold):
  - `Rand FSK` — Random FSK noise
  - `FM tone` — FM-modulated tone
  - `CW sweep` — Continuous wave sweep
  - **`Rand CW`** — Random amplitude CW (default)
  - `Sine` — Sine wave
  - `Square` — Square wave
  - `Sawtooth` — Sawtooth wave
  - `Triangle` — Triangle wave
  - `Chirp` — Linear frequency chirp
  - `Gauss` — Gaussian noise
  - `Brute` — Broadband noise
- **Speed**: Modulation speed or symbol rate within each channel:
  - `10 Hz`, `100 Hz`, `1 kHz`, **`10 kHz`** (default), `100 kHz`
- **Hop**: Dwell time per channel before switching:
  - **`0ms !!`** — Fastest possible; **causes the HackRF UI to freeze**. Press the hardware RESET button to recover.
  - `1ms`, `10ms`, **`50ms`** (default), `100ms`, `1s`, `2s`, `5s`, `10s`
- **TX** (1–180 seconds): How long the app transmits before entering the sleep/pause period.
- **Sle3p** (1–60 seconds): Duration of the pause (cooldown) between TX bursts. During the pause the transmitter is disabled. This TX→pause cycle repeats continuously while running.
- **Jitter** (1–60, unit: /60 seconds): A random offset applied to the TX and pause timers using an LFSR sequence, preventing a perfectly regular duty cycle. Higher values mean more timing variation.
- **Gain** (0–47): TX gain.
- **A** (0/1): RF amplifier on/off.

195 Frequency Counter

The display shows the current channel number and total number of active channels (e.g. `03/05`) as the baseband cycles through them.

> **NOTE** Hopper has no settings persistence — frequency lists must be reloaded from SD card after each app launch.

195.1 Jammer

> **WARNING** Intentional radio jamming is illegal in most countries, including all EU member states and the United States. Use only for authorised testing on equipment you own in a shielded environment. You are solely responsible for compliance with all applicable laws.

Jammer TX transmits interference signals across up to three independently configured frequency ranges simultaneously. Each range is split into 1 MHz segments (hops), and the jammer cycles through them sequentially. All three ranges are active at the same time when enabled.

This is an external app (`.ppma` file on the SD card).

196 Range Tabs (Range 1 or Range 2 or Range 3)

Each tab configures one independent frequency range. The three ranges operate together when transmitting.

- **Enable range** (checkbox): Activates this range. At least one range must be enabled to start.
- **Load range**: Opens the Freqman frequency manager to load a saved range from the `FREQMAN/` directory on the SD card.
- **Start or Stop buttons**: Set the lower and upper bound of the range directly. Can also be set by loading a range.
- **Center or Width buttons**: Alternative way to define the range by center frequency and bandwidth.

The combined bandwidth across all enabled ranges must not exceed **80 MHz** in total (80 × 1 MHz segments). Exceeding this limit shows an error and prevents transmission from starting.

The hop counter (e.g. `03/12`) next to the Type field shows which 1 MHz segment is currently being transmitted and the total number of segments across all active ranges.

197 Settings

- **Type**: Signal type generated in each hop segment:
- Option: Rand FSK; Description: Random FSK noise
- Option: FM tone; Description: FM-modulated tone
- Option: CW sweep; Description: Continuous wave sweep
- Option: Noise; Description: Random amplitude noise
- Option: Sine; Description: Sine wave
- Option: Square; Description: Square wave
- Option: Sawtooth; Description: Sawtooth wave
- Option: Triangle; Description: Triangle wave
- Option: Chirp; Description: Linear frequency chirp
- Option: Gauss; Description: Gaussian noise
- Option: Brute; Description: Broadband noise
- **Speed**: Modulation rate within each hop segment: `10 Hz`, `100 Hz`, `1 kHz`, `10 kHz`, `100 kHz`
- **Hop**: Dwell time per 1 MHz segment before moving to the next: `Off`, `10ms`, `50ms`, `100ms`, `1s`, `2s`, `5s`, `10s` — `Off` disables segment hopping; the jammer stays on each segment indefinitely.
- **TX** (1–180 seconds): How long the jammer transmits before entering the sleep pause.
- **Sleep** (0–60 seconds): Pause duration between TX bursts. When **Sleep = 0**, the TX/pause cycle is disabled entirely and the jammer transmits continuously without stopping. When Sleep > 0, the jammer alternates between TX and pause, with the button label changing to `PAUSED` during the sleep period.
- **Jitter** (0–60, unit: /60 seconds): Adds a random LFSR-based offset to the TX and sleep timers to prevent a perfectly periodic duty cycle. `0` = no jitter.
- **Gain** (0–47): TX gain.
- **A** (0/1): RF amplifier on/off.

198 Transmission Parameters

- Parameter: Baseband bandwidth; Value: 28,000,000 Hz (max TX LPF)
- Parameter: Sampling rate; Value: 3,072,000 Hz

199 Example Configuration (2.4 GHz Wi-Fi)

To target Wi-Fi channels 1, 6, and 11 (2.400–2.483 GHz):

- Range 1: Start `2400000000`, Stop `2483000000`
- Type: `Noise`
- Speed: `10 kHz`
- Hop: `10ms`
- TX: 30 Secs, Sleep: 0 (continuous)
- Jitter: 0

Note that the HackRF's low transmit power (~5–10 dBm) limits effective jamming range to a few metres without an external amplifier.

NOTE Jammer TX has no settings persistence — all settings reset when the app is closed.

199.1 KeeLoq TX

This is an external app (`.ppma` file on the SD card).

Legal warning: Transmitting KeeLoq signals to garage doors, gates, or vehicles you do not own or have no authorisation to access is illegal in most jurisdictions. This app is intended for testing your own equipment and for security research on systems you are authorised to evaluate.

KeeLoq TX retransmits KeeLoq rolling-code signals loaded from `.keeloq` capture files. These files are created by the SubGhzD receiver app when the Save button is used on a received KeeLoq or Superrollo packet. The app reconstructs and transmits the encrypted rolling-code payload based on the captured serial number, counter value, button code, and the KeeLoq learning variant used by the target device.

Supported transmission modes:

- **KeeLoq simple learning** — standard 64-bit OOK frame at Te=400 µs

- **KeeLoq normal learning** — standard 64-bit OOK frame at Te=400 µs
- **Superrollo (GW60/HCS361)** — 67-bit OOK frame at Te=450 µs, automatically selected when a file with `mf_name=Superrollo` is loaded

For all modes, the correct manufacturer key must be present in `KEELOQKEYS/MFCODES` to reconstruct the encrypted rolling-code payload. Without the key the transmission cannot be generated correctly.

200 Settings

- **Open file**: Opens the file manager to select a `.keeloq` file from the `KEELOQREMOTES/` directory on the SD card.
- **Serial**: Displays the 28-bit serial number read from the loaded file.
- **Counter**: The 16-bit synchronisation counter. Editable — incrementing this value moves the counter forward in the rolling-code sequence.
- **Button**: The button code (0–15) to encode in the transmission.
- **Repeat**: Number of transmission repetitions. Default: 4.
- **Payload**: Live preview of the resulting encrypted frame in hexadecimal.
- **Transmit or Stop**: Starts or stops the transmission.

201 Manufacturer Key File (MFCODES)

KeeLoq encryption requires a 64-bit manufacturer key specific to the device brand. Keys are loaded from `KEELOQKEYS/MFCODES` on the SD card. The file is not included in the firmware — it must be created manually.

File format — one entry per line, semicolon-delimited:

```
mf_name;key_hex;type
```

- Field: mf_name; Description: Manufacturer name as recognised by SubGhzD (e.g. Superrollo)
- Field: key_hex; Description: 64-bit manufacturer key in hexadecimal (16 characters)
- Field: type; Description: Learning mode: 1 = simple, 2 = normal

Example: `Superrollo;XXXXXXXXXXXXXXXX;1`

Replace `XXXXXXXXXXXXXXXX` with the actual 64-bit key for your device. The key is not publicly distributed and must be obtained independently (e.g. via hardware analysis of the transmitter IC).

202 Settings Persistence

Settings are saved to `tx_keeloq.ini`.

202.1 Key fob TX

> **Legal warning:** Transmitting remote keyless entry signals to vehicles you do not own is illegal in most jurisdictions and may constitute attempted vehicle theft or unlawful interference with a radio device. Use only on your own vehicle for testing and research purposes.

Key fob TX transmits fixed-code remote keyless entry (RKE) signals for compatible vehicle remotes. Currently only **Subaru** is supported. The transmitted frame is a fixed OOK-encoded signal — it does not use rolling codes (KeeLoq). It is transmitted 4 times in sequence at 433.920 MHz.

This is an external app (`.ppma` file on the SD card).

203 Settings

- **Make**: Vehicle manufacturer. Currently: **Subaru** only.
- **Command**: The remote command to transmit:
  - **Lock**
  - **Unlock**

▸ **Trunk**

- **Payload A or Payload B**: The raw hexadecimal frame bytes, updated automatically when Make or Command is changed. Can also be edited manually for experimentation.
- **Transmit or Stop**: Starts or stops the 4-repeat transmission sequence.

204 Settings Persistence

Settings are saved to `tx_keyfob.ini`.

204.1 LGE Tool

LGE Tool transmits proprietary RF control frames used by certain laser tag systems operating in the 868 MHz ISM band. The system uses RFM69-based FSK packets with a fixed network ID and sync word. The tool was originally written by Furrtek for the portapack-havoc project.

> **WARNING** This tool is intended for research and testing on equipment you own or operate. Transmitting control commands on third-party systems without authorisation may be illegal in your jurisdiction.

This is an external app (`.ppma` file on the SD card).

205 Radio Parameters

The system uses three fixed channels in the 868 MHz band. The default frequency is **868.067 MHz**.

- Channel: 1; Frequency: 868.067 MHz
- Channel: 2; Frequency: 868.183 MHz
- Channel: 3; Frequency: 868.295 MHz
- Parameter: Modulation; Value: FSK
- Parameter: Baud rate; Value: 9600
- Parameter: Deviation; Value: 4,000 Hz
- Parameter: Bandwidth; Value: 2,500,000 Hz
- Parameter: Packet format; Value: RFM69, Network ID 5, Sync 0x2DD4

206 Controls

- **Frame**: Selects the type of command frame to transmit (see below).
- **All ch.** (checkbox, default: on): When enabled, the frame is transmitted sequentially on all three channels (868.067 → 868.183 → 868.295 MHz). When disabled, only the currently selected frequency is used.
- **Room** (1–2): Room/arena number.
- **Text**: Player or team nickname, up to 15 characters. Opens the alpha keyboard on press.
- **Team** (1–6): Team number.
- **Player** (1–50): Player number.

207 Frame Types

- Frame: Key; Description: Simulates a hit/shot on the target player; Fields used: Room, Player
- Frame: Set nickname; Description: Assigns a nickname to a player; Fields used: Room, Player, Team, Text
- Frame: Set team; Description: Sets the name for a team; Fields used: Team, Text
- Frame: Brdcst nick; Description: Broadcasts a player's nickname to all units; Fields used: Player, Team, Text
- Frame: Start; Description: Sends a game start command; Fields used: Room
- Frame: Game over; Description: Sends a game end command; Fields used: Room
- Frame: Set vest; Description: Configures a vest/sensor unit; Fields used: See below

207.1 Set vest (vest configuration)

The **Set vest** frame configures the behaviour of an individual vest unit:

- **ID** (1–2): Vest unit ID.
- **Pow** (1–10): Zap power level (duty cycle).
- **Time** (1–25, × 100 ms): Zap duration. A value of `2` = 200 ms.
- **Heartbeat** (checkbox, default: on): Enables heartbeat signal from vest.
- **RX tick** (checkbox, default: on): Enables RX tick signal from vest.

The console area below the controls shows the raw hex bytes of each generated frame.

208 Settings Persistence

Settings are saved to `tx_lge.ini`.

208.1 MDC-1200 TX

MDC-1200 TX transmits MDC-1200 signaling packets, used in some two-way radio systems for unit identification, emergency signaling, selective calling, and remote control operations. The protocol is AFSK at 1200 baud transmitted as FM subaudible data.

> **WARNING** Transmitting MDC-1200 commands on operational radio systems without authorisation is illegal and may interfere with radio communications. Use only on equipment you own or have explicit permission to test.

This is an external app (`.ppma` file on the SD card).

209 Transmission Parameters

- Parameter: Default frequency; Value: 462.562500 MHz
- Parameter: AFSK Mark; Value: 1800 Hz
- Parameter: AFSK Space; Value: 1200 Hz
- Parameter: Baud rate; Value: 1200
- Parameter: FM deviation; Value: 2500 Hz
- Parameter: Bandwidth; Value: 1,750,000 Hz
- Parameter: Default TX gain; Value: 30
- Parameter: Encoding; Value: NRZI + convolutional + bit interleaving
- Parameter: CRC; Value: CRC-16 with bit reversal, XOR 0xFFFF

210 Controls

- **Op**: Operation type to transmit. Selecting an operation automatically sets the default **Arg** value where applicable (see table below).
- **ID** (two fields, 0–255 each): The 16-bit target unit ID, entered as two separate decimal bytes (high byte and low byte). For example, ID 256 is entered as `001 000`.
- **Arg** (0–255): The argument byte for the operation. Automatically set when the Op is changed. Can be overridden manually.
- **Src** (two fields, 0–255 each): The 16-bit source unit ID, used only for double-packet operations (Call Alert, Sel Call 1, Sel Call 2). Entered the same way as the target ID.
- **Status bar**: Shows `Ready`, `TX...`, or `Done!` depending on transmission state.
- **Frequency or Gain or AMP**: Standard TX controls in the bottom bar.
- **Start or Stop**: Transmit the current frame. Single transmit only — no loop mode.

211 Operations

- Operation: PTT ID; Op code: 0x01; Default Arg: 0x00; Packet type: Single
- Operation: Emergency; Op code: 0x00; Default Arg: 0x00; Packet type: Single
- Operation: Radio Check; Op code: 0x63; Default Arg: 0x85; Packet type: Single
- Operation: Stun; Op code: 0x2B; Default Arg: 0x00; Packet type: Single
- Operation: Unstun; Op code: 0x2B; Default Arg: 0x0C; Packet type: Single
- Operation: Status Req; Op code: 0x22; Default Arg: 0x00; Packet type: Single
- Operation: Status Resp; Op code: 0x06; Default Arg: 0x00; Packet type: Single
- Operation: Rem Monitor; Op code: 0x11; Default Arg: 0x8A; Packet type: Single
- Operation: Message; Op code: 0x07; Default Arg: 0x00; Packet type: Single
- Operation: Sel Call 1; Op code: 0x80; Default Arg: —; Packet type: Double (uses Src)
- Operation: Call Alert; Op code: 0x81; Default Arg: —; Packet type: Double (uses Src)
- Operation: Sel Call 2; Op code: 0x82; Default Arg: —; Packet type: Double (uses Src)

Double-packet operations send two codewords. The first codeword contains the target ID; the second contains the source ID from the **Src** field.

212 Frame Structure

Each transmission consists of: 3 × 7-byte preamble (`0xAA`) + 5-byte sync word + one or two 14-byte codewords.

Each codeword carries: Op, Arg, ID high byte, ID low byte, CRC (2 bytes), padding (1 byte), plus 7 bytes of convolutional-encoded and bit-interleaved data.

213 Settings Persistence

Settings are saved to `tx_mdc_tx.ini`.

213.1 Morse TX

Morse TX encodes a text message into Morse code and transmits it in either CW (carrier on/off keying) or FM (tone-modulated) mode.

This is an external app (`.ppma` file on the SD card).

214 Controls

- **Foxhunt** (checkbox): When enabled, disables the custom message field and instead uses one of 11 predefined fox hunt identification codes, selected with the adjacent number field (1–11). The codes are standard ARDF (Amateur Radio Direction Finding) sequences:
- #: 1; Code: MOE; Pattern: – – – – – ·
- #: 2; Code: MOI; Pattern: – – – – – · ·
- #: 3; Code: MOS; Pattern: – – – – – · · ·
- #: 4; Code: MOH; Pattern: – – – – – · · · ·
- #: 5; Code: MO5; Pattern: – – – – – · · · · ·
- #: 6; Code: MON; Pattern: – – – – – – ·
- #: 7; Code: MOD; Pattern: – – – – – – · ·
- #: 8; Code: MOB; Pattern: – – – – – – · · ·
- #: 9; Code: MO6; Pattern: – – – – – – · · · ·
- #: 10; Code: MO; Pattern: – – – – –
- #: 11; Code: S; Pattern: · · ·
- **Speed** (10–45 wpm): Transmission speed in words per minute. The time unit per dot/dash is calculated as `1200 ÷ wpm` ms. Default: 15 wpm.
- **Tone** (100–9999 Hz, step 20): The audio tone frequency used in FM mode. In CW mode the tone determines the sidetone pitch only. Default: 700 Hz.
- **Modulation**: `CW` (carrier keying, default) or `FM` (tone-modulated carrier).
- **Loop**: Automatically retransmits the message after the selected interval: `Off`, `5 sec`, `10 sec`, `30 sec`, `1 min`, `3 min`, `5 min`
- **Set message**: Opens the text keyboard to enter a custom message. The message is shown on the line above. Default message: `PORTAPACK`.
- **Frequency or Gain or AMP**: Standard TX controls in the bottom bar. Frequency is stored in persistent memory.
- **TX will last**: Displays the calculated transmission duration in seconds based on the current message, speed, and tone settings, updated live.
- **Progress bar**: Shows transmission progress. When Loop is active, shows the countdown to the next retransmission.
- **Start or Stop**: Starts transmission. Press again to abort.

215 Known Issue

Due to a timing issue in the transmitter, the first 1–2 characters of a message may be clipped or not transmitted correctly. As a workaround, prefix the message with two short characters, for example `E` (letter E followed by a space), before the actual content.

216 Settings Persistence

Settings are saved to `tx_morse.ini`, including message text, foxhunt selection, speed, tone, modulation, and loop interval.

216.1 OOK

> **WARNING** This application is intended solely for experimental purposes on equipment you own or have permission to test. You are responsible for complying with all applicable laws.

OOK TX transmits On-Off Keying signals using pre-defined encoder chip protocols. It is commonly used for testing remote controls, gate openers, and similar 315 MHz or 433.92 MHz OAM devices. The app has two tabs: **Config** (manual frame construction) and **de Bruijn** (automated code scanning).

217 Config Tab

- **Type**: Encoder chip protocol. 14 types are supported:
- Name: 2260-R2; Chip: PT2260-R2; Address symbols: 0, 1, F; Data symbols: 0, 1; Word length: 12; Default clk: 150 kHz
- Name: 2260-R4; Chip: PT2260-R4; Address symbols: 0, 1, F; Data symbols: 0, 1; Word length: 12; Default clk: 150 kHz
- Name: 2262; Chip: PT2262; Address symbols: 0, 1, F; Data symbols: 0, 1, F; Word length: 12; Default clk: 20 kHz
- Name: 16-bit; Chip: —; Address symbols: 0, 1; Data symbols: 0, 1; Word length: 16; Default clk: 25 kHz
- Name: 1527; Chip: RT1527; Address symbols: 0, 1; Data symbols: 0, 1; Word length: 24; Default clk: 100 kHz
- Name: 526E; Chip: HK526E; Address symbols: 0, 1; Data symbols: 0, 1; Word length: 12; Default clk: 20 kHz
- Name: 12E; Chip: HT12E; Address symbols: 0, 1; Data symbols: 0, 1; Word length: 12; Default clk: 3 kHz
- Name: 5026; Chip: VD5026; Address symbols: 0, 1, 2, 3; Data symbols: 0, 1, 2, 3; Word length: 12; Default clk: 100 kHz
- Name: UM3750; Chip: UM3750; Address symbols: 0, 1; Data symbols: 0, 1; Word length: 12; Default clk: 100 kHz
- Name: UM3758; Chip: UM3758; Address symbols: 0, 1, F; Data symbols: 0, 1; Word length: 18; Default clk: 160 kHz
- Name: BA5104; Chip: BA5104; Address symbols: 0, 1; Data symbols: 0, 1; Word length: 9; Default clk: 455 kHz
- Name: 145026; Chip: MC145026; Address symbols: 0, 1, F; Data symbols: 0, 1; Word length: 9; Default clk: 455 kHz
- Name: HT6*; Chip: HT6xxx; Address symbols: 0, 1, F; Data symbols: 0, 1; Word length: 18; Default clk: 80 kHz
- Name: TC9148; Chip: TC9148; Address symbols: 0, 1; Data symbols: 0, 1; Word length: 12; Default clk: 455 kHz
- **Repeat** (1–99): Minimum number of times the frame is repeated per transmission. Automatically set to the encoder's default when the type changes.
- **Clk** (1–1000 kHz): Encoder oscillator clock frequency. Changing this updates **Frame** automatically. Step size is set by the adjacent **Step** selector (1 or 10 or 100 kHz).

- **Frame** (300–99,999 µs): Frame duration in microseconds. Changing this updates **Clk** automatically. Step size is set by its adjacent **Step** selector (1 or 10 or 100 or 1000 µs). Clock and frame are interlinked: `frame_µs = clk_period_µs × word_length`.
- **Symbols**: The bit pattern to transmit. Each position corresponds to a symbol in the word format shown on the line below (A = Address, D = Data, S = Sync). Symbols are selectable per encoder type — for example `0`, `1`, and `F` (float) for tri-state encoders.
- **Waveform**: Live preview of the generated OOK waveform for the current symbol pattern.
- **Progress bar or status**: Shows repeat count (`n/total`) and transmission progress.

218 de Bruijn Tab

Transmits a de Bruijn sequence — a cyclic bit sequence that contains every possible n-bit combination exactly once. This is used to scan all possible codes for a given bit length without needing to enumerate them individually.

- **Length** (3–24 bits): The code word length `n`. A length-`n` de Bruijn sequence covers all 2^n combinations.
- **Bit length** (two fields): The duration of each bit in microseconds, entered as tens + units (e.g. `40` + `0` = 400 µs per bit). Range: 10–889 µs.

219 Common Controls

- **Frequency or Gain or AMP**: Standard TX controls in the bottom bar. Default frequency: 433.920 MHz.
- **Start or Stop**: Begins or aborts transmission. When the de Bruijn tab is active, starts a scan; when the Config tab is active, transmits the configured frame.

220 Settings Persistence

Settings are saved to `tx_ook.ini`.

220.1 OOK Brute

> **Legal warning:** Brute-forcing remote control codes to operate devices you do not own or are not authorised to control is illegal in most jurisdictions. This includes garage doors, gates, barriers, and building access systems belonging to others. Use only on your own equipment for legitimate security testing.

OOK Brute sequentially transmits all OOK remote control codes within a configurable range for a selected protocol. It is intended for testing the code space of fixed-code OOK remote systems — for example, verifying that your own receiver responds to the correct code, or auditing the security of a system you own.

This is an external app (`.ppma` file on the SD card).

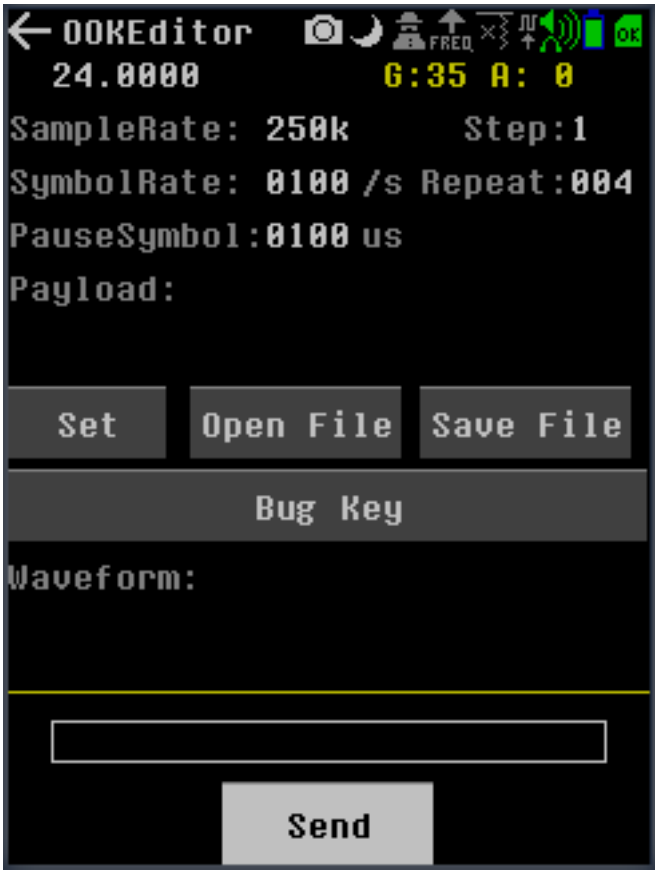
221 Settings

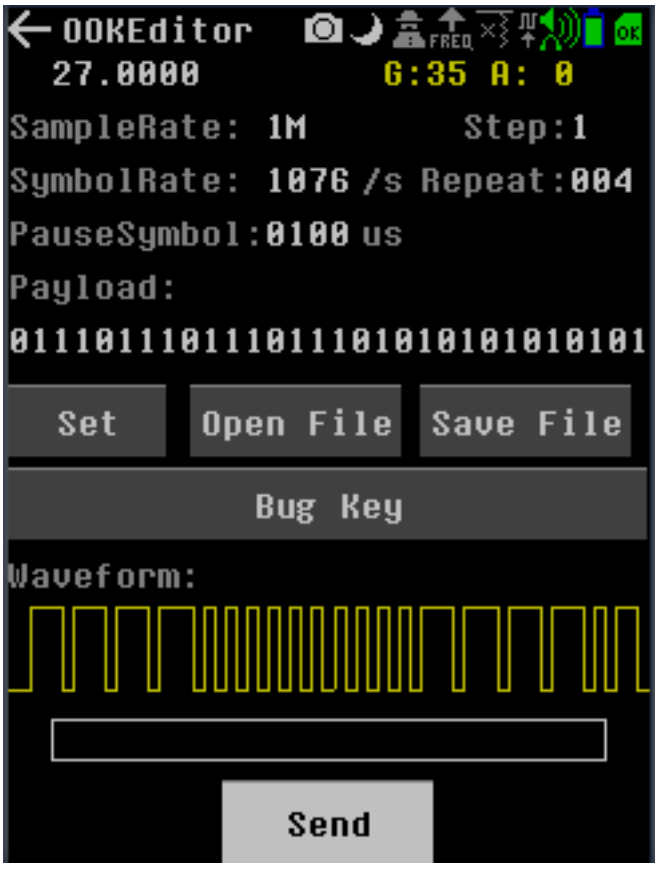
- **Protocol**: The OOK encoding standard to brute-force:
  - **Came12** — CAME 12-bit fixed code
  - **Came24** — CAME 24-bit fixed code
  - **Nice12** — Nice 12-bit fixed code
  - **Nice24** — Nice 24-bit fixed code
  - **Holtek12** — Holtek HT12E 12-bit fixed code
  - **Princeton24** — Princeton PT2262 24-bit fixed code
- **Start**: First code to transmit (decimal). Can be entered via the on-screen keypad button.
- **Stop**: Last code to transmit (decimal). The app iterates from Start to Stop inclusive.
- **Frequency**: Default 433.920 MHz. Editable.
- **Start or Stop button**: Begins or halts the brute-force sequence.

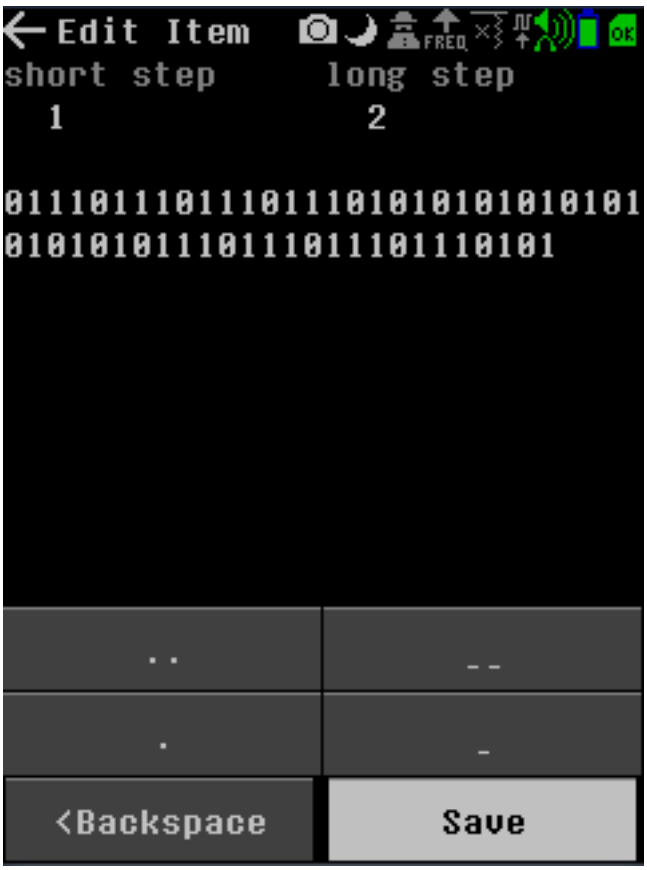
222 Settings Persistence

Settings are saved to `tx_ookbrute.ini`.

222.1 OOK Editor







OOK Editor is a free-form On-Off Keying transmitter. Unlike the OOK TX app, it does not use predefined encoder chip protocols — instead you define the payload bit-by-bit, either by typing it directly, using the Bug Key composer, or loading a `.OOK` file from the SD card.

This app is a lightly enhanced version of the ROOK application by @sasaga.

This is an external app (`.ppma` file on the SD card).

223 Controls

- **Frequency or Gain or AMP**: Standard TX controls. Default frequency: **433.920 MHz**.
- **SampleRate**: The baseband sample rate used for transmission. Options: `250k`, `1M`, `2M`, `5M`, `10M`, `20M` Hz.
- **Step**: Increment used when adjusting **SymbolRate** and **PauseSymbol** fields with the encoder. Options: `1`, `10`, `100`.
- **SymbolRate** (0–9999): Number of symbols per second.
- **Repeat** (1–999): Number of times the payload is repeated per transmission.
- **PauseSymbol** (0–9999 µs): Pause duration between repetitions.
- **Payload**: Displays the current bit string to be transmitted (`0` = carrier off, `1` = carrier on).
- **Set**: Opens the text keyboard to type a payload manually.
- **Open File**: Opens a file browser in `OOKFILES/` to load a `.OOK` configuration file.
- **Save File**: Saves the current settings to a `.OOK` file in `OOKFILES/`.
- **Bug Key**: Opens the Bug Key composer view (see below).
- **Waveform**: Live preview of the current payload's OOK waveform.
- **Send or Stop**: Starts or stops transmission.

224 Bug Key

The Bug Key is a quick pattern composer. It appends repeated `0` or `1` bits to the payload using configurable step sizes.

- **Primary Step** (0–550): Number of bits appended by the short buttons (`0` and `1`).
- **Secondary Step** (0–550): Number of bits appended by the long buttons (`00` and `11`).
- Button: 0; Action: Appends Primary Step × 0 bits (carrier off)
- Button: 00; Action: Appends Secondary Step × 0 bits (carrier off)
- Button: 1; Action: Appends Primary Step × 1 bits (carrier on)
- Button: 11; Action: Appends Secondary Step × 1 bits (carrier on)

| `

224.1 P25 TX

> **WARNING** P25 TX transmits a simulated trunked radio control channel. Transmitting on frequencies you are not licensed for is illegal and may interfere with public safety communications. Use only on licensed frequencies, in RF-shielded environments, or with a dummy load.

P25 TX generates a continuous P25 Phase 1 trunked control channel (TSCC). It cycles through TSBK (Trunked System Block) packets that identify the network and assign voice channels to talkgroups, allowing P25 radios and decoders to lock onto and follow the simulated system. Frames are transmitted continuously in a loop until stopped.

This is an external app (`.ppma` file on the SD card).

225 Transmission Parameters

- Parameter: Default frequency; Value: 155.000 MHz
- Parameter: Bandwidth; Value: 12,500 Hz (12.5 kHz)
- Parameter: Sample rate; Value: 2,400,000 Hz
- Parameter: Modulation; Value: C4FM, 9600 baud
- Parameter: Frame type; Value: TSBK (DUID 0x07)

226 Controls

All fields are hexadecimal.

- Field: NAC; Digits: 3 hex; Description: Network Access Code; Default: 293
- Field: SYS; Digits: 3 hex; Description: System ID; Default: 001
- Field: WACN; Digits: 5 hex; Description: Wide Area Communication Network ID; Default: BEEF0
- Field: RFSS; Digits: 2 hex; Description: RF Subsystem ID; Default: 01
- Field: SITE; Digits: 2 hex; Description: Site ID; Default: 01
- Field: TG; Digits: 4 hex; Description: Talkgroup ID included in voice channel grant messages; Default: 0001
- Field: VCH; Digits: 3 hex; Description: Voice Channel number assigned in group voice grant messages; Default: 001
- **Frequency or Gain or AMP**: Standard TX controls. The configured frequency is also embedded into the IDEN_UP TSBK packet as the channel centre frequency.
- **Start or Stop**: Begins or halts continuous TSBK transmission. Status shows `TX TSCC` while running.

227 TSBK Sequence

The app transmits a repeating 5-packet cycle:

1. `IDEN_UP` (0x3D) — Channel identifier with frequency
2. `NET_STATUS` (0x3B) — WACN and System ID
3. `RFSS_STATUS` (0x3A) — RFSS and Site identification
4. `GRP_V_GRANT` (0x00) — Group voice channel grant for the configured TG or VCH
5. `NET_STATUS` (0x3B) — repeated

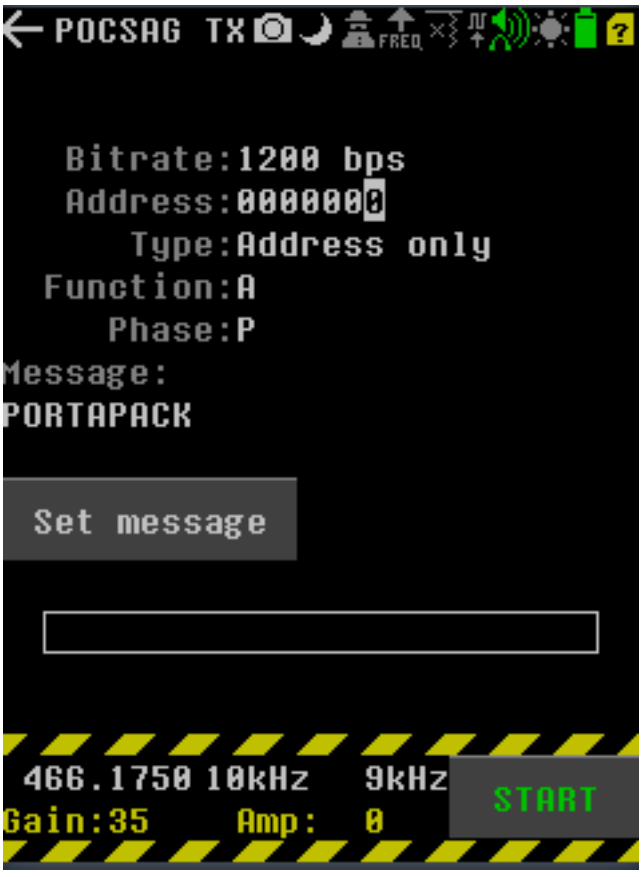
Each packet is BCH(63,16)-encoded, 1/2-rate trellis-encoded, interleaved, and status-symbol-inserted per the P25 Phase 1 standard before transmission.

228 Settings Persistence

Settings are saved to `tx_p25_tx.ini`.

228.1 POCSAG TX

228.2 Overview



This is an external app (`.ppma` file on the SD card).

228.3 Introduction

This application allows you to transmit messages using the **POCSAG** protocol with the PortaPack running **Mayhem** firmware. POCSAG (Post Office Code Standardisation Advisory Group) is a one-way paging protocol that uses 2-FSK modulation on narrowband FM channels.

It is intended for testing, experimentation, and educational use with licensed amateur radio equipment or on frequencies for which you hold the appropriate authorization.

228.4 Frequency

The default frequency is **466.175 MHz**, which is a commercial paging frequency used in some regions. This must be changed to match your intended use:

- Frequency: 439.9875 MHz; Use: Amateur Radio DAPNET (worldwide)
- Frequency: 466.075 MHz; Use: Commercial paging (region-dependent)
- Frequency: 466.175 MHz; Use: Commercial paging (default, region-dependent)
- Frequency: Other; Use: Varies by country and operator

For Amateur Radio use with DAPNET, use **439.9875 MHz** at **1200 bps**. See DAPNET for registration details.

228.5 Message Controls

These settings define the content and format of the POCSAG message:

228.5.1 Bitrate

Select the transmission bitrate:

- **512 bps** — older pagers, longer range
- **1200 bps** — most common, used by DAPNET and most commercial systems
- **2400 bps** — faster, shorter range, fewer compatible pagers

228.5.2 Address

The destination pager address. This is a 7-digit decimal field (0000000 – 9999999). The POCSAG protocol supports 21-bit addresses (maximum 2,097,151). Values above this are outside the protocol specification. Address 0 is a broadcast address received by all pagers.

228.5.3 Type

The message type determines how the payload is encoded:

- Type: Address only; Description: No message payload — sends address and function code only. Used to trigger tone-only pagers.
- Type: Numeric only; Description: Numeric characters only. Valid characters: 0–9, S, U,], [, -, space.
- Type: Alphanumeric; Description: Full ASCII text message, up to 80 characters.

228.5.4 Function

The function code is a 2-bit field (A–D) transmitted with every POCSAG message. Its meaning is defined by the receiving pager and operator, but conventional usage is:

- Code: A; Conventional meaning: Tone only or alert
- Code: B; Conventional meaning: Numeric message
- Code: C; Conventional meaning: Alphanumeric message
- Code: D; Conventional meaning: Voice call or operator-defined

Most modern pagers accept any function code for alphanumeric messages. When in doubt, use **A**.

228.5.5 Phase

Selects the signal polarity of the transmitted bitstream:

- **P (Positive)** — bits are inverted before transmission (`~codeword`). Some pager systems expect this polarity.
- **N (Normal)** — bits are transmitted as-is. This is the standard POCSAG polarity per the specification.

If a pager does not respond to transmission with one phase setting, try the other.

228.5.6 Message

The text payload to transmit. Maximum length is **80 characters**.

- For **Numeric only**, only `0–9 S U ] [ -` and space are valid. Other characters will trigger an error.
- For **Alphanumeric**, standard printable ASCII is supported.
- Use the **Set message** button to enter the text.

228.6 Transmit Controls (Lower Bar)

228.6.1 First Line

- **Frequency** — Transmission frequency. Adjust with the encoder wheel, on-screen keypad, or load from SD card.
- **Frequency Step** — Step size when adjusting frequency with the encoder.
- **Channel Bandwidth** — RF bandwidth. Fixed at 1.75 MHz (narrowband FM, suitable for 12.5 kHz and 25 kHz POCSAG channels).

228.6.2 Second Line

- **Gain (IF)** — TX gain in dB (0–47). See Description of gain settings for details.

- **AMP** — RF amplifier: `1` = enabled (+14 dB), `0` = disabled.

228.7 Transmission

Press the **START** button to begin transmitting. A progress bar shows transmission progress. Transmission stops automatically when the message is complete.

228.8 USB Serial Console

The POCSAG TX app can be triggered remotely via the USB serial console using the `sendpocsag` command:

```
sendpocsag   [baud] [type] [function] [phase]
```

- Parameter: addr; Values: 0–2097151; Description: Destination pager address
- Parameter: msglen; Values: 0–80; Description: Message length in bytes. Use 0 for address-only.
- Parameter: baud; Values: 512, 1200, 2400; Description: Bitrate (optional, default 1200)
- Parameter: type; Values: 0, 1, 2; Description: 0=ADDRESS_ONLY, 1=NUMERIC_ONLY, 2=ALPHANUMERIC
- Parameter: function; Values: A, B, C, D; Description: Function code
- Parameter: phase; Values: P, N; Description: Signal polarity

After sending the command, the console will prompt: `send  bytes`. Send exactly that number of bytes as the message payload. If `msglen` is 0, transmission starts immediately without a message.

Example — send alphanumeric message to address 1234567 at 1200 bps:

```
sendpocsag 1234567 12 1200 2 A N
```

Then send: `Hello World!`

See the USB Serial Console page for connection instructions.

228.9 Testing

A reliable way to verify your transmission is to use the POCSAG Receiver app on a second PortaPack, or use an SDR receiver with software such as PDW (Windows) or multimon-ng (Linux).

228.10 Legal notice

POCSAG operates on licensed frequencies. Transmitting on commercial paging frequencies without authorization is illegal in all jurisdictions. Transmitting on amateur radio frequencies (such as 439.9875 MHz for DAPNET) requires a valid amateur radio license.

The HackRF One is a test and development device. The licensed operator is responsible for ensuring that all transmissions comply with local regulations regarding frequency, power, and authorized use.

228.11 Further reading

- POCSAG Receiver
- POCSAG protocol overview
- USB Serial Console — sendpocsag command
- DAPNET Amateur Radio Paging Network
- multimon-ng POCSAG decoder

228.12 RDS

RDS TX transmits Radio Data System (RDS) data on the 57 kHz subcarrier of an FM broadcast signal. It allows broadcasting of station name, program type, program ID, traffic flags, and scrolling text to RDS-capable FM receivers.

The app uses 4 tabs: **Name**, **Text**, **Time**, and **Audio**. Only the Name and Text tabs are currently functional; Time and Audio display "Not yet implemented."

229 RF Parameters

- Parameter: Default frequency; Value: Stored in persistent memory
- Parameter: Frequency step; Value: 50 kHz
- Parameter: Bandwidth; Value: 1.75 MHz
- Parameter: Sample rate; Value: 2.28 MSps

- Parameter: TX gain; Value: 0–47 dB (LNA), 0/14 dB (AMP)

Settings are saved to `tx_rds.ini`.

230 Common Controls (all tabs)

These controls appear at the top of the screen regardless of the active tab:

- **Program type (PTY):** Dropdown with 32 options (codes 0–31): None, News, Affairs, Info, Sport, Educate, Drama, Culture, Science, Varied, Pop, Rock, Easy, Light, Classics, Other, Weather, Finance, Children, Social, Religion, PhoneIn, Travel, Leisure, Jazz, Country, National, Oldies, Folk, Docs, AlarmTst, Alarm.
- **Program ID (PI):** 4-digit hex value (0000–FFFF). Identifies the station globally.
- **TP (Traffic Program):** Checkbox — signals that the station regularly broadcasts traffic information.

231 Name Tab

- **Transmit PSN** (checkbox): Enables transmission of the Program Service Name.
- **PSN:** The station name shown on FM receivers. Maximum 8 characters. Default: `TEST1234`. Press **Set** to open the text entry keyboard.
- **Stereo:** Checkbox — sets the DI (Decoder Identification) bit to indicate stereo audio.
- **Music:** Checkbox — sets the MS (Music/Speech) bit.
- **Traffic announcement (TA):** Checkbox — signals an active traffic announcement on this channel.

232 Text Tab

- **Transmit Radiotext** (checkbox): Enables transmission of the RadioText (RT) message.
- **Text:** Free-form scrolling text displayed on receivers. Maximum 28 characters in this implementation. Default: `Radiotext test ABCD1234`. Press **Set** to open the text entry keyboard.

233 Time Tab

Displays “Not yet implemented.” Date and time transmission (RDS group type 4A or CT) is not functional.

234 Audio Tab

Displays “Not yet implemented.” Audio-related RDS data transmission is not functional.

235 Starting Transmission

Press **Start** at the bottom of the screen to begin transmitting. Press it again to stop. All enabled tabs (PSN and/or Radiotext) are transmitted in rotation.

235.1 RTTY TX

RTTY TX transmits Radio TeleTYpe (RTTY) signals using Baudot/ITA2 encoding. It generates an FSK signal with configurable baud rate, frequency shift, and tone placement.

This is an external app (`.ppma` file on the SD card).

236 Controls

Baud — Transmission speed. Options: 45, **45.45** (default), 50, 75, 100, 110, 150, 200 baud.

Shift — Frequency separation between Mark and Space tones. Options: 85, **170** (default), 200, 450, 850 Hz.

Inverted (checkbox, default: on) — Swaps Mark and Space tone assignments. Enabled by default to match the most common RTTY convention.

Mark t — Sets the absolute frequency offset of the Mark tone. Options:

- Label: CNT (default); Meaning: Centered: Mark = −shift/2, Space = +shift/2
- Label: 0; Meaning: Mark at 0 Hz offset
- Label: 2125; Meaning: Mark at +2125 Hz (US HF AFSK standard)
- Label: −2125; Meaning: Mark at −2125 Hz
- Label: 1275; Meaning: Mark at +1275 Hz
- Label: −1275; Meaning: Mark at −1275 Hz

The computed Mark and Space tone values (in Hz, relative to the carrier) are shown on the line below these controls. They update automatically when Shift, Mark t, or Inverted are changed.

Set message — Opens the text entry keyboard. The message text is displayed below the button. Default: `PORTAPACK`. The message is encoded using Baudot/ITA2 (USOS disabled for maximum receiver compatibility) and transmitted once per START press.

237 RF Parameters

- Parameter: Frequency; Value: Stored in persistent memory — typical HF RTTY: 14.080 MHz (20 m)

- Parameter: Bandwidth; Value: 2.048 MHz
- Parameter: Sample rate; Value: 2.048 MSps
- Parameter: Stop bits; Value: 1.5 (fixed)
- Parameter: TX gain; Value: 0–47 dB (LNA), 0/14 dB (AMP)

238 Transmission

Press **START** to transmit. The message is sent once and transmission stops automatically when complete. Press **STOP** to abort early.

239 Settings Persistence

Settings are saved to `tx_rtty.ini`.

239.1 SAME TX

SAME TX generates and transmits Specific Area Message Encoding (SAME) digital headers — the protocol used by the US Emergency Alert System (EAS) and NOAA Weather Radio. SAME-capable receivers use these headers to selectively activate for alerts targeting their programmed county.

This is an external app (`.ppma` file on the SD card).

240 Controls

- Field: Org; Description: Originator code; Range or Options: WXR, EAS, CIV, PEP; Default: WXR
- Field: Evt; Description: Event type (index 0–25); Range or Options: See table below; Default: RWT (index 0)
- Field: FIPS State; Description: 2-digit US state FIPS code; Range or Options: 00–99; Default: 39 (Ohio)
- Field: FIPS County; Description: 3-digit county FIPS code; Range or Options: 000–999; Default: 007 (Ashtabula)
- Field: Dur H; Description: Alert duration — hours; Range or Options: 00–99; Default: 00
- Field: Dur M; Description: Alert duration — minutes; Range or Options: 00–59; Default: 30

A live preview of the message string updates on screen as fields are changed.

241 Originator Codes

- Code: WXR; Meaning: National Weather Service
- Code: EAS; Meaning: EAS Participant (broadcaster or cable system)
- Code: CIV; Meaning: Civil Authorities
- Code: PEP; Meaning: Primary Entry Point System (presidential-level)

242 Event Codes

- #: 0; Code: RWT; Event: Required Weekly Test
- #: 1; Code: RMT; Event: Required Monthly Test
- #: 2; Code: NPT; Event: National Periodic Test
- #: 3; Code: NST; Event: National Silent Test
- #: 4; Code: NMT; Event: National Periodic Test
- #: 5; Code: EAN; Event: Emergency Action Notification
- #: 6; Code: EAT; Event: Emergency Action Termination
- #: 7; Code: NIC; Event: National Information Center
- #: 8; Code: ADR; Event: Administrative Message
- #: 9; Code: AVA; Event: Avalanche Watch
- #: 10; Code: AVW; Event: Avalanche Warning
- #: 11; Code: BZW; Event: Blizzard Warning
- #: 12; Code: CFW; Event: Coastal Flood Watch
- #: 13; Code: CFS; Event: Coastal Flood Statement

- #: 14; Code: DSW; Event: Dust Storm Warning
- #: 15; Code: EQW; Event: Earthquake Warning
- #: 16; Code: EVI; Event: Evacuation Immediate
- #: 17; Code: FFW; Event: Flash Flood Warning
- #: 18; Code: FFS; Event: Flash Flood Statement
- #: 19; Code: FFH; Event: Flash Flood Watch
- #: 20; Code: FRW; Event: Fire Warning
- #: 21; Code: HLS; Event: Hurricane Statement
- #: 22; Code: HUW; Event: Hurricane Warning
- #: 23; Code: HUH; Event: Hurricane Watch
- #: 24; Code: SVR; Event: Severe Thunderstorm Warning
- #: 25; Code: TOR; Event: Tornado Warning

243 Message Format

The transmitted header has the form:

`ZCZC-ORG-EVT-SS0CCC+HHMM-0010000-SAMETX--`

where `SS` is the state FIPS, `0CCC` is the county FIPS (the leading `0` is literal, part of the SAME county encoding), `HHMM` is the duration, and `-0010000-SAMETX-` are fixed placeholder fields for issue time and station ID. The full message is preceded by 16 preamble bytes and transmitted **3 times** automatically, as required by the SAME standard.

244 RF Parameters

- Parameter: Default frequency; Value: 162.550 MHz (NOAA Weather Radio)
- Parameter: Bandwidth; Value: 1.75 MHz
- Parameter: Sample rate; Value: 1.536 MSps
- Parameter: Modulation; Value: AFSK or FM
- Parameter: Mark tone; Value: 2083 Hz
- Parameter: Space tone; Value: 1563 Hz
- Parameter: Baud rate; Value: ~520.83 bps
- Parameter: FM deviation; Value: 5 kHz
- Parameter: Repeats; Value: 3×

245 Settings Persistence

Settings are saved to `tx_same.ini`.

245.1 Signal gen

The signal generator found under Utilities can be used to generate a carrier which can be FM modulated with the following waveforms as defined by the Shape function.

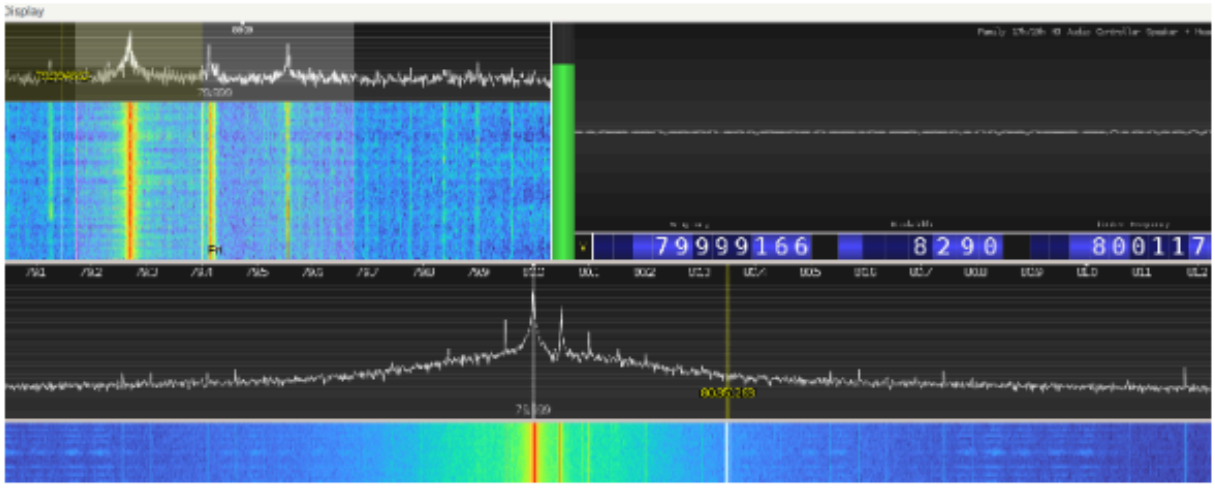
- CW : unmodulated carrier
- Sine signal
- triangle signal
- sawtooth up signal
- sawtooth down signal
- square signal
- noise signal

This is an external app (`.ppma` file on the SD card).

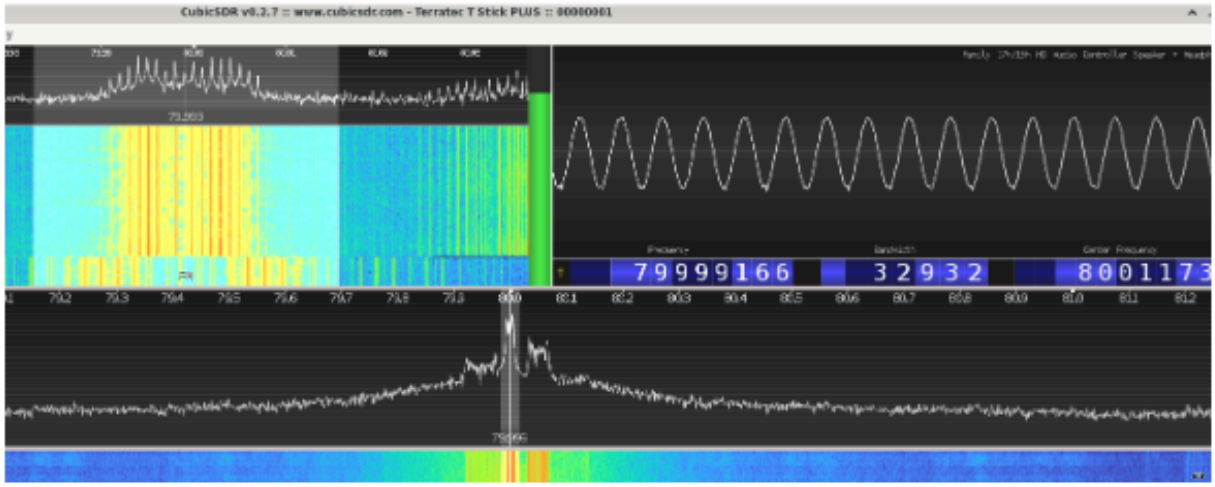
245.2 * Sine or Triangle or Sawtooth up or Sawtooth down or Square

We can select the periodic signal to modulate the carrier , in FM .(and we can adjust below the delta FM deviation)

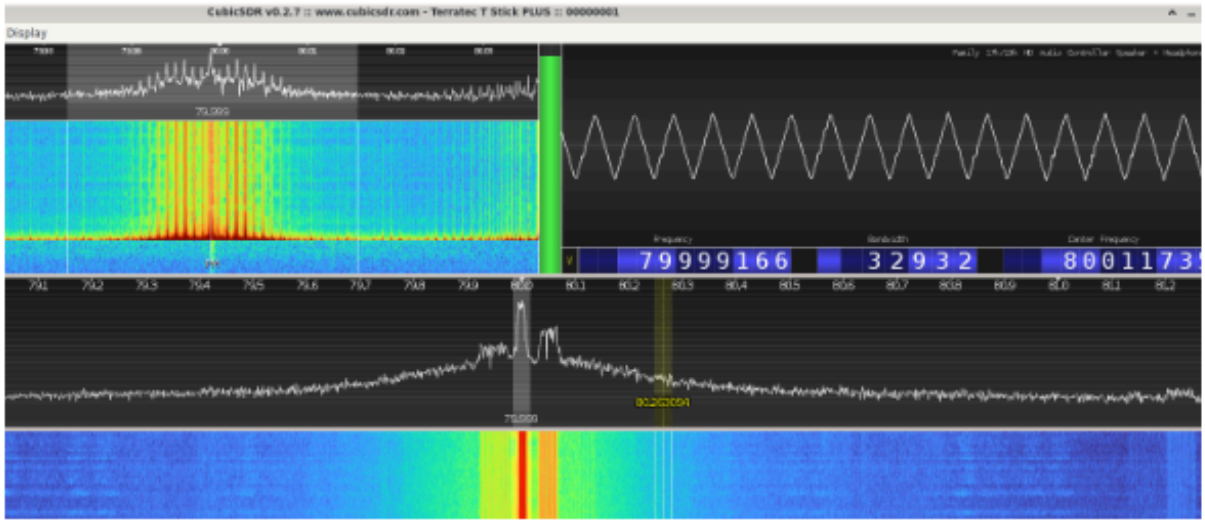
CW (just carrier no modulation)



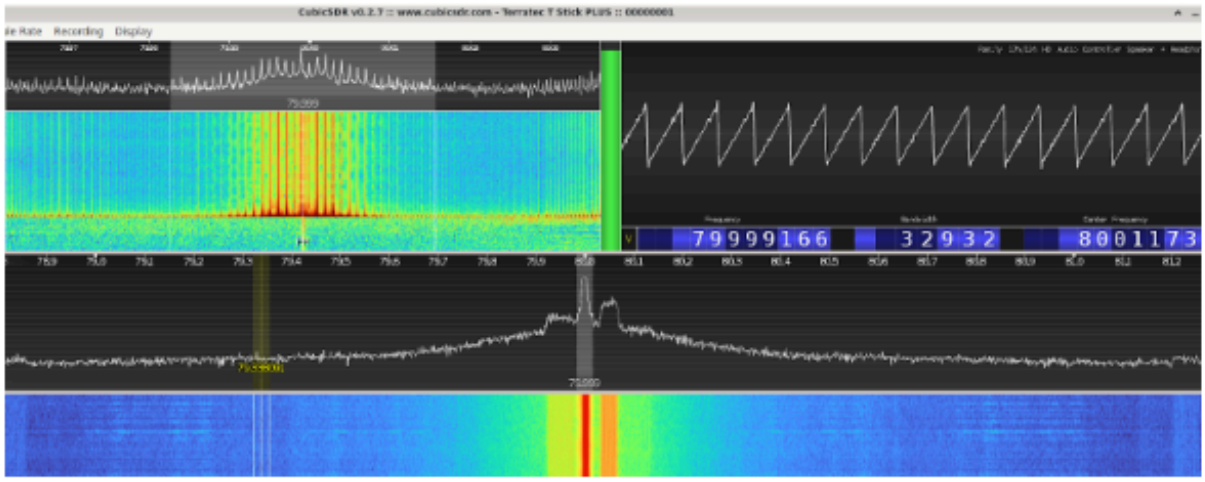
SINE (we can select throught the GUI the tone frequency)



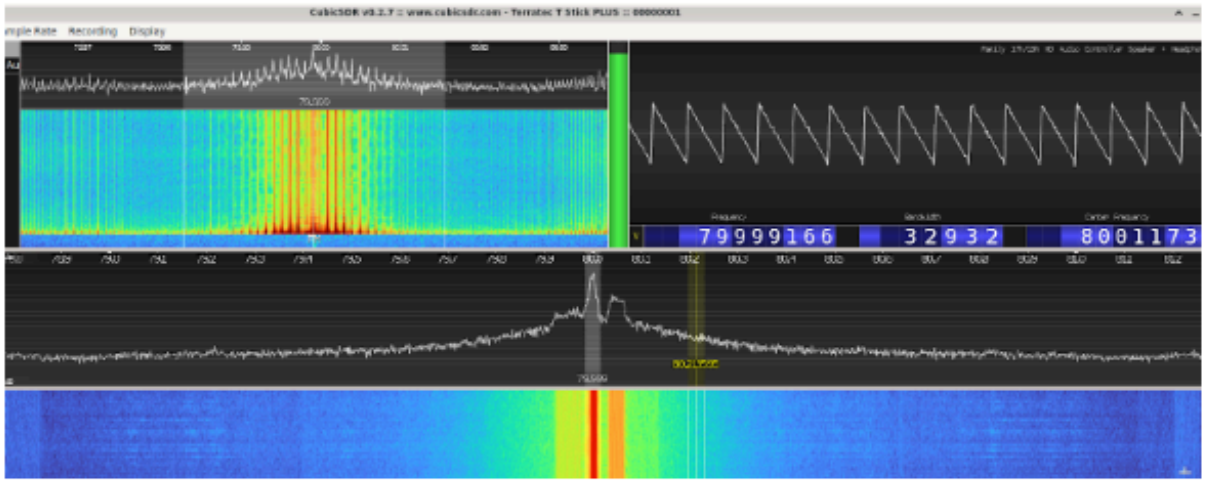
TRIANGLE , (we can select throught the GUI the tone frequency)



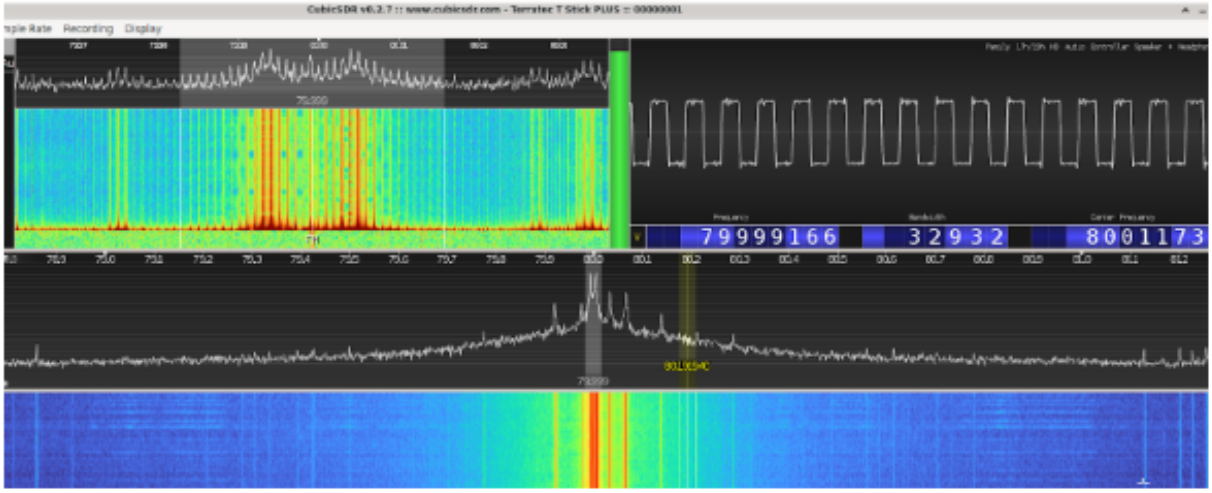
SAW UP. (we can select throught the GUI the tone frequency)



SAW DOWN , (we can select throught the GUI the tone frequency)



SQUARE , (we can select throught the GUI the tone frequency)



Tone defines the frequency of modulation.

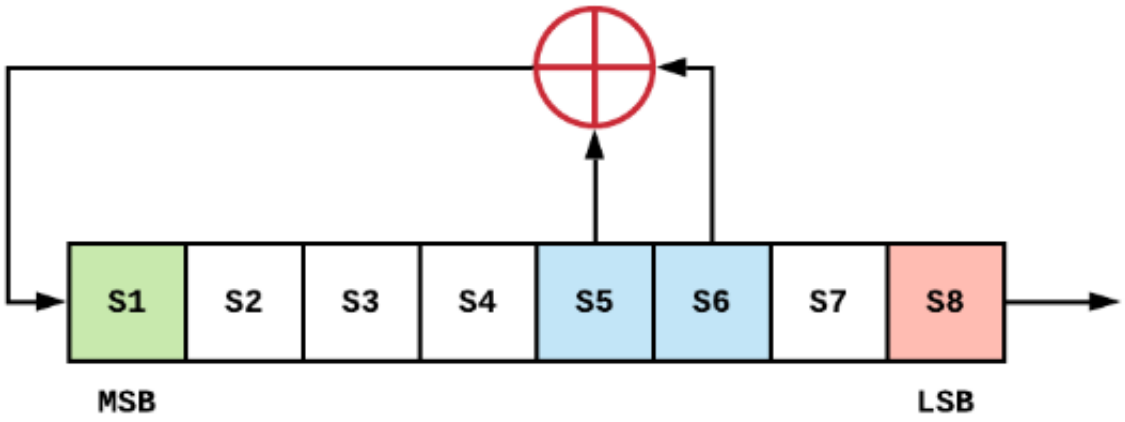
Update when unchecked requires the user to drop carrier, then re-start carrier in order for a change of modulation Shape or Tone frequency to take effect. When checked, values will update on the fly.

Stop after 1s limits carrier duration to 1 second

245.3 * White Noise generator options:

We are generating simulated White Noise ,using pseudo random noise generator, 8 bits linear-feedback shift register (LFSR) algorithm, variant Fibonacci. (Following this wiki link)

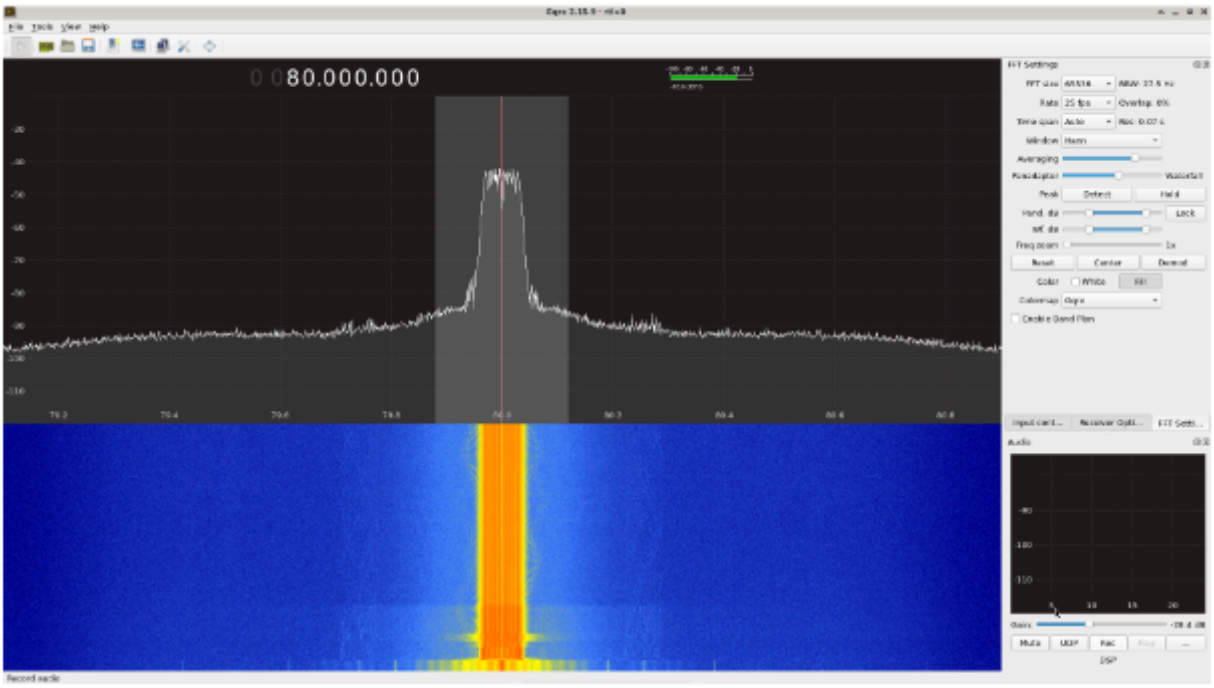
Example of the implementation of a 8 bite Noise option



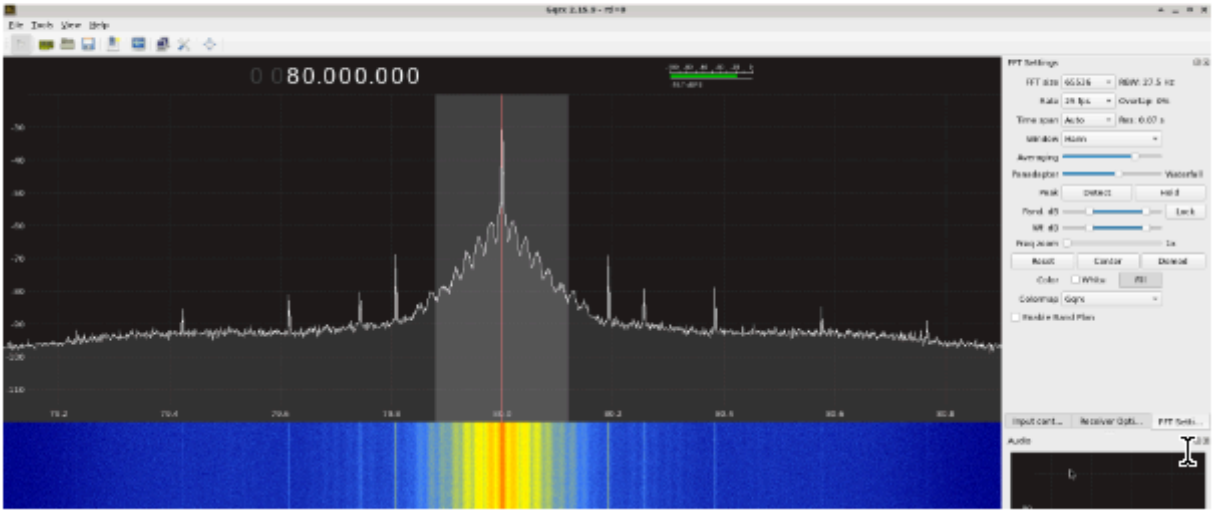
$$P(x) = x^6 + x^5 + 1$$

We have introduced only one user option, about Noise generator , using the best LFSR that we tested : 16 bit polynomial feedback LFSR , that has longer random sequence period and produces more continuous spectrum.Using noise generator, we can hear in any FM receiver the random noise (and we can also see the random demodulated signal, as it is attached below) and the peak radiated spectrum power is now just −10 dB's compared to the triangle or saw signal , that is fine (acceptable) . (not −40 or −50 dB's as it was before, really too small) .

TX gain 43 , reference triangle signal

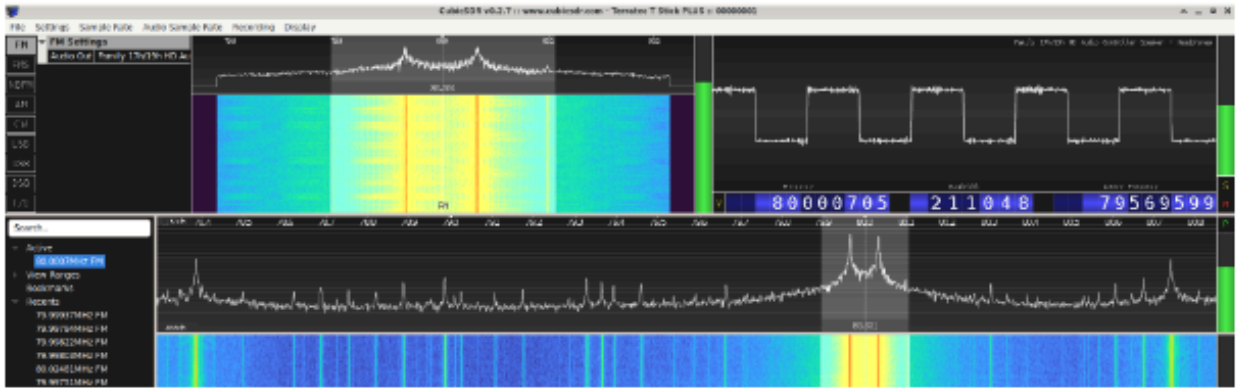


This PR Noise signal , radiated power spectrum

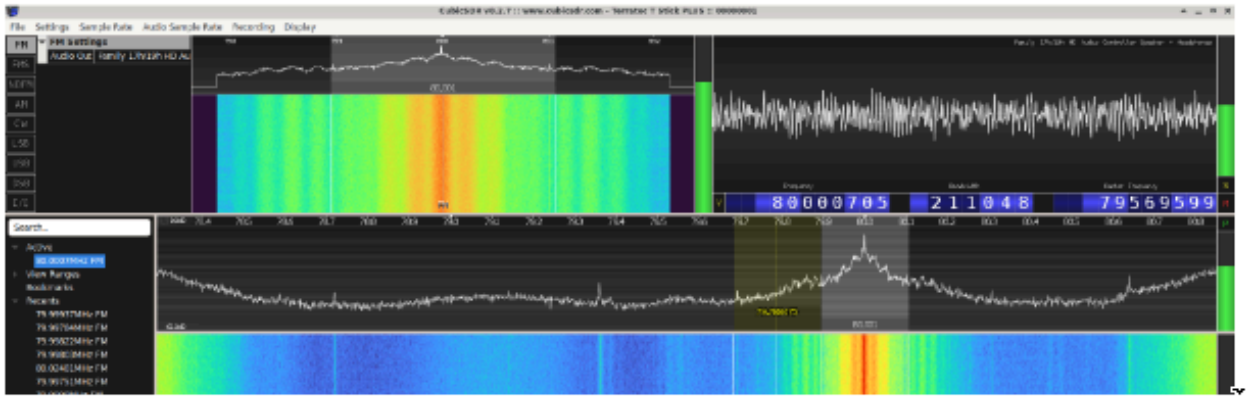


See on top right corner the pseudo random noise demodulated sinal

example, reference square signal , demodulated top right corner



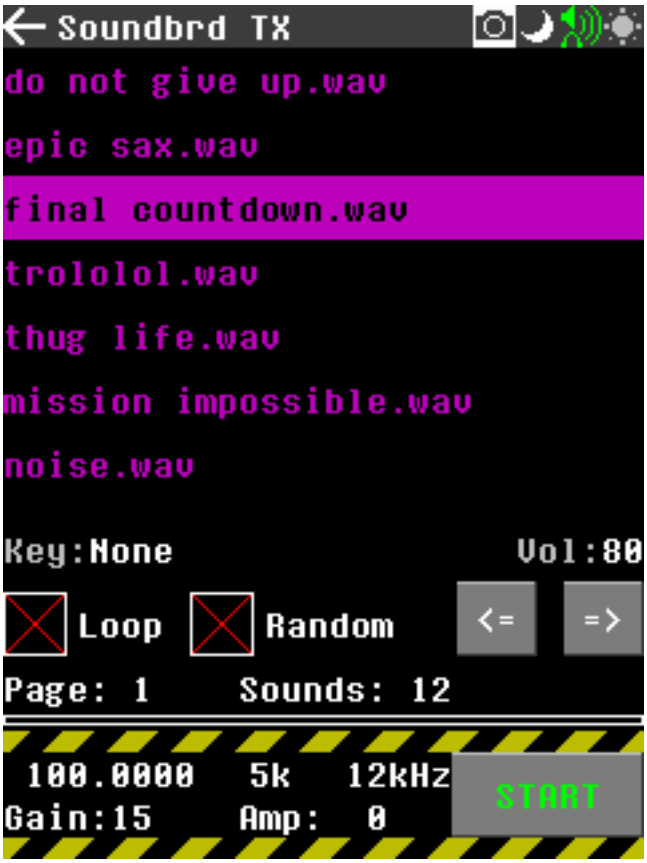
Noise signal , this PR using 16 bit , we can see the demodulated baseband random noise.



Update when unchecked requires the user to drop carrier, then re-start carrier in order for a change of modulation Shape or Tone frequency to take effect. When checked, values will update on the fly.

Stop after 1s limits carrier duration to 1 second

245.4 Soundboard



Soundboard transmits `.WAV` files from the SD card as FM audio. Select a file from the list and press START, or enable Random mode to play files in random order.

This is an external app (`.ppma` file on the SD card).

246 File List

WAV files are loaded from the `WAV/` directory on the SD card. Up to 50 files are shown per page; use the `<=` and `=>` buttons to navigate between pages. The current page number and total files on that page are shown above the list.

Files must be mono and either 8-bit unsigned PCM or 16-bit signed PCM. The sample rate is read automatically from the WAV header — files sampled between 24,000 and 48,000 Hz work reliably. Stereo files and other bit depths are silently skipped.

247 Controls

- **Loop** (checkbox) — Repeats the currently selected file continuously until stopped.
- **Random** (checkbox) — Plays a randomly chosen file from the list each time START is pressed. When the file ends, a new random file is selected and played automatically.
- **Key** — Selects a subaudible tone to encode on the carrier. The full list includes standard CTCSS tones (labeled by EIA code, e.g. “1 XZ” through “50 ZZ”) followed by wireless microphone pilot tones: Shure 19 kHz, Axient 28 kHz, Sennheiser 32.000 kHz, Sony 32.382 kHz, and Sennheiser 32.768 kHz. Select **None** (default) to disable.
- **Vol** — Controls the audio output volume to the speaker. Disabled (shown as `--`) when a Key tone is active, since the pilot tone signal would be audible on the speaker.
- **START or STOP** — Starts or stops transmission.
- **Progress bar** — Shows playback position within the current file.

248 RF Parameters

- Parameter: Modulation; Value: FM
- Parameter: Bandwidth; Value: 1.75 MHz (baseband)
- Parameter: Sample rate; Value: Read from WAV header
- Parameter: TX gain; Value: 0–47 dB (LNA), 0/14 dB (AMP)

The bandwidth field in the TX bar sets the FM channel bandwidth (deviation), not the WAV sample rate. The sample rate is applied automatically from the file header.

249 Preparing WAV Files

Files must be mono, 8-bit unsigned or 16-bit signed PCM, sampled at 24,000–48,000 Hz. The included sample files are 48,000 Hz or 8-bit. A quick conversion with `sox`:

```
for f in ./*.wav; do sox "$f" -r 48000 -c 1 -b 8 --norm "conv/$f"; done
```

In Audacity: mix down to mono via **Tracks → Mix → Mix Stereo Down to Mono**, then export as **Other Uncompressed Files** with Header **WAV (Microsoft)** and Encoding **Unsigned 8-bit PCM**.

250 Settings Persistence

Settings are saved to `tx_soundboard.ini`.

250.1 Spectrum Painter

Spectrum Painter transmits an image or text in a way that makes it visible as a pattern on a waterfall display. The image or text is encoded as the instantaneous frequency of the signal over time, painting each row of pixels as a horizontal line in the spectrum.

The app has two tabs: **Image** and **Text**.

This is an external app (`.ppma` file on the SD card).

251 Controls (common to both tabs)

- **Frequency** — Center frequency of the transmission.
- **GAIN** — TX gain, 0–47 dB. Default: 47.
- **A:** — RF amplifier, 0 or 14 dB. Default: off (0).
- **Loop** (checkbox, default: on) — Repeats the transmission continuously. When active, the Pause field sets a gap between repetitions.
- **BW** — Transmission bandwidth. This determines how wide the painted spectrum appears. Options: 12k5, 16k, 25k, 32k, 50k, 75k, **100k** (good starting point), 150k, 250k, 500k, 600k, 750k, 1000k, 1250k, 1500k, 1750k, 2000k, 2250k, 2500k, 3000k. Default: 12k5. For visible results on a multi-MHz waterfall span, use at least 500k–2000k.
- **Du:** — Duration of each transmission in seconds (1–999). Default: 10.
- **P:** — Pause between transmissions in seconds when Loop is active (0–99). Default: 5.
- ▸ / ■ — Start or stop transmission. A progress bar shows the current position within the transmission.

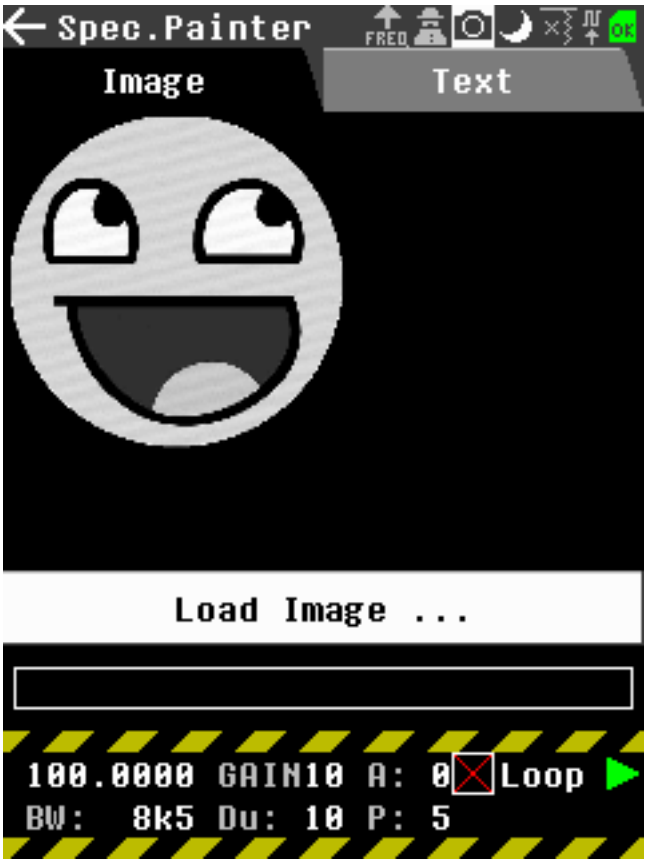
The internal baseband bandwidth is fixed at 1.75 MHz regardless of the BW setting.

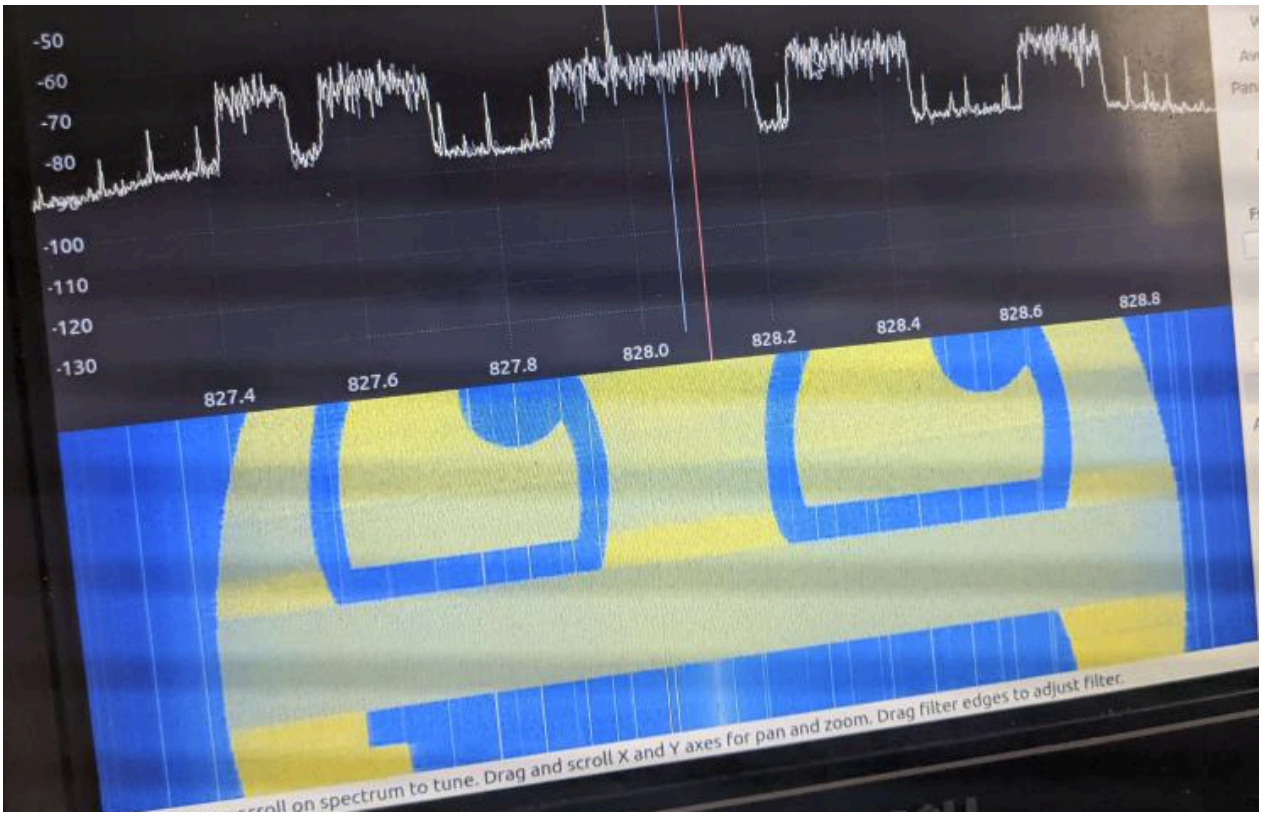
252 Image Tab

Place `.BMP` files in the `SPECTRUM/` directory on the SD card. Press the file selector to choose a file — a scaled preview is shown on screen before transmission. The selected file is transmitted row by row, with each pixel row becoming one horizontal line in the spectrum.

Supported formats: 16-bit, 24-bit, and 32-bit uncompressed BMP (no compression, single plane). All three bit depths are fully supported.

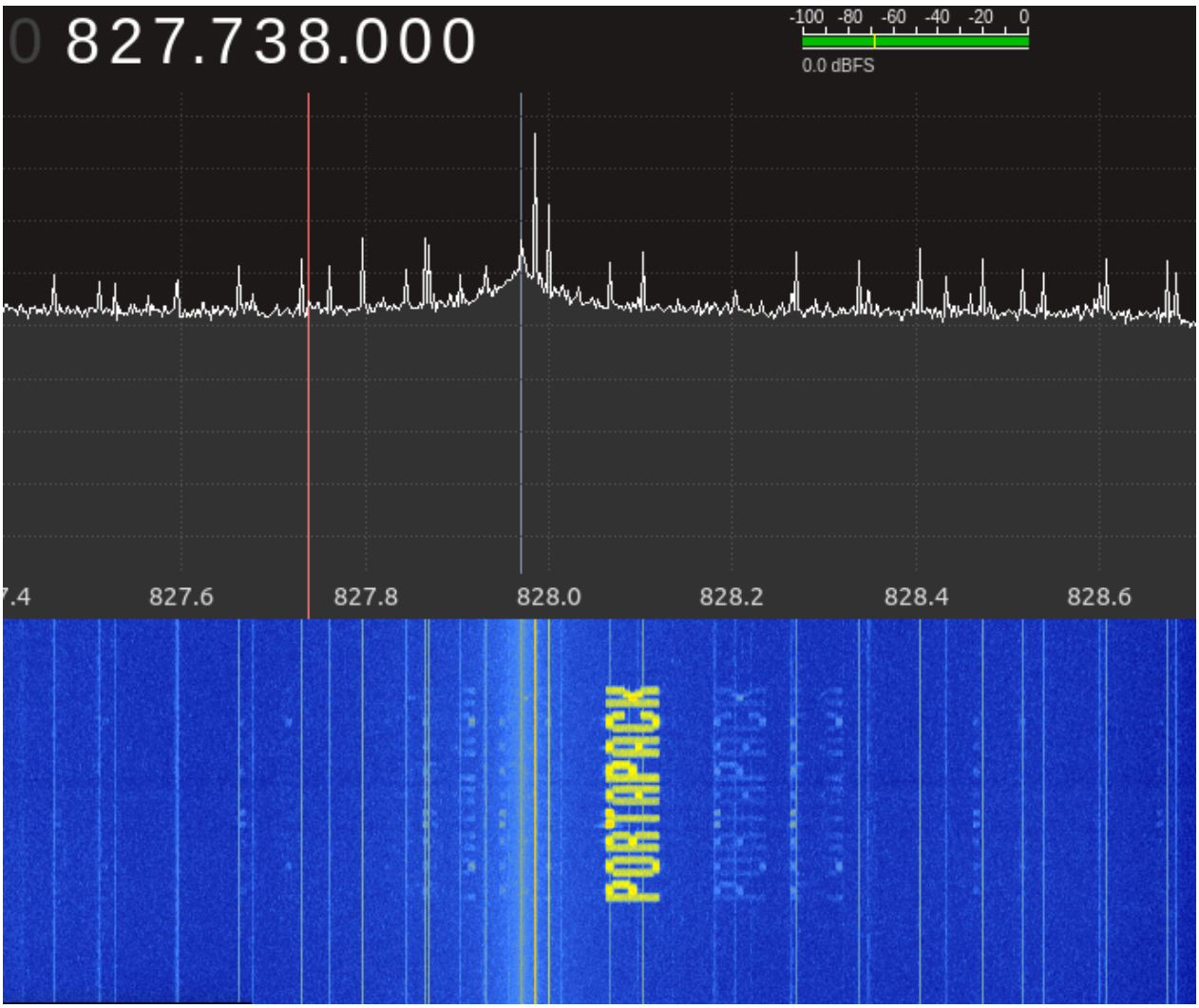
For best results, use a high horizontal resolution. A width of 2048 pixels works well. Using a power of 2 for the width (512, 1024, 2048, ...) gives optimal results.





253 Text Tab

Enter text using the **Set message** button (opens the keyboard). The message is displayed across up to 10 lines of 30 characters each on screen. Maximum message length is 300 characters.



254 Settings Persistence

Settings are saved to `tx_painter.ini`.

254.1 SSTV

SSTV TX transmits Slow Scan Television images from `.BMP` files stored on the SD card. The signal is transmitted as FM audio tones in the standard SSTV frequency range (1,500–2,300 Hz).

This is an external app (`.ppma` file on the SD card).

255 Controls

- **File** — Selects the `.BMP` file to transmit from the `SSTV/` directory on the SD card.
- **Mode** — Selects the SSTV transmission mode. Default: Scottie 2.
- **Frequency, Gain, AMP** — Standard RF controls. Default gain: 12 dB.
- ▸ — Starts transmission. A progress bar shows the current scanline.

256 Supported Modes

- Mode: Scottie 1; VIS Code: 60; Lines: 256; µs/pixel: 432.0
- Mode: Scottie 2 (default); VIS Code: 56; Lines: 256; µs/pixel: 275.2
- Mode: Scottie DX; VIS Code: 76; Lines: 256; µs/pixel: 1080.0
- Mode: Martin 1; VIS Code: 44; Lines: 256; µs/pixel: 457.6
- Mode: Martin 2; VIS Code: 40; Lines: 256; µs/pixel: 228.8
- Mode: SC2-180; VIS Code: 55; Lines: 256; µs/pixel: 734.4

All modes transmit 320 pixels per line and 256 lines (320×256 total).

257 Image Requirements

- Format: 24-bit uncompressed BMP (`.bmp`)
- Size: exactly **320×256 pixels**
- Location: `SSTV/` directory on the SD card

Images that do not match the required resolution will produce incorrect output.

258 RF Parameters

- Parameter: Sample rate; Value: 3.072 MSps
- Parameter: Baseband bandwidth; Value: 1.75 MHz (fixed)
- Parameter: Frequency step; Value: 10 kHz

259 Settings Persistence

Settings are saved to `tx_sstv.ini`.

259.1 TEDI/LCR

TEDI/LCR TX generates and transmits LCR (Langage de Commande Routier — Road Command Language) frames, a French protocol used to control dynamic road equipment: variable message signs (PMV), tunnel systems, traffic cameras, emergency phones, weather stations, and related infrastructure. It uses AFSK modulation (Bell 202-compatible) over FM.

This is an external app (`.ppma` file on the SD card).

260 Controls

EC — Éclairage (lighting) mode. Options: Auto (`A`), Jour/Day (`J`), Nuit/Night (`N`), S? (`S`).

RGSB — The 4-character destination address code (e.g. `AI10`). Press the **RGSB** button to enter a custom address via the keyboard. In scan mode, this is overridden by the selected address list.

AM1–AM5 — Up to 5 address message slots. Each slot has a checkbox (to include it in the frame) and a **Set** button to enter the 7-character payload text. Only checked slots are included in the transmitted frame.

Modem Setup — Opens the modem configuration screen where you set the baudrate and repeat count. These settings are shared with other AFSK apps and persist in memory.

Scan (checkbox) — When enabled, automatically cycles through all RGSB addresses in the selected list, transmitting the configured frame to each address in sequence.

List — Selects the preconfigured RGSB address list used in scan mode. Contains a list of known addresses for the Lille road network (36 entries).

Clear — Clears all AM slot content and unchecks all checkboxes.

START or STOP — Starts or stops transmission. A progress bar and status line show repeat and scan progress (`repeat/total scan_index/total`).

261 Message Format

The transmitted LCR frame has the following structure:

```
<7 sync bytes> RGSB AM=1 [AM=2 ...] EC=
```

Text in each AM slot is padded to 7 characters. The checksum is an XOR of all characters from the RGSB field to the EOM byte, masked to 7 bits. The complete frame is transmitted the number of times set in Modem Setup.

262 Preconfigured RGSB Lists

Lille (36 addresses): AI10–AI70, AJ10–AJ70, AK10, EAA0–EAD0, EbA0–EbS0

Reims (20 addresses): AI10–AI70, AJ10–AJ70, AK10–AK70 (selected), AP10

263 RF Parameters

- Parameter: Modulation; Value: AFSK or FM (Bell 202-compatible)
- Parameter: Mark frequency; Value: 1200 Hz
- Parameter: Space frequency; Value: 2200 Hz
- Parameter: Baud rate; Value: Configurable via Modem Setup (default: 1200 bps)
- Parameter: Bandwidth; Value: 1.75 MHz
- Parameter: Sample rate; Value: 1.536 MSps
- Parameter: Frequency step; Value: 10 kHz

264 Settings Persistence

Settings are saved to `tx_lcr.ini`. Baudrate and repeat count are stored in shared persistent memory via Modem Setup.

264.1 TouchTunes

TouchTunes is a jukebox platform found in bars, pubs, and restaurants across North America. TouchTunes jukeboxes (Gen 2 and above) use a wireless remote that transmits NEC-encoded messages via ASK/OOK at 433.92 MHz. These remotes are used for basic jukebox functions such as power, skip, volume, etc. The wireless messages are addressed using PINs 000–255 to prevent multiple units from interfering with each other. **The default PIN is 000 in most deployments.**

This app emulates TouchTunes wireless remotes (Gen 2 and newer) and supports both single-address and brute-force (scan) modes.

265 How To Use

- **Transmit a command:** Press any button on the touchscreen, or highlight and select with the physical controls. Each command is transmitted 4 times automatically.
- **Change the PIN:** Adjust the 3-digit PIN field in the top-left using the encoder knob. Default is 000. Range: 000–255.
- **Brute force (Scan mode):** Enable the **Scan** checkbox, then press any button. The selected command will be transmitted 4 times for each PIN from 000 to 255 in sequence, covering all 256 possible addresses.
- **EW Mode (Electronic Warfare, Experimental):** Transmits a continuous CW carrier at 433.92 MHz to jam the jukebox's receiver, preventing others from sending commands. You retain control — the jamming signal automatically pauses when you transmit a command, then resumes. Enable with the **EW Mode** checkbox.

266 Message Structure

266.0.1 Basic Info

- **Frequency:** 433.92 MHz
- **Modulation:** ASK/OOK
- **Protocol:** NEC (without address complement)

- **Sample rate:** 2.28 MSps
- **Symbol rate:** 1766 baud (measured; nominal NEC rate is 1786 baud)
- **Symbol period:** 566 µs
- **Repeats per command:** 4

266.0.2 NEC Symbol Encoding

- **Short (0):** `10` (OOK: ON OFF)
- **Long (1):** `1000` (OOK: ON OFF OFF OFF)

266.0.3 Frame Structure

`{PREAMBLE} {SYNC} {PIN} {COMMAND} {TAIL}`

- **Preamble (literal symbols):** `0xFFFF00`
- **Sync (decoded NEC):** `0x5D`
- **PIN 000–255 (decoded NEC):** `0x00–0xFF` (LSB first)
- **Tail (literal symbols):** `0x8`

266.0.4 Commands (pre-NEC encode)

Each command byte is doubled with the second half bit-reversed before NEC encoding. For example, Pause (`0x32`) becomes `0x3223` before encoding.

- Code: 0x32; Button: Pause
- Code: 0x78; Button: On/Off (Power)
- Code: 0x70; Button: P1
- Code: 0x60; Button: P2 Edit Queue
- Code: 0xCA; Button: P3
- Code: 0x20; Button: F1 Restart
- Code: 0xF2; Button: Up (^)
- Code: 0xA0; Button: F2 Key
- Code: 0x84; Button: Left (<)
- Code: 0x44; Button: OK
- Code: 0xC4; Button: Right (>)
- Code: 0x30; Button: F3 Mic A Mute
- Code: 0x80; Button: Down (V)
- Code: 0xB0; Button: F4 Mic B Mute
- Code: 0xF0; Button: 1
- Code: 0x08; Button: 2
- Code: 0x88; Button: 3
- Code: 0x48; Button: 4
- Code: 0xC8; Button: 5
- Code: 0x28; Button: 6
- Code: 0xA8; Button: 7
- Code: 0x68; Button: 8
- Code: 0xE8; Button: 9
- Code: 0x18; Button: * (Music or Karaoke)
- Code: 0x98; Button: 0
- Code: 0x58; Button: # (Lock Queue)
- Code: 0xD0; Button: Zone 1 Vol+
- Code: 0x90; Button: Zone 2 Vol+
- Code: 0xC0; Button: Zone 3 Vol+
- Code: 0x50; Button: Zone 1 Vol-
- Code: 0x10; Button: Zone 2 Vol-
- Code: 0x40; Button: Zone 3 Vol-

266.0.5 Example

Command: PIN 000 – On/Off

Literal symbols (hex): `ffff00 a2888a2aaaa8888aa2aa2220`

(PREAMBLE SYNC PIN COMMAND TAIL)

266.1 TPMS TX

TPMS TX transmits simulated Tire Pressure Monitoring System (TPMS) sensor signals. It is the companion transmitter to TPMS Rx and can replay sensor data previously saved by the receiver app.

> **Note:** Transmitting fake TPMS signals in the 315 or 433 MHz band may interfere with vehicle dashboard readings. Use only on your own vehicles and equipment in a controlled environment.

This is an external app (`.ppma` file on the SD card).

267 Settings

- **Frequency**: **314.9 MHz**, **315.0 MHz**, or **433.9 MHz**. Default: 314.9 MHz.
- **Sensor type**: **Schrader**, **FLM_64**, **FLM_72**, **FLM_80**, or **GMC_96**. Determines the encoding and signal type used.
- **Pressure unit**: kPa, PSI, or BAR.
- **Temperature unit**: °C or °F.
- **Transponder ID**: 24-bit or 32-bit hexadecimal sensor ID depending on sensor type.
- **Pressure**: Pressure value to transmit. Maximum is 340 kPa for Schrader/FLM types, 701 kPa for GMC_96. A `!MAX` warning is shown if the value exceeds the type's encoding limit.
- **Temperature**: Temperature value to transmit.
- **Flags**: Additional status flags byte.
- **Repeat**: Number of transmission repetitions. Default: 5.
- **Load**: Load a previously saved sensor file from `TPMS/TPMS_.TXT` (files created by the TPMS RX app's Save button).
- **Save**: Save the current settings to a file in the `TPMS/` directory for later replay.
- **Transmit or Stop**: Starts or stops the transmission.

268 Settings Persistence

Pressure unit and temperature unit are saved to `tx_tpms.ini`.

268.1 Transceivers

268.2 Microphone Transceiver

Checking Portapack-Havoc repository , we can see , that this excellent app was initially developped by Furrtek in 2017.

- "**Microphone FM transmit with CTCSS**", providing support to Narrow Band FM Transmitter + CTCSS & Receiver in Half Duplex operation, like a walkie-talkie (two-way directional voice communication but one at a time).

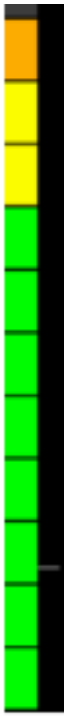
Later on , thanks to many other great sw developers , gradually it has been added many more nice functionalities (VOX control, Roger Beep,...) ,and improving it day by day ... And from Sept −2020, it was also added the support of multi analogue mic Modulation types in half duplex TX or RX, highly appreciated specially by all ham amateur radio community. Those mod types are widely used in LF, HF , VHF, 2m band , maritime communications ,airport airband communications, UHF PMR446,... And since them we are currently supporting those following ones (valid from , Nightly Release - 2022-10-17 onwards) :

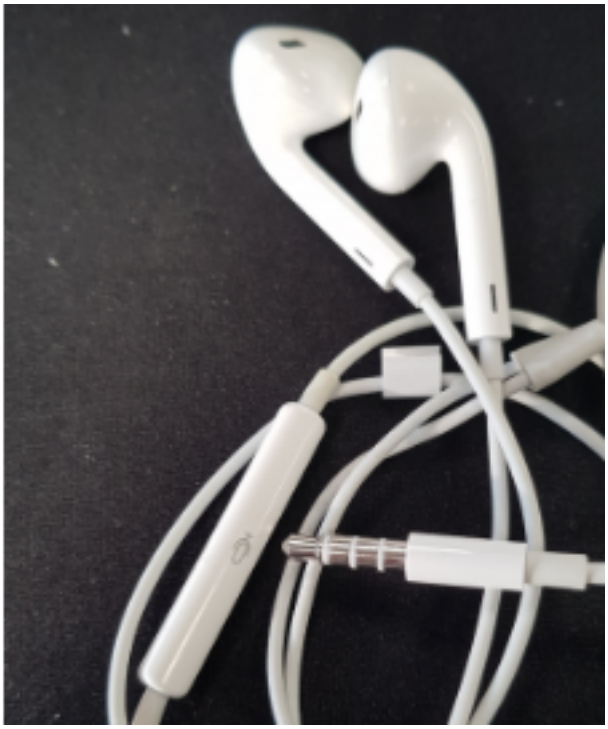
- Narrow and normal band FM (NBFM/FM),
- Wide band FM (WBFM),
- Double side band AM with carrier,(DSB-C,AM),
- Upper Side Band (USB),
- Low Side Band (LSB)
- Double Side Band with suppressed carrier (DSB-SC).

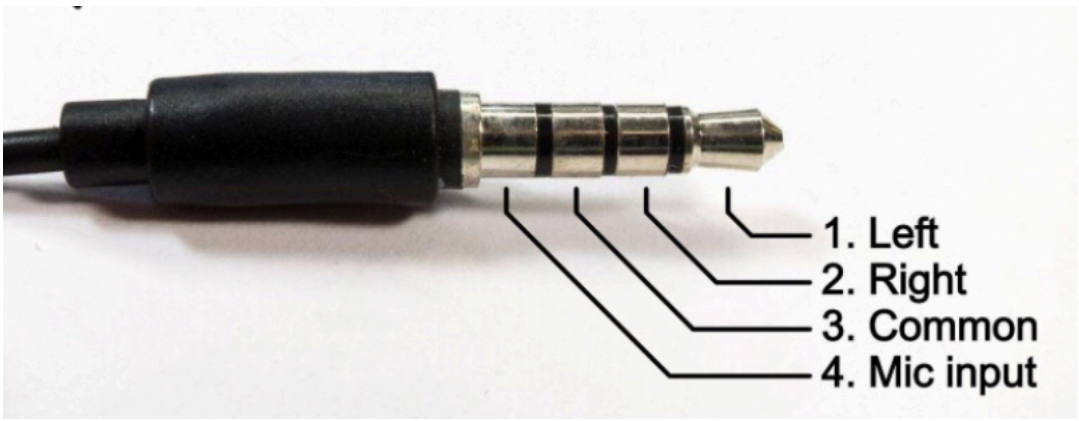
268.3 # Key Controls

- **VU-meter:** Just launching that Mic App, the left side peak audio VU-meter is active, and you can plug your head mic set and confirm its sensitivity and correct operation. The sensitivity of the microphone is

shown on the lefthand side of the LCD screen, and should be configured so the range is green for most of the audio and never hits red to limit over deviation of the signal. Note, this feature is not working once we activate below “Rx audio listen” ; but in any case, when pressing PTT, in TX mode, it always works.



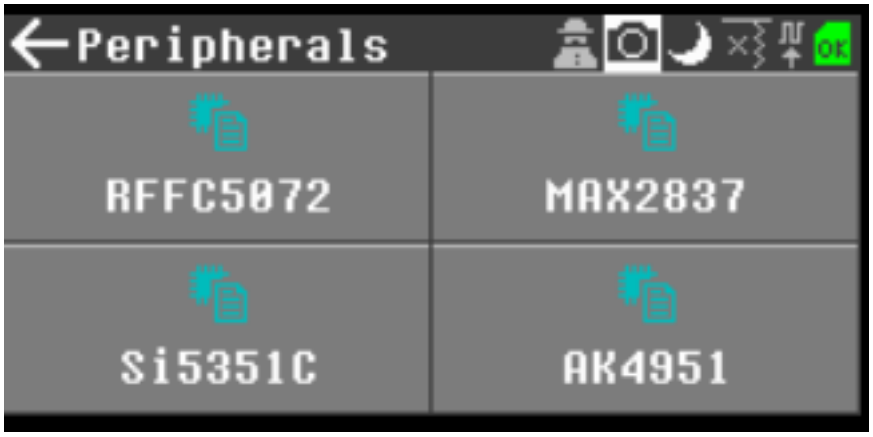


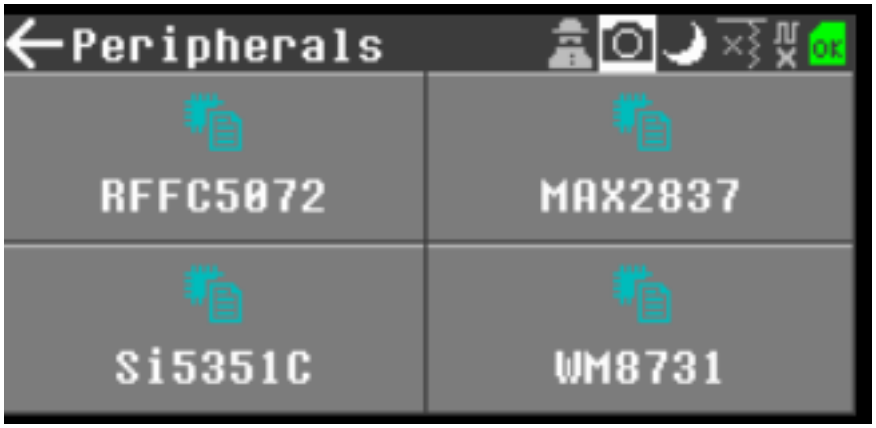


- **MIC. GAIN:** Cursor selection and use rotary encoder is used to select a fixed gain of x0.5, x1.0, x1.5, x2.0. The setting needs to be selected based, on the Microphone used which is connected via the Headset/ Microphone socket (standard smartphone 4 segment 3.5mm connector).

That mic gain adjustment is in fact a post scaling by above factors (x0.5, x1.0, x1.5, x2.0) , of each captured mic voice data . Then ,that adjustment should be only used,to make mic gain fine tuning of the non distorted voice . But if the captured data is already distorted , due to too much mic sensitivity , or to close mic-mouth distance, or too loud voice , the ADC saturation , should be addressed by other means (ALC or Boost adjustment (*1) , or increasing mic-mounth cms distance , or reducing the voice loudness.)

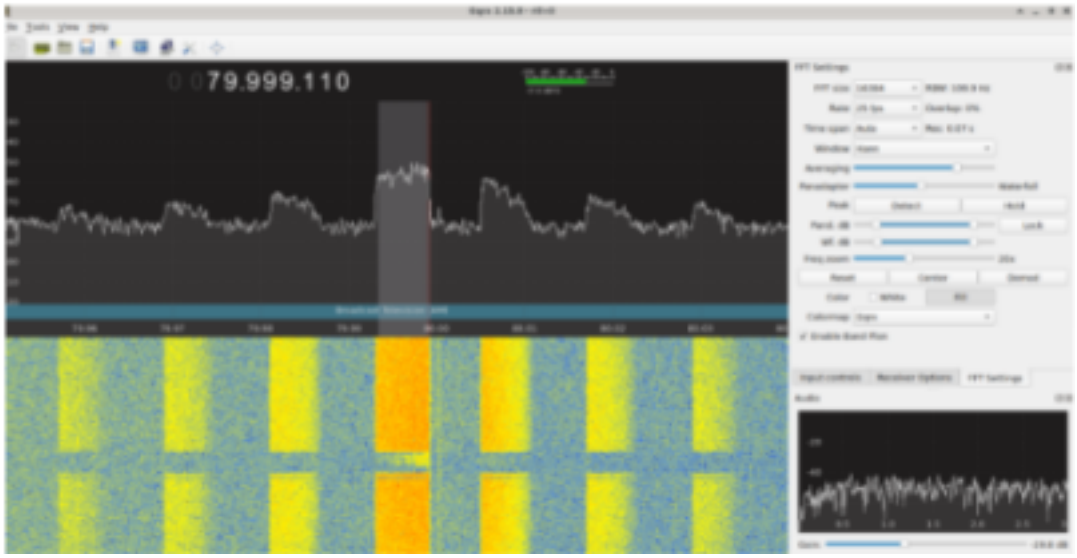
If you want to know the audio codec IC of your Portapack board, you do not need to dissassembly the boards, just go to the “debug” -> peripherals menu , and you will see which peripheral device, you can explore, looking its register map values. In the audio block icon , you will see the audio codec IC name , mounted on your Portapack board : AK4951 or WM8731. (see annexed photo of both two devices)

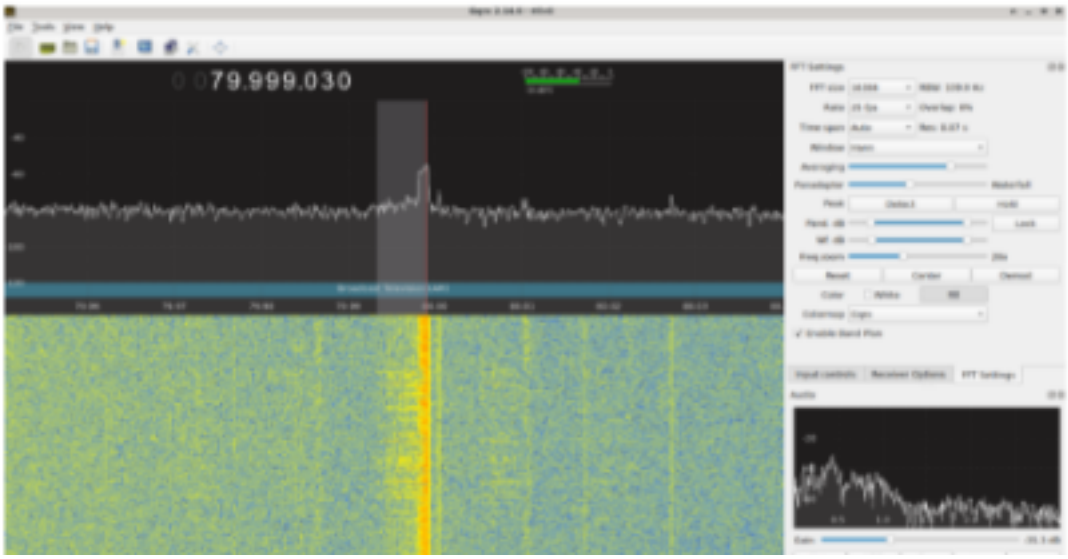
- 



Automatic mic Level Control “ALC” (AK4951 sound codec IC) or “Boost” pre-amplifier mic control adjustment (WM8731 audio codec IC) : (*1)

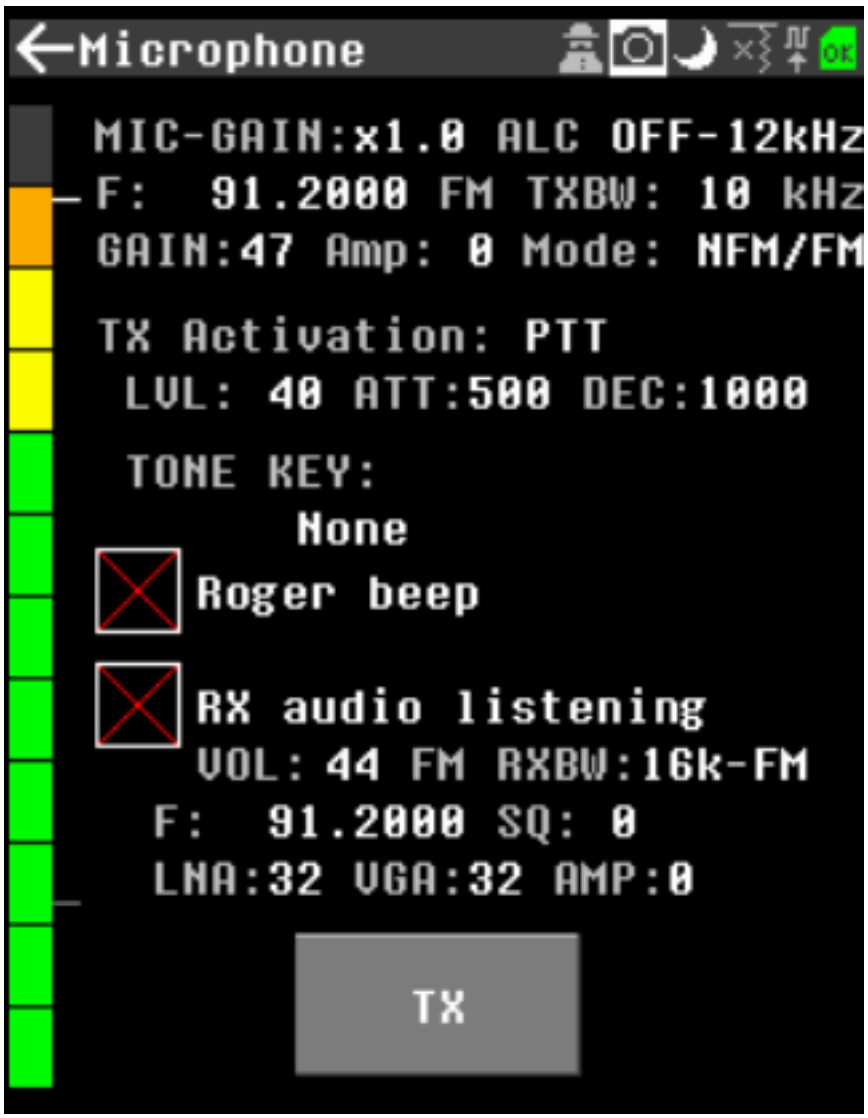
To minimize that mic ADC saturation (and consequently, audio distortion , and spectrum harmonics radiation), we introduced ALC(AK4951) and Boost(WM8731) control options.

- 



With proper adjustment , we can solve that mic ADC saturation, and adapt correctly to our mic gain sensitivity , and the distance mic-mouth from the speaker , and the user voice loudness … That proper settings , can be done, selecting the best ALC or Boost option (checking that the mic voice sound Vumeter (left mic volume peak bar indication) is below 80-90%, in the green area, see annexed pictures) (example LSB with low modulation index)

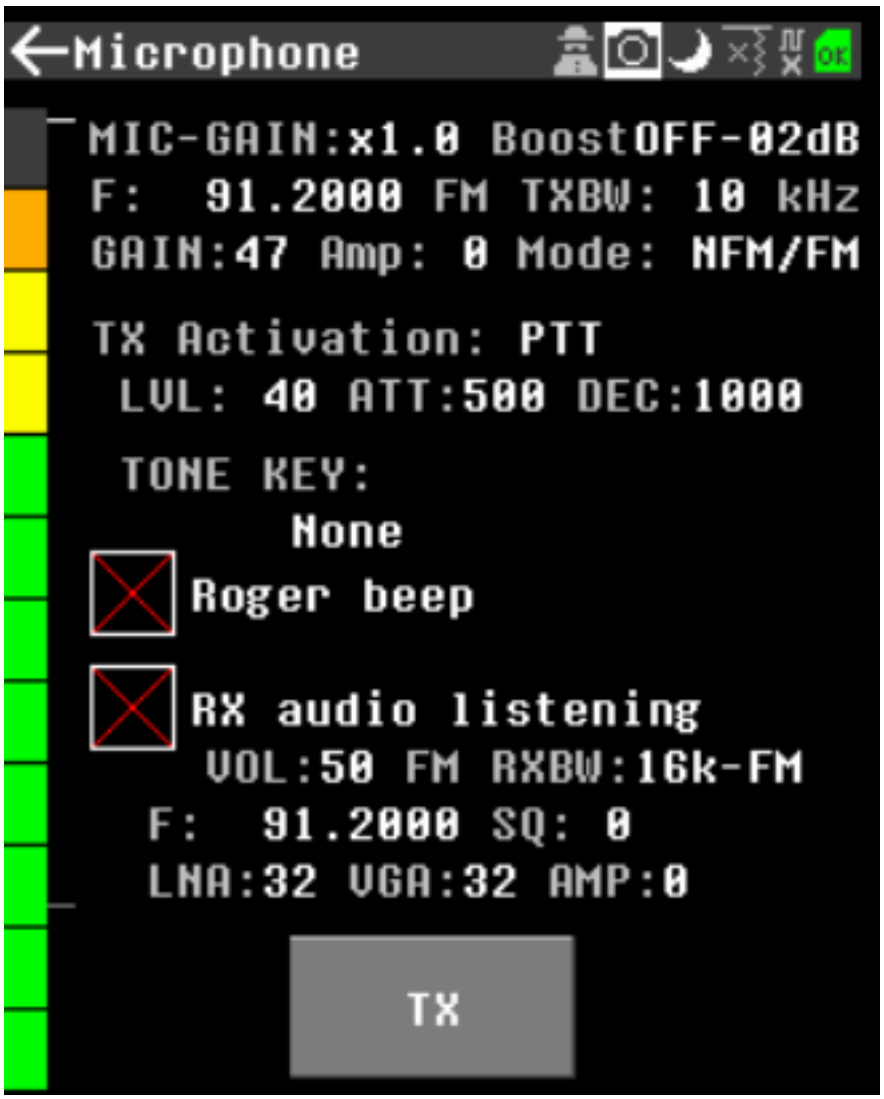
1. • **ALC Automatic mic volume Level Control** (available in Portapack boards that uses AK4951 audio codec platform) : With those “ALC” GUI options, user can adjust from +12 dB’s to −12 dB’s mic gain. For instance , if we want ,to talk , like handsfree mode , and pick up all the voices and sounds inside our room (from 50 cms till several meters distance from the mic, we can select +12 dB’s) , If we want to talk around 15 to 20 cms , we can select +0dB’s or −3dB’s depending on the loudness of our voice … If we want to avoid saturation when shouting , or talking touching the mic with the lips , you can reduce it til −9, −12 dB’s If you leave the default OFF settings , you will have exactly same Mic settings as in previous Mayhem FW versions <=1.4.3 , without activating any digital ALC block of in the AK4951. When we saiy , “OFF - 12Khz” it just means that the audio base band output from the IC 4951 has this initial max bandwith, but obviously ,later , when it is post-processed and used for modulation , it is also reduced it. We tried to use and combine several nice programmable digital features, like , ALC , Wind Nose Filter, LPF , and boosting peaking Equalizer, and also Pre-amp mic level. Ex. +12dB-6Khz , means activated ALC with maximum gain of 12dB’s and with digital Low Pass Filter of 6kHz .

- 
Microphone
MIC-GAIN:x1.0 ALC OFF-12kHz
F: 91.2000 FM TXBW: 10 kHz
GAIN:47 Amp: 0 Mode: NFM/FM
TX Activation: PTT
LVL: 40 ATT:500 DEC:1000
TONE KEY:
None
Roger beep
RX audio listening
VOL: 44 FM RXBW:16k-FM
F: 91.2000 SQ: 0
LNA:32 VGA:32 AMP:0
TX

USER GUI LABEL : ALC - Bandith :	PRE-AMP MIC	DIG-ALC VOL& REF (max gain at ALC recovery operation)	DIG -WIND NOISE FILTER (Auto detected dynamic HPF)	DIG EQUALIZER (f0, BW) *3dBs peaking	DIG 1st Order LPF (fcut)
OFF-12kHz (Default options)	+21dB's (current)	OFF	OFF	OFF	OFF
+12dB-6kHz	+21dB's (current)	+12dB's	ON	OFF	6 khz
+09dB-6kHz	+21dB's (current)	+09dB's	ON	OFF	6 khz
+06dB-6kHz	+21dB's (current)	+06dB's	ON	OFF	6 khz
+03dB-2kHz	+21dB's (current)	+03dB's	ON	EQ1 ~2 kHz (f0:1.150Hz, fb:1.700Hz)	3,5khz
+03dB-4kHz	+21dB's (current)	+03dB's	ON	EQ2 ~2700 Hz (f0:1.475Hz, fb:2.450Hz)	4khz
+03dB-6kHz	+21dB's (current)	+03dB's	ON	OFF	6 khz
+00dB-6KHz	+21dB's (current)	+00dB's	ON	OFF	6 khz
-03dB-6KHz	+21dB's (current)	-03dB's	ON	OFF	6 khz
-06dB-6KHz	+21dB's (current)	-06dB's	ON	OFF	6 khz
-09dB-6KHz	+18dB's	-09dB's	ON	OFF	6 khz
-12dB-6KHz	+15dB's	-12dB's	ON	OFF	6 khz

2. • **Boost mic** (in Portapack boards that uses WM8731 audio codec platform) :

With those "Boost · GUI options, user can adjust from +12 dB's to −08 dB's mic gain. We have added five user "**Boost**" options , activating on/off , the mic-boost pre-amplifier (+20 dB's) and playing with internal captured data, to allow smaller steps. In order to avoid mic distortion, We only recommend to use the pre-amplifier boost on , when we are using low sensitivity mics or when we are keeping some distance (example >15 cms) to the mic :

- 
Microphone
MIC-GAIN:x1.0 BoostOFF-02dB
F: 91.2000 FM TXBW: 10 kHz
GAIN:47 Amp: 0 Mode: NFM/FM
TX Activation: PTT
LVL: 40 ATT:500 DEC:1000
TONE KEY:
None
Roger beep
RX audio listening
VOL:50 FM RXBW:16k-FM
F: 91.2000 SQ: 0
LNA:32 VGA:32 AMP:0
TX

WM8731 GUI FM relative Gain Mode	Mic-boost (on/off) +20dB/0dB	Internal effective audio capture gain	Relative mic gain dB's Respect current fw ref (+20dB's)
+12 dBs	+20 dB's (**Mic boost ON**)	+12 dBs	+20+12=32 dB's 32-20 (fw 1.5.x) = +12 **(+12dB ref fw 1.5x)** (No sat. > 15 cms)
+06 dBs	+20 dB's (**Mic boost ON**)	+6 dBs	+20+6=26 dB's 26-20 (fw 1.5.x) = +06 **(+06dB ref fw 1.5x)** (No sat. > 15 cms)
+04 dB's	+0 dB's (Mic boost OFF)	+24 dBs	-0+24= +24 dB's 24-20 (fw 1.5.x) = +04 (+04 vs dB ref fw 1.5x) (No sat. > 2-3 cms)
-02 dB's (Default option)	+0 dB's (Mic boost OFF)	+18 dBs	-0+18 = +18 dB's 18-20 (fw 1.5.x) =-02 (-02 dB's vs ref fw 1.5x) (No sat. > 2-3 cms)
-08 dB's	+0 dB's (Mic boost OFF)	+12 dBs	+0+12 = +12 dB's 12-20 = -08 (-08 dB's vs ref fw 1.5x)

We also added several new features : a new check box (next to Roger beep) to the user to be able to select Separated (default)) / Common freq. RX tuning control respect TX freq., another one “Hear Mic” (see attached new updated GUI pictures), and small new TX feature, new GUI about IQ TX phase calibration adjustment . (It is in the bottom right part of the screen) : “TX-IQ-CAL”, and a TXBW selection in SSB mode :

- **F:** This field set the TX Frequency for the transmitter.(*1) Use rotary encoder dial to adjust by Step value (it uses a default step value, or the step value saved in SETTINGS\tx_mic.ini file if “Load App Settings” is enabled). Also , be aware, that like in all other App’s, it is available the the called tuning feature , by some users : “freq field trick” : a long press of the Select button when you are in the Frequency field , it will be highlight in blue color selected frequency digit ,it allows adjusting frequency digits individually (you can move the cursor left/ right till selecting the frequency digit decimal that you want to increase or decrease with the rotary encoder).
- **F_RX:** This field set the RX Frequency for the receiver. (It also applies all previous (*1)).
- **TXBW:** This field sets the FM BandWidth of the transmission under normal conditions from 0-150 kHz. (In fact,in FM mode it is the +/- FM deviation in Khz that we will apply to the modulator.). Recently , also in SSB mode (USB or LSB) , we allow to the user to set up audio baseband BW to be transmitted (2KHz, 3kHz). (sse annexed example spectrum)

- **GAIN:** Is set for TX LNA(IF) and (0-47)
- **AMP:** RF TX amplifier , 0dB or 14dB.
- **MOD:** to set the selected analogue TX & RX modulation type : NFM-FM // WFM // AM // USB // LSB // DSB

  - Narrow band FM (NFM-FM) TX ,and supporting NFM-FM RX : BW: 8K5 (8K50F3E) , 11Khz (11K0F3E) ,and FM RX BW :16Khz (16K0F3E)

  - Wide band FM (WBFM) TX , and supporting WFM RX of the following BW: 200kHz (Emissions Designator 200KF3E), 180khz and 40khz (see more details below)

  - Amplitude Modulation (AM) TX fixed to the standard AM-6K00A3E, but supporting AM RX with two selectable receiver bandwidth filters, BW : 9 and 6kHz (covering both side lateral bands, AM DSB with carrier AM-9K00A3E or AM-6K00A3E)
  - Upper Side Band (USB) TX , supporting USB RX with BW : 3kHZ (SSB-3K00J3E)

  - Lower Side Band (LSB) TX, supporting LSB RX with BW : 3kHZ (SSB-3K00J3E)

  - Double Side Band with suppressed carrier (DSB), supporting DSB RX with BW : 6 kHZ (covering both side lateral bands)

- **TX Activation:** Field can be selected for one of three settings. Off, PTT, AUTO. In the PTT setting, the PortaPack will transmit when the TX button and the bottom of the App screen is pressed on the touch screen. The AUTO is also known as a VOX control (automatic PTT transmission activated by voice). In AUTO the Receiver section below ("Rx audio listening") is not available , but the triggering is set by the following 3 settings.

  - **LVL:** The level that triggers the transmission is set by the audio level going over the threshold value set (0-255). This can be seen as a grey dash marked next to the vu-meter and shows its current setting compared to the microphone level.

  - **ATT:** This is the attack time of the microphone audio level must be above the set threshold to start transmitting (0 to 999mS). Higher attack helps avoid false triggers but might cut off the first words you say.

  - **DEC:** This is the decay time of the microphone audio signal falls below the threshold level be for the transmission is stopped (0 to 9999mS). Lower decay avoids silence at the end of the message but might cut you off in the middle of a sentence. Adjust levels depending on your speaking habits.

- **TONE KEY (CTCSS) :** Continuous Tone-Coded Squelch System or CTCSS is one type of in-band signaling that is used to reduce the annoyance of listening to other users on a shared two-way radio communication channel, sometimes referred to as tone squelch.
- Note1 the level of the Tone % compared to the Microphone setting is set in Options >Audio.
- Note2, that feature is only available in both FM modes (NFM-FM or WFM).
- **Check box Roger beep:** This tick is activating an ending sequence of six consecutive digital synthesized audio tones , to indicate that the operator has concluded speaking.
- Note: that feature is not available in SSB modes (USB , LSB)
- **Check box Hear Mic:** This feature will allow to the user to check his Mic through the combined headphone and mic headset , without the need to use an additional receiver.It works for both Portapack

audio codec platforms : AK4951 / WM8731 . You can adjust the Mic sound volume with the same below receiver VOL .

- Notes: to avoid user confusions, that audio sound feature is disabled when user selects to hear the receiver demodulated sound . In both audio codec platforms, when clicking that “Hear Mic” , we are changing the default pre-setting ALC or BOOST option to the best that has maximum audio codec volum sound …
- **Check box F = F_RX:** (Added from [Nightly Release - 2023-04-05] It allows to the user to select Separated or Common tuning frequency from Receiver(RX) to the Transmitter(TX). Once is marked , we have a common same frequency (F = F_TX) and therefore, to not confuse to the user , we hide the bottom independent F_RX field . (as you could see in the two above GUI pictures)
- **Check box RX audio listening:** The Tick box to select this item is difficult to select with cursor having to go down and up though item, and with some luck you may enable the box. It is better to select from the touch screen. If this is enabled then it will turn on the Audio Receiver and with the following setting will allow you to listen to a receive channel when not transmitting. The TX-RX timing gap is not known. The frequency can be set separately or Common to that of the transmitter section above. The following setting can be applied to the receiver.
- Note : this feature is not available when we are in AUTO VOX control .
- **VOL :** Audio receiver volume control.
- **RX BW:** This GUI option , allows several analogue demodulation options :

in NFM-FM mode to set up the receiver Bandwidth setting. It can be set to either

- 8k5kHz- NFM (delta FM deviation +/- 1,25 khz, FM index modulation:0,4)
- 11Khz - NFM (delta FM deviation +/- 2,50 khz, FM index modulation:0,75)
- 16kHz - FM (delta FM deviation +/- 5,0 khz, FM index modulation:1,6)

In WFM mode

- 200Khz FM bandwidth , for commercial FM stations with soft band pass transition. And recently also added the bandwiths BW: 180khz and 40khz (that last one, mainly for NOAA analog FM APT - 137 Mhz reception), both with sharp stop band transition.

In AM

- DSB1 −9Khz, DSB2-6Khz bandwidth.

In USB , upper +3khz single side band.

In LSB , lower −3khz single side band.

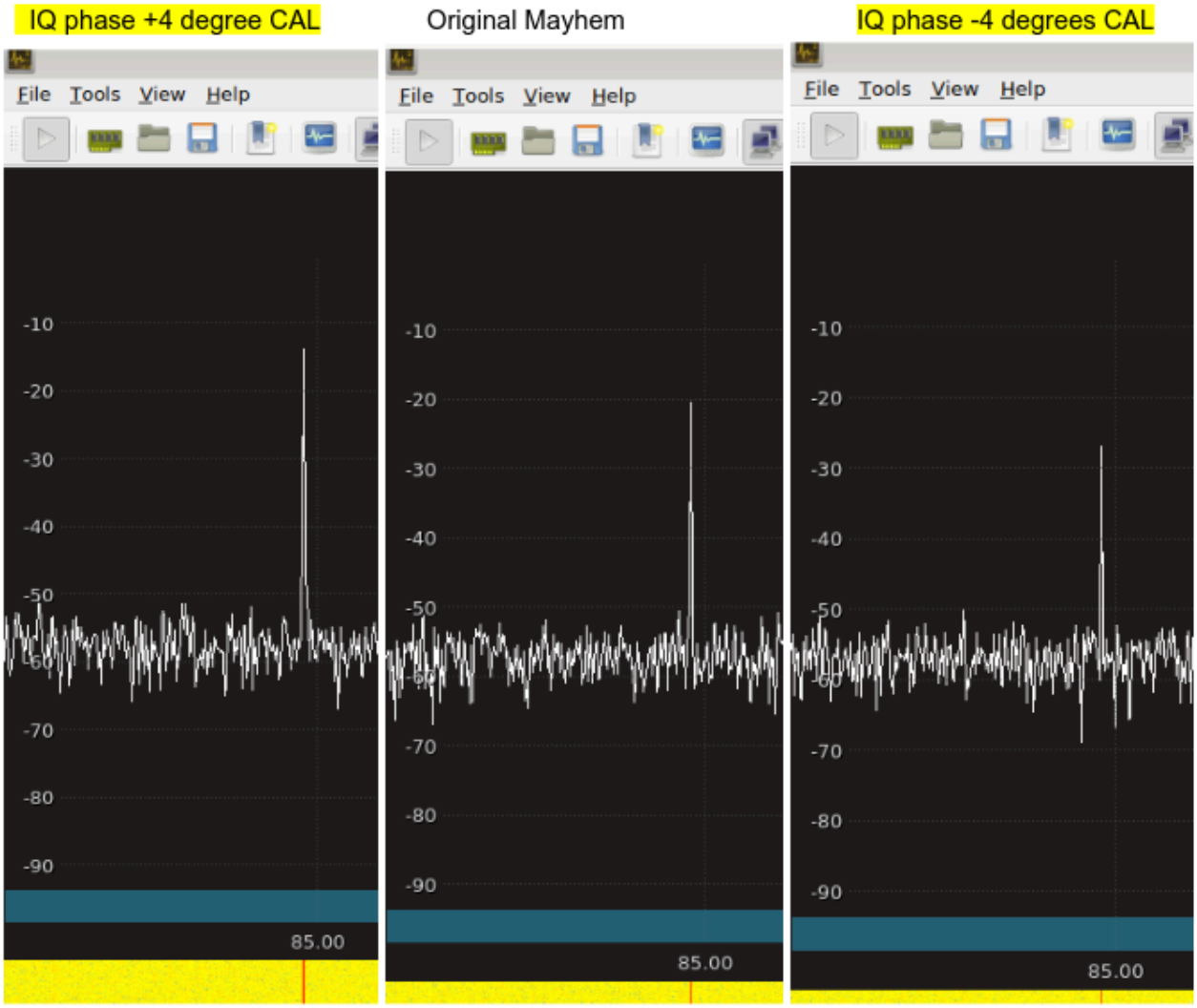
In DSB-SC , we can select to demodulate any of both SSB bands (LSB, USB). In DSB-SC mod type , both side bands has exactly the same modulated content, but sometimes we may have more particular interference signals in one side than the other.

- **SQ:** This can be set between 0 and 99 typically 50 is threshold.
- **GAIN:** The RX gain setting of LNA(IF) (0-40)
- **VGA:** RX (Baseband)(0-62)
- **AMP:** RX LNA pre-amplifer 0=0dB or 1=14dB.

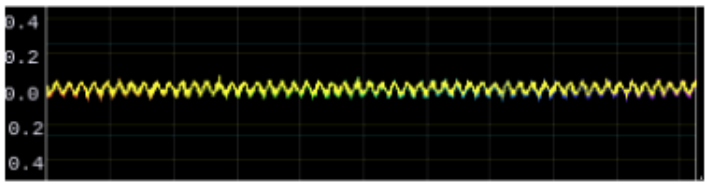
- **TX-IQ-CAL:** TX I/Q channel phase calibration, to adjust the imbalance IQ modulator phase error. The adjustment range is (+4º ...-4º). Specially usefull to optimize the SSB (USB-LSB) or DSB-SC , Suppressed carrier . In Hackrf versions r1 to r8 , we are using max2837 , that has a calibration register settings of 5 bits , in Hackrf r9 , we are using max2839, that has 6 bits to cover the same range (+4º,..., −4º phase adj).

At the moment we do not have any automatic IQ imbalance phase error calibration process. You will need some other SDR receiver , and check the suppressed carrier level when you push PTT _TX in SSB (USB or LSB) without any modulation, and then go to the right modify the “TX-IQ-CAL” value , and press again PTT_TX till getting the lower attenuated “suppressed carrier.

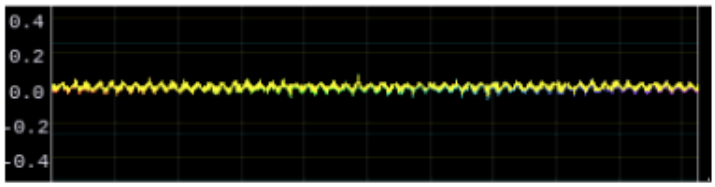
As example , see the attenuation of the residual SSB carrier effect, when adjusting that IQ Phase calibration value in an H1R1 with max2837,



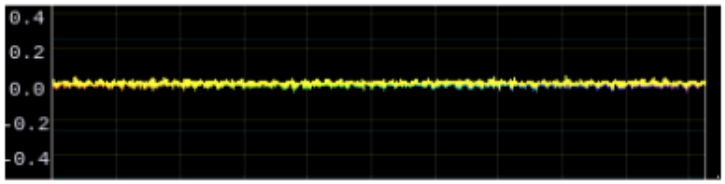
Checking in that above sample ,temporal demodulated residual oscilloscope simulation by SDRangel :
00 (+4deg (Q lags I by 94degs)



15 (+0deg)



31 (-4deg (Q lags I by 86degs)



Note this Mic application while performing a transmit function may well over deviate and generate unwanted harmonics.

268.4 KISS TNC

The **KISS TNC** app turns the PortaPack into a bidirectional **KISS TNC** (Terminal Node Controller) over the USB serial port. It receives APRS/AX.25 packets from radio and forwards them to a connected PC as KISS frames, and transmits KISS frames received from the PC over radio as AFSK 1200 baud AX.25.

This is an external app (`.ppma` file on the SD card).

268.5 What is KISS TNC?

KISS (Keep It Simple, Stupid) is a protocol that allows a PC application to send and receive AX.25 packet radio frames through a TNC over a serial connection. With this app, the PortaPack acts as the TNC, and the USB serial port is the link to the host PC. Any APRS or packet radio software that supports a serial KISS TNC can be used — for example Direwolf (as a client), APRSISCE/32, APRSdroid, or custom scripts.

268.6 Controls

- Field: Frequency; Description: RX/TX frequency. Default: 144.390 MHz (APRS standard)
- Field: RF Amp; Description: RF amplifier on/off
- Field: LNA; Description: LNA gain
- Field: VGA; Description: VGA gain
- Field: RSSI; Description: Signal strength indicator
- Field: USB; Description: Shows Connected or Disconnected — whether the PC has opened the USB serial port
- Field: RX; Description: Count of AX.25 packets received from radio and forwarded to the PC
- Field: TX; Description: Count of frames received from the PC and transmitted over radio
- Field: Console; Description: Live log of received packets, shown as SOURCE>DESTINATION

The status line at the top toggles between **Listening** (RX active) and **Transmitting** (TX in progress).

268.7 Connecting a PC

Connect the PortaPack to the PC via USB. The USB serial port appears as a standard CDC serial device. Open it in your packet radio software and configure it as a KISS TNC at any baud rate (the USB CDC connection is not baud-rate limited).

NOTE On Linux the device typically appears as `/dev/ttyACM0`. On Windows it appears as a COM port. No driver installation is needed on Linux or macOS; Windows may require the standard CDC driver.

268.8 Operation

- **RX**: The app continuously listens on the configured frequency. When a valid AX.25 packet is decoded, it is wrapped in a KISS frame and sent to the PC over USB serial. The RX counter increments and the packet's source and destination callsigns appear in the console.
- **TX**: When the PC sends a KISS data frame (command byte `0x00`) over USB serial, the app switches to transmit mode, sends the AX.25 frame as AFSK 1200 baud Bell 202, then automatically returns to RX mode. The TX counter increments for each transmitted frame.

WARNING The maximum transmittable frame payload is **200 bytes**. Frames larger than this are silently discarded.

268.9 Technical details

- Parameter: Modulation; Value: AFSK Bell 202
- Parameter: Baud rate; Value: 1200 baud
- Parameter: Mark or Space; Value: 1200 Hz or 2200 Hz
- Parameter: Default frequency; Value: 144.390 MHz
- Parameter: Max TX frame size; Value: 200 bytes
- Parameter: KISS framing; Value: Standard KISS (0xC0 FEND, 0xDB FESC escaping)

268.10 Recon

269 Recon App

- Recon App
  - Important note
  - Introduction

    - In AM/NFM/WFM mode
    - In SPEC mode
  - Limitations
- TODO first
  - Common defaults options to set
- Main Screens
- CONFIG
  - Main RECON settings page
  - More CONFIG settings page
- Recon/Search/Manual
  - RECON mode
  - SCAN mode
  - MANUAL-S
  - Wait/lock Coloring conditions:
- Color meaning for main freq
- Wait modes
- Lock counts
- Matching modes
  - Continuous
  - Sparse
- Frequencies to scan
- Modulation Mode
- Persistant settings
- Freqman file format
- HamRadio type
- Repeater type
  - In Recon, when current type is Repeater and repeater mode is DISABLED
  - In Recon, when current type is Repeater and repeater mode is ENABLED
  - Advanced Repeater configuration
    - Repeater half duplex dual direction example
- Workflow and tips
  - Classic workflow
  - Repeater workflow:
  - Tips:
- Troubleshooting
- Scan speed vs Chosen modulation mode
- Power consumption

269.1 Important note

Repeater and all associated repeat modes are under development. The documentations related to it are for devs/nightly testers. You will not yet find it in stable.

269.2 Introduction

The Recon app is full rework of the Scanner app, offering different possibilities and customisations.

Both are using all the frequencies in their hand and pause on a frequency when matching criteria (like modulation, amplitude,…)

The Recon is using a different approach than the Scanner app it is originated from: it does not use a thread for frequency shifting and else rely on the statistics update event message to do the work.

This allow perfect matching between 'shifting to a new frequency' and 'waiting for the first statistic update of that frequency'. The Recon app will stop into any frequency carrying a signal strong enough. You can adjust the signal power threshold with the SQUELCH

269.2.1 In AM/NFM/WFM mode

The statistics only update once each 100ms, the minimum lock waiting time is also 100 ms, and the quickest frequency scanned rate is 10/s.

269.2.2 In SPEC mode

The statistics are updated differently in regard of the bandwidth used.

The average locking times or scan rates are in the following the table (they may change when there are baseband or record process modifications)

- average of 10/s from 12.5k to 1500k
- average of 5/s from 1750k to 3000k
- average of 3/s from 3000k to 5000k

269.3 Limitations

The portapack hardware is limited, and so is the list of elements that you can load.

A maximum number of 115 elements is allowed. One more element is allowed to detect that the file was truncated.

If a file was loaded and truncated, the displayed list name and current entry description are in yellow instead of other colors.

270 TODO first

If you are launching the Recon app for the first time, chances are high that you do not have the wanted options selected by default.

We highly encourage you to go and check what's under the "CONFIG" button and get yourself used to the 'Main' and 'More' pages under it.

270.1 Common defaults options to set

Go into "CONFIG":

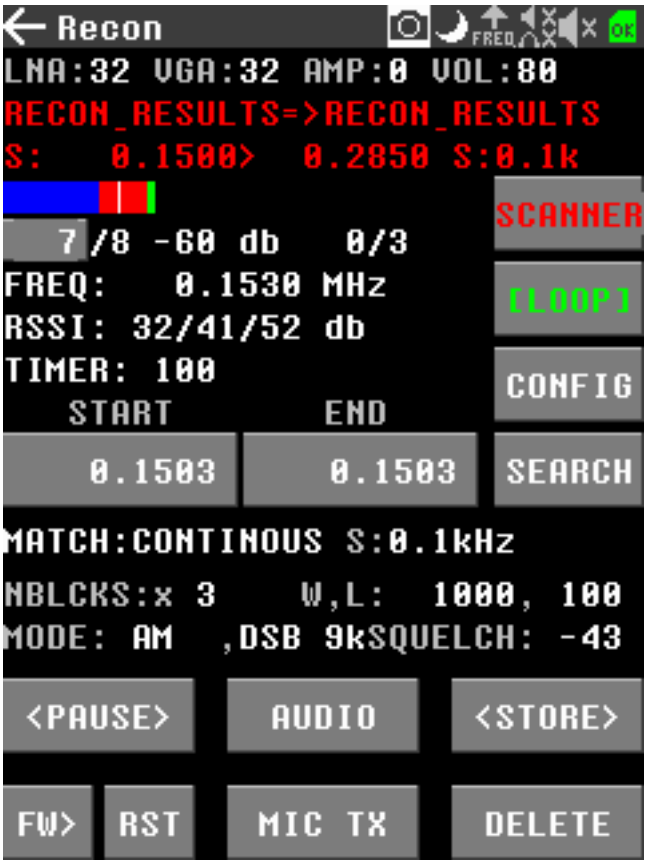
Main:

- Check input file and output file name
- Check options under it. A lots are complaining that the search is not starting by itself, but they do not have 'autostart recon' checked.

More:

- Check that 'input: load freqs' and 'input: load ranges' are checked

271 Main Screens



Buttons and information description, from top to bottom, and left to right. [NAME] is used to mark a button or gui element that the user can change, else it's a description of an onscreen information.

- [LNA] , [VGA] , [AMP] , [VOL] => gains, amplification on/off , volume
- Current input file name => will be in yellow if file contain too much lines or red if an error description is shown, or follow mode colors
- Current entry description => input name, will be in yellow if file contain too much lines or red if an error description is shown, or follow mode colors
- [RssiGraph], [MODE] => radio levels indication, jump to level app on click. Mode is one of [RECON/ SCAN/MANUAL] and show actual mode, and switch to next mode on click
- [XXX] / XXX , XX db, XX/XX value => index of the current frequency in the loaded list (move with encoder, or set a value by clicking), number of frequencies in the list, actual DB value, number of locks or number of needed locks for a match, button to the settings page. Will be in red if file contain too much lines or if an error description is shown
- FREQ: XXXX , [LOOP] => Current frequency , [LOOP] button to control continuous mode or not (green is on, white is off)
- RSSI: XXX/XXX/XXX => Current min/med/max RSSI values in db
- TIMER, CTCSS, [CONFIG] => elapsed timer for current frequency lock (0 if no lock) , detected tone if NFM is selected, and CONFIG button to set things like input/output files
- [START], [END], [SEARCH] => set the manual start and end of a range and launch a search on it. These values will be updated by the search if auto update m-ranges is checked. If highlighted, you can use the rotary encoder to adjust start or end
- [MatchMode] , [STEP] => Matching mode, manual range steps
- [NBLocks] , [W] , [L] => NBLocks is the numbers of locks we need to have a match. [W] (wait after match) is the time we will stay on the frequency if it's reaching nb_locks during lock_wait (continuously or sparsely). If wait is a negative number, then it represent the time we are staying on a matched frequency waiting for new activity, and a new lock during the wait restart the counters (you keep staying on it until

a full wait without a lock is reached). [L] (lock duration) is the maximum time we stay on a freq waiting for all locks in SPARSE match mode, and the time we are waiting for the first lock in CONTINUOUS match mode.

- [MODULATION] , [BW] , [SQUELCH] => actual modulation and bandwidth ,squelch is the level of DB needed to start to lock on a signal
- [PAUSE] , [AUDIO] , [STORE] => pause or resume the search. If highlighted, you can use the rotary encoder to manually step in the frequencies/ranges, jump to Audio App, store actual frequency in output file. Audio button is becoming red in auto record modes to indicate that a record in ongoing
- [FW], [RST], [MIC TX] , [REMOVE/DELETE] => forward/reverse , reset search (it's restarting from the beginning of input file), jump to Mic app, Remove a frequency from active/loaded list in RECON mode or Delete from both active/loaded list and output file in SCAN mode

272 CONFIG

272.1 Main RECON settings page

- input file => File from which we will load frequencies or ranges to search (default FREQMAN/RECON.TXT)
- output file => File into which we will save frequency using STORE button or autosave mode (default FREQMAN/RECON_RESULTS.TXT)
- output file name => The name of the current output file. Can be edited when clicking on it so you can set a new filename
- autosave freqs => During the search, matching frequencies will be saved without clicking on STORE. No duplicates will be made
- auto start search => Search will start when entering the app, or when going in and out of CONFIG
- continuous => If checked then the search will loop when reaching boundaries of the loaded input file or the manual range search
- clear output at start => If checked then the output file is blanked at app start. If you're using that feature and want to keep one of your search results, do not forgot to go into filemanager to rename the file before starting the Recon app one more time

272.2 More CONFIG settings page

One of the first two options have to be checked else nothing will be loaded at all and only manual range search will be available

- load freq => allow load of frequencies
- load repeater => allow load of frequencies
- load range => allow load of ranges
- load hamradio => allow load of ham radios
- auto update m-ranges => if checked then the manual range start and stop values are updated using the actual searched range values. If it's actually searching a frequency, manual ranges are untouched
- record locked period=> activate auto record during locked wait (green) state. In audio modes, a wav file with actual time of capture as filename is created in AUDIO directory. In SPEC mode, a raw c16 with actual time of capture as filename is created in CAPTURES directory.
- repeater,[keep or delete] => activate repeat function, keep or delete repeated files. nb => number of repetitions
- amp => enable/disable amp when TX repeat is on. gain => gain to use when TX repeat is on

!! WARNING: YELLOW COLORED OPTION ARE FOR TX CONFIG !!

!! WARNING: USING REPEATER ACTIVATE RECON TX ABILITIES !!

!! WARNING: BE CAUTIOUS WITH 'auto record locked period' OPTION. DON'T FORGET CLEANING UP FROM TIMES TO TIMES, TOO MUCH FILES WILL CRASH THINGS !!

273 Recon/Search/Manual

When using the Recon app, 3 modes are accessibles.

You can switch from RECON to SEARCH using the GUI button, and to Manual search using the Manual-S button (just under the RECON/SEARCH button).

273.1 RECON mode

Mode button is in blue, and the label is 'RECON'.

In that mode it takes what you choose in "CONFIG/select input file" and use it as the list to search. The matched frequencies (if autosave is checked or 'add' is used) are put in what you choose in "CONFIG/select output file".

REMOVE is removing current entry from active/loaded list. No files are touched.

STORE is adding current entry to output file.

In that mode you have an input search/recon list, and an output file. You can temporary 'disable' some entries by a hit on REMOVE. Clicking on RST is reloading the fully populated list.

273.2 SCAN mode

Mode button is in red, and the label is 'SCAN'.

In that mode it takes what you choose in "CONFIG/select output file" and use it as the list to search.

The matched frequencies (if autosave is checked or 'add' is used) are put in what you choose in "CONFIG/select output file".

DELETE is removing current entry from active/loaded list and from output file.

STORE is adding current entry to output file.

In that mode you have the same file used as an input list and an output file.

273.3 MANUAL-S

In that mode RECON or SEARCH mode are invalidated. Values in GUI are used as a single range entry.

The matched frequencies (if autosave is checked or 'add' is used) are put in what you choose in "CONFIG/select output file".

STORE and REMOVE are working on the output file but changes are not reflected in the used list since it's a loaded range.

In that mode you have a single range as an input list, and an output file.

273.4 Wait/lock Coloring conditions:

Lock coloration (only in SPARSE matching mode):

- if lock duration is lower or equal to (min_lock_duration x nb_locks) => value is in yellow to indicate that there is not enough time to match nb_locks. Coloration are not used in CONTINUOUS as lock timer is disarmed at first lock count.

Wait coloration:

- if wait is between [−500,500] ms and not 0, it's red because the audio will have hard time start and stop that quick during consecutive matches
- if wait it > 500 it's the time we are staying on a matching freq before skipping, wait value is in white
- if wait < −500 it's the time we are waiting for new activity before skipping. Any new activity (db>squelch) is resetting the timer to 'abs(wait)', wait value is in green
- if wait == 0 it's recon mode. No wait after match. No audio start/stop. Matching freqs are auto saved according to the options in CONFIG, wait value is in blue

274 Color meaning for main freq

- white => pause or reading signal
- yellow => got a lock, trying to achieve a match (when we have nb_locks counted during lock_wait, continuously or sparsely)
- green => during a search : locked , during a pause: current freq is matching squelch level and nb_locks have been reached

275 Wait modes

- wait is the timer started after a matched frequency

- wait > 0 : after a lock stay on freq for 'wait' msecs before skipping to next
- wait == 0 => immediately leave freq after a lock. Use in conjonction of autosave option to make a quick map of matching frequencies
- wait < 0 : after a lock stay on freq for 'wait' msecs before skipping to next, each new lock (db>squelch) reset the counter to 'wait'

276 Lock counts

Lock count is reset to 0 after using one of the following buttons:

- PAUSE
- FW,RW
- CONFIG
- SQUELCH

277 Matching modes

277.1 Continuous

- measurement is done in lock_wait msecs
- at each loop if the mean db is > squelch, it's a lock
- in that mode you need consecutive nb_locks to match a frequency
- it's quicker because we do not wait lock_wait to get nb_locks, it's matching as soon as the counter is reached. It can also miss signals since it's rigorous and skip if there is a gap between matches

277.2 Sparse

- measurement is done in lock_wait msecs
- at each loop if the mean db is > squelch, it's a lock
- we need nb_locks before lock_wait expires, but we can have some that are not matching, the lock_count is kept until nb_locks or the timer is set
- it may take longer: any first match is going to trigger a delay of lock_wait, which wait until it reach 0 or nb_locks is reached. It's less sensible to bad reception
- the lock duration timer can be yellow colored to indicate that you're not leaving enough time for a SPARSE match of nb_locks to happen

278 Frequencies to scan

The application parses `FREQMAN\RECON.TXT` by default. If you set something else in the CONFIG menu under input, then the specified file will be used You can use the Frequency manager app (Tools -> **Freq manager**) to add more entries to that list

Alternatively, you are able to manually input a search range "on the fly" by keying in START and END frequencies. It will uses the selected step on screen

If highlighted you can also use the rotary encoder on start or end buttons to adjust the frequency

279 Modulation Mode

You can select between **AM** , **NFM** and **WFM** modulation modes

On each modulation you can select bandwidth according to modulation value

Step can also be changed, and is only used in range search (manual or not)

If a custom modulation/bandwidth/step is set on the actual freqman entry (from freqman file), it's used as new defaults for next entries

280 Persistant settings

In the idea to be more user friendly the Recon app is keeping some settings in memory and on the sd card. All the settings that you can set in CONFIG menus will be saved and restored upon poweroff/on or app restart

Squelch is saved between runs or updates

281 Freqman file format

See Freqman Manager page

282 HamRadio type

HamRadio type is special case.

It represents a Ham Radio relay: relay RX freq for frequency_a (r=) , relay TX freq for frequency_b (t=)

In Recon, when current type is HamRadio:

- Both RX and TX frequencies are scanned for activity as two single frequencies
- When clicking 'MIC' while on one of the HamRadio entry frequencies, Mic TX frequency is set to relay RX frequency, and Mic RX frequency is set to relay TX frequency

283 Repeater type

Repeater type is special case. It's an entry that allows the portapack to act as a half duplex Repeater.

The frequency_a (l=) is the Repeater listening frequency, and frequency_b (t=) is the Repeater TX frequency

!! WARNING: These functions are activating TX in Recon. You have to know what you do !!

If you want to use the Repeater functionality:

- in CONFIG, 'record locked period' and 'repeat' have to be checked
- while you're at it, set the number of reps and the delay before each TX
- the modulation type have to be SPEC and use the target relay signal bandwidth

283.1 In Recon, when current type is Repeater and repeater mode is DISABLED:

- only listening freq is used to check on activity (l=)
- entry will be scanned as single frequency

283.2 In Recon, when current type is Repeater and repeater mode is ENABLED:

- listening freq is used to check on activity (l=)
- entry will be scanned as single frequency
- on squelch level match raw record is started
- on timeout or inactivity timeout the raw record is stopped
- tx delay is played if any
- raw record is replayed using the Repeater TX (t=) frequency

Raw Record, when used in conjunction of Repeater, is using the same file to store the record: /CAPTURES/RECON_REPEAT.C16

283.3 Advanced Repeater configuration

The Repeater, in addition of Recon, can be used as a half duplex multiplexed relay.

Here is a graph showing how Alice and Bob could talk using a Repeater list in Recon

283.3.1 Repeater half duplex dual direction example

```
flowchart LR
subgraph Repeater half duplex dual direction
    AliceTX(Alice TX)
    AliceRX(Alice RX)
    BobTX(Bob TX)
    BobRX(Bob RX)
    R1TX(Repeater 1 TX)
    R1RX(Repeater 1 RX)
    R2TX(Repeater 2 TX)
    R2RX(Repeater 2 RX)
    AliceTX-->R1RX--->R1TX--->BobRX
    BobTX-->R2RX--->R2TX--->AliceRX
    subgraph Alice
        AliceTX
        AliceRX
    end
    subgraph Bob
        BobRX
        BobTX
    end
    subgraph Recon Repeater's list
        subgraph "Repeater 1"
            R1RX
            R1TX
        end
        subgraph "Repeater 2"
            R2RX
            R2TX
        end
    end
end
```

In our example:

- a single portapack is used
- in a freqman file, we are using two Repeater entries: R1 and R2
- R1 is to repeat Alice TX to Bob RX
- R2 is to repeat Bob TX to Alice RX
- Recon is set in autostart, continuous scan
- Wait timeout is set to −1500 (which in our case mean stop recording after 1500ms of inactivity)
- Lock timeout is set to 100
- Match Mode is set to continuous
- NB match is set to 3
- auto record locked, repeater are checked
- nb repeat is set to 1 and the delay to 0

What is going to happen:

- Alice is going to use any tone/start delay to open Repeater 1
- Recon is matching a squelch level on Repeater 1 and is going to start record
- Alice have stopped talking. 1500 ms later, the record is stopped
- Recon transmit the record to Bob, using Repeater 1 TX frequency
- Bob hear Alice. He wants to answer, and send to Repeater 2
- Recon is matching a squelch level on Repeater 2 and is going to start record
- Bob have stopped talking. 1500 ms later, the record is stopped
- Recon transmit the record to Alice, using Repeater 2 TX frequency

What are the possibilities:

- as long as Alice and Bob are not talking simultaneously, they can use the Repeater as much times each as they want

- you can add more Repeaters in the list. Keep in mind that it's 100ms more scan time for each entry

284 Workflow and tips

284.1 Classic workflow

- Select an input an output file, adjust the settings in CONFIG while you're at it
- Start the RECON, match frequencies, add them to your output file using autosave or STORE button
- Switch to SCAN mode to refine your results

284.2 Repeater workflow:

- Select a prepared input file or use the classic workflow to set the frequencies you want in the scanner list
- In CONFIG/MORE, be sure that repeaters options are set (enable/disable, nb rep, tx amp, tx gain, load repeater entries)
- For a repeater mode, Recon entries have to use SPEC mode. Set the bandwidth accordingly, it's used as record/replay bandwidth
- Select the lowest lock time value (l) : 100
- Select a negative wait locked value, it means on freq as long as there no more than 'wait locked' msec of inactivity. −1500 proved to be good for conversations, for signals you may reduce the wait locked. It's also the minimum recorded time by the way.
- Set a squelch level that allow to only catch what you want. As an example, here I have a background noise of −40 db, when I hit the remote the level is going up to +5 db. To filter sparse signals from my remote, I set squelch of −10db, and adjust it down to catch more, up to catch less
- If you're not in 'autostart' you'll have to press 'RESUME' at start and after a record. Don't forget to set it in 'CONFIG' if needed
- Having 'LOOP' enabled is a good idea whatever your list is (it will allow you to avoid pressing RESUME each time too)
- Set number of matching packet from 1 to 3 as you want to quickly start recording and not wait for precisely long signal
- Set match type to continuous match
- Let it run
- Hit the remote or Make a MicTX com or whatever you're allowed to record and repeat
- auto record should start (RAW button should go red)
- at the end of record, progress bar of replay is shown or played or hidden

284.3 Tips:

- When a file is loaded and checkbox 'autostart searching' have been checked, the search is starting as soon as the apps open up, using last launch settings
- If 'autostart searching' is not used, then at the app start the last input list is loaded and the search is set in pause on the first frequency of the list
- To start a search using the input file, use the PAUSE/CONTINUE/RESTART or RST button
- To start a search using the manual entry fields on the main screen, fill START/END and use the SEARCH button
- The rotary encoder can be used on the PAUSE/CONTINUE/RESTART and START/END buttons to adjust the frequencies values
- On a frequency match, if the focus is given to the PAUSE/CONTINUE/RESTART button, press the button or use the rotary encoder to allow you to continue the search. It allows you to go to next freq, and trough both direction, changing the search direction on the fly. Pressing the button while in frequency match (locked mode continue the search using actual direction
- Recon app is doing it's best when using ranges
- Use a low squelch value and step down little by little to start to catch stronger frequencies first. If you're using a too low squelch value then the search will stop on a lot of unwanted but matching criteria frequencies
- Input and output file should not be the same files. While it's not advised it's working but you loose the ability to delete while in recon mode
- Both STORE and autosave will save frequency with current searched frequency or range description, or "ADD FQ" if no description

285 Troubleshooting

Most of the time if the Recon app is not working as you expected it's coming from a SDCARD problem. You need a SDCARD for the Recon app to save settings between runs and between settings menu or main gui

If not you will not be able to load a file from FREQMAN directory or save settings

if no frequencies are loading, check that by default the 'input: load X' fields in 'Recon app -> CONFIG -> More' are all checked. And yes, you HAVE to click save in ordre to save the settings

If by some magic you somewhat trashed the configuration and the app isn't starting anymore, try to delete the SETTINGS/recon.ini and SETTINGS/recon.cfg and SETTINGS/recon.ini files using the tools/ file manager app

If you're not having your frequencies searched in both direction, maybe you miswrote 'a=' for a single frequency entry

Using the same input and output file is not going to pause a problem until you forgot to uncheck 'clear output at start'. Yeah, you've just clear what you wanted to scan

286 Scan speed vs Chosen modulation mode

Detection of power level to match against the given squelch value is made at each firmware statistic updates. This determine how long we are staying on a frequency to have a corresponding squelch value.

In all AM/NFM/WFM modes, it's an average of 100ms per statistic update.

In SPEC mode it's somewhat consistent around 100ms from 12.5k up to 1500k included, from which the statistics updates are taking longer, from 300ms to a bit under 5000ms the wider the bandwidth

287 Power consumption

A continuous search of 63H41M43S was run by user @vag3d, using the default antenna. Settings were a range from 10MHz to 6GHz, WFM, 5kHz steps, 1s wait

Results: 8002 matching frequencies in the output list, and a consumption as following: 63:41:43 4.8659V 0.4191A 120.448Wh 24.796Ah

The search results were having this form: f=14670000,d=R 10.0000>6000.0000 S 0.0050

287.1 Capture

Capture App is designed to capture I/Q data to a file on the SD card. Captured data is stored as pairs of 16-bit signed values in a .C16 file (complex 16 bit), or optionally 8-bit signed values in a .C8 file (complex 8 bit). C16 values are stored in little-endian format. The Metadata ("Center frequency" and Bandwidth) is stored in a .TXT file with the same name.In GRC, the "file source" and "ishort to complex" blocks can be used to process the data.

The Sampling rate used may not be supported by some MicroSD cards, it is best to use high speed class SD cards. The HackRF PortaPack Capture/Replay functionality is based on Havoc firmware version.

The Key Items on the App that can be selected with the cursor and changed with the encoder knob are:

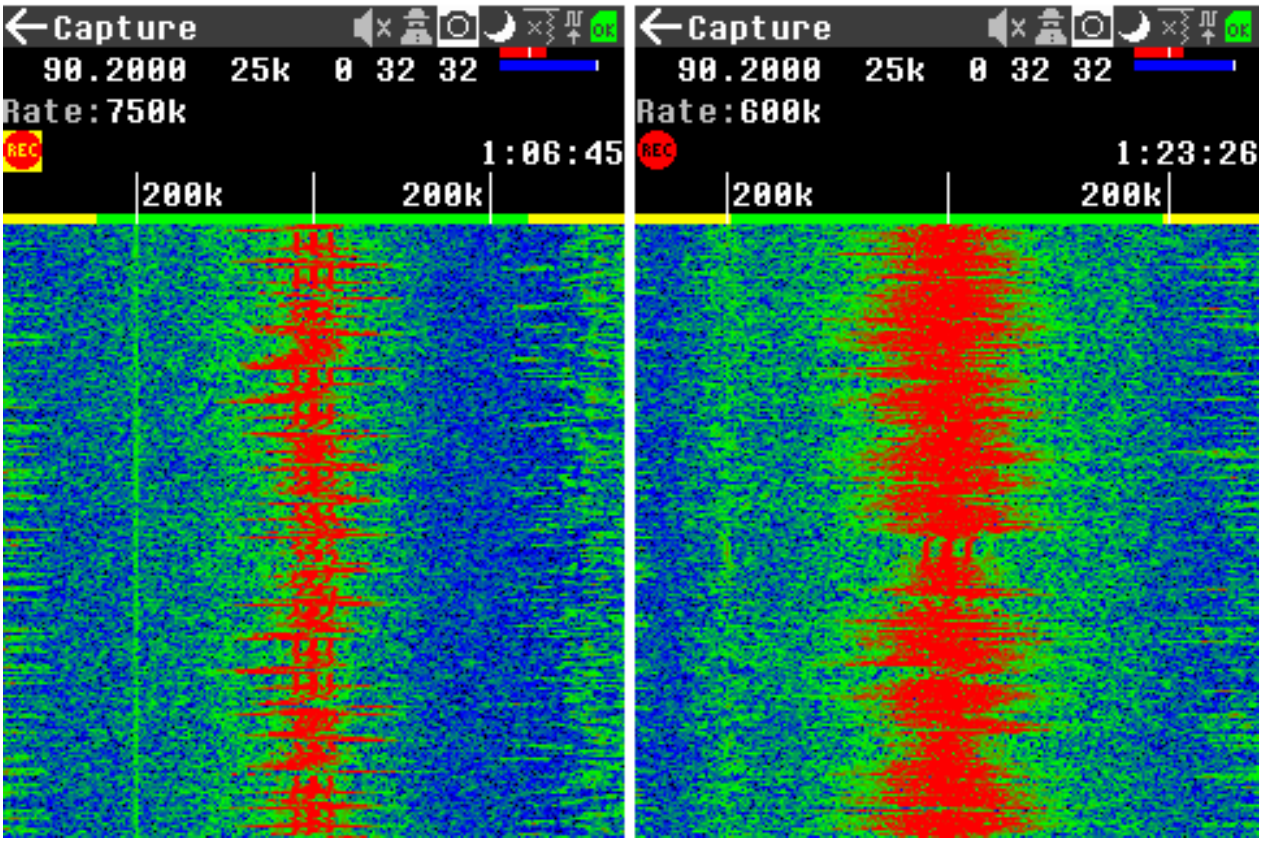
- **Title bar:** The usual Items may be changed and displayed.
- **Frequency:** The setting of the frequency using the keypad can be completed and is stored in persistant memory so can be returned to when the App is used again. But if you have load/save App Settings enabled from SD card, it will continue to use what it was last set at. And you can always edit the file in the SD card, /SETTINGS/rx_capture.ini and edit those related default parameters, according to your needs in that App.
  - Note : Additionally , to be more user friendly , from version 1.7.4+ holding in the Select button on the Frequency field for a second until a digit turns blue, then you can use Left/Right select which digit you'd like to adjust, and then you can use the Encoder Dial to adjust any digit up/down by 1 to tune more precisely. (Press Select again to exit this tuning mode.

- **Step Size:** The selected step size of frequency adjustment carried out by the Rotary encoder.
- **Gain:** Amp (0dB or 14dB), LNA(IF) (0-40),VGA(Baseband)(0-62)
- **Rate:** The sample rate, which by its nature set the set bandwidth of capture. This is shown in the markers around the centre line of the waterfall display. The sample rate is variable from 12k5 to 5500K (*) in many steps. You need to ensure that the sample rate is more than twice the bandwidth of the signal you want to capture see (Nyquist Principle or also called Shannon's Law)
- **Format:** The user can select two file formats, of the recorded IQ data 16 or 8 bits : C16 (complex 16) or C8 (complex 8), and its related file extension ".C16" or "C8". (By default we are preselecting C16).
- **Trim:** If checked, the capture will attempt to be automatically trimmed to only contain the signal part. This is not reversable. Consider using the IQ Trim utility instead if you're not sure you can capture the signal again.

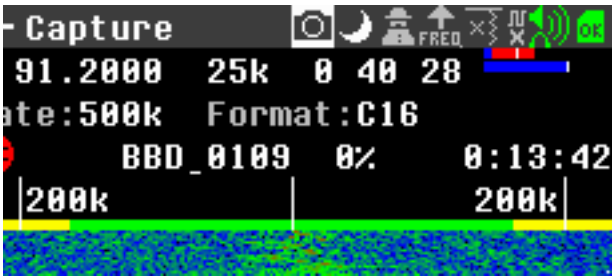
Note (*) : Currently , **for correct reliable Replay application ,you should ONLY use Capture App selecting any Bandwidth capture <= 1Mhz (but 500Khz is the recommended for majority micro SD cards compatibility because it requires a quite common average write speed in our system >2MB/sec,C16)**. From 600khz till 1Mhz (and 1.25 MHz) , you will need more fast and good quality micro SD card (with min average write speed in our system >3MB/sec (for BW=750khz,C16) , and >4MB/sec (for BW=1Mhz ,C16), (>5MB/sec for BW=1.25Mhz ,C16) and with as small as possible write random latency. (In the GUI , those correct bandwidth capture options appear with the Normal usual "REC" icon Background color, as user recommended BW capture options). If you face too much % dropped samples when recording , you can retry it reducing C16 to C8 , that also reduces the needed average write speed :2 (example 1Mhz rec will need average sd interface speed of 4MB/sec in C16 or 2MB/sec in C8).

Recently, thanks to x4 Oversampling and /4 Decimation introduction , we have also added 1.25Mhz BW REC , but it is in the limit of our hardware System and it starts to have some M4 sample drops. (It is still experimental).

Above 1.25Mhz till 5.5Mhz bandwith options (with YELLOW REC button background Icon), in current fw version , the recorded files have periodical sample drops , and therefore it can not record all full original samples content and therefore it is NOT useful for the Replay App , just useful to check the spectrum image, example using the linux tool "inspectrum" or "Audacity") . Anyway when replaying those captured files , the replay time will be shorter than real . but normally you should still get a correct modulation, but in case of voice contents, it would sound , unnatural , "like with accelerated playing speed" , because in fact,due to Hackrf One System (HW+SW) limitations, those recorded files (under those high speed BW with YELLOW icon) have decimated its content - when recorded there were skipped some real time samples.

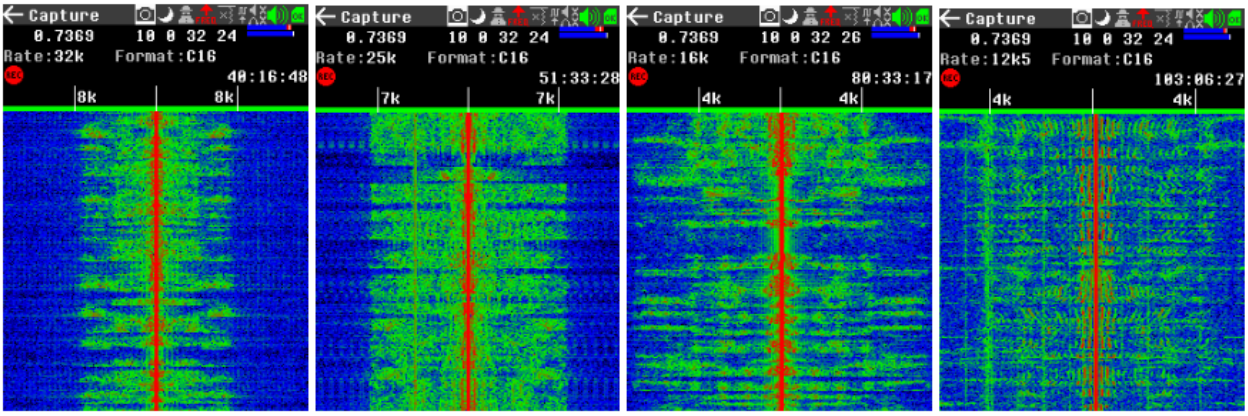


- **Record Button:** The red button shows as Rec or Stop. If record is selected then it will record the I/Q file. To the side of the Record Button is additional information that is shown for the recording file, % of Dropped Samples, total Recording Time Remaining. (As we mentioned above, for correct Replay operatons, please make sure to select a proper Capture bandwith option , with normal black background "REC" icon , not the yellow one.)




```
 * File Name (as above example , BBD_0109.C16, based on the selected format C16)
 * % of Dropped Samples (recording error rate, due to SD card write latency, above sample
0%)
 * Total Recording Time Remaining (based on available SD card capacity , above example 0h:
13min: 42 secs)
```

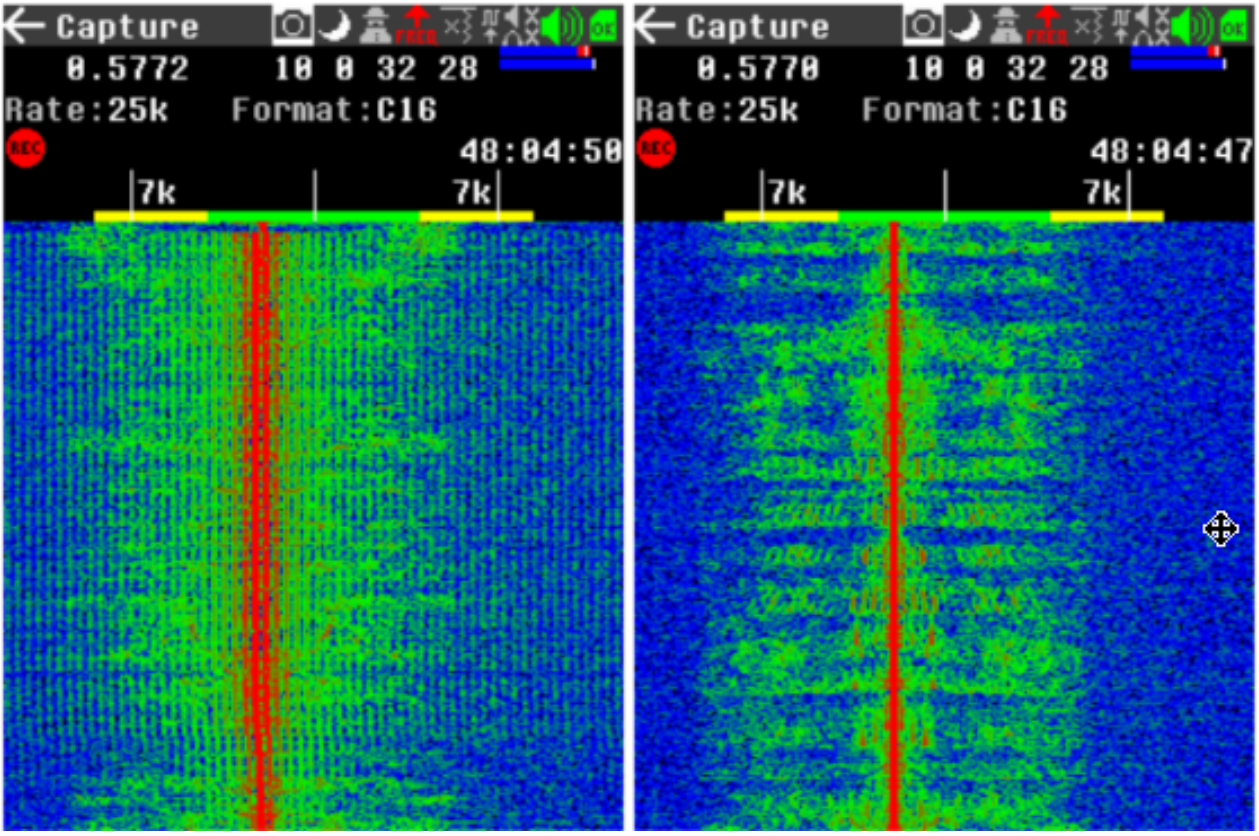
From nightly 23-08-19 onwards, we revised and extended a good functionality of all low bit rate bandwith REC options If you adjust correctly the GAIN and LNA and center freq. it is possible to capture good bit streams in C16 or C8 format , Pls. find attached some screenshot examples , capturing the same tuned AM MF band broadcasting with a BW aprox of 16Khz , using an upconverter ,



287.2 Note : (this below retuning process description, it is not longer necessary) :

About +- n x 10 Hz it is not necessary anymore , after that PR : Capture: fix discontinuous IQ samples in low-rate waterfall FFTs- #3390 In fact was an small bug, now solved.)

(Old process description :) Depending on the tuned frequency, (mainly visible in 25k, 32k, 50k and 75k) we may got a random strange left picture effect -on the screen, it seems not affecting to the recorded files.C16 or .C8 - with “vertical stripes” , this can be easily corrected, just readjusting the center frequency some Hz up or down , till trying to have full convergence of these vertical red carriers to the central unique one , with +-10Hz steps.



Please check the video below for HackRF PortaPack Capture/Replay functionality.

HackRF PortaPack Capture/Replay functionality

NOTE (1) If you have an external GPS module attached, and after starting the Capture app you receive a valid GPS signal the CAPTURE will contain GPS meta data too. The filename will get _GEO appended. If you share this recording, it could contain sensitive info. You can remove it by editing the TXT, and removing the GPS data.

(2) : Recording weak signals and avoiding DC-related artifacts

When recording weak signals, especially AM signals, it is recommended to capture the desired signal with a small frequency offset from the IQ center frequency. This helps prevent the signal of interest from overlapping the DC spike that may become visible when the recording is transmitted using Replay. Choose an offset large enough to separate the wanted signal from the DC artifact, while keeping the complete signal bandwidth inside the recorded IQ spectrum.

Note: This technique avoids placing the wanted signal directly at DC; it does not remove the DC component itself.

287.3 C16 Format

The [[Capture]] app records into C16 files, and the Replay app read those files. *(Support for C8 format has since been added to firmware, but this chapter assumes that files were captured in C16 format; most instructions below apply to C8 files as well by just replacing the 16 with 8.)*

As described in an issue in PortaPack's repo, this format consist of a tuple of 16 bits signed integers. The first number is I and second Q, forming a complex number. As a result, you get a tridimensional representation of the capture: the real and the imaginary parts in the file (I and Q) versus the time (defined by the sample rate, in this case in an adjacent TXT file with the same filename as the C16).

287.4 Metadata file (.TXT)

Each capture file is accompanied by a `.TXT` file with the same name. The file contains at minimum:

```
center_frequency=<FREQ IN HZ>
sample_rate=<RATE IN HZ>
```

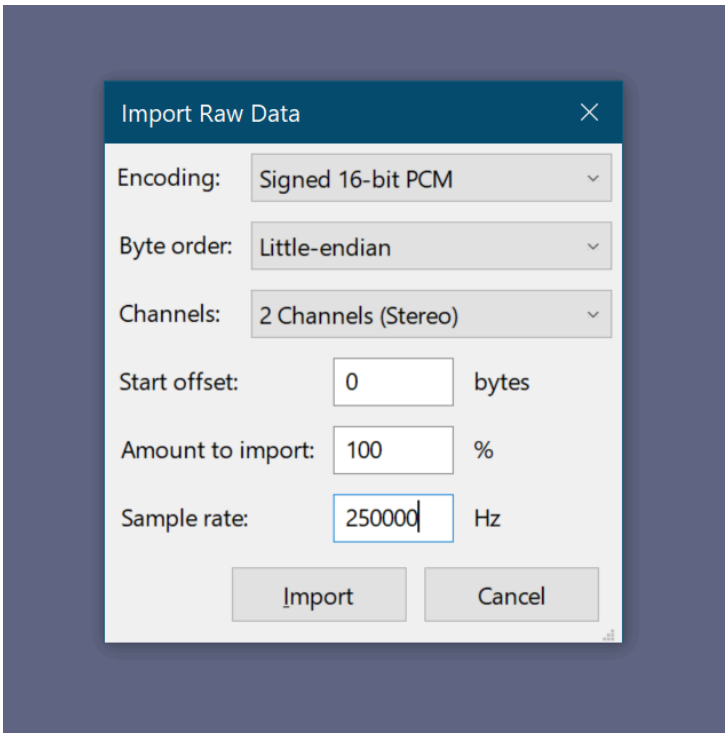
If a GPS module was connected and provided a valid fix during capture, three additional fields are written:

```
latitude=<DECIMAL DEGREES>
longitude=<DECIMAL DEGREES>
satinuse=<SATELLITES IN USE>
```

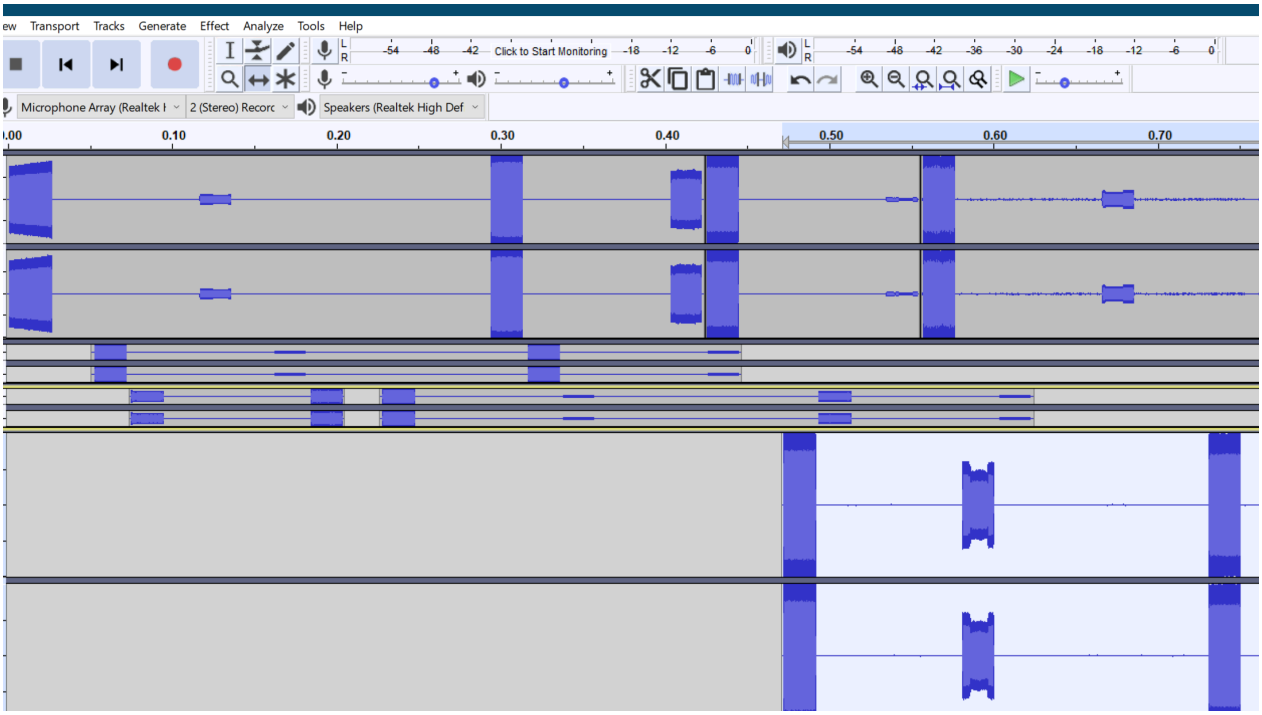
These fields are optional — the Replay app reads them but does not require them.

287.4.1 Capture Manipulation (Audacity)

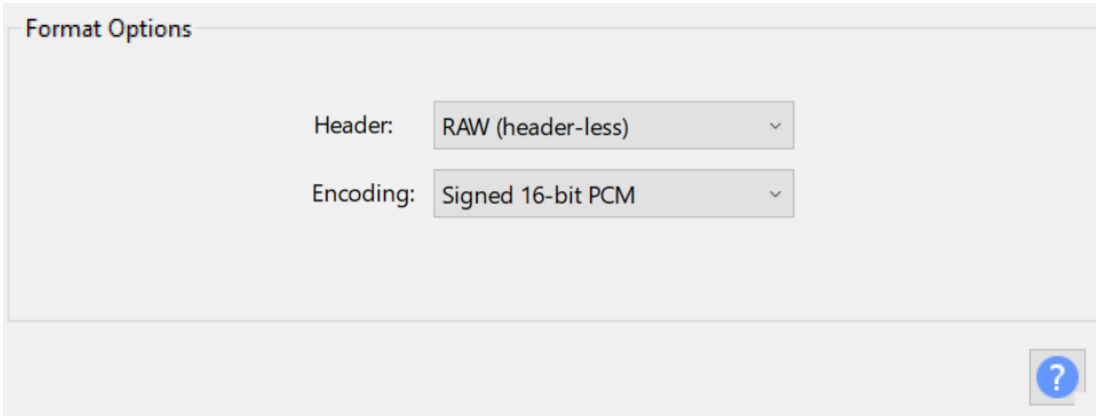
You can open the C16 file in Audacity importing it as *Raw data*. Consider the sample rate you used when you did the capture.



It is possible to manipulate the data as if it was audio. When importing the file as signed 16 bit PCM with two channels, the upper channel contains the I and the second Q, as explained above. You can apply filters, add silences, trim, mix and merge.

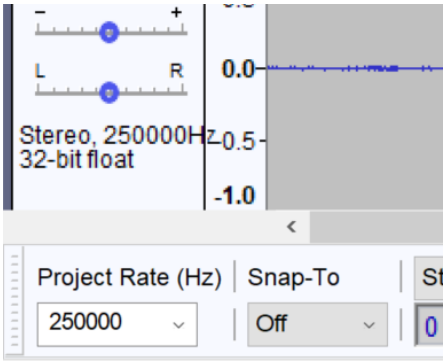


To export the data as a C16 file, select Export, Export Audio … and then select RAW as a 16 bit signed PCM. After saving, change the extension and you will be able to use the file in the Replay app.



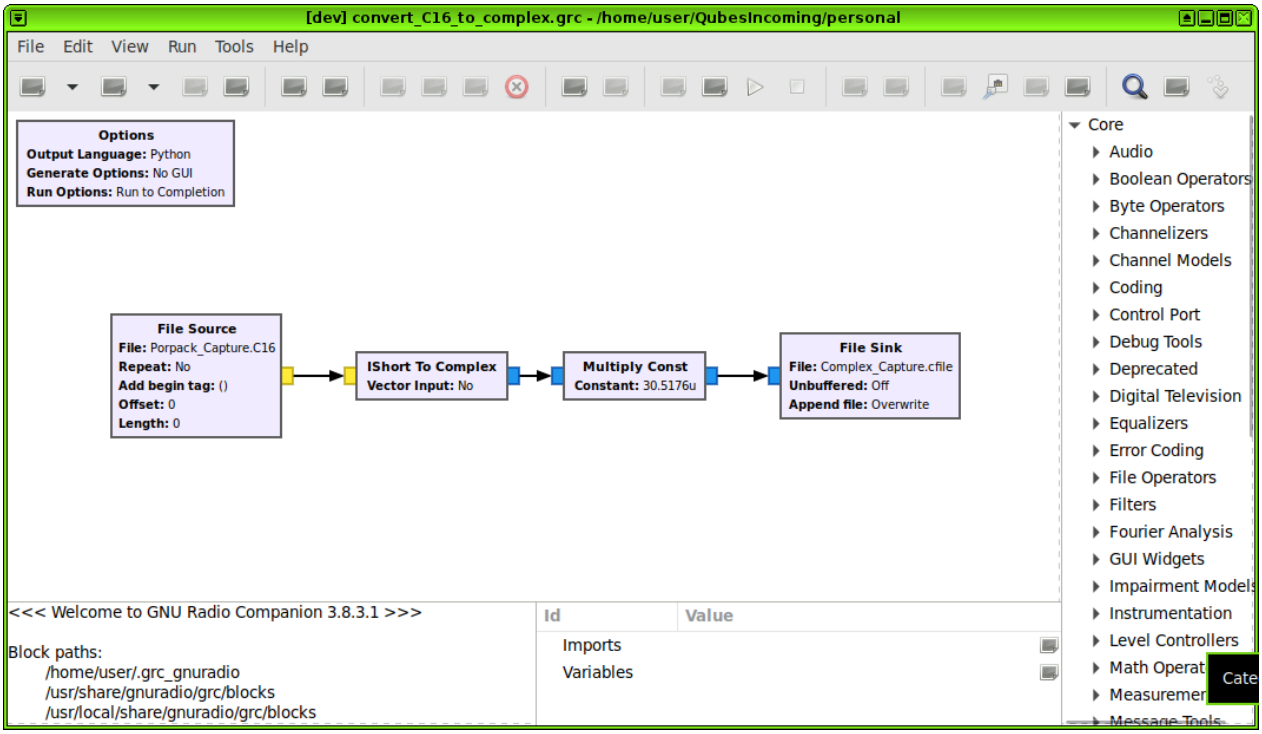
For using the new C16 file in the Replay app, you will also need to include a TXT file with the metadata as described above.

As a final remark, if you created a new file, be sure that the Project Rate matches your capture:



287.4.2 Capture Manipulation (GNU Radio)

You can use GNU Radio Companion (GRC) to convert the C16 short into a complex IQ file.



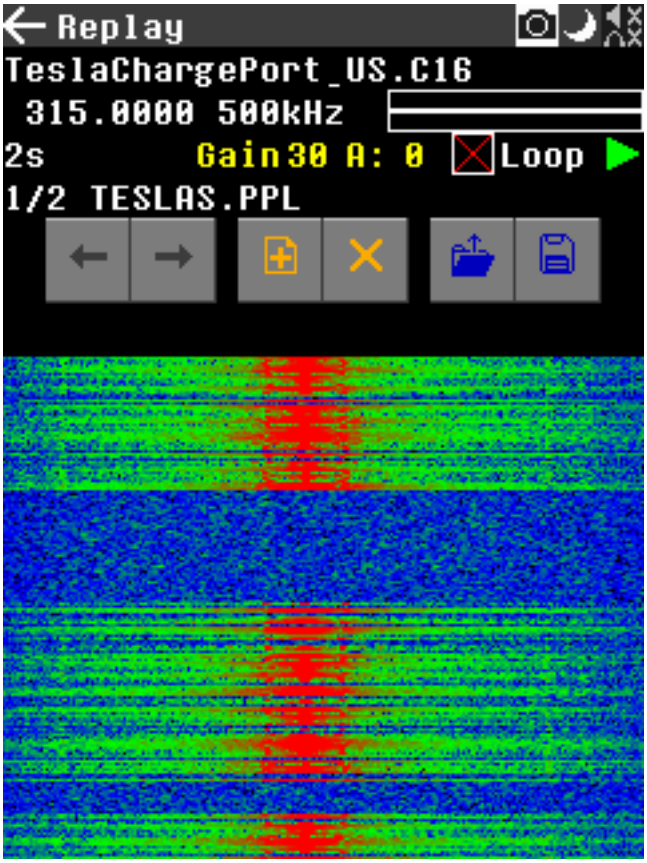
As seen in the image above you'll need to bring in the original C16 file using the GRC block `File Source` setting the output type to short, pipe it to `IShort To Complex`, then `Multiply Constant` using the magic number 1.0 or 32768.0 for the constant, and finally export it with `File Sink`.

The reason that the value 32768.0 is used to normalize the C16 capture is because int16 has a range of −32768 to +32767. 2^15 is 32768.0, so dividing int16 value by 2^15 gives a number that is normalized between -1 and +1.

This GRC script can be found at `firmware/tools/convert_C16_to_complex.grc`

287.5 Replay

288 The UI



Replay transmits captured IQ files (`.C16` or `.C8` or `.CU8`) over the air. Files can be played individually or organized into a playlist (`.PPL`) for sequential playback with optional delays between entries.

289 Display

- **Line 1**: Name of the currently loaded capture file.
- **Line 2**: Frequency (editable) and sample rate of the current track. Two stacked progress bars on the right — the top shows position within the playlist, the bottom shows transmit progress of the current file.
- **Line 3**: Duration of the current track in seconds. TX gain (0–47), TX amp (0/14), Loop checkbox (default: on), and the play/stop button.
- **Line 4**: Current track number or total tracks (e.g. `1/2`) on the left, current playlist filename on the right.

A live waterfall is displayed below the controls during transmission.

290 Controls

- ◄ / ►: Navigate to the previous or next track in the playlist.
- 📄 **(Add file)**: Opens the file picker to select a `.C16` or `.C8` or `.CU8` capture file and append it to the end of the playlist.
- 🗑️ **(Delete)**: Removes the currently selected track from the playlist.
- 📂 **(Open)**: Opens the file picker to load a `.PPL` playlist file. This replaces the current playlist.
- 💾 **(Save)**: Saves the current playlist to the active `.PPL` file.
- **Loop** (checkbox, default: on): When enabled, the playlist restarts from the beginning after the last track finishes.
- ▸ / ■: Start or stop playback.

The frequency field is editable per-track and is not saved to the playlist file.

291 Playing a Single File

When the app opens with an empty playlist, the Add file button is automatically focused. Press it, select a `.C16` or `.C8` or `.CU8` file, and the play button receives focus immediately — allowing fast two-button playback.

You can also open a capture or playlist file directly from the File Manager app, which will launch Replay with that file pre-loaded.

292 Playlist Files

Playlist files use the extension `.PPL` and are stored in the `PLAYLIST/` directory on the SD card. They are plain text, comma-delimited files.

Format:

```
ABSOLUTE_PATH_TO_FILE,DELAY_MS
# Comment lines starting with # are ignored
```

The delay (in milliseconds) is optional. If omitted, no pause is inserted between tracks.

> **WARNING** The pre-delay is implemented as a blocking sleep on the UI thread. Setting a long delay will cause the device to appear frozen for that duration. There is no way to interrupt it other than resetting the device.

Example:

```
# Playlist file
/SAMPLES/TeslaChargePort_US.C16
/SAMPLES/TeslaChargePort_EU_AU.C16,100
```

This plays the US file, then pauses 100 ms, then plays the EU/AU file.

When a new capture file is added to an empty playlist, a new playlist file is created automatically with the name `PLAY_XXXX.PPL` in `PLAYLIST/`. The playlist is not saved to disk until you press the Save button.

Metadata fallback: If a capture file has no matching `.TXT` metadata file, the current transmit frequency and a sample rate of 500,000 Hz are assumed.

293 Settings Persistence

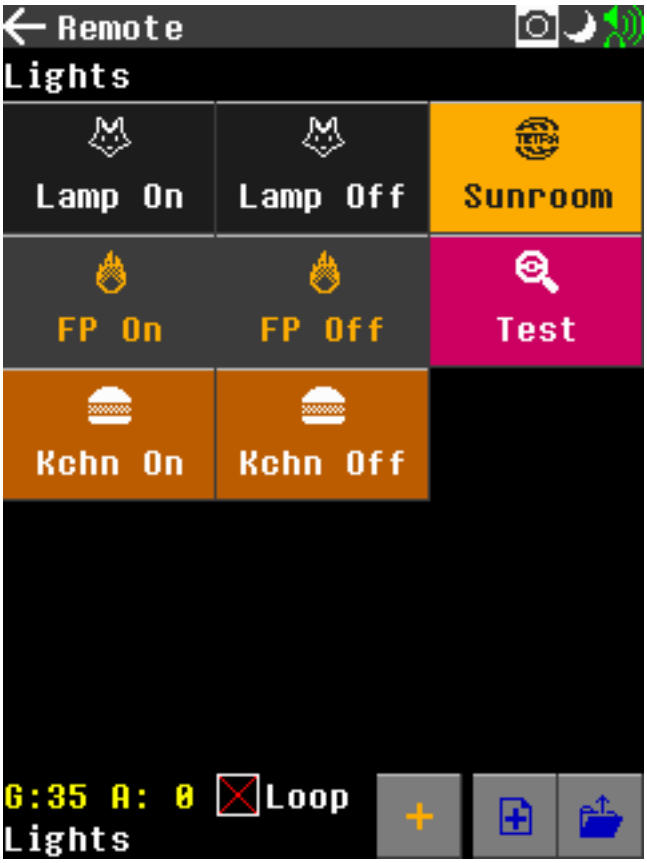
Settings are saved to `tx_replay.ini`.

294 High sample-rate Replay and SD-card performance

High sample-rate Replay requires sufficiently fast SD-card read performance. The required data throughput depends on the IQ sample format:

Source sample rate	C8 IQ	C16 IQ
1 MHz	2 MB/s	4 MB/s
2 MHz	4 MB/s	8 MB/s
3 MHz	6 MB/s	12 MB/s
5 MHz	10 MB/s	20 MB/s

294.1 Remote



The Remote app lets you create a custom button panel and bind each button to a capture file for one-touch playback.

Remote files (`.REM`) are stored in the `REMOTES/` folder on the SD card. New remotes are automatically named `REMOTE_0001.REM`, `REMOTE_0002.REM`, etc.

This is an external app (`.ppma` file on the SD card).

294.2 Main UI

- **Title** — The top line shows the remote's name. Select it to rename it (max 30 characters).
- **Buttons** — Up to 12 customizable buttons arranged in a 4×3 grid. Select a button to transmit its capture file. Select the same button again to stop. Long-press Select to open the button edit screen.
- **Waterfall** — A small waterfall display shown between the button grid and the bottom bar while transmitting.
- **Gain or Amp** — Transmitter gain and amplifier control (via the TX bar at the bottom). The indicator color reflects transmitter power.
- **Loop** — When checked, continuously retransmits the capture until stopped.
- **Filename** — Shows the current remote filename (without extension). Select it to rename the file.
- **Add button (+ icon)** — Adds a new empty button to the grid. New buttons open in edit mode the first time they are selected.
- **New remote (new-file icon)** — Saves the current remote and starts a new empty one.
- **Open remote (folder icon)** — Saves the current remote and opens an existing `.REM` file from the `REMOTES/` folder.

Remotes are automatically saved when the app exits.

294.3 Button Edit UI

- **Name** — The text label shown on the button (max 30 characters).
- **Path** — Select to pick a capture file (`.C16` / `.C8`) from the `CAPTURES/` folder.
- **Freq** — The center frequency to transmit on. Defaults to the capture file's metadata frequency if a `.met` file is found, otherwise falls back to the current TX frequency.
- **Rate** — The sample rate read from the capture file's metadata, or 500 ksps if no metadata is found. Display only — cannot be changed here.
- **Icon** — Selects an icon to show on the button (index 0 = no icon; indices 1–24 select from the available icon set).
- **FG Color** — Foreground (text/icon) color index (0–20).
- **BG Color** — Background color index (0–20).
- **Preview** — Live preview of the button as it will appear in the main UI.
- **Trash (red)** — Deletes this button after a confirmation prompt and returns to the main screen.
- **Done** — Saves changes and returns to the main screen.

294.3.1 Available colors (index → name)

- Index: 0; Color: Black; Index: 7; Color: Orange; Index: 14; Color: Dark orange
- Index: 1; Color: White; Index: 8; Color: Yellow; Index: 15; Color: Dark yellow
- Index: 2; Color: Darker grey; Index: 9; Color: Green; Index: 16; Color: Dark green
- Index: 3; Color: Dark grey; Index: 10; Color: Blue; Index: 17; Color: Dark blue
- Index: 4; Color: Grey; Index: 11; Color: Cyan; Index: 18; Color: Dark cyan
- Index: 5; Color: Light grey; Index: 12; Color: Magenta; Index: 19; Color: Dark magenta
- Index: 6; Color: Red; Index: 13; Color: Dark red; Index: 20; Color: Purple

294.4 Technical details

- Parameter: Baseband bandwidth; Value: 2.5 MHz
- Parameter: Sample rate; Value: From capture metadata (or 500 ksps fallback)
- Parameter: Metadata fallback frequency; Value: Current TX frequency
- Parameter: Settings file; Value: tx_remote.ini

294.5 REM file format

Remote files are plain text and can be edited in any text editor. Empty lines and lines beginning with `#` are ignored.

- The first non-ignored line is the remote **title**.
- Each subsequent line (up to 12) defines one button in this comma-separated format:

```
capture_path,button_name,icon_index,bg_color_index,fg_color_index,center_frequency_hz,sample_rate_sps
```

Example:

```
My Remote
CAPTURES/door.C16,Door,6,3,1,433920000,500000
CAPTURES/lock.C16,Lock,0,17,1,433920000,500000
```

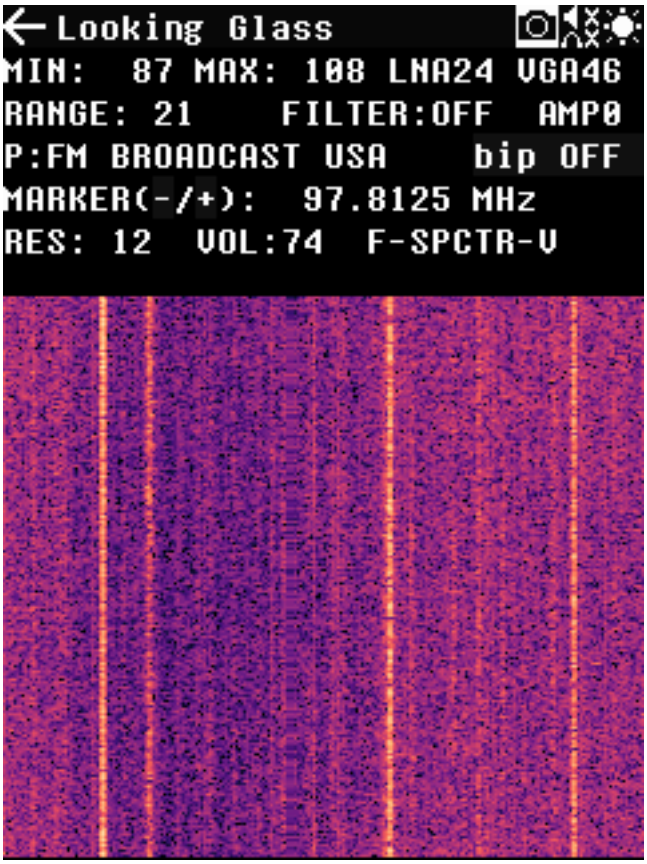
The order of buttons on screen matches the order of lines in the file, so buttons can be rearranged by reordering lines.

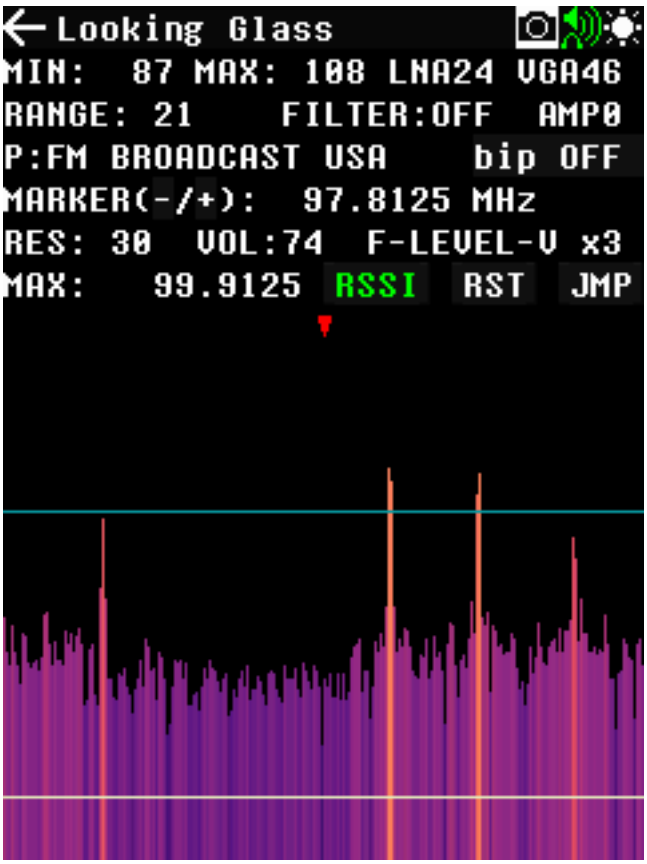
294.6 Tips

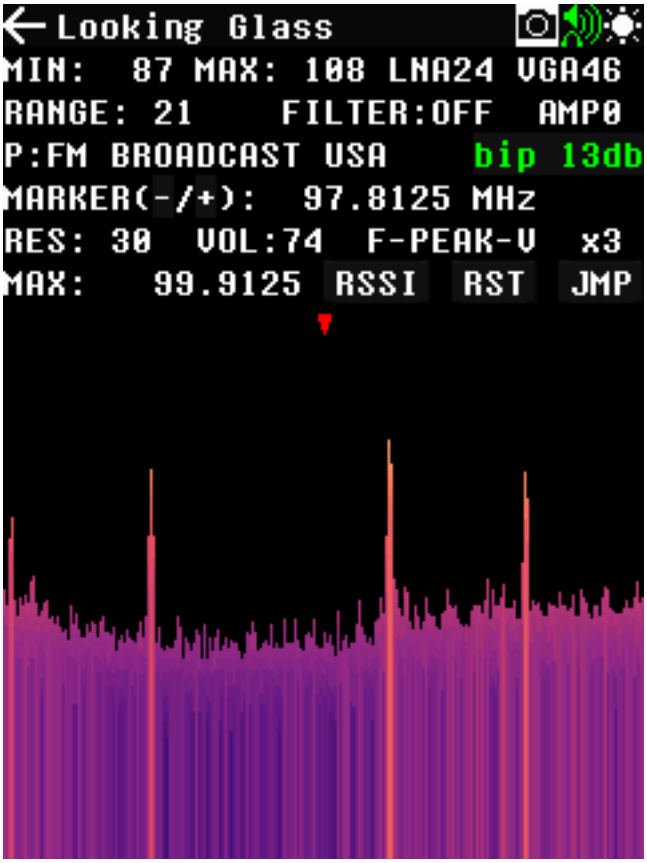
- Use the IQ Trim utility to trim silence from captures so they start transmitting instantly.
- If a `.txt` metadata file exists alongside a capture, its frequency and sample rate are used automatically.
- Button positions can be rearranged by editing the `.REM` file directly.

294.7 Looking Glass

295 Looking Glass







296 Introduction

Looking Glass provides a wide-band spectrum waterfall view covering a large frequency range in a single scrolling display. The HackRF steps through the configured range in slices, and each completed full sweep updates the waterfall with a new row.

For wide scan ranges the sweep time is long and the display may appear frozen — this is normal. Keep the scan range as small as practical to get fast, responsive updates. Despite the HackRF's modest sensitivity, Looking Glass is a useful tool for spotting local RF activity across a broad span when paired with an appropriate antenna.

Frequency presets are loaded from all `.TXT` files in the `LOOKINGGLASS/` folder on the SD card. Multiple preset files are supported — all valid entries across all files are merged into the preset list. Keep preset ranges small to limit memory usage.

297 Controls

- **MIN or MAX:** Place the cursor on the `MIN` or `MAX` field and use the rotary encoder to adjust the lower or upper scan frequency in steps of the selected step size. The `RANGE` label shows the total scan span.
- **Range lock:** Click on the range value label to lock or unlock the range width when adjusting MIN/MAX. When locked, moving MIN also moves MAX so the span stays fixed.
- **PRESET:** Move the cursor to the `PRESET:` field to select one of the pre-set frequency ranges from `LOOKINGGLASS/PRESETS.TXT` on the SD card.
- **LNA or VGA or AMP:** Gain controls. LNA (IF) adjustable 0–40 dB; VGA (baseband) 0–62 dB; AMP 0 = off, 1 = on (+14 dB).
- **FILTER:** Selects a display smoothing filter — `OFF`, `MID`, or `HIGH`. The filter affects the visual contrast of detected signals. Adjust to get the best contrast for the signals you are trying to see; the optimal setting depends on scan speed and gain.
- **F- / S-:** Scan mode — `F-` is fast but less accurate; `S-` is slow but more accurate.
- **SPECTR or LIVE-V or PEAK-V:** Selects the display mode:
  - `SPECTR` — scrolling waterfall (default)
  - `LIVE-V` — live frequency power level bars
  - `PEAK-V` — peak frequency power level bars
- **RES:** Resolution (FFT trigger point), adjustable 2–128. Default is 32. Increase for finer frequency resolution; decrease for faster updates. Adjust together with gain and FILTER for best results.
- **MARKER:** Rotate the encoder while the cursor is on this field to move a red frequency marker arrow across the waterfall display. The marker's frequency and the per-pixel frequency step are shown. Pressing the encoder knob or the button jumps to the Audio app at the marker frequency with 1 MHz steps and a 10 MHz view. Note: returning from the Audio app resets Looking Glass to its default settings.
- **BEEP:** Click to enable or disable the audio squelch beep. When enabled, a beep is emitted whenever signal power at the marker exceeds the configured squelch threshold (adjustable from −100 to +20 dB; default is 20, which disables it).
- **VOL:** Controls the volume of the beep when beep is enabled.

297.1 LIVE-V and PEAK-V additional controls

When LIVE-V or PEAK-V mode is selected, additional UI elements appear:

- **x0 − x9** *(LIVE-V only)*: Integration multiplier. `x0` gives the fastest response with more noise; `x9` gives the slowest, smoothest response with fewer spikes.
- **▸/■ (Play/Pause)** Toggle button. When the sweep is running the button shows a stop icon (■) in green. Pressing it pauses the sweep — the display freezes and the button turns red showing a play icon (▸). Pressing again resumes. Useful for inspecting a snapshot of the spectrum without the display updating.
- **RSSI** Toggle button. When active (shown in green), two horizontal lines are drawn across the display indicating the current min and max RSSI power levels. When inactive (shown in dark), the lines are hidden. Disabled by default.
- **MAX: VALUE** Displays the frequency with the highest reception power.

- **RST** Resets the max-hold value and clears the display.
- **JMP** Jumps to the Audio app at the frequency shown in `MAX`.

297.2 Utilities

297.3 Antenna length

The **Antenna Length** calculator computes the optimal antenna length for any given frequency, displayed in both metric and imperial units. It also shows how many elements to extend for each of your saved telescopic antennas.

This is an external app (`.ppma` file on the SD card).

> **WARNING** Selecting the right ("tuned" or "matched") antenna for each frequency is of paramount importance — otherwise RX or TX will be less than optimal. Under some circumstances, such as a grossly mismatched antenna during high power TX, you could end up damaging your equipment. This applies to all radio TX equipment.

297.4 Controls

- Field: Frequency; Description: Target frequency. Adjust with the encoder; step size is 1 MHz.
- Field: Wave; Description: Antenna type or fraction of a wavelength to calculate.
- Field: Metric; Description: Calculated length in meters and centimeters.
- Field: Imperial; Description: Calculated length in feet and inches.

297.5 Wave Type Options

- Option: Full; Fraction: 1; Notes: Full wavelength
- Option: Half; Fraction: 1/2; Notes: Common dipole
- Option: Quarter; Fraction: 1/4; Notes: Default — common whip/monopole
- Option: 3/4; Fraction: 3/4
- Option: 1/8; Fraction: 1/8
- Option: 3/8; Fraction: 3/8
- Option: 5/8; Fraction: 5/8; Notes: Popular for VHF/UHF gain
- Option: 7/8; Fraction: 7/8

297.6 Telescopic Antenna Guidance

Below the calculated length, the app shows how many **elements** to extend on each of your saved telescopic antennas to reach the target length.

For example: `ANT500: 3 1/4 elements` means extend 3 full sections plus roughly a quarter of the next.

When the target length falls outside an antenna's physical range, the app reports it instead of a section count: **`retracted`** when the required length is shorter than the fully-collapsed antenna (frequency above the antenna's range), and **`fully ext.`** when it is longer than the fully-extended antenna (frequency below the antenna's range).

If no `ANTENNAS.TXT` file is found on the SD card, the app falls back to the built-in **ANT500** antenna only.

297.7 Adding Your Own Antennas

Place a file at `WHIPCALC/ANTENNAS.TXT` on the SD card. Lines starting with `#` are comments.

Format: `label,element1_mm,element2_mm,...`

Each element value is the total extended length in millimeters when that section is fully extended.

297.7.1 Example ANTENNAS.TXT

```
#antenna label,elements length in mm, separated by commas
ANT700,95,134,175,206,230,245
ANT500,185,315,450,586,724,862
TELESCOPIC,118,183,253,326,399,476
Cheap 11cm,118,183,253,326,399,476
Comet W100RX,200,300,400,510,615,800,986
```

The SD card image ships with a ready-made `ANTENNAS.TXT` that includes several presets such as the **Comet W100RX**.

298 References

Episode 44: telescopic antenna of the HackRF (The RF Noob) https://www.youtube.com/watch?v=MzbAEBghcPM

298.1 Calculator

The **Calculator** is a FORTH-programmable scientific RPN (Reverse Polish Notation) calculator based on the IVT (IVEE-TINY) engine by zooxo/deetee.

This is an external app (`.ppma` file on the SD card).

298.2 What is RPN?

In RPN, operands are entered before the operator. Instead of `3 + 4`, you enter `3`, then `4`, then `+`. Numbers are pushed onto a stack; operations consume values from the top of the stack and push the result.

- **D** (DUP) pushes/duplicates the current number onto the stack, or confirms number entry.
- **C** (DROP) removes the top of stack, or clears the current number entry.

298.3 Keyboard Layout

The keypad has two modes. Press **F** to toggle the shifted layer.

298.3.1 Normal mode

- Col 1: F (shift); Col 2: 7; Col 3: 8; Col 4: 9
- Col 1: E (10^x); Col 2: 4; Col 3: 5; Col 4: 6
- Col 1: N (negate); Col 2: 1; Col 3: 2; Col 4: 3
- Col 1: C (DROP/clear); Col 2: 0; Col 3: . (decimal); Col 4: D (DUP/enter)

298.3.2 F-shifted mode (press F first)

- Col 1: MENU; Col 2: SUM+; Col 3: PRG; Col 4: /
- Col 1: SWAP; Col 2: DICT; Col 3: USR; Col 4: *
- Col 1: ROT; Col 2: RCL; Col 3: STO; Col 4: -
- Col 1: CA; Col 2: PI; Col 3: INT; Col 4: +

298.4 Key Reference

- Key: 0–9, .; Description: Enter digits and decimal point
- Key: E; Description: Scientific notation exponent (Y × 10^X)
- Key: N; Description: Negate (change sign)
- Key: D; Description: DUP — duplicate top of stack or confirm number entry
- Key: C; Description: DROP — remove top of stack or clear current entry
- Key: F; Description: Shift to function layer; double-press opens user MENU
- Key: + − × ÷; Description: Basic arithmetic
- Key: SWAP; Description: Swap top two stack values (1 2 → 2 1)
- Key: ROT; Description: Rotate stack (1 2 3 → 2 3 1)
- Key: STO or RCL; Description: Store/recall a number to memory slot 0–9 (enter slot number first)
- Key: CA; Description: Clear all — clears stack and statistics memories
- Key: PI; Description: Push π onto the stack
- Key: INT; Description: Integer part of top value
- Key: SUM+; Description: Add X (or X,Y) data point for statistics or linear regression
- Key: DICT; Description: Browse the full dictionary of all commands
- Key: USR; Description: Assign a dictionary entry to the user menu
- Key: MENU; Description: Open the user-defined menu
- Key: PRG; Description: Edit a user program (enter program number 0–15 first)

298.5 Dictionary Functions

Access all commands via **DICT** (F+4). Functions include:

Math: `SQRT`, `POW` (y^x), `INV` (1/x), `EXP`, `LN`, `SIN`, `COS`, `TAN`, `ASIN`, `ACOS`, `ATAN`, `GAMMA` (ln Γ)

Stack: `DUP`, `DROP`, `SWAP`, `ROT`, `OVER`

Statistics: `SUM+`, `SUMclr`, `MEAN/STDEV`, `LR` (linear regression)

Other: `PV` (present value), `ND` (normal distribution PDF/CDF), `P>R` / `R>P` (polar/rectangular), `nPr`, `nCr`, `SOLVE` (root finder)

Control flow (for programs): `IF`, `ELSE`, `THEN`, `BEGIN`, `UNTIL`, `<`, `=`, `<>`, `>`

298.6 Programming

IVT supports up to **16 user programs** (numbered 00–15) with up to 440 total steps.

To edit a program:

1. Enter the program number (e.g. `0`, `5`)
2. Press **F** + **PRG**
3. Navigate steps with **E** (up) and **N** (down)
4. Insert steps by pressing a key or selecting from DICT
5. Delete a step with **.**
6. Save and exit with **C**

NOTE User program **00** is reserved for the **SOLVE** (root finder) function.

298.7 Limits

- Item: Stack depth; Limit: 24
- Item: Significant digits; Limit: 7
- Item: Memory slots (STO/RCL); Limit: 10 (0–9)
- Item: User programs; Limit: 16
- Item: Total program steps; Limit: 440
- Item: User menu slots; Limit: 32
- Item: Number precision; Limit: IEEE 754 32-bit float (~6–7 decimal digits)

298.8 Example Programs

```
ABS:    DUP 0 > IF THEN NEG
SINH:   EXP DUP INV NEG + 2 /
LOG:    LN 1 0 LN /
%:      OVER or 1 0 0 *
C<>F:   DUP 1 . 8 * 3 2 + SWAP 3 2 - 1 . 8 /
```

298.9 Cart Lock

Shopping Cart lock/unlock app by RocketGod (@rocketgod-git).

This is an external app (`.ppma` file on the SD card).

WARNING This app plays audio at **maximum volume** automatically.

298.10 Requirements

Two WAV files must be placed in the `WAV/` folder on the SD card:

- Filename: shopping_cart_lock.wav; Function: Played when pressing Lock
- Filename: shopping_cart_unlock.wav; Function: Played when pressing Unlock

If either file is missing, the Lock and Unlock buttons will not appear.

298.11 Usage

- Button: Lock; Action: Plays the lock WAV file on loop
- Button: Unlock; Action: Plays the unlock WAV file on loop
- Button: Stop; Action: Stops playback

The audio loops continuously until **Stop** is pressed or the app is closed.

The log panel at the top shows diagnostic output: file scan results, playback state, and any errors.

298.12 Notes

- Original `.cu8` signal files by @jimilinuxguy
- Audio is output through the headphone jack or internal speaker at maximum volume
- The app requires WAV files to be present — it does **not** generate tones internally

298.13 Debug Menu

The Debug menu provides hardware diagnostics, memory inspection tools, and developer utilities. Some items are built-in; others are external apps (`.ppma`) that load dynamically from the SD card and appear in the Debug menu automatically when the SD card is present.

298.14 Audio Test

NOTE This is an external app. It requires the `audio_test.ppma` file to be present on the SD card.

Generates a sine wave tone of a configurable frequency and duration to test the audio output chain. Duration of 0 plays the tone continuously. The sample rate and frequency step can also be adjusted. A speaker toggle controls whether the built-in speaker (if present) is used.

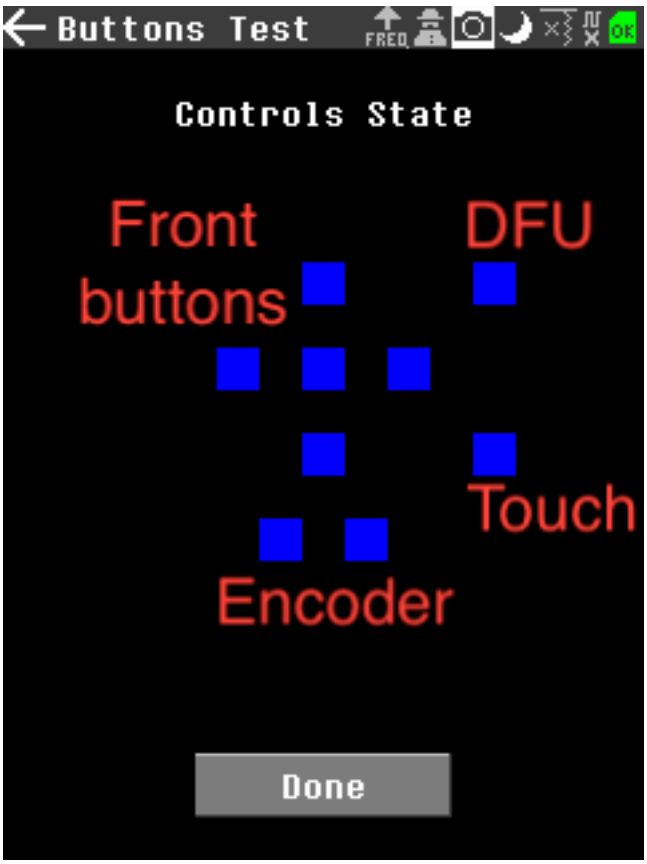
298.15 Battery

NOTE Only shown when a MAX17055 battery management IC is detected (H4M hardware only).

Displays the raw register contents of the MAX17055 fuel gauge IC, including voltage, current, capacity, state of charge, and other battery parameters. Values are shown in decoded form with units. Use the encoder dial to scroll through pages of registers.

298.16 Buttons Test

Shows real-time input from all hardware controls: directional buttons, encoder dial, and touchscreen. Useful for verifying that buttons register correctly and that the encoder steps cleanly without skipping or double-stepping. The test screen also includes a long-press test for the directional keys and the DFU switch.



298.17 Debug Dump

NOTE This is an external app. It requires the `debug_pmem.ppma` file to be present on the SD card.

Writes a file containing debug information (persistent memory contents, firmware version, hardware info) to the `DEBUG/` folder on the SD card. Useful for reporting bugs or diagnosing configuration issues.

298.18 Ext Module

Displays information about a connected PortaPack expansion module (PPMod) communicating over I2C. Shows the module name, firmware version, and a list of apps provided by the module. If no module is connected, the screen shows "No module connected". The display updates once per second.

298.19 ExtSensor

NOTE This is an external app. It requires the `extsensors.ppma` file to be present on the SD card.

Scans for I2C-connected external sensors and displays their readings. Supports environmental sensors that report temperature, humidity, and pressure (e.g. BME280). Found I2C device addresses are listed on startup.

298.20 Font Viewer

NOTE This is an external app. It requires the `font_viewer.ppma` file to be present on the SD card.

Displays the built-in 5×8 and 8×16 font character sets used by the Mayhem UI.

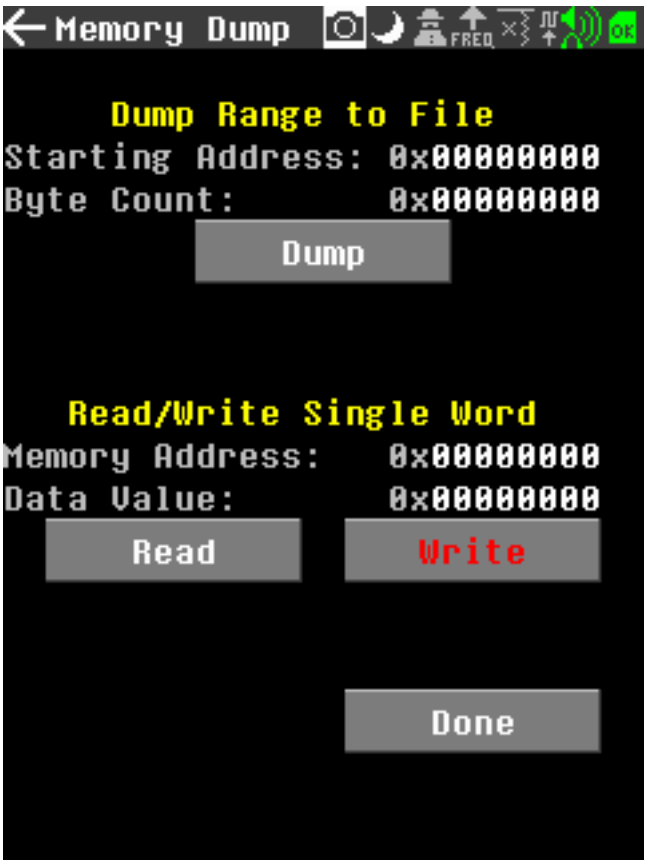
298.21 M0 Stack Dump

Writes a file containing the M0 core stack contents to the `DEBUG/` folder on the SD card. Used for diagnosing crashes or stack overflows in the M0 processor.

298.22 Memory Dump

Allows a region of memory to be saved to a file in the `DEBUG/` folder in hexadecimal ASCII format. Also allows direct read and write access to specific memory addresses for debugging purposes.

> **WARNING** Memory addresses must be multiples of 4 to avoid causing a fault. Writing to some addresses may cause hardware damage. Use with care.



298.23 Peripherals

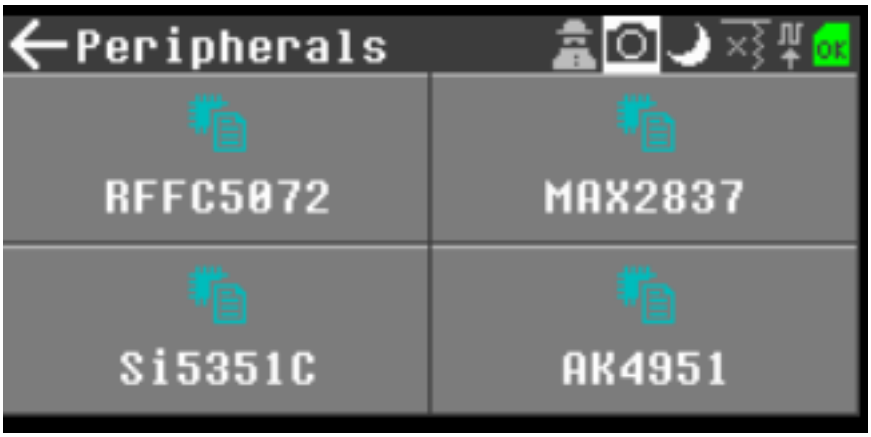
Displays the raw register contents of the HackRF radio hardware ICs, which vary by hardware revision:

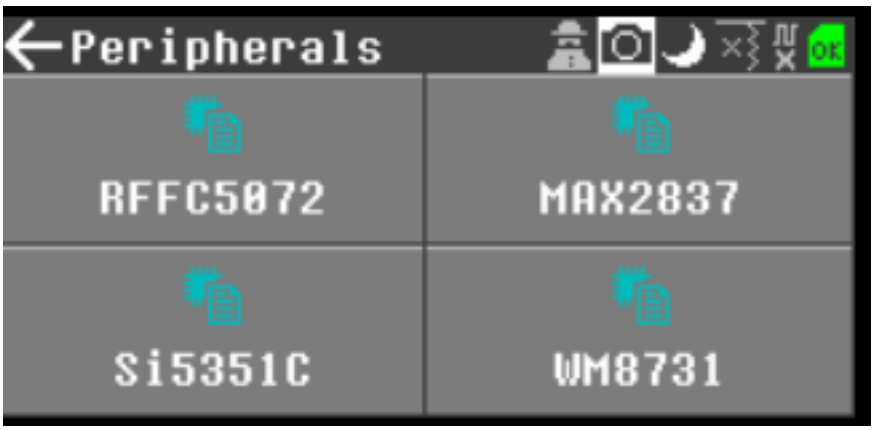
- Hardware: R1–R8 (OG); ICs shown: RFFC5072, MAX2837, Si5351C, audio codec, SGPIO
- Hardware: R9 or R10C; ICs shown: RFFC5072, MAX2839, Si5351A, audio codec, SGPIO
- Hardware: HackRF Pro; ICs shown: RFFC5072, MAX2831, Si5351A, iCE40 FPGA, audio codec, SGPIO

If a MAX17055 battery IC is detected (H4M), it also appears here.

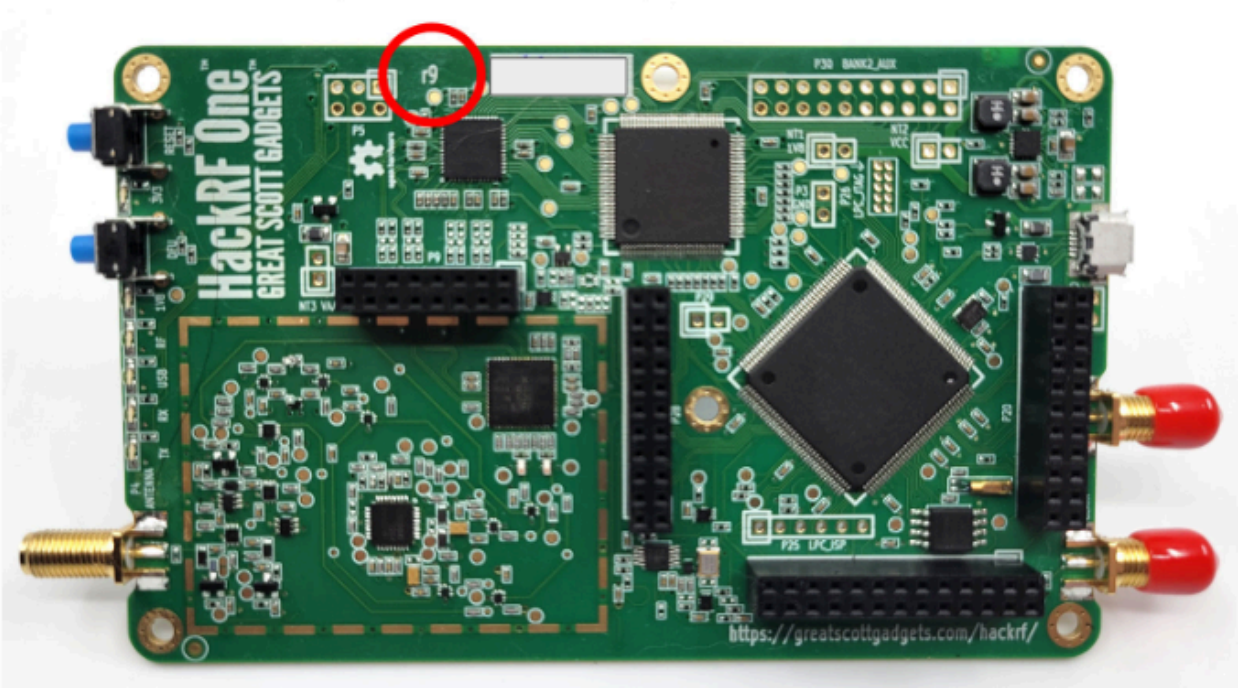
Individual registers can be read or written. If more registers exist than fit on the screen, turn the encoder dial when the Update button is highlighted to scroll through them.

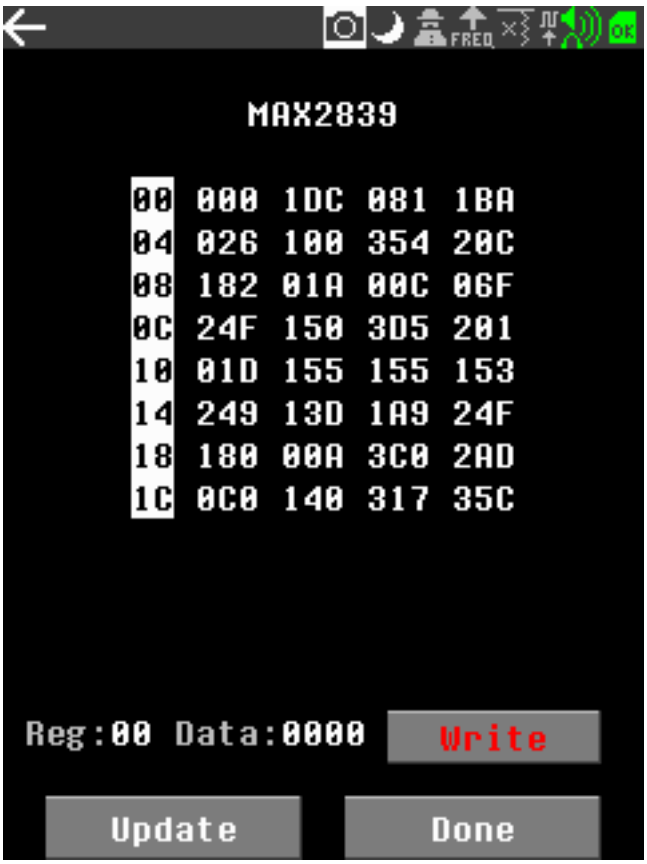
WARNING Writing to some registers may cause hardware damage. Use with care.

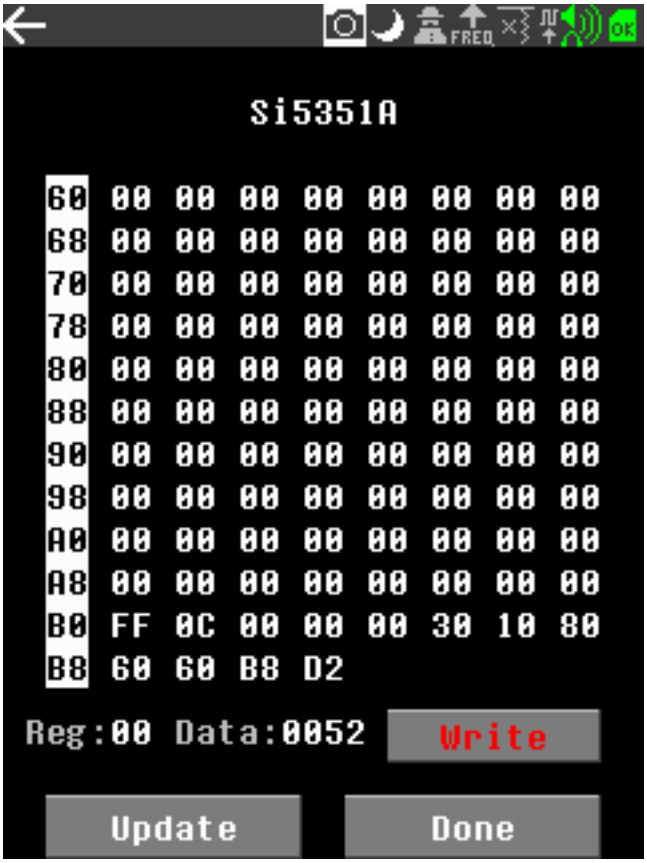




Starting with HackRF One r6, hardware revisions are detected by firmware and reported by `hackrf_info`. The R9 board can be identified without disassembly via the Peripherals view:



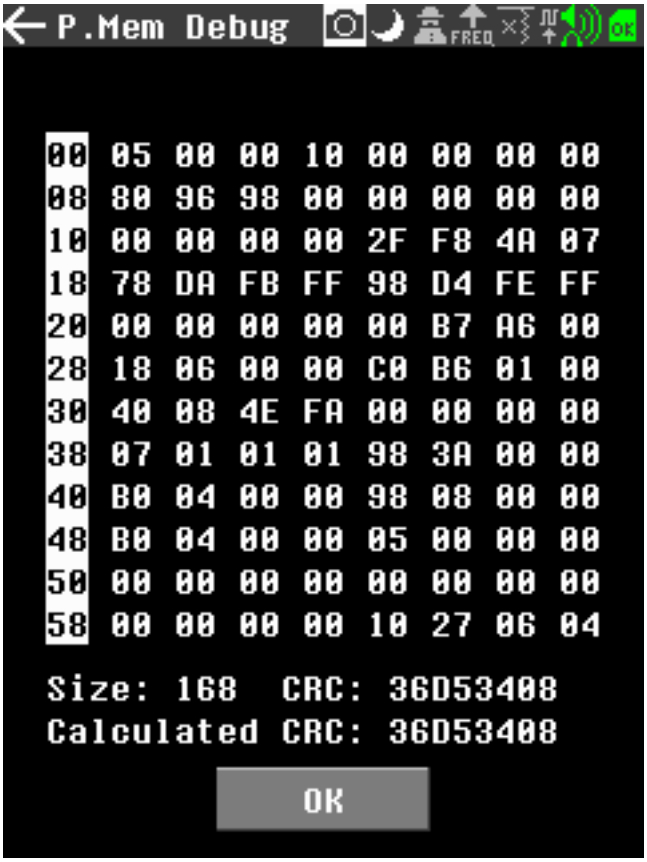




298.24 Persistent Memory

Displays the 256-byte persistent memory (p.mem) area as a hex dump, split across pages. Use the encoder dial to scroll between pages. The current byte offset from the start of p.mem is shown in the left column.

At the bottom, the current size of the `data_t` struct and the stored checksum are displayed. The checksum is calculated from the first 252 bytes and stored in the last 4 bytes. The config version is visible as the first 4 bytes of the dump.



298.25 Pro Debug

Only shown on HackRF **Pro** hardware (PRALINE build). It does not appear on H1/H2/H4/H4M.

A submenu of low-level diagnostics for the HackRF Pro's radio front-end and clocking. Intended for developers and hardware bring-up, not everyday use.

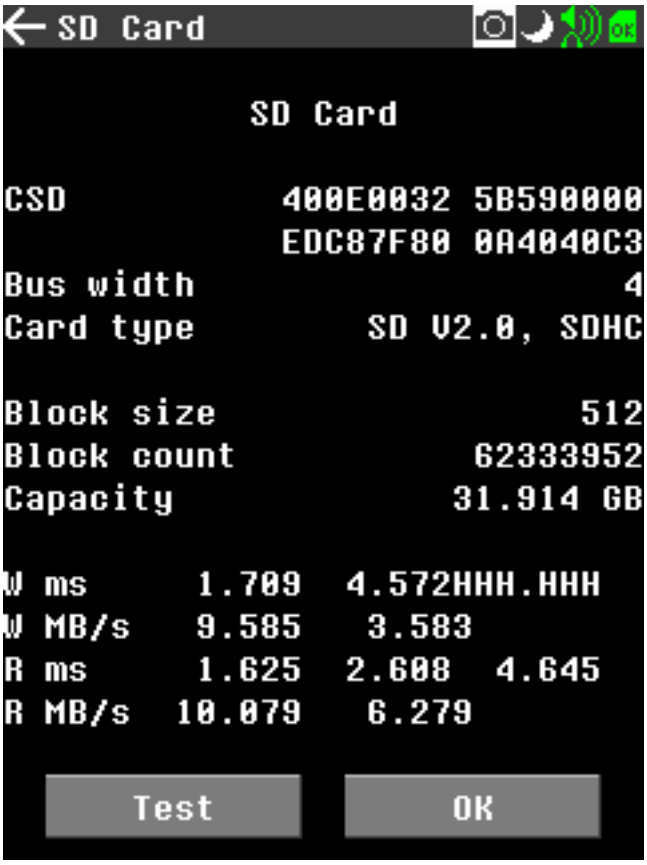
- Tool: WFM Audio; Purpose: WFM audio-path debug view
- Tool: Clocks; Purpose: Pro clock-domain status
- Tool: MSynth; Purpose: Si5351 MultiSynth debug
- Tool: Radio Diag; Purpose: Radio diagnostics overview
- Tool: Radio Debug; Purpose: Pro radio register/debug
- Tool: Signal Path; Purpose: RX/TX signal-path status
- Tool: System Diag; Purpose: System diagnostics
- Tool: PLL A; Purpose: Si5351 PLL A debug
- Tool: PLL B; Purpose: Si5351 PLL B debug
- Tool: GPIO; Purpose: GPIO state debug
- Tool: RFFC Status; Purpose: RFFC5072 mixer/synth status
- Tool: RFFC Tuning; Purpose: RFFC5072 tuning debug
- Tool: MAX2831; Purpose: MAX2831 transceiver registers
- Tool: Si5351; Purpose: Si5351 clock-generator registers
- Tool: SGPIO Clk; Purpose: SGPIO clock detector
- Tool: Baseband; Purpose: Baseband status
- Tool: SGPIO Live; Purpose: Live SGPIO monitor
- Tool: RX Test; Purpose: Radio RX test

298.26 Reboot

Immediately reboots the device.

298.27 SD Card

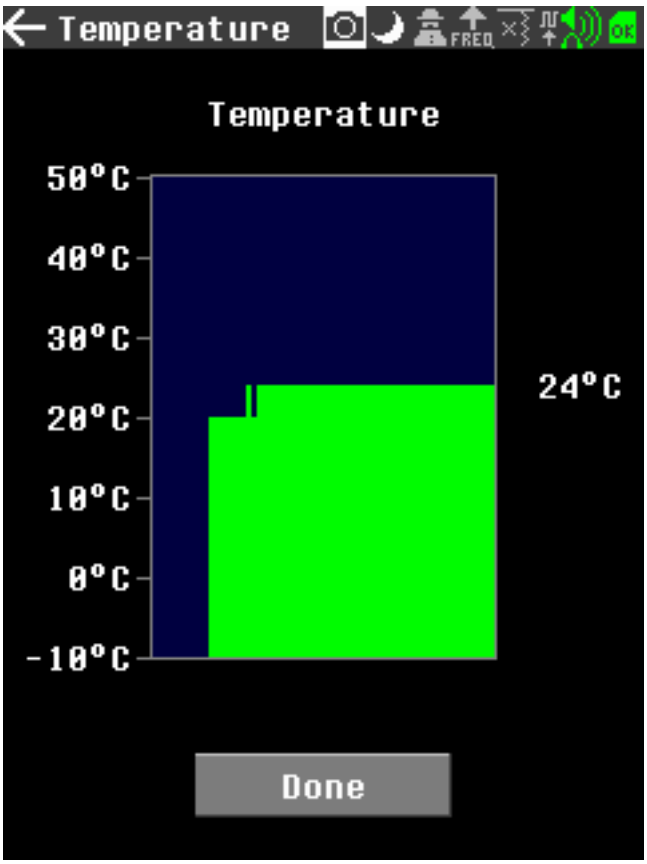
Displays information about the inserted SD card and allows it to be tested for read/write integrity.



298.28 Temperature

NOTE This is an external app. It requires the `mcu_temperature.ppma` file to be present on the SD card.

Displays the current temperature as reported by the on-chip digital temperature sensor of the MAX2837 (R1–R8) or MAX2839 (R9/R10C). Accuracy is approximately ±4.33 °C per step.



298.29 Touch Test

Allows drawing on the screen with a stylus to test touchscreen calibration and accuracy.

Controls:

- **Select** — return to the Debug menu

- **Left** — change the pen to a random color (hold to cycle faster)
- **Down** — clear the screen to a random color
- **Encoder dial** — change the pen size

The screenshot icon remains functional but is hidden; press the exact spot on the title bar to activate it. For improving touchscreen calibration, see Settings → Calibration.

298.30 File manager

Used to manage files on the SD Card.

298.31 Controls

- Use the encoder or up and down buttons to move between items.
- Use the select (center) key to enter a folder or to select an item to take an action on.
- Use the right button to select a folder to take an action on (e.g. rename a folder).
- Use the left button or select the ".." folder to go up a directory.
- Click on "–>" to get to next page, you will generally find it at the bottom of the list.
- Click on "<–" to get to previous page, you will generally find it at the top of the list.

A small, blinking arrow on the right indicates that there are more items if you scroll down.

Items with filenames longer than the display width will automatically scroll horizontally when highlighted.

298.32 Pagination

When there are too many elements, the sorting is disabled (not enough memory to do it) but the pagination is turned on, allowing to view a lot more elements.

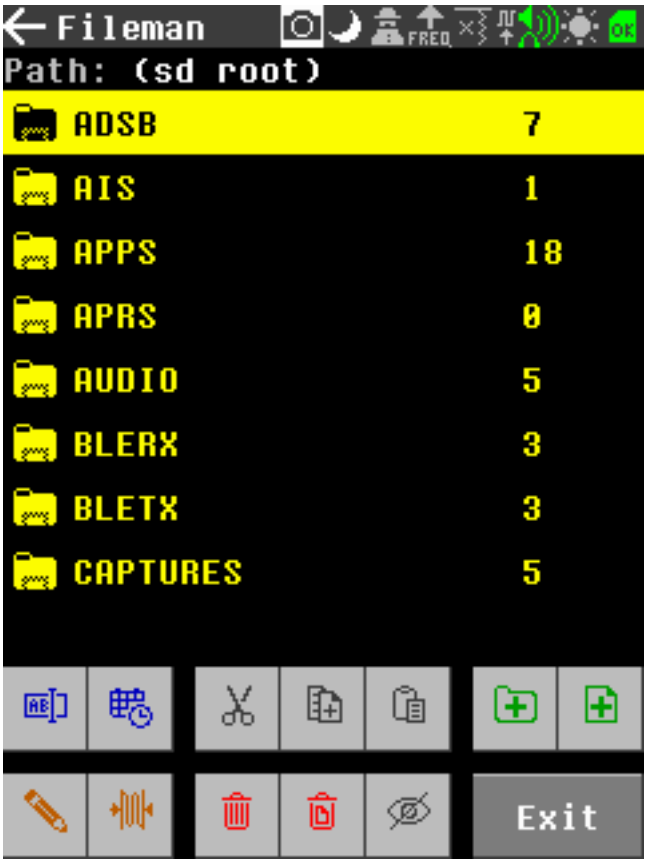
On pagination button lines and in place of the size, there will be a page number indicating to which page you will be going.

298.33 File type handlers

Selecting certain types of files launches a viewer for the content. Use the right button to enter the toolbar for these file types. Exiting the viewer when launched this way will return back to Fileman.

- **BMP files** will open in the Splash viewer if located in the SPLASH folder (and can be set as the splash screen). BMP files in any other folder open in the BMP File Viewer.
- **TXT files** will open in Notepad. Exception: TXT files in the WATERFALLS folder open the waterfall copy dialog instead.
- **PPL, C8, CU8 and C16 files** will open in Replay in the correct mode.
- **PNG files** will open the Screenshot viewer (this can only view screenshots generated on the device). When viewing a screenshot, press any key to return back to Fileman.

298.34 Buttons



When an item has been selected, the following commands are available in the toolbar.

- **Rename** — rename the file or folder.
- **Timestamp Rename** — rename the file or folder with the date appended.
- **Cut** — add the selected file or folder to the clipboard to be moved (by using Paste).
- **Copy** — add the selected item to the clipboard to be copied.
- **Paste** — move/copy the content of the clipboard into the current folder.
- **New Folder** — create a new folder in the current folder.
- **New File** — create an empty file in the current folder.
- **Notepad** — open the file in the text editor.
- **Trim** — open a capture file in the IQ Trim app.
- **Delete** — delete the file or folder.
- **Clean** — erase all the files in the folder.
- **Show Hidden** — shows hidden files.
- **Exit** — exit the file manager.

298.35 Partner files

Capture files (.C8, .CU8 or .C16) will have a “partner” .TXT file containing metadata. Renaming or deleting one of the pair will prompt the same action to be applied to the other.

298.36 Known folders

- **/ (root)** — holds the current splash screen file (splash.bmp), and the folders below.
- **ADSB** — holds map and airline data for the ADSB RX app.
- **AIS** — holds data for the AIS app.
- **APPS** — holds external apps such as PacMan and Calculator (app version on SD card must exactly match firmware version or it cannot be executed).
- **APRS** — holds captures from the APRS app.
- **AUDIO** — holds captures from the Audio app.

- **BLERX** — holds captures from the BLE RX app.
- **BLETX** — holds packet files for the BLE TX app (.TXT format).
- **BMP** — holds `.BMP` images saved by the NOAA APT app (`noaa_YYYYMMDDHHMMSS.bmp`). This folder is created automatically by the app if it does not exist.
- **CAPTURES** — holds captures from the Capture app.
- **CVSFILES** — holds files for the CVS Spam app.
- **DEBUG** — holds debug dump files created from the Debug Dump app.
- **EPIRB** — holds files for the EPIRB RX and TX apps. EPIRB RX reads `FREQ.TXT` for the frequency selector. EPIRB TX reads `BEACONS.TXT` for beacon playback in file mode.
- **FIRMWARE** — recommended directory to store alternate Mayhem firmware images.
- **FREQMAN** — holds frequency list files for the Freqman, Scanner, Recon, and Looking Glass apps.
- **GPS** — holds data for the GPS Sim app.
- **HOPPER** — holds files for the Hopper app.
- **KEELOQKEYS** — holds the `MFCODES` key file for the KeeLoq TX and SubGhzD apps. `MFCODES` is a plain text file with one entry per line in the format `mf_name;key_hex;type` (e.g. `Superrollo;XXXXXXXXXXXXXXXX;1`). Without a matching entry, KeeLoq and Superrollo signals cannot be decrypted or retransmitted.
- **KEELOQREMOTES** — holds `.KEELOQ` remote files created by SubGhzD when saving a decoded KeeLoq or Superrollo signal. These files are loaded by the KeeLoq TX app for retransmission.
- **LOGS** — holds logs from various apps including ADSB RX, ERT, TPMS, POCSAG, Radiosonde, SubGhzD (`SUBGHZDLOG_*.CSV`), and SubCar (`SubCarLOG_*.CSV`).
- **LOOKINGGLASS** — holds preset frequency ranges for the Looking Glass app. All `.TXT` files in this folder are loaded automatically.
- **MACADDRESS** — holds MAC address files.
- **OOKFILES** — holds signal files for the OOK Editor app.
- **PLAYLIST** — holds playlist files (.PPL) for the Replay app.
- **REMOTES** — holds remote files for the Remote app.
- **SAMPLES** — holds example capture files.
- **SCREENSHOTS** — holds screenshots.
- **SECPLUS** — holds `.SECPLUS` remote files for the Security+ TX app.
- **SETTINGS** — holds saved App Settings (.ini files), PMEM_FILEFLAG and pmem_settings (only if persistent memory is being saved on the SD card), DATE_FILEFLAG (for incrementing pseudo-date only when coin cell battery is dead), and blacklist (list of apps to be disabled).
- **SPECTRUM** — holds files for the Spectrum Painter app.
- **SPLASH** — holds example splash screen files. To select a new splash screen, open a BMP file with FileMan and press the Right button to copy it to the root directory as your new splash.bmp file ("Show Splash" also needs to be enabled on the Settings → User Interface page).

- **SSTV** — holds images for the SSTV app.
- **SUBGHZ** — holds `.sub` signal files for the FlipperTX app.
- **TPMS** — holds log files from the TPMS RX app.
- **WATERFALLS** — holds preset files for the Waterfall Designer app. TXT files in this folder open the waterfall copy dialog instead of Notepad.

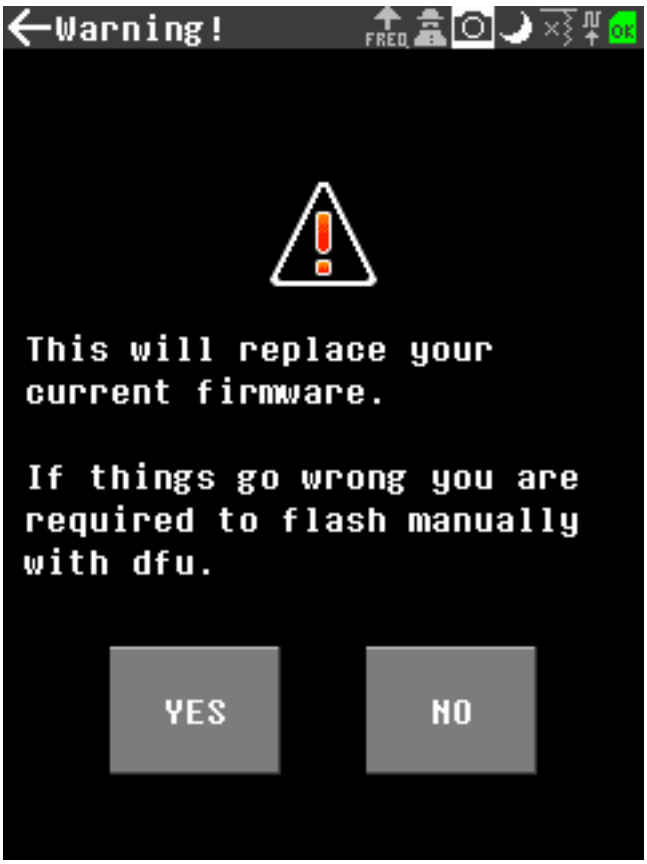
- **WAV** — holds WAV audio files for the Soundboard app.
- **WHIPCALC** — holds data for the Antenna Length calculator.

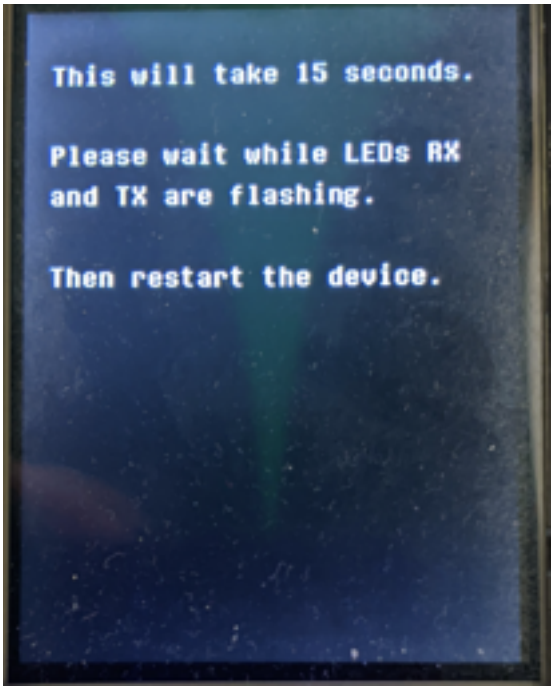
298.37 Known Issues

- Items are listed unsorted after a fixed number of elements (hardware limitation, not enough memory to sort all the files).
- Copy & Paste does not work on folders. (Note that Cut & Paste *is* now supported on folders.)
- Folders must be empty in order to be deleted (use the Clean button to empty a folder first).
- Cut/Copy/Paste does not work with "partner" files.

Tip: Use the SD-over-USB utility for more advanced file and folder editing via an attached computer.

298.38 Flash Utility





The **Flash Utility** allows you to install new firmware directly on your PortaPack — without needing a PC. It is the preferred update method for users in the field or those with OS compatibility issues.

298.39 Preparing the firmware file

Download the latest release from the Releases page and place the file(s) in the `FIRMWARE/` folder on the SD card.

Two file types are supported:

- File: mayhem_vX.X.X.ppfw.tar; Description: Recommended. Contains both the firmware image and all external apps. The Flash Utility extracts and installs everything automatically.

- File: portapack-h1_h2-mayhem.bin; Description: Firmware image only. External apps in the APPS/ folder are not updated — old apps may stop working if their version no longer matches the new firmware.

TIP The easiest method is to unzip `mayhem_vX.X.X_COPY_TO_SDCARD.zip` directly onto your SD card. This places the firmware in `FIRMWARE/` and also updates the external apps in `APPS/` and any other required data files.

NOTE The SD card can be accessed either by physically removing it or — while the PortaPack is connected via USB — by using the [[SD-Over-USB|SD Over USB]] app.

298.40 Flashing

1. Open **Flash Utility** from the Utilities menu.
2. The file browser shows all firmware files found in `FIRMWARE/`:
   - `.bin` files are shown in **red**
   - `.tar` files are shown in **cyan**
3. Select the desired file.
4. A confirmation dialog appears:

   "This will replace your current firmware. If things go wrong you are required to flash manually with dfu."

   Select **Yes** to proceed.
5. If a `.tar` file was selected, it is automatically extracted first. The display shows the filenames as they are unpacked.
6. The firmware file is validated before flashing. If the file is corrupt or the wrong size, the error `BAD FIRMWARE FILE OR W/R ERR` is shown and the process is aborted — your current firmware remains intact.
7. Flashing begins. The display shows:

   "This will take 15 seconds. Please wait while LED RX is on and TX is flashing."

   Do not power off the device during this step.
8. The device automatically restarts when flashing is complete.

298.41 External apps

The version of any external apps must match the version of the currently running firmware. If only the main firmware `.bin` is updated without also updating the `APPS/` folder, old external apps will not run. Using the `.ppfw.tar` bundle (see above) avoids this problem entirely, as it updates both the firmware and all external apps in a single step.

298.42 If things go wrong

Sometimes something goes wrong and the PortaPack refuses to turn on again. No matter what happens, you can always recover from this situation — the PortaPack has a DFU (Device Firmware Upgrade) mode built into hardware that can never be deleted or overwritten. See [[Update-firmware-troubleshooting]] for recovery instructions.

298.43 Freq manager

The **Freq Manager** (Freqman) is a frequency database manager. It lets you browse, create, edit, and delete frequency entries organized into named lists (categories). These lists are shared with other apps such as [[Recon]], [[Scanner]], and the [[Audio-Receivers|Audio]] receiver, which can load them directly.

298.44 SD Card location

Freqman files are stored in the `FREQMAN/` folder on the SD card as plain `.TXT` files. Each file is one category.

298.45 Using the app

The top bar shows the current category (file). Use the encoder to cycle through categories.

- Button: 📄 (new file icon); Action: Create a new category
- Button: 🗑️ (red trash icon); Action: Delete the current category
- Button: Edit; Action: Open the full edit screen for the selected entry
- Button: Frequency; Action: Quick-edit just the frequency of the selected entry
- Button: Description; Action: Quick-edit just the description of the selected entry
- Button: ➕ (add icon); Action: Add a new entry to the current category
- Button: 🗑️ (red delete icon); Action: Delete the selected entry
- Button: Exit; Action: Exit the app

The edit screen lets you set: Entry Type, Frequency A, Frequency B, Modulation, Bandwidth, Step, Tone (CTCSS), and Description.

298.46 Freqman file format

Files are plain text, one entry per line. Fields are comma-separated `key=value` pairs.

- Empty lines are ignored
- Lines starting with # are comments and are ignored
- Maximum description length: 30 characters
- Maximum entries per file: 150 (file is truncated if exceeded)

WARNING Any content that does not match a recognized field is silently ignored.

298.46.1 Entry type fields

Every entry must begin with one of these type-defining key pairs:

- Keys: f=freq; Type: Single; Description: One frequency
- Keys: a=start,b=end; Type: Range; Description: Frequency range to scan
- Keys: r=rx_freq,t=tx_freq; Type: HamRadio; Description: Ham relay: RX and TX frequencies. TX is scanned as a single frequency in scanner mode.
- Keys: l=listen_freq,t=tx_freq; Type: Repeater; Description: Repeater: listen frequency (also used as single freq when repeater mode is inactive) and TX offset

298.46.2 Optional fields

- Key: m=modulation; Description: Modulation mode
- Key: bw=bandwidth; Description: Bandwidth (must appear after m= in the line)
- Key: s=step; Description: Frequency step — only meaningful for Range entries
- Key: c=tone; Description: CTCSS tone frequency — only meaningful for HamRadio entries
- Key: d=description; Description: Free-text description (max 30 characters)

298.46.3 Possible values for modulation

- Value: AM
- Value: NFM
- Value: WFM
- Value: SPEC; Notes: Spectrum or raw IQ
- Value: AMFM; Notes: Special: AM+FM demod for HF WeatherFax (used by [[WeatherFax]] app)
- Value: FMAM; Notes: Special: FM+AM demod for NOAA 137 MHz APT (used by [[NOAA]] app)

298.46.4 Possible values for bandwidth

AM: `DSB 9k`, `DSB 6k`, `USB+3k`, `LSB-3k`, `CW`

NFM: `8k5`, `11k`, `12k5`, `16k`

WFM: `80k`, `180k`, `200k`

SPEC: `12k5`, `16k`, `25k`, `32k`, `50k`, `75k`, `100k`, `150k`, `250k`, `500k`, `600k`, `750k`, `1000k`, `1250k`, `1500k`, `1750k`, `2000k`, `2250k`, `2500k`, `3000k`, `3500k`, `4000k`, `4500k`, `5000k`, `5500k`

298.46.5 Possible values for step

`10Hz`, `50Hz`, `0.1kHz`, `1kHz`, `5kHz`, `6.25kHz`, `8.33kHz`, `9kHz`, `10kHz`, `12.5kHz`, `15kHz`, `25kHz`, `30kHz`, `50kHz`, `100kHz`, `250kHz`, `500kHz`, `1MHz`

298.47 Example file

```
# My frequency list
f=468000000,d=Single Freq
f=468000000,m=AM,d=Single Freq AM
f=468000000,m=NFM,bw=12k5,d=Single Freq NFM
f=468000000,m=WFM,bw=200k,d=Single Freq WFM
f=468000000,m=AM,bw=USB+3k,d=Single Freq AM USB
a=87000000,b=110000000,m=WFM,bw=200k,s=50kHz,d=WFM radio search
r=430150000,t=430550000,d=HAM relay
r=430150000,t=430550000,m=NFM,bw=12k5,c=88.5,d=HAM with CTCSS
l=430150000,t=431150000,m=SPEC,bw=1000k,d=1MHz repeat txoffset
```

298.48 Additional freqman files

Community-maintained and country-specific freqman files are available at: https://github.com/portapack-mayhem/mayhem-freqman-files

298.49 IQ Trim

IQ Trim allows you to trim “radio silence” from the beginning and end of a C16, C8, or CU8 capture file.

298.50 UI Components

- Field: Capture File; Description: Tap to open a .C16, .C8, or .CU8 file from the CAPTURES/ folder.
- Field: IQ Display; Description: Shows the signal power across 240 buckets. The green line marks the Start position, the red line marks the End position.
- Field: Start; Description: Starting sample of the trim region. Adjustable manually. Step size scales with file length.
- Field: End; Description: Ending sample of the trim region. Adjustable manually. Step size scales with file length.
- Field: Samples; Description: Total number of samples in the capture file.
- Field: Max Pwr; Description: Power (complex magnitude) of the strongest sample in the capture file, after amplification. Shown in red if the amplification setting would cause clipping.
- Field: Cutoff; Description: Signal-to-noise cutoff as a percentage of Max Pwr (1–100%). Default: 7%. Used to automatically determine the Start/End trim points.

- Field: Amplify; Description: Amplification factor applied to the trimmed file (1–9×). Default: 1× (no amplification). Useful if the captured signal was weak.
- Field: Trim; Description: Applies the trim (and amplification) to the file. Overwrites the file in place — no backup is made.

298.51 Trimming Capture Files

1. Open a capture file by tapping the file path field. You should see the signal power displayed in the IQ Display.
   - If the display shows no visible block, the captured signal was too weak for the tool to detect automatically.
2. Either adjust **Cutoff %** to automatically set Start/End, or set **Start** and **End** manually.
3. Optionally increase **Amplify** if the signal was weak. Watch the **Max Pwr** field — it turns red if clipping would occur.
4. Press **Trim**. The file will be edited in place.

> **WARNING** Trim overwrites the original file without making a backup. Use the [[File-manager|File Manager]] to make a copy first if you want to preserve the original.

298.52 Splitting Capture Files

Capture files can be split into multiple files by combining File Manager and IQ Trim:

1. Use [[File-manager|File Manager]] to make copies of the original capture file.
2. Open each copy in IQ Trim and trim to the desired Start/End range.
3. The resulting files can be trimmed further if needed.

298.53 Metronome

The **Metronome** is a utility app that produces audible click tracks at a configurable tempo and rhythm. It is useful for musicians practicing with the PortaPack's speaker or headphone output.

This is an external app (`.ppma` file on the SD card).

298.54 Controls

- Field: BPM; Range: 1 – 1000; Description: Beats per minute (tempo). Default: 120.
- Field: Tap Tempo; Range: —; Description: Opens the Tap Tempo screen to set BPM by tapping.
- Field: Accent Beep Tune; Range: 380 – 24000 Hz; Description: Frequency of the accented beat (first beat of each bar). Default: 880 Hz.
- Field: Unaccent Beep Tune; Range: 380 – 24000 Hz; Description: Frequency of all other beats. Default: 440 Hz.
- Field: Rhythm; Range: 1–99 or 1–99; Description: Time signature as beats per bar or note value (e.g. 4/4, 3/4, 6/8).
- Field: Beep Flash Duration; Range: 10 – 999 ms; Description: How long each beep lasts. Default: 100 ms.
- Field: Volume; Range: 0 – 99; Description: Audio output volume.
- Field: Play or Stop; Range: —; Description: Large button at the bottom. Starts or stops the metronome.

298.55 Progress Bar

The progress bar shows the current beat position within the bar:

- 🔴 **Red** — accented beat (first beat of the bar)
- 🟢 **Green** — unaccented beats

298.56 Tap Tempo Screen

Pressing **Tap Tempo** opens a sub-screen where you can:

- **Tap BPM** — tap the large button repeatedly in rhythm; the app averages the last 4 taps to calculate the BPM.
- **Input BPM** — type a BPM value directly via the keyboard.
- **Apply** — confirm and return to the main screen.
- **Cancel** — discard changes and restore the previous BPM.

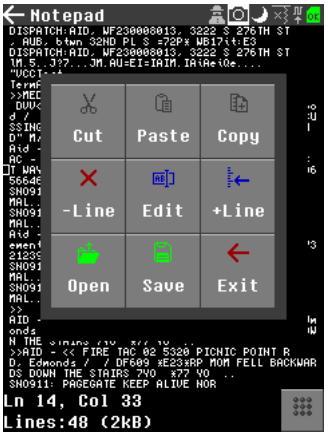
298.57 Notes

- The metronome uses the PortaPack's audio output — connect headphones via the 3.5 mm jack or use the internal speaker.
- The accent beat is always the **first beat of each bar**; all other beats are unaccented.
- The Rhythm field follows standard music notation: the left number is beats per bar, the right number is the note value that counts as one beat (e.g. 6/8 plays six eighth-note beats per bar).

298.58 Notepad

Views text files and supports very basic editing.

298.59 The interface



The menu button on the bottom right opens and closes the main menu. On start, this button will immediately show the file picker.

The file can be navigated with the direction buttons. Additionally, the encoder can be used to scroll. The encoder direction is set by the last direction button that was pressed. If you want scroll horizontally, first press the left or right button. When scrolling vertically, the encoder moves 16 lines per step in order to make scrolling through large files easier.

The center (select) button will show the main menu. To close the menu again, click the menu button on the bottom right.

298.60 Opening a file

There are two ways to open a file. If you first open Notepad, select the menu button and the file picker will be displayed. You can also open a file from within Fileman by selecting a file and clicking the Notepad icon in the toolbar.

298.61 Editing a file.

The following edit modes have been implemented.

- Delete line (- Line) - deletes the line the cursor is on.
- Add line (+ Line) - adds a new line above the line the cursor is on. If the cursor is at the very end of the file, adds the newline below the cursor.
- Edit - opens the text entry view to edit the current line. If the line is very long, this may crash due to memory constraints. Using "back" will cancel the edit. Selecting "Ok" will commit the edit to the file.

298.62 Known limitations

- Exiting Notepad with the "back" button (top left) will *not* prompt you to save. Your changes are not lost. Any in-progress edits are saved in the temp file "filename.ext~". You can recover this file using "Rename" in Fileman.
- Edits on larger files are slow. It currently requires multiple passes over the file contents because all operations are performed directly on the file contents (vs. in-memory) in order to work within the memory constraints of the device.
- The first edit is additionally slow because it copies the source file on first write. This is to allow changes to be made that can be discarded.

- There's currently no UI indication that the program is performing IO. As long as you don't see a fault, it should be working. Just be patient.
- Cut/Copy/Paste is not implemented yet.
- The cursor doesn't automatically get updated after edits. Navigation will snap the cursor back to the text.

298.63 Playlist Editor

This is an external app (`.ppma` file on the SD card).

299 Introduction

That application allows you to create/modify playlists.

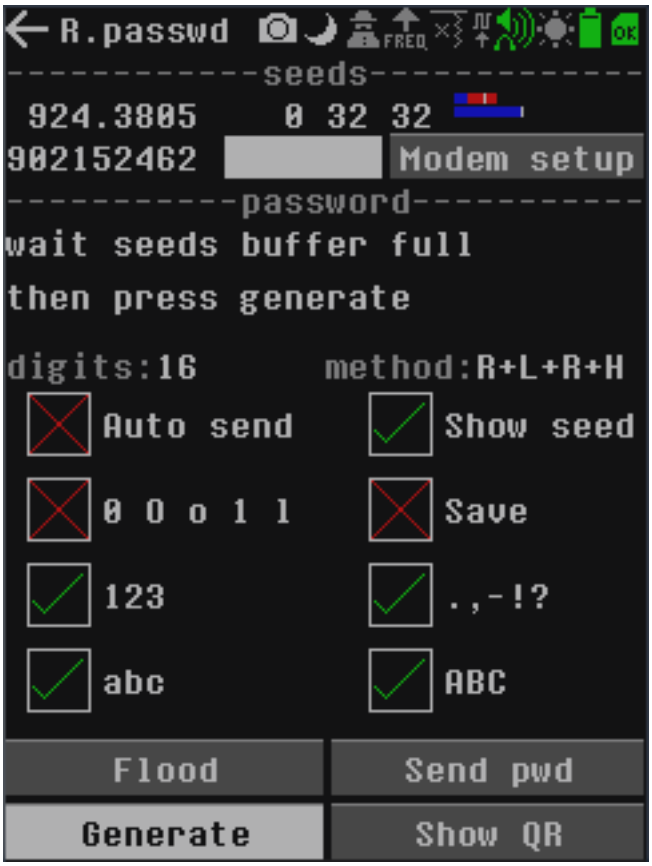
300 How to use

- Open an existing playlist, or create a new playlist from scratch
- Edit items and/or insert some new ones
- Save playlist

301 Edit/Insert fields

- Delay: use rotary encoder to adjust delay
- Browse: select capture file to replay
- Input Delay: set a custom delay using the keyboard instead of using the rotary encoder

301.1 Rand Pwd



This is an external app (`.ppma` file on the SD card).

This app use AFSK demodulated data as random seeds for each char, use LCG + one more random layer as PRNG algorithms to generate passwords.

In theory if the quality of seeds (which is from mostly radio noise) is good enough, then the password randomization would be good enough. But usually they are not, or we don't know, or they do but not stable. So even if this is more secure than many other PRNG generators, in the best case it even can be considered as TRNG generators, don't use this at high security scenes, for example money-related things.

All the "send" button/checkbox in this app means send via serial async messages, check serial page in this wiki for more details.

Flood mode is from streaming generated code, this will always streaming into serial, alternatively you can check the savin checkbox to save generated codes and seeds. Keep in mind that with seeds, you can generated same code with proper PRNG, so they are same level of security, don't leak them.

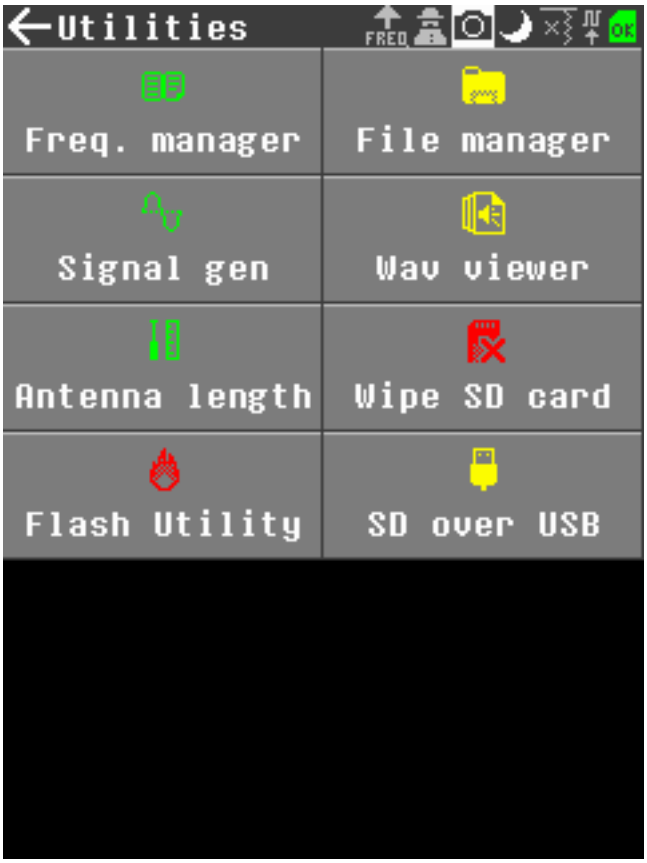
302 Each shuffle algo and what it can bring you

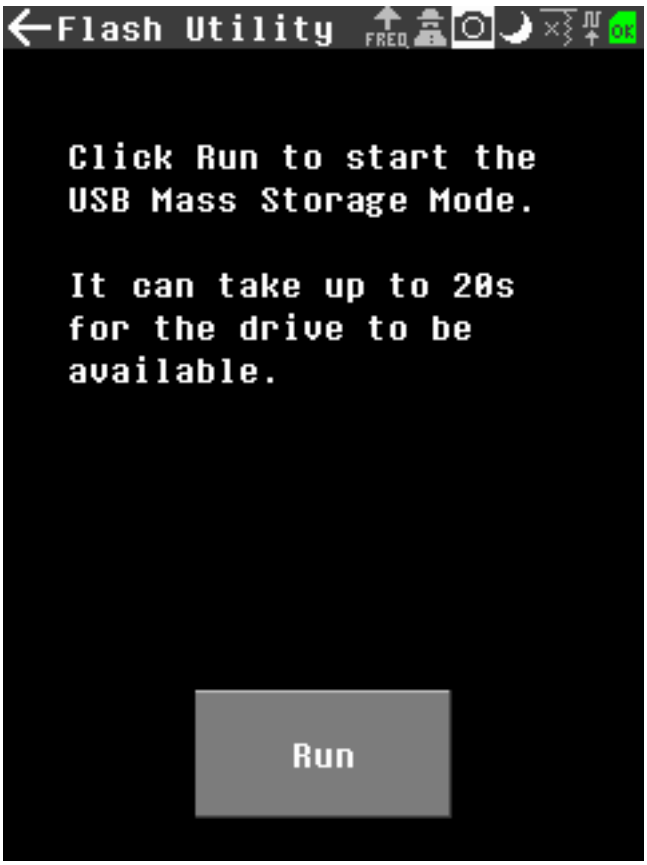
There are two dimensions of password safety:

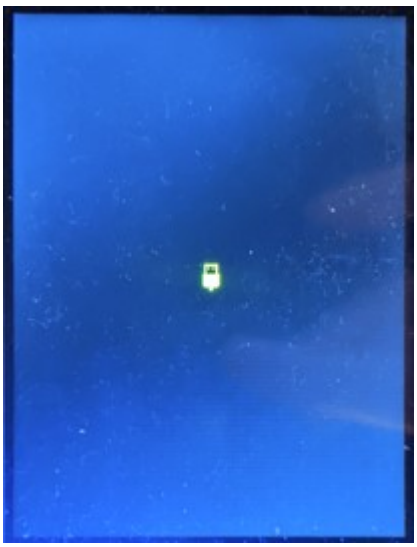
- Entropy: When you generate mass of password, it should split evenly in the possible space, that controls password quality. In this app, the only entropy provider is the demodulated AFSK data. the LCG/ sha-512/ shuffle algo won't bring more entropy.
- Randomness: It controls how many total possibilities can be generated. In this app, the only randomness provider is the "each digits using different seed in the buffer" algorithm, which spread the total possible combination into the max value of possibility: (CHARNUMBER)^{DIGITS}.
- Algo or Source: Use time as seeds to pick a random frequency with LCG to fetch AFSK data; Bring Entropy ?: 🚫; Bring Randomization ?: ✅; Bring difficult to reverse calculation ?: 🚫; Made it harder to Brute-Force ?: 🚫; Time Complexity and Space Complexity: O(1) - O(1)

- Algo or Source: The original demodulated AFSK data as seeds; Bring Entropy ?: ✅; Bring Randomization ?: 🚫; Bring difficult to reverse calculation ?: 🚫; Made it harder to Brute-Force ?: 🚫; Time Complexity and Space Complexity: O(1) - O(1)
- Algo or Source: Each digits using different seed in the buffer; Bring Entropy ?: ✅; Bring Randomization ?: 🚫; Bring difficult to reverse calculation ?: 🚫; Made it harder to Brute-Force ?: ✅; Time Complexity and Space Complexity: O(n^DITIGS_NUMBER) - O(n^DITIGS_NUMBER)
- Algo or Source: The LCG PRNG algo inside of cpp STL; Bring Entropy ?: 🚫; Bring Randomization ?: ✅; Bring difficult to reverse calculation ?: 🚫; Made it harder to Brute-Force ?: 🚫; Time Complexity and Space Complexity: O(n) - O(1)
- Algo or Source: Shuffle with two groups of seeds; Bring Entropy ?: 🚫; Bring Randomization ?: ✅; Bring difficult to reverse calculation ?: 🚫; Made it harder to Brute-Force ?: ✅; Time Complexity and Space Complexity: O(n) - O(n)
- Algo or Source: The SHA-512 hash algo; Bring Entropy ?: 🚫; Bring Randomization ?: ✅; Bring difficult to reverse calculation ?: ✅; Made it harder to Brute-Force ?: 🚫; Time Complexity and Space Complexity: O(n) - O(1)
- Algo or Source: Result; Bring Entropy ?: ✅ x 2; Bring Randomization ?: ✅ * 4; Bring difficult to reverse calculation ?: ✅ * 1; Made it harder to Brute-Force ?: ✅ * 2

302.1 SD over USB







The **SD Over USB** app exposes the SD card as a USB Mass Storage device, allowing a connected PC to read and write files directly — without physically removing the card.

This is an external app (`.ppma` file on the SD card).

> **NOTE** When SD Over USB is active, normal PortaPack operation is completely suspended. The device runs a dedicated USB Mass Storage firmware — no other apps or functions are available until the PortaPack is restarted.

302.2 Usage

1. Connect the PortaPack to a PC via USB.
2. Open **SD Over USB** from the Utilities menu.
3. Press the **Run** button.
4. The screen clears and the HackRF icon appears on a black background — this indicates USB Mass Storage mode is active.
5. It can take **up to 20 seconds** for the drive to appear on the PC.
6. Use the SD card as you would any external drive — copy, delete, or modify files freely.

302.3 Exiting USB Mass Storage mode

There is no exit button. To return to normal PortaPack operation, first **safely eject or unmount** the drive on your PC, then **power cycle the device**. On the H4M use the power switch; on older models (H1, H2) double-press the encoder knob to turn off, then press once to turn on.

> **WARNING** Always unmount the drive on your PC before power cycling — failing to do so may cause filesystem corruption on the SD card.

302.4 Performance

Transfer speed is approximately **400–500 KB/s**. This is a hardware limitation of the USB connection. For large file transfers, physically removing the SD card and using a card reader may be significantly faster.

For example, copying the 2 GB `world_map.bin` file to the SD card via SD Over USB can take up to 2 hours. Using a card reader directly is much faster.

302.5 Stopwatch

The **Stopwatch** is a utility app for measuring elapsed time with lap support and configurable millisecond precision display.

This is an external app (`.ppma` file on the SD card).

302.6 Display

The stopwatch shows two rows in large digits:

- Row: TOTAL; Label: Total elapsed time since last reset; Color: Minutes: 🔴 red, Seconds: 🟢 green, Milliseconds: 🟡 yellow
- Row: LAP; Label: Time of the last lap mark; Color: White

Time is displayed in the format: `MM:SS.mmm`

302.7 Buttons

- Button: START or STOP; While stopped: Starts the timer; While running: Stops (pauses) the timer
- Button: RESET or LAP; While stopped: Resets the timer to zero; While running: Records a lap time
- Button: EXIT; While stopped: Exits the app; While running: Exits the app

302.8 Milliseconds Display Level

When the timer is stopped, an option field in the top right controls how many millisecond digits are shown:

- Setting: & - -; Digits shown: 1 digit (hundreds)
- Setting: & & -; Digits shown: 2 digits (hundreds + tens)
- Setting: & & &; Digits shown: 3 digits (full milliseconds)

This option is hidden while the timer is running to prevent accidental changes.

302.9 Notes

- The timer continues accumulating time correctly after a pause — resuming does not reset the elapsed time.
- Lap time is fixed at the moment the **LAP** button is pressed; the TOTAL counter keeps running.
- Only one lap time is displayed at a time; pressing **LAP** again overwrites the previous lap.

302.10 Tuner

The **Tuner** is a utility app that plays reference tones through the PortaPack's audio output to help tune musical instruments by ear.

This is an external app (`.ppma` file on the SD card).

302.11 Controls

- Field: Instrument; Description: Select the instrument preset: Guitar, Violin, or Pitch Fork.
- Field: Note; Description: Select the string or note to tune to. Changes dynamically based on the selected instrument.
- Field: Note Frequency; Description: Displays the frequency (Hz) that will be played.
- Field: Note Octave Shift; Description: Displays the octave shift applied. A value of −1 means the PortaPack plays one octave lower than the actual target pitch (hardware limitation for very low frequencies).
- Field: Volume; Description: Audio output volume (0–99).
- Field: Play or Stop; Description: Large button at the bottom. Starts or stops the reference tone.

302.12 Instruments & Notes

302.12.1 Guitar (Folk Guitar)

- String: E2; Frequency played: 165 Hz; Octave shift: −1; Note: PortaPack plays E3; actual string is E2
- String: A2; Frequency played: 110 Hz; Octave shift: 0; Note: —
- String: D3; Frequency played: 147 Hz; Octave shift: 0; Note: —
- String: G3; Frequency played: 196 Hz; Octave shift: 0; Note: —
- String: B3; Frequency played: 247 Hz; Octave shift: 0; Note: —
- String: E4; Frequency played: 330 Hz; Octave shift: 0; Note: —

302.12.2 Violin (440 Standard, 12ET)

- String: G3; Frequency: 196 Hz
- String: D4; Frequency: 294 Hz
- String: A4; Frequency: 440 Hz
- String: E5; Frequency: 659 Hz

302.12.3 Pitch Fork

- Standard: 12ET A4; Frequency: 440 Hz
- Standard: Sarti's A4; Frequency: 436 Hz
- Standard: 1858 A4; Frequency: 435 Hz
- Standard: Verdi's A4; Frequency: 432 Hz

302.13 Octave Shift Indicator

When playing a note with a non-zero octave shift, the display shows:

- The **played frequency** (left, in red) — what the speaker actually outputs
- The **real target frequency** (right, in red) — what the actual string should sound like
- The **shift value** in octaves (e.g. `-1 * 8ev`)

When no shift is applied, these indicators are hidden.

302.14 Notes

- Audio output goes to the 3.5 mm headphone jack or the internal speaker.
- The tone plays continuously until the **Stop** button is pressed or the note is changed.
- Changing the note while a tone is playing immediately switches to the new frequency without stopping.
- Very low frequencies (e.g. guitar E2) are played one octave higher due to PortaPack hardware speaker limitations.

302.15 WardriveMap

The **WardriveMap** displays geotagged RF captures and Flipper Zero `.sub` files on a map. If a GPS module is attached, the map follows your position live.

You can also pan manually by changing the Lat or Lon fields — once you do, GPS will no longer auto-move the map.

Each marker on the map shows the filename of the capture above it.

> **NOTE** Maximum 30 captures can be viewed at once. Use the `<-` / `->` buttons to page through all geotagged files, or pan/zoom the map to see captures in a different area.

This is an external app (`.ppma` file on the SD card).

302.16 Requirements

- A world map installed on the SD card (see [[Create-a-Custom-Map-with-Offline-Map-Maker]]).
- Geotagged capture files in the `CAPTURES/` folder, **or** Flipper Zero `.sub` files with GPS coordinates in the `subghz/` folder.

302.17 Supported File Formats

- Format: .txt metadata; Folder: CAPTURES/; Description: Created automatically by the [[Capture]] app when GPS data is present
- Format: .sub Flipper Zero; Folder: subghz/; Description: Flipper Zero sub-GHz captures with embedded GPS coordinates

302.18 Controls

- Control: <-; Description: Page to the previous set of markers
- Control: ->; Description: Page to the next set of markers
- Control: Lat or Lon fields; Description: Manually pan the map to a specific coordinate

The `<-` and `->` buttons are hidden when there is nothing further to page to.

302.19 Notes

- Only files with valid, non-zero GPS coordinates are shown as markers.
- The map automatically centers on the first geotagged file found when opening the app.
- If a GPS and orientation sensor are connected, your current position and heading are shown on the map.

302.20 Waterfall Designer

Waterfall Designer is a utility for creating and editing custom waterfall colour palettes. A live waterfall display is shown while editing so you can see the effect of your changes in real time. Profiles are saved as files in the `WATERFALLS/` directory on the SD card and can be applied to replace the active waterfall colour scheme.

Note: Changes made while editing are previewed live but are automatically reverted when you leave the app — unless you explicitly press **Apply**. Only after pressing Apply is the palette permanently set as the active colour scheme.

This is an external app (`.ppma` file on the SD card).

303 Workflow

1. Press **New** to start a fresh profile, or **Open** to load an existing one from `WATERFALLS/`.
2. Select a signal level entry from the list (0–255, representing signal strength).
3. Press **Edit color** to open the colour picker and assign an RGB colour to that level.
4. Use **Add level** / **Remove level** to insert or delete colour stops in the palette.
5. Press **Save** to write the current profile back to its file in `WATERFALLS/`.
6. Press **Apply** to set this profile as the active waterfall colour scheme and exit.

304 Controls

- **New**: Creates a new empty profile.
- **Open**: Opens the file manager to load a profile from `WATERFALLS/`.
- **Save**: Saves the current profile to disk. The filename is shown after saving.
- **Add level** (+): Adds a new colour stop to the palette.
- **Remove level** (−): Removes the currently selected colour stop.
- **Edit color**: Opens the colour picker sub-view for the selected level.
- **Apply**: Applies the current profile as the active waterfall palette and exits the app.

305 Colour Picker

When editing a colour stop the following fields are available:

- **Index**: The signal level this colour applies to (0–255). 0 = weakest signal (background), 255 = strongest.
- **R or G or B**: Red, Green, Blue components, each 0–255.
- **Step**: Increment size for encoder knob adjustments on the R/G/B fields.
- **Save**: Confirms the colour and returns to the main view.

306 RF Settings

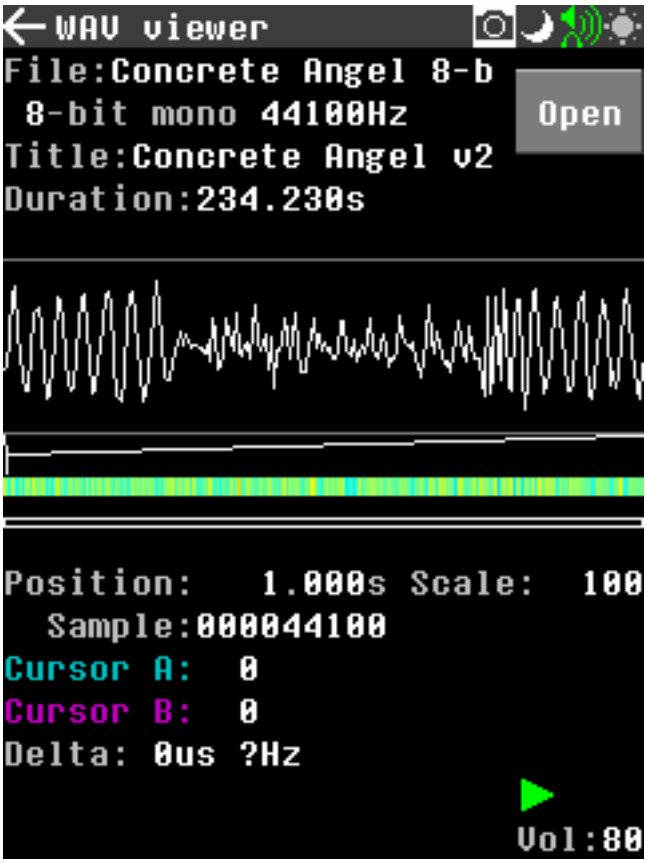
The live waterfall display is fully functional while editing — you can tune to a real signal to judge how your palette looks on actual data.

- **Frequency** and **frequency step**: Standard RX frequency controls.
- **RF Amp**, **LNA**, **VGA**: Standard gain controls.

307 Files

Profile files are stored in `WATERFALLS/` on the SD card. Settings are saved to `rx_wtf_designer.ini`.

307.1 Wav Viewer



308 Wav Viewer

The Wav Viewer app can be used to view and play back 8-bit unsigned or 16-bit signed mono WAV files. Sample WAV files may be found in the `WAV/` folder of the SD card, and WAV files captured using the Audio app will be placed in the `AUDIO/` folder.

This is an external app (`.ppma` file on the SD card).

308.1 Creating Compatible WAV Files

Using a PC, WAV files can be converted from existing audio using Audacity or ffmpeg:

```
# 8-bit unsigned mono, 48000 Hz
ffmpeg -i lovely_music.mp3 -ar 48000 -ac 1 -acodec pcm_u8 lovely_music.wav

# 16-bit signed mono, 48000 Hz
ffmpeg -i lovely_music.mp3 -ar 48000 -ac 1 -acodec pcm_s16le lovely_music.wav
```

> **NOTE** Only **mono** WAV files with **8-bit unsigned** or **16-bit signed** PCM encoding are supported. Stereo or other formats will show an error on load.

308.2 File Information Displayed

Once a file is opened, the app shows:

- Field: File; Description: Filename
- Field: xx-bit mono; Description: Bit depth (8 or 16) and channel count
- Field: Sample rate; Description: e.g. 48000Hz
- Field: Title; Description: Embedded title tag from the WAV file header (if present)
- Field: Duration; Description: Total length in seconds

308.3 Waveform Display

- The main **waveform** panel shows the audio samples for the current view window.

- Below the waveform, a compressed **amplitude overview** bar shows the overall signal level across the entire file using a color spectrum (darker = quieter, brighter = louder).
- A small **window indicator** shows which portion of the full file is currently visible in the waveform.

308.4 Controls

- Control: Open; Description: Open a .WAV file from the SD card
- Control: ▸ / ■; Description: Start or stop audio playback
- Control: Vol; Description: Audio output volume
- Control: Position (s or ms); Description: Navigate to a specific time position in the file
- Control: Sample; Description: Navigate by sample number
- Control: Scale; Description: Zoom the waveform view (1 = 1 sample per pixel, higher = zoomed out)
- Control: Cursor A; Description: Position of cursor A (cyan) on the waveform (0–239 pixels)
- Control: Cursor B; Description: Position of cursor B (magenta) on the waveform (0–239 pixels)
- Control: Delta; Description: Time difference between Cursor A and Cursor B, shown as duration and equivalent frequency

308.5 Playback

Pressing ▸ starts audio playback from the current position. A progress bar tracks playback through the file. The position fields update live during playback. Waveform view changes made during playback are applied once playback finishes.

308.6 Wipe SD card

The **Wipe SD Card** app overwrites the FAT (File Allocation Table) region of the SD card with pseudo-random data, destroying the filesystem structure. This makes the card appear blank and unreadable without reformatting.

This is an external app (`.ppma` file on the SD card).

> **WARNING This is not a secure data erasure tool.** Only the FAT area (~256 KB) is overwritten — the actual file data on the card remains physically intact and may be recoverable with data recovery software. Do not rely on this tool to securely destroy sensitive captures or recordings.

> **WARNING After wiping, the SD card must be reformatted on a PC** before it can be used again. The PortaPack itself cannot reformat the card.

308.7 What it does

When launched, a confirmation dialog appears:

> *"Wipe FAT of SD card?"*

Selecting **Yes** starts the wipe. The app writes 512 blocks of pseudo-random data (512 bytes each = 256 KB total) over the FAT base sectors of the filesystem. A progress bar is shown during the process. The app exits automatically when done.

Selecting **No** returns to the menu without making any changes.

308.8 When to use it

This tool is useful when you want to quickly render the card's contents unnavigable — for example, to prevent casual browsing of captures and recordings. It is **not** suitable as a privacy or security measure if the data must not be recoverable.

308.9 After wiping

The SD card will need to be reformatted (FAT32) on a computer before the PortaPack can use it again. See [[How-to-Format-SDCard]] for instructions.

308.10 Games

308.11 Battleship

Battleship is a two-player naval strategy game played over RF between two PortaPack devices. Each player places their ships on a grid, then takes turns calling coordinates to try to sink the opponent's fleet. Move data is transmitted and received wirelessly using the POCSAG2 baseband, making it the only multiplayer game in Mayhem that requires two devices and an RF link.

This is an external app (`.ppma` file on the SD card).

309 Setup

1. Both players select a frequency and gain settings — these must match on both devices.
2. Each player selects a team colour: **Red** or **Blue**.
3. Ships are placed on the grid using the touchscreen or encoder knob.
4. Once both players are ready, the game begins.

310 Controls

- **Touchscreen**: Tap to place ships and select coordinates during play.
- **Encoder knob**: Navigate the grid.
- **Left or Right or Up or Down buttons**: Move cursor.
- **Select**: Confirm action.

311 RF Settings

- **Frequency**: Configurable — both devices must use the same frequency.
- **RF Amp**, **LNA**, **VGA**: Standard gain controls for TX and RX.

312 Scoring

Wins and losses are tracked and displayed as `W:x L:x` on screen.

312.1 Blackjack

Blackjack is the classic casino card game against an automated dealer. The goal is to reach a hand value as close to 21 as possible without exceeding it. Face cards count as 10, Aces as 1 or 11. The dealer draws until reaching at least 17.

This is an external app (`.ppma` file on the SD card).

313 Controls

- **Encoder knob / Left or Right buttons**: Navigate between Hit and Stand options.
- **Select button**: Confirm the chosen action.

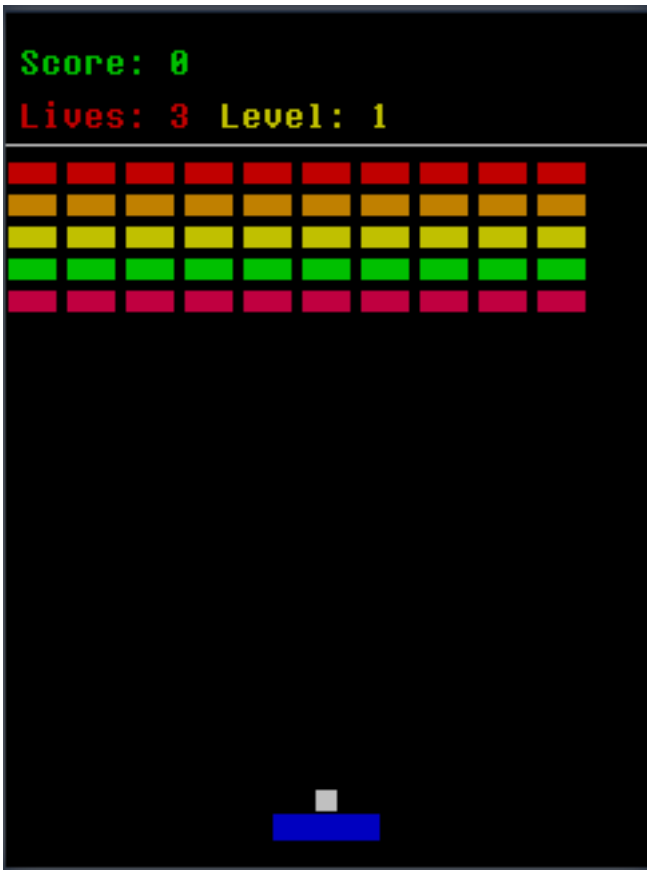
314 Gameplay

At the start of each round two cards are dealt to the player and two to the dealer, with one dealer card face-down. The player can:

- **Hit** — draw another card
- **Stand** — end their turn and let the dealer play

The dealer then reveals the hidden card and draws until reaching 17 or higher. The higher hand without busting wins. A high score is saved persistently.

314.1 Breakout



This is an external app (`.ppma` file on the SD card).

315 Introduction

- Breakout game by RocketGod (@rocketgod-git) https://betaskynet.com

316 Controls

- 'select' to start the game
- 'select' to launch the ball
- 'left' and 'right' keys or rotary encoder to move the paddle
- Simultaneously press key 'left' and key 'up' => put the focus on the hidden 'go back to main menu' icon on the top right part of the screen, click 'select' key to exit the game or optionally tap the top left corner of screen to exit.

316.1 2048

2048 is a sliding tile puzzle. The 4×4 grid starts with two tiles. Each move slides all tiles in one direction — tiles with equal values merge into their sum. The goal is to create a tile with the value 2048, though play can continue beyond that.

This is an external app (`.ppma` file on the SD card).

317 Controls

- **Up or Down or Left or Right buttons**: Slide all tiles in that direction.
- **Encoder knob**: Also moves tiles (rotating slides direction).

318 Scoring

- **Score**: Points are earned each time two tiles merge (equal to the merged tile's value).
- **Best score**: The highest score reached is tracked and displayed.

318.1 Digital Rain

Digital Rain is a visual demo that displays a Matrix-style falling character effect on the PortaPack screen. Columns of characters cascade from top to bottom on a black background, fading as they fall. There are no controls and no interaction — press the back button to exit.

318.2 Dino Game

Dino Game is an endless runner inspired by the offline dinosaur game. The dino runs automatically and the player must dodge oncoming obstacles — cacti and flying pterodactyls. Speed increases over time. The game also features a difficulty selector before starting.

This is an external app (`.ppma` file on the SD card).

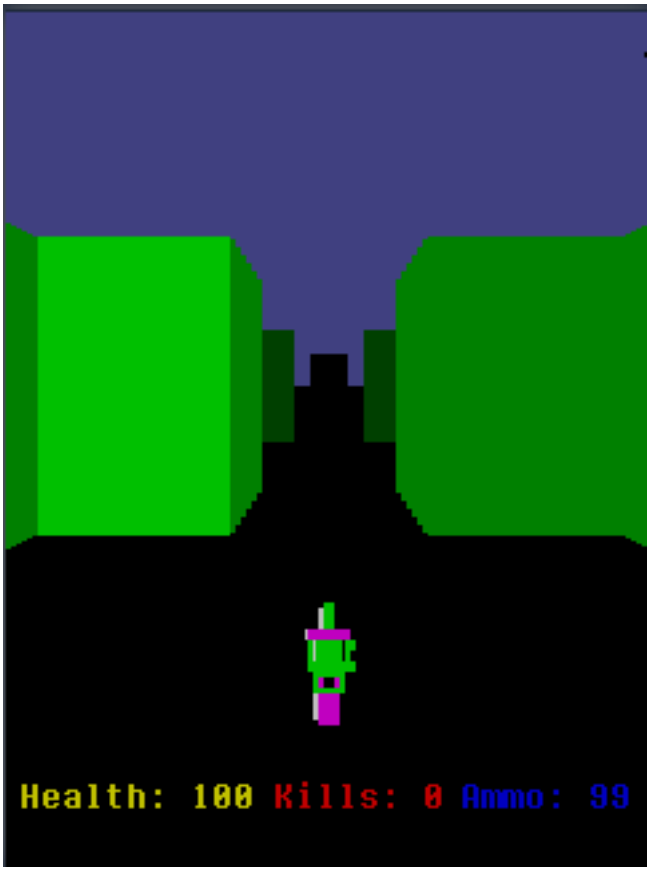
319 Controls

- **Select button** or **Up button**: Jump. Tap once for a standard jump.
- **Down button**: Duck (crouch under low-flying obstacles).
- **Encoder knob**: Also triggers jump.

320 Scoring

Score increases continuously while the dino is alive. A high score is tracked between sessions.

320.1 Doom



This is an external app (`.ppma` file on the SD card).

321 Introduction

-Doom game by RocketGod (@rocketgod-git) https://betaskynet.com Very simple Doom game, but it's already hard for our tiny device :-)

322 Controls

- 'select' to start the game
- arrow keys to move
- 'select' to shoot
- Tap the top left corner of the screen to exit

322.1 Morse P

Morse Practice is a Morse code keying trainer. It lets you practise sending Morse code by pressing and holding a key button, then decodes what you entered and displays the corresponding character. Audio feedback plays a tone through the speaker while the key is held down.

This is an external app (`.ppma` file on the SD card).

323 Controls

- **KEY button** (centre of screen): Hold to key — release to decode. Works with both the touchscreen and the Select hardware button.
- **CLR button**: Clears the decoded text display.
- **Volume**: Audio output level for the keying tone.

324 How it works

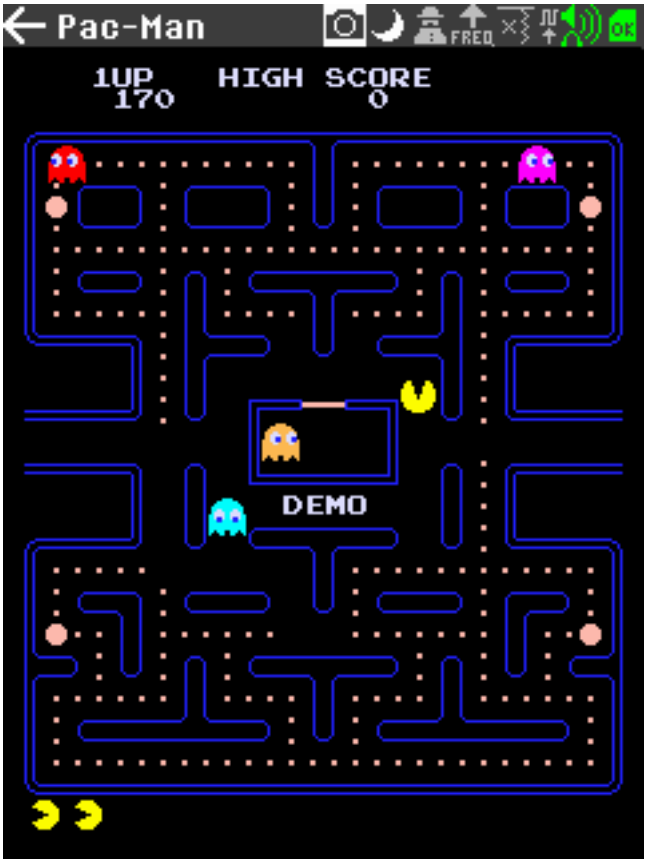
The app measures the duration of each key press and gap:

- A short press is decoded as a **dit (·)**

- A long press is decoded as a **dah (−)**
- A gap between presses ends the current symbol and looks it up in the Morse table
- The dot/dash threshold is determined automatically from your keying rhythm using a statistical split of recent pulse durations

The current dot-dash sequence is shown on screen, and the decoded character appears below it when the inter-character gap is detected. Decoded characters accumulate in the scrolling text area below.

324.1 Pac-Man



325 Introduction

It's a game. You might have heard of it. When the pac-man eat the big dots, the ghosts become sad face, and you can eat them.

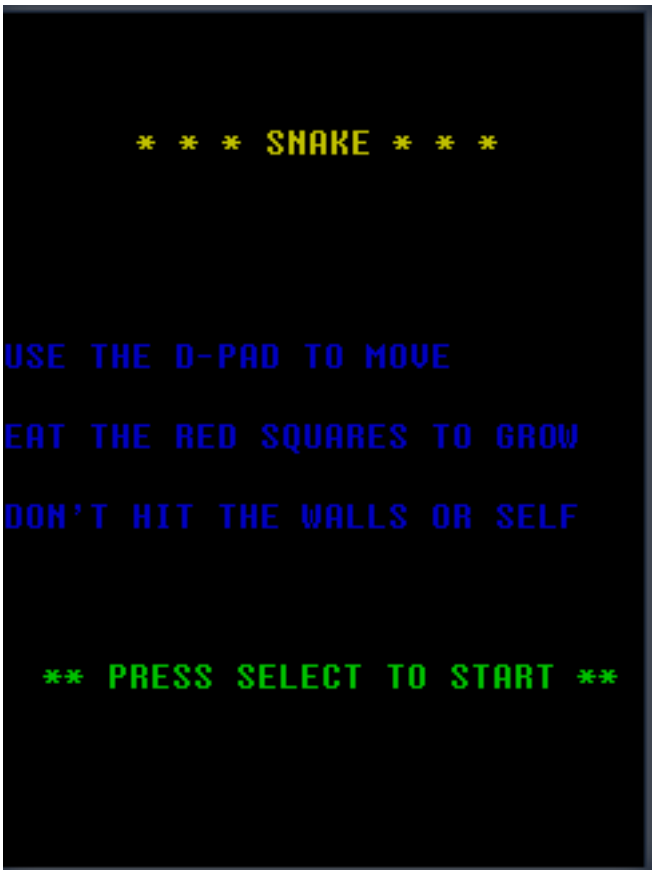
326 Controls

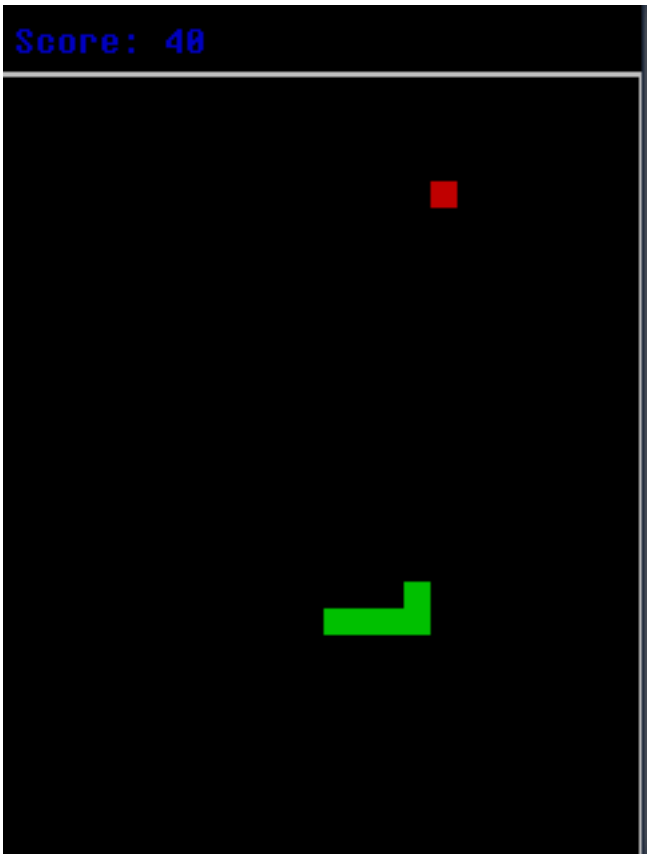
- Hit Select button to start
- Use arrow keys to move
- Long press back button or long press the title to exit it

327 Cheat

We're having a cheat mode, find it out yourself!

327.1 Snake





This is an external app (`.ppma` file on the SD card).

328 Introduction

- Snake game by RocketGod (@rocketgod-git) https://betaskynet.com

It's like the name says: the snake game :-)

You are starting as a single block snake. Eat the fruits to get it longer!!

329 Controls

- 'select' to start the party
- Use the arrow keys to change the direction of the snake
- Simultaneously press key 'left' and key 'up' => put the focus on the hidden 'go back to main menu' icon on the top right part of the screen, click 'select' key to exit the game or optionally tap the top left corner of the screen to exit.

329.1 Space Invaders

Space Invaders is a classic fixed-shooter arcade game. Waves of alien invaders descend from the top of the screen while the player's cannon moves along the bottom. The player shoots upward to eliminate all aliens before they reach the bottom or shoot the player. A difficulty mode toggle is available on the title screen.

This is an external app (`.ppma` file on the SD card).

330 Controls

- **Left or Right buttons**: Move the cannon horizontally.
- **Select button**: Fire.
- **Encoder knob**: Also moves the cannon.

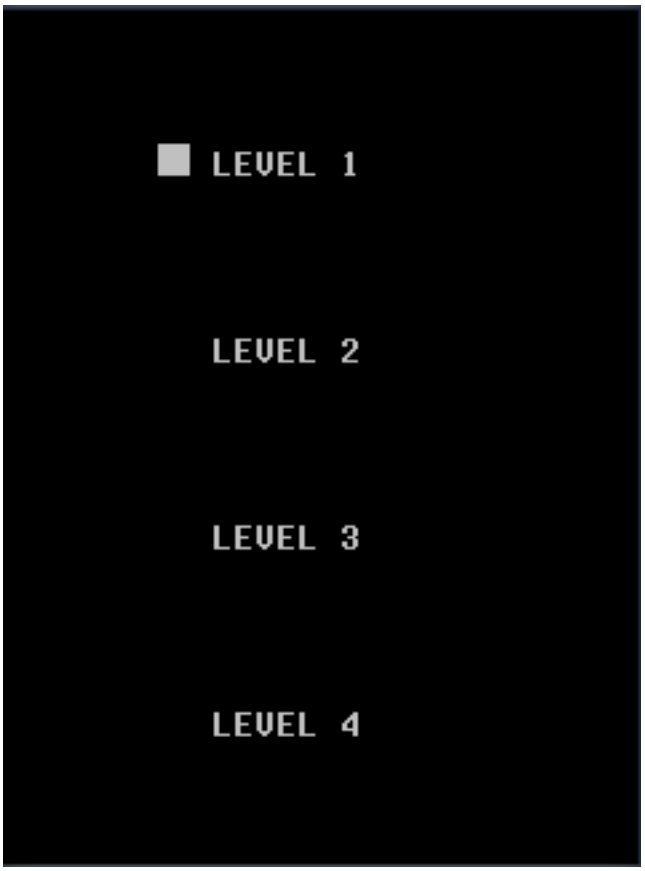
331 Gameplay

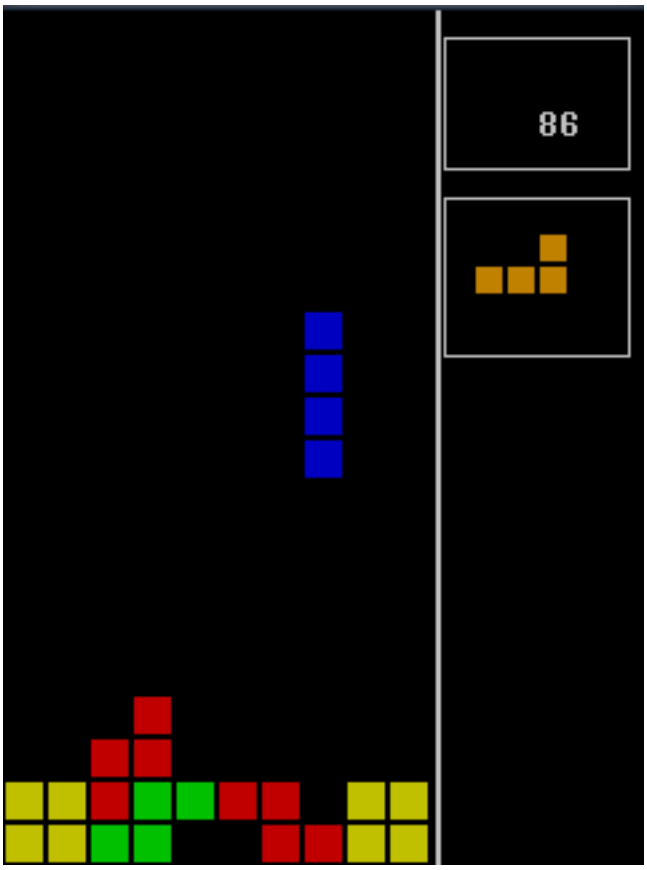
- The player starts with **3 lives**.
- Aliens fire back at the player — up to 3 enemy bullets can be active simultaneously.
- Clearing all aliens advances to the next wave at increased speed.
- **Mode: EASY or HARD** toggle on the title screen — saved persistently.

332 Scoring

Score increases with each alien destroyed. A high score is saved between sessions.

332.1 Tetris





This is an external app (`.ppma` file on the SD card).

333 Introduction

The classic Tetris game ! Select the level using the arrow keys or the rotary encoder, press 'select' to start !

334 Controls

- Move the blocks using the arrow keys
- Rotate the blocks using 'select' key or the rotary encoder
- Simultaneously press key 'left' and key 'up' => put the focus on the hidden 'go back to main menu' icon on the top right part of the screen, click 'select' key to exit the game

334.1 Settings

334.2 HackRF Mode

The **HackRF** entry in the main menu switches the device from Mayhem firmware into standard HackRF mode. In this mode the PortaPack operates as a plain HackRF One, fully compatible with GNU Radio, GQRX, SDR#, and any other software that supports the HackRF via USB — without any additional flashing or modification.

334.3 Switching to HackRF mode

Select **HackRF** from the main menu. A confirmation dialog appears:

> *"This mode enables HackRF functionality. To return, press the reset button."*

Select **Yes** to confirm. The Mayhem UI shuts down and the HackRF firmware takes over.

> **NOTE** All PortaPack inputs — the encoder dial, buttons, and touchscreen — are inactive in HackRF mode. The display shows a static block diagram at startup, and adds live values once an SDR application connects via USB.

334.4 Firmware version in HackRF mode

When in HackRF mode, `hackrf_info` still reports the Mayhem firmware version, for example:

```
—$ hackrf_info
hackrf_info version: 2024.02.1
libhackrf version: 2024.02.1 (0.9)
Found HackRF
Index: 0
Serial number: 0000000000000009xxxxxxxxxxxxxx3
Board ID Number: 2 (HackRF One)
Firmware Version: n_250606 (API:1.08)
Part ID Number: 0xa000cb3c 0x0066476c
Hardware Revision: r10
Hardware does not appear to have been manufactured by Great Scott Gadgets.
Hardware supported by installed firmware:
    HackRF One
```

This is expected — the HackRF USB interface is provided by the Mayhem firmware image, so the version string reflects Mayhem, not the original HackRF firmware.

334.5 Driver requirements

Linux or macOS: The HackRF is supported natively by the `libhackrf` package. Install it via your package manager (e.g. `sudo apt install hackrf`, `brew install hackrf`) and the device will be recognized automatically.

Windows: Windows does not include a HackRF driver by default. You need to install the WinUSB driver using Zadig:

1. Download and run Zadig from https://zadig.akeo.ie
2. Select **Options → List All Devices**.

3. Select **HackRF One** from the dropdown.
4. Select **WinUSB** as the driver and click **Install Driver** or **Replace Driver**.

Once installed, the HackRF will be recognized by GQRX (via GNU Radio), SDR#, and `hackrf_transfer`.

> **NOTE** If you previously used the PortaPack with the [[SD-Over-USB|SD Over USB]] app on Windows, a different driver may already be assigned to the device. Zadig will let you replace it.

334.6 Returning to Mayhem

To return to Mayhem firmware, press the **RESET button** — it is on the top of the PortaPack, the outer (right) one. See [[Hardware-overview]] for its exact location.

> **NOTE** There is no software way to exit HackRF mode. Only the RESET button will return the device to Mayhem.

334.7 Use cases

HackRF mode is useful when you need to use the HackRF with PC-based SDR software directly — for example to use GQRX or SDR# for wideband monitoring, run GNU Radio flowgraphs, or use `hackrf_transfer` for raw capture and replay from the command line.

To leave this mode, press the reset button. It is on the top of the PortaPack the outer (right) one.

- PortaPack in HackRF Mode:

HackRF Mode connected to gqrx:

1 090 000 000
16 dB
6.00 MHz
20 dB
EXT
PORTAPACK (H4M)
MIC / EXT
POWER